SUCCESS 4SURE

AP® ENVIRONMENTAL SCIENCE

COMPLETE STUDY GUIDE

Seven units · eleven chapters · every concept explained simply, with original diagrams and unsolved classroom examples



7 Units · 11 Chapters	Original Diagrams & Models	Aligned to the 2026 CED

Prepared by

Mr. Waseem Allabadi

Published by Success4Sure Academy

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Book One teaches the complete course in the academy's seven-unit, eleven-chapter order. Classroom examples are intentionally unsolved; complete worked solutions are in Book Three.

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The chapter order follows the Success4Sure Academy textbook structure (7 units, 11 chapters). The complete content of the official AP Environmental Science Course and Exam Description is covered; an internal cross-reference map links every College Board topic to its chapter.

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UNIT I
EARTH SYSTEMS AND RESOURCES



IN THIS UNIT

Chapter 1 The Earth
Chapter 2 The Atmosphere
Chapter 3 Global Water Resources and Use

CHAPTER
1

UNIT I — EARTH SYSTEMS AND RESOURCES
The Earth

1.1 · Geologic Time Scale
1.2 · Earth's Structure
1.3 · Plate Tectonics
1.4 · Earthquakes
1.5 · Tsunamis
1.6 · Volcanoes
1.7 · Solar Intensity, Seasons, and Latitude–Longitude
1.8 · Soil
1.9 · The Rock Cycle
1.10 · Practice Questions and Answers

LEARNING OBJECTIVES

- Place major events on the geologic time scale and use half-lives to date rocks.
- Describe Earth's layers and how plate motion produces earthquakes, tsunamis and volcanoes.
- Explain how latitude, sun angle and tilt control sunlight and seasons, and read latitude and longitude.
- Describe how soil forms, its horizons, texture and properties, and how erosion is reduced.
- Trace rock through the rock cycle and name the three rock families.

1.1 Geologic Time Scale

CONCEPT MADE SIMPLE

Earth is about 4.6 billion years old. Geologists split that span into named blocks, as a historian splits history into ages. Squeeze all of Earth's history into one day and modern humans appear in the last few seconds.

The **geologic time scale** is the calendar of Earth history. From longest to shortest its units are **eons, eras, periods** and **epochs**. The four eons are the Hadean, Archean and Proterozoic (together the Precambrian, nearly 90% of time) and the Phanerozoic ("visible life"). Boundaries fall where the fossil record changes sharply, and several mark **mass extinctions**, when a large share of species vanishes. Five major ones occurred, about 445, 370, 252, 201 and 66 million years ago (Ma).

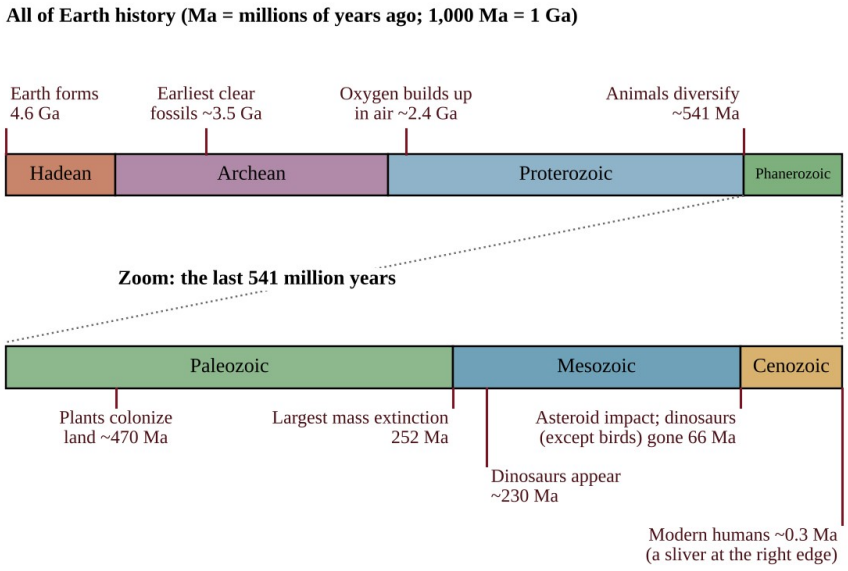


Figure 1.1 — The geologic time scale: the four eons (top) and the three eras of the last 541 million years (bottom)

Era (began)	Periods	Defining features
Paleozoic (541 Ma)	Cambrian to Permian	Marine animals diversify; fish, land plants, amphibians, reptiles. Ends with the Permian extinction (about 90% of marine species).
Mesozoic (252 Ma)	Triassic, Jurassic, Cretaceous	Dinosaurs, first mammals and birds. Ends with an asteroid impact.
Cenozoic (66 Ma)	Paleogene, Neogene, Quaternary	Mammals diversify; ice ages. The Holocene epoch began about 11,700 years ago.

Life began in the oceans by about 3.5 billion years ago (Ga), and photosynthetic bacteria raised oxygen in the air about 2.4 Ga.

Dating rocks

Relative dating orders events: in undisturbed layers (**strata**) older rock lies below younger rock, and a fault is younger than the rock it cuts. **Radiometric dating** gives ages in years. Unstable **parent isotopes** decay into **daughter isotopes** at a fixed rate called the **half-life**, the time for half of the atoms to decay. The clock starts when a mineral crystallizes, so igneous rock is dated directly and sedimentary layers are dated by igneous layers above and below. Carbon-14 (half-life 5,730 years) dates wood and bone up to about 50,000 years old; potassium-40 (1.25 billion years) and uranium-238 (4.47 billion years) date rocks.

FORMULA & MATH SKILL

Fraction remaining = $(½)^n$, n = elapsed time ÷ half-life. **Age = n × half-life.** 100% → 50% → 25% → 12.5% after 1, 2, 3 half-lives.

REAL-WORLD EXAMPLE

Fossil fuels formed over millions of years and are burned in centuries, so they are non-renewable on human time scales, and a rapid CO_2 rise is a sharp break from the slow geologic record.

COMMON STUDENT MISTAKES

- Thinking a sample is "all decayed" after two half-lives. 25% remains.
- Using carbon-14 on very old rock; too little is left after about 50,000 years.

CLASSROOM EXAMPLE SG-C1-E001

A charcoal sample from an ancient hearth contains 12.5% of the carbon-14 found in living wood. The half-life of carbon-14 is 5,730 years. (a) Calculate how many half-lives have passed. (b) Calculate the age of the charcoal. (c) A volcanic rock contains 25% of its original potassium-40 (half-life 1.25 billion years); assume no loss of the daughter product matters. Calculate the rock's age in billion years. (d) Explain why carbon-14 cannot be used to date a 100-million-year-old fossil.

QUICK CONCEPT SUMMARY

- Earth is about 4.6 Ga old; time is divided into eons, eras, periods and epochs, and boundaries often mark mass extinctions.
- Radiometric dating uses half-lives: fraction remaining = $(½)^n$.

1.2 Earth's Structure

CONCEPT MADE SIMPLE

Earth is like a peach: a thin skin (crust), thick flesh (mantle) and a dense stone (core of iron and nickel). Going down, everything gets hotter, denser and more squeezed.

Early Earth was partly molten, so dense iron and nickel sank to the center and light rock rose (**differentiation**). Layers by composition:

- **Crust**: thin, solid. **Oceanic crust** is 5–10 km thick, dense basalt (about 3.0 g/cm³) and young. **Continental crust** is 30–70 km thick, lighter granite-type rock (about 2.7 g/cm³) and much older.
- **Mantle**: about 84% of Earth's volume, hot solid rock to 2,900 km depth that flows very slowly. Density 3.3–5.7 g/cm³.
- **Core**: iron and nickel, 10–13 g/cm³. The **outer core** is liquid and its motion makes Earth's magnetic field; the **inner core** is solid because of the enormous pressure.

By strength: the **lithosphere** (crust plus the rigid top of the mantle, about 100 km thick) is cool and brittle and is broken into **plates**. The **asthenosphere** below it is soft and flows slowly, so plates can move over it. Thick continents ride higher than ocean basins (**isostasy**).

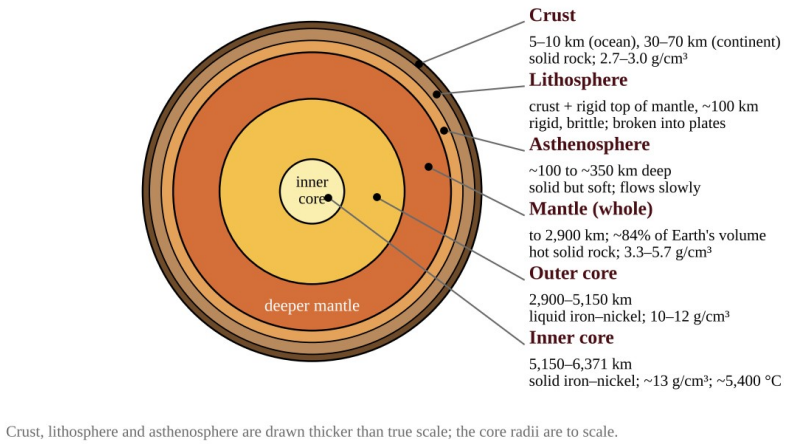


Figure 1.2 — Earth's layers by composition and by strength

Nobody has drilled beyond about 12 km. The layers are mapped by **seismic waves**: they speed up or bend where material changes, and S waves cannot pass through liquid, so they vanish behind the liquid outer core.

COMMON STUDENT MISTAKES

- Saying the mantle is liquid. It is flowing solid; only the outer core is liquid.
- Mixing up crust and lithosphere. The lithosphere includes the rigid top of the mantle.

QUICK CONCEPT SUMMARY

- Crust, mantle, core: density increases with depth; the outer core is the only liquid layer.
- The rigid lithosphere (in plates) floats on the soft asthenosphere.

1.3 Plate Tectonics

CONCEPT MADE SIMPLE

Earth's rigid shell is cracked into giant plates that creep about as fast as fingernails grow. Where plates push together, pull apart or slide past each other, the crust crumples, splits or snaps. Mountains, volcanoes and earthquakes cluster along those cracks.

The engine: moving plates

The lithosphere is broken into about a dozen large plates and many small ones. Heat from Earth's interior drives slow mantle convection, and the pull of cold, dense plate edges sinking into the mantle moves plates a few centimeters per year. **Plate motion is powered by Earth's internal heat, not by the Sun.** Evidence includes matching coastlines, rocks and fossils across oceans, seafloor that gets older with distance from a ridge, and GPS. Boundaries concentrate geologic activity, so a map of boundaries is also a map of risk.

Three boundary types

At a **convergent boundary** plates move toward each other.

- *Oceanic meets continental.* The denser oceanic plate sinks in a **subduction zone**, forming a deep **trench**. Water squeezed out of the slab at about 100 km depth helps rock above it melt, so volcanoes lie roughly 100–200 km inland of the trench, not at it, and magma rises to build volcanoes and mountains (the Andes, the Cascades). Powerful earthquakes occur along the slab.
- *Oceanic meets oceanic.* One sinks, forming a trench and a curved chain of volcanic islands, an **island arc** (Japan, the Aleutians). Oceanic plate cools and thickens as it ages away from a ridge, so the **older, cooler, denser plate** is the one that sinks.
- *Continental meets continental.* Neither sinks easily, so crust buckles into high mountains (the Himalayas). Earthquakes are common; volcanoes are rare.

At a **divergent boundary** plates move apart and magma rises to make new crust: a mid-ocean ridge with **seafloor spreading** under the sea, or a **rift valley** on land (East African Rift), with volcanoes and shallow earthquakes.

At a **transform boundary** plates slide past each other. Crust is neither created nor destroyed, but friction locks the plates and the result is earthquakes (the San Andreas Fault).

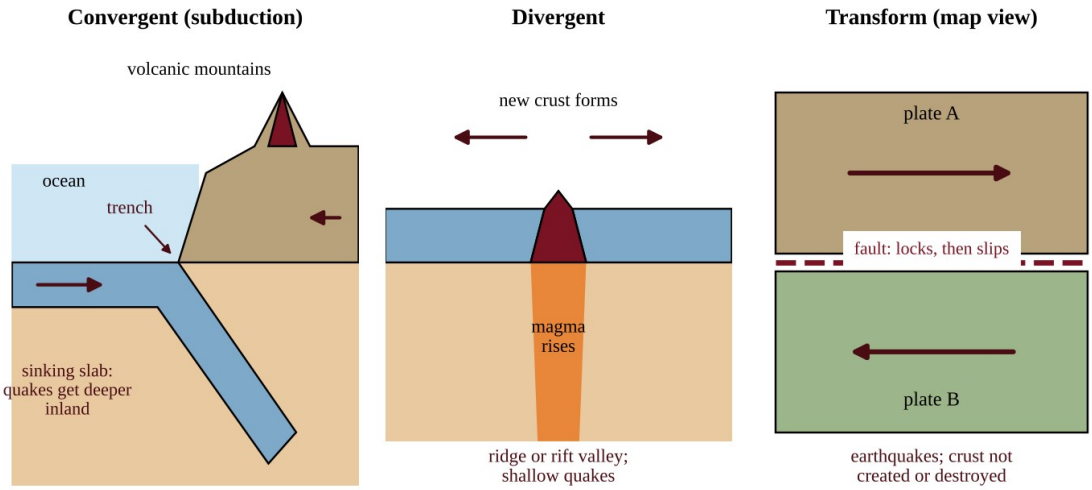


Figure 1.3 — Convergent, divergent and transform plate boundaries

Hot spots and reading global maps

A **hot spot** is a plume of hot rock rising from deep in the mantle, far from a plate edge. The plate drifts over it, leaving a line of volcanoes: the active one over the plume, older extinct ones farther along the direction of plate motion (the Hawaiian Islands).

To read a boundary map: a long narrow line of volcanoes and earthquakes marks a boundary; volcanoes just inland from a trench mean oceanic–continental subduction; a curved island chain beside a trench means an island arc; earthquakes without volcanoes along a straight line mean a transform fault or continental collision; a lone volcano far from a boundary suggests a hot spot. The Pacific "Ring of Fire" has most of the world's earthquakes and active volcanoes because so many subduction zones lie there. Near a trench earthquakes are shallow, and farther **inland** they are progressively **deeper**, tracing the sinking slab; a graph of depth against distance from the trench that slopes down to the right signals

subduction. Transform faults give only shallow earthquakes. Rifts and hot spots heat groundwater, giving hot springs and geothermal energy.

FORMULA & MATH SKILL

Plate speed = distance between two volcanoes ÷ difference in their ages. The active volcano has age 0. Shortcut: 1 km per million years = 0.1 cm per year.

CLASSROOM EXAMPLE SG-U4-E011

Three extinct volcanic islands lie along a chain that ends at an active hot-spot volcano. Island P is 140 km from the active volcano and 2.0 million years old; Island Q is 350 km away and 5.0 million years old; Island R is 630 km away and 9.0 million years old. (a) Calculate the plate's speed from the data for Island R, in km per million years and in cm per year. (b) Check it with Island P. (c) Predict the age of an island 1,050 km from the active volcano. (d) State which direction the plate is moving relative to the hot spot and explain why the volcanoes are not found at a plate boundary.

COMMON STUDENT MISTAKES

- Saying plates "float on liquid magma." The asthenosphere is solid but flows slowly.
- Saying transform boundaries make volcanoes. They make earthquakes only.
- Saying all convergent boundaries make volcanoes. Continent collisions build mountains with few or none.

AP EXAM STRATEGY

For a boundary question name the **motion** (toward, apart, sideways), the **landform** (trench, ridge, mountains) and the **hazard** (earthquake, volcano), using the CED words: island arcs, seafloor spreading, rift valleys, hot spots, faults.

CLASSROOM EXAMPLE SG-U4-E001

A geologist studies three sites. Site 1 is a line of active volcanoes 150 km inland, parallel to a deep ocean trench offshore. Site 2 is a long underwater mountain chain where new basalt is found and the seafloor gets older with distance on both sides. Site 3 is a straight valley where a road has been offset 4 m sideways by an earthquake, with no volcanoes nearby. (a) Identify the type of plate boundary at each site. (b) For Site 2, describe what happens to the plates and name the process that adds new crust. (c) State which site is most likely to produce a tsunami, and justify your choice in one sentence.

QUICK CONCEPT SUMMARY

- Plates move on the asthenosphere, driven by internal heat; boundaries concentrate quakes and volcanoes.
- Convergent: subduction, trenches, volcanic mountains, island arcs, or collision mountains. Divergent: ridges, rifts. Transform: earthquakes.
- Hot spots leave chains of volcanoes inside plates.

1.4 Earthquakes

CONCEPT MADE SIMPLE

Bend a ruler until it snaps: it stores energy, then releases it all at once. Rock along a fault does the same. Plates keep moving, the fault stays stuck, strain builds, and finally the rock slips and sends out shaking.

PROCESS & CAUSE-AND-EFFECT

Plate motion → fault locks and stress builds → stress exceeds friction → sudden slip → stored energy released as seismic waves → shaking at the surface.

An earthquake occurs when stress overcomes a locked fault, releasing stored energy. The underground point where slip begins is the **focus**; the surface point directly above is the **epicenter**. Smaller **aftershocks** follow. Most earthquakes occur at plate boundaries: transform faults, subduction zones (down to about 700 km) and ridges and rifts (shallow). Shallow quakes (under 70 km) do the most damage.

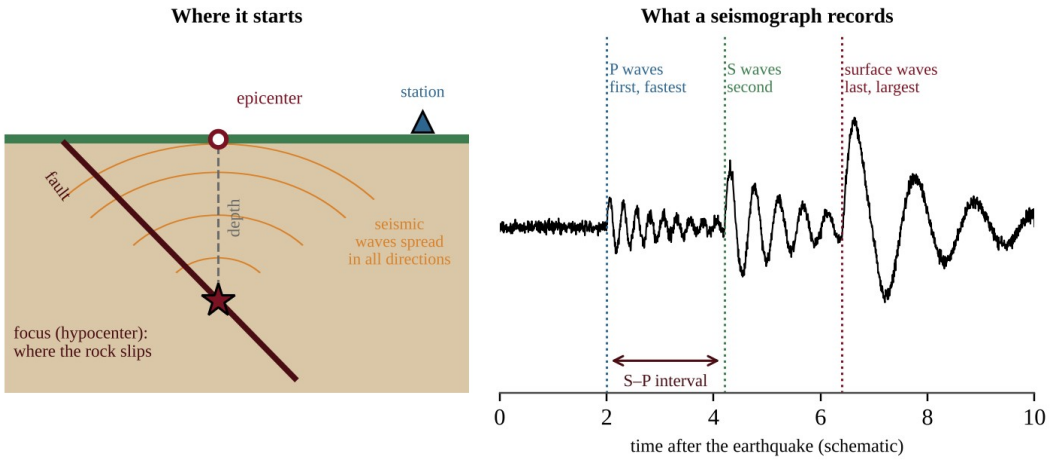


Figure 1.4 — Focus and epicenter, and the wave arrivals on a seismogram

Wave	Motion	Speed (crust)	Passes through
P	Push–pull	about 6 km/s (fastest)	Solids and liquids
S	Side-to-side	about 3.5 km/s	Solids only
Surface	Rolling along the ground	slowest	Surface; most damage

A **seismograph** records P waves first, then S, then surface waves. The longer the S–P gap, the farther the station is from the quake; distances from three stations place the epicenter where three circles meet.

Magnitude measures the energy released at the source (one value per earthquake) on the logarithmic **moment magnitude** scale: each whole-number step is a tenfold larger wave amplitude and about 32 times more energy. **Intensity** describes shaking and damage at one place on the **Modified Mercalli** scale (I–XII); it falls with distance and rises on soft ground.

FORMULA & MATH SKILL

Amplitude ratio = 10^ΔM; energy ratio ≈ 32^ΔM (ΔM = magnitude difference, which may be a decimal; use a calculator). **Distance = (S–P time) × vP·vS ÷ (vP − vS)**; for 6.0 and 3.5 km/s, 8.4 km per second of S–P time.

Hazards: shaking, surface rupture, **liquefaction** (saturated sand and fill act like liquid and buildings tilt), landslides, fires, dam failure and tsunamis. Damage depends on magnitude, depth, distance, ground type and construction; soft sediment amplifies shaking.

REAL-WORLD EXAMPLE

Risk is cut by flexible reinforced building codes, retrofitting, keeping critical facilities off soft or filled ground, and drills. Early-warning systems detect fast P waves and alert people seconds before S and surface waves arrive. Earthquakes cannot be predicted, only forecast as probabilities.

COMMON STUDENT MISTAKES

- Confusing magnitude (energy at the source) with intensity (local effects).
- Putting the epicenter underground. The focus is underground; the epicenter is on the surface.

CLASSROOM EXAMPLE SG-C1-E002

A seismograph station records the first P wave at 08:14:10 and the first S wave at 08:14:35. Assume P waves travel at 6.0 km/s and S waves at 3.5 km/s. (a) Calculate the S–P interval in seconds. (b) Calculate the distance from the station to the earthquake. (c) Calculate how many seconds after the earthquake began the P wave arrived. (d) Explain why data from at least three stations are needed to locate the epicenter.

CLASSROOM EXAMPLE SG-C1-E003

Earthquake A has a moment magnitude of 6.1 and earthquake B has a magnitude of 8.1. (a) Calculate how many times larger the ground-motion amplitude of B is than that of A. (b) Using a factor of 32 per magnitude unit, calculate how many times more energy B releases. (c) State which measure, magnitude or Modified Mercalli intensity, can differ between two towns affected by the same earthquake, and explain why.

QUICK CONCEPT SUMMARY

- Stress builds on a locked fault; slip releases P, S and surface waves.
- Magnitude (logarithmic energy) differs from intensity (local effects). Most quakes occur at plate boundaries.

1.5 Tsunamis

CONCEPT MADE SIMPLE

A tsunami is a push through the whole depth of the ocean, not a wind wave. A seafloor jolt shoves a huge volume of water, and the push crosses the sea as a very long, fast, low wave that becomes huge only in shallow water.

A **tsunami** is a series of very long ocean waves caused by sudden displacement of water. The usual cause is a large undersea earthquake at a subduction zone that lifts or drops the seafloor by meters; others are underwater landslides, volcanic eruptions and asteroid impacts. Sideways (transform) quakes seldom make large tsunamis.

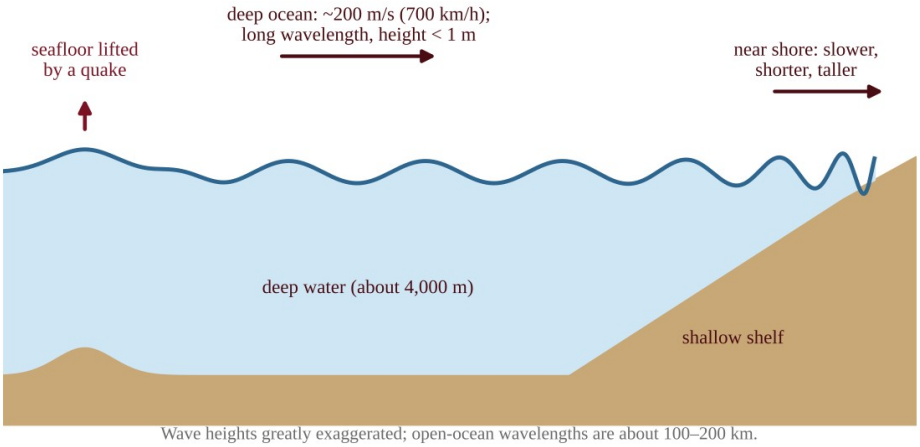


Figure 1.5 — A tsunami: seafloor displacement, a fast low wave in deep water, a slow tall wave near shore

The wavelength is about 100–200 km, far more than the ocean depth, so the wave's speed depends only on depth.

FORMULA & MATH SKILL

$v = \sqrt{(g \times d)}$, g = 9.8 m/s², d = depth in m, v in m/s (× 3.6 for km/h). **Travel time = distance ÷ speed.** At 4,000 m depth v ≈ 198 m/s (about 710 km/h).

In the open ocean the wave is under 1 m high, so ships do not notice it. In shallow water it slows, its wavelength shortens and its energy is squeezed into less water, so height grows (**shoaling**) and water runs far inland (**run-up**). Several waves arrive, minutes to an hour apart, and the first is not always the largest; the sea may first drain away.

Tsunamis kill by drowning and debris, destroy ports and power plants (Fukushima, 2011) and wetlands and reefs, and salt water contaminates soil and freshwater aquifers. The 2004 Indian Ocean tsunami (magnitude 9.1) killed over 200,000 people. Warning combines seismometers, deep-ocean pressure buoys (DART) and tide gauges. Across an ocean there are hours of warning; near a quake there are minutes, so strong shaking, a sudden sea retreat or a roar means *move to high ground or far inland now*. Mangroves and reefs reduce wave energy somewhat; zoning and drills help.

COMMON STUDENT MISTAKES

- Calling a tsunami a "tidal wave." It has nothing to do with tides.
- Thinking it is tall at sea. It is low and long there and grows only in shallow water.

CLASSROOM EXAMPLE SG-C1-E004

A tsunami crosses an ocean 4,000 m deep. Use $v = \sqrt{(g \times d)}$ with $g = 9.8$ m/s². (a) Calculate the wave speed in m/s and in km/h. (b) Calculate the time, in hours, to reach a coast 2,500 km away, assuming constant depth. (c) Calculate the speed where the depth is 10 m. (d) Explain why the wave becomes taller as it approaches shore even though it slows down.

QUICK CONCEPT SUMMARY

- Undersea quakes at subduction zones (also landslides, eruptions) displace water and cause tsunamis.
- Speed = $\sqrt{(g \cdot d)}$: fast in deep water, slower and taller in shallow water. Sensors and evacuation plans save lives.

1.6 Volcanoes

CONCEPT MADE SIMPLE

Think of a shaken soda bottle. Thin, runny magma lets gas bubble out gently, like opening it slowly. Thick, sticky magma traps the gas until it blasts out, like soda spraying across the room.

Rock melted inside Earth is **magma**; at the surface it is **lava**. Volcanoes form at **subduction zones** (water from the slab lowers the melting point of the mantle above; most explosive volcanoes), at **divergent boundaries** (pressure drops as plates part and mantle melts to basalt; Iceland) and at **hot spots** (Hawaii, Yellowstone).

Eruption style depends on the **silica (SiO_2) content** of the magma, which sets its **viscosity** (resistance to flow) and how much gas it traps.

Magma	Silica	Temperature	Viscosity and gas	Eruption
Basaltic	about 50%	1,000–1,200 °C	Low; gas escapes	Quiet lava flows
Andesitic	57–63%	intermediate	Medium–high	Often explosive
Rhyolitic	over 68%	650–800 °C	Very viscous; traps gas	Very explosive

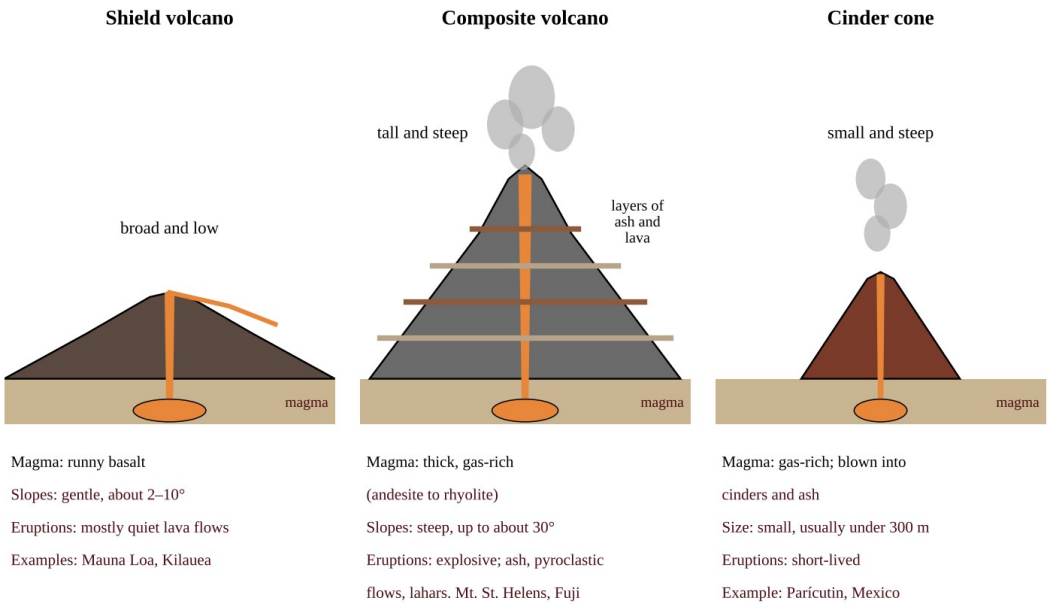


Figure 1.6 — Three common volcano types

A **caldera** is a large basin left when an emptied magma chamber collapses (Crater Lake). The **Volcanic Explosivity Index** (0–8) rises about tenfold per step with erupted volume.

Hazards: **lava flows** (usually slow); **pyroclastic flows**, glowing clouds of gas, ash and rock up to 700 °C moving over 100 km/h, the deadliest; **ash fall** that collapses roofs, grounds aircraft and harms lungs; **lahars**, volcanic mudflows down valleys; gases (SO_2 forms acid rain; CO_2 can pool and suffocate); and landslides and tsunamis.

Climate. A big explosive eruption puts SO_2 into the stratosphere, where it forms sulfate **aerosols** that reflect sunlight and cool Earth for one to three years (Pinatubo 1991, about −0.5 °C; Tambora 1815, "the year without a summer"). Volcanic CO_2 is far less than human emissions, so volcanoes do not explain modern warming. Volcanoes also give fertile ash soils, geothermal energy and new land. Scientists monitor swarms of small quakes, ground tilt, gas and heat, and authorities evacuate danger zones.

COMMON STUDENT MISTAKES

- Saying all eruptions are explosive. Basaltic ones are usually gentle.
- Saying volcanoes warm the climate. A big eruption cools it for a few years.

AP EXAM STRATEGY

Chain the logic: composition → viscosity → gas trapping → eruption style, then match volcano type to boundary or hot spot, hazards and a monitoring method.

CLASSROOM EXAMPLE SG-C1-E005

A town lies 18 km from a volcano's vent. A second town lies 45 km from the vent down a river valley. A pyroclastic flow moves at an average of 120 km/h and a lahar at an average of 36 km/h. (a) Calculate the time, in minutes, for the pyroclastic flow to reach the first town. (b) Calculate the time, in minutes, for the lahar to reach the second town. (c) State which hazard allows more time to evacuate and give one monitoring method that could provide warning. (d) Explain why rhyolitic magma tends to erupt more explosively than basaltic magma.

QUICK CONCEPT SUMMARY

- Volcanoes form at subduction zones, rifts and hot spots.
- High-silica, viscous, gas-rich magma erupts explosively; basalt flows quietly. Aerosols from big eruptions cool climate briefly.

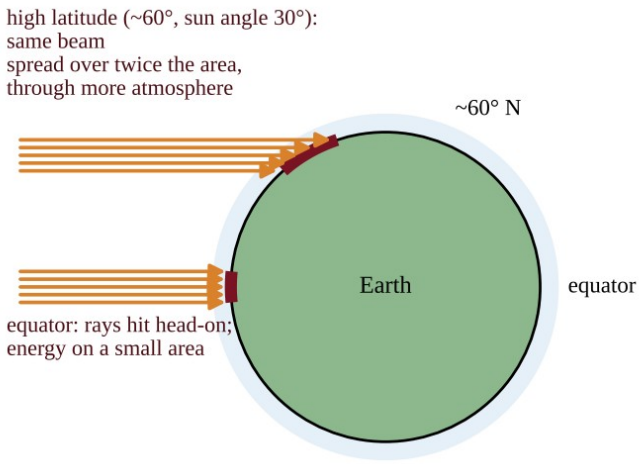
1.7 Solar Intensity, Seasons, and Latitude–Longitude

CONCEPT MADE SIMPLE

Shine a flashlight straight down on a table, then tilt it. The straight beam makes a small bright spot; the slanted beam spreads the same light over a larger, dimmer area. Earth's curved surface does the same, so the equator is hot and the poles are cold.

Insolation, angle and latitude

Insolation (incoming solar radiation) is Earth's main energy source and depends on **season and latitude**. The **angle of the sun's rays** determines intensity. Because Earth is a sphere, the rays strike at 90° at only one latitude at a time, the place directly under the sun, where the energy is spread over the smallest area and passes through the least atmosphere. At higher latitudes rays arrive at a slant, spread over more area and cross more atmosphere, where more is absorbed, scattered or reflected. The highest radiation per unit area is at the equator and decreases toward the poles.



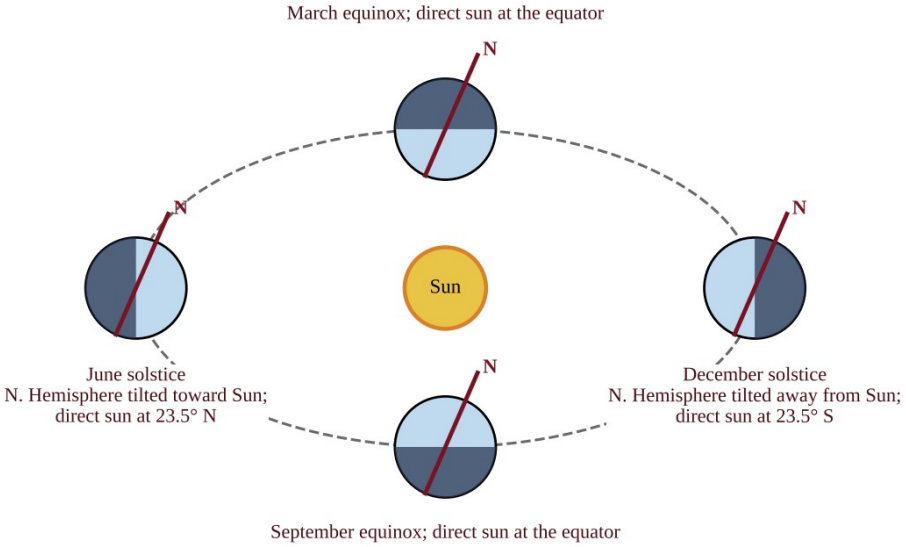
Parallel rays from the Sun. Red arcs: ground area the beam covers (shown at an equinox). The atmosphere is the thin light ring.

Figure 1.7 — The same beam covers a small area at the equator and about twice the area at 60° latitude

The tilt makes the seasons

Earth's axis is tilted about **23.5°** and points in the same direction in space all year, so the Northern Hemisphere leans toward the sun for half the orbit and away for the other half.

- **June solstice (about June 21):** sun overhead at the Tropic of Cancer (23.5° N); Northern days longest, sun highest.
- **December solstice (about December 21):** sun overhead at 23.5° S; Northern days shortest, sun lowest.
- **Equinoxes (about March 20 and September 22):** sun overhead at the equator; day and night nearly equal everywhere.



The 23.5° tilt always points the same way in space. Dark half of each Earth = night. Not to scale.

Figure 1.8 — The fixed tilt points the Northern Hemisphere toward the Sun in June and away in December

The tilt also sets **day length**: about 12 hours all year at the equator, longer days in the summer hemisphere at higher latitudes, and 24 hours of daylight poleward of 66.5° in the summer hemisphere (north in June, south in December). A place receives the most radiation on its longest summer day and the least on its shortest winter day, because high sun angle and long daylight work together. On a day-length graph, a

small swing means near the equator, a large swing a high latitude, and June maxima the Northern Hemisphere. With no tilt there would be no seasons.

FORMULA & MATH SKILL

Noon sun angle = 90° − |latitude − latitude of the direct sun| Direct-sun latitude: +23.5° (June solstice), 0° (equinoxes), −23.5° (December solstice); north positive. Intensity on flat ground is proportional to the **sine** of the sun angle (the cosine of the angle from vertical): sin 60° ÷ sin 30° = 0.866 ÷ 0.500 ≈ 1.7. **Energy absorbed = incoming sunlight × (1 − albedo)**; **albedo** is the fraction reflected (0–1). Extra energy absorbed when a bright surface is replaced by a dark one = incoming × (old albedo − new albedo), in W/m².

Latitude and longitude

Latitude is the angle north or south of the equator (0° to 90°); lines of equal latitude (**parallels**) run east–west. **Longitude** is the angle east or west of the **prime meridian** through Greenwich (0° to 180°); lines of equal longitude (**meridians**) run pole to pole. Give latitude first, with N/S and E/W. The **Tropics** (23.5° N and S) mark the farthest the sun can be overhead, and the **Arctic and Antarctic Circles** (66.5°) mark where 24-hour day or night occurs, giving tropical (0–23.5°), temperate (23.5–66.5°) and polar (66.5–90°) zones.

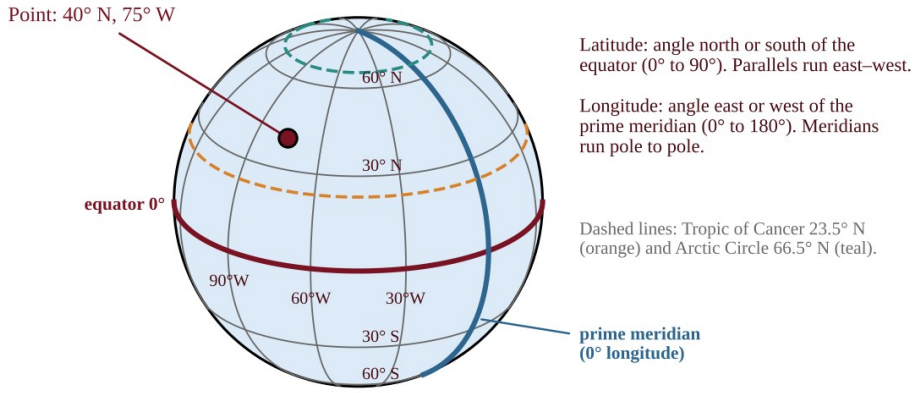


Figure 1.9 — Latitude and longitude; the point is 40° N, 75° W

FORMULA & MATH SKILL

1° of latitude ≈ 111 km. 1° of longitude ≈ 111 km × cos(latitude). Earth turns 15° per hour, so places 15° apart in longitude differ by 1 hour of solar time; the one farther east is ahead.

COMMON STUDENT MISTAKES

- Saying seasons occur because Earth is closer to the sun in summer. Earth is closest in early January; tilt, not distance, causes seasons.
- Saying the tilt changes during the year. Only Earth's position in orbit changes.

AP EXAM STRATEGY

Give **two** linked reasons for seasonal change, sun angle (intensity per area) and day length (hours of exposure), and trace both to the tilt.

CLASSROOM EXAMPLE SG-U4-E016

A city is at 52° N. (a) Calculate the noon sun angle on the June and December solstices. (b) Calculate how many times more intense the noon sunlight is in June than in December, using the sine of each angle. (c) A snow-covered field with albedo 0.80 receives 180 W/m² and then melts to bare soil with albedo 0.15. Calculate the increase in energy absorbed per square meter. (d) Explain why a city at 5° N has much smaller seasonal temperature changes than this city.

CLASSROOM EXAMPLE SG-U4-E008

A city lies at 40° N. (a) Calculate the noon sun angle at the city on the June solstice and on the December solstice using the formula in this guide. (b) Identify which date gives more intense radiation per unit area at the city, and explain why with reference to area and atmospheric path length. (c) State a second reason, other than sun angle, that the city receives more total energy on the June solstice than on the December solstice. (d) At which latitude is the sun directly overhead on the March equinox?

CLASSROOM EXAMPLE SG-C1-E006

Two research stations are both at 52° N. Station 1 is at 100° W and Station 2 is at 70° W. Use 1° of latitude = 111 km and 15° of longitude = 1 hour of solar time. (a) Calculate the difference in solar time between the stations and state which is ahead. (b) Calculate the distance between the stations along the 52° N parallel, using cos 52° = 0.616. (c) Calculate the distance from Station 1 north to the Arctic Circle (66.5° N) along its meridian. (d) Explain why 1° of longitude is a shorter distance at 52° N than at the equator.

QUICK CONCEPT SUMMARY

- Intensity depends on the sun's angle and is greatest at the equator; Earth's 23.5° tilt, not distance, causes seasons and day length.
- Latitude (north–south) and longitude (east–west) locate places; 1° latitude ≈ 111 km.

1.8 Soil

CONCEPT MADE SIMPLE

Soil is crushed rock mixed with dead plant and animal matter, air, water and living things. A centimeter takes hundreds of years to form, but one storm can wash it away. Soil also filters water like a coffee filter. Coarse sand drains like gravel; fine clay holds water like a hard-to-wring sponge; loam, a balanced mix, gives the best of both.

How soil forms

Soil begins as **parent material**, bedrock or loose sediment, broken down by weathering. **Physical weathering** (freeze–thaw, root wedging, temperature swings, abrasion) breaks rock without changing its chemistry. **Chemical weathering** (weak acids, oxidation, hydrolysis) changes minerals. Lichens, roots, burrowers and microbes add to both and supply organic matter. Weathered material may stay put or be **transported** by water, wind, ice or gravity and **deposited** elsewhere (floodplains, windblown loess, glacial till). Soil depends on parent material, climate, organisms, slope and time. Warmth and moderate rain speed weathering and decomposition; cold or dryness slow them. On new ground such as cooled lava, **primary succession** begins as lichens and mosses release acids and leave organic remains. **Humus**, the dark, stable end product of decomposition, gives fertile soil its crumbly structure and nutrient-holding ability.

Soil horizons

Soil develops layers called **horizons**; a vertical cut through them is a **soil profile**.

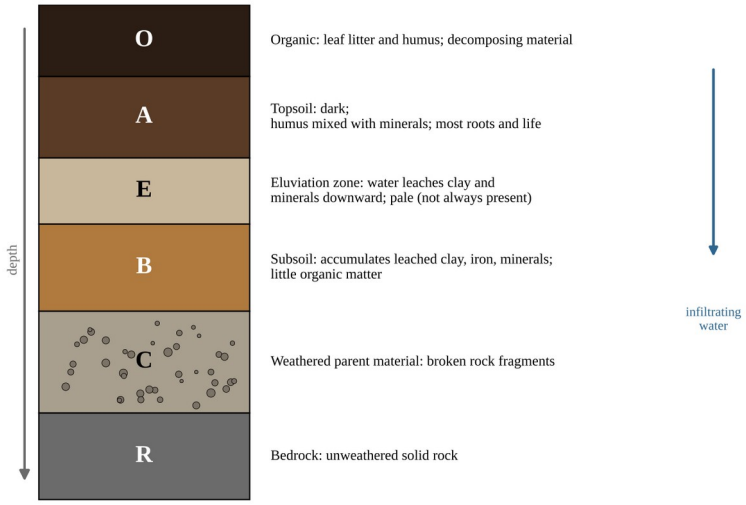


Figure 1.10 — A generalized soil profile with its horizons

Percolating water carries dissolved minerals and fine clay out of upper layers (**leaching**) into the **B horizon**. The **A horizon** is topsoil, the most biologically active layer, with **biotic** parts (roots, earthworms, fungi, bacteria) and **abiotic** parts (sand, silt, clay, minerals, water, air). O-horizon litter shields soil from raindrops, holds moisture and recycles nutrients, so losing it dries and erodes the soil. Young and dry-climate soils may lack an E horizon.

Erosion and soil protection

Erosion moves soil: wind lifts loose, dry, bare soil (the 1930s Dust Bowl); water erodes by raindrop splash, **sheet** erosion, **rills** and **gullies**. It is worst on bare, steep, loose soil, and topsoil, with its nutrients and humus, goes first. Eroded sediment and attached fertilizer, pesticides and bacteria cloud streams, smother fish eggs and fuel algal blooms. Continuous cropping removes plant matter and speeds decomposition, so soil organic matter (humus) falls unless residue, manure or cover crops return it. Intact soil does the opposite: it traps sediment, clay and humus hold pollutants and nutrient ions, and microbes

break down contaminants. **Protecting soils protects water quality.** Soluble nitrate is not held tightly, so excess fertilizer leaches to groundwater. Conservation methods keep cover (cover crops, perennials), use no-till, plow along contours, terrace, strip-crop, plant windbreaks and leave vegetated buffer strips; each slows wind or water, binds soil with roots or covers the surface.

FORMULA & MATH SKILL

Net soil change = formation rate − erosion rate (e.g., t/ha/yr). **Total = net rate × area (ha) × time (yr). Depth lost = mass lost ÷ (bulk density × area)**; 1 ha = 10^4 m²; × 1,000 for mm. Loss is sustainable only if depth lost per year does not exceed depth formed per year.

CLASSROOM EXAMPLE SG-U4-E012

A 25-ha field loses soil at 12 t/ha/yr while new soil forms at 1.0 t/ha/yr. The soil's bulk density is 1.25 t/m³. (a) Calculate the net loss per hectare per year. (b) Calculate the total net loss from the whole field over 30 years. (c) Calculate the depth of soil lost per year, in mm, from the erosion rate. (d) New soil forms at about 0.08 mm per year. Justify whether the erosion is sustainable, and propose one practice to bring it closer to the formation rate.

CLASSROOM EXAMPLE SG-U4-E002

A soil pit shows these layers from the top down. P (0–3 cm): loose, dark, partly decayed leaves. Q (3–15 cm): dark brown, crumbly, many roots, rich in humus. R (15–28 cm): pale gray, sandy, few nutrients. S (28–60 cm): reddish brown, sticky, clay-rich. T (60–110 cm): angular broken rock pieces with a little clay. U (below 110 cm): solid, unbroken rock. (a) Match each layer to its horizon letter (O, A, E, B, C or R). (b) Explain why layer R is pale and nutrient-poor while layer S is clay-rich. (c) Identify the layer that a farmer loses first when heavy rain erodes a bare field, and describe one practice that would reduce this loss.

Sand, silt and clay; porosity and permeability

Sand (0.05–2.0 mm) is gritty, **silt** (0.002–0.05 mm) smooth like flour, **clay** (under 0.002 mm) sticky and ribbon-forming. **Texture** is their proportion. **Porosity** is the percent of a soil's volume that is pore space; **permeability** is how quickly water moves through; clay can be highly porous yet have low permeability because its pores are tiny. **Water-holding capacity** is the total water a soil holds against gravity; it varies with soil type, and water retention supports productivity and fertility. Particle size and composition of each horizon affect porosity, permeability and fertility; for example, a sandy A horizon over a clay B horizon drains quickly then perches water above the clay. **Fertility** depends on N, P, K, pH, organic matter and soil life.

Property	Sandy	Silty	Clay
Pore size	Large	Medium	Very small
Permeability	High	Medium	Low

Property	Sandy	Silty	Clay
Water-holding capacity	Low	Medium–high	High
Nutrient retention	Poor (leaches)	Moderate	Good
Problems	Droughty	Easily eroded	Waterlogged, hard to till

The soil texture triangle

A **soil texture triangle** plots clay, silt and sand percentages (summing to 100%) and names the soil. Find clay on the left scale and follow its horizontal line; find sand on the bottom scale and follow its line up and left; the lines cross in the soil's class (silt is a check). Two components suffice. **Loam**, a blend of clay, silt and sand with a specific range, not equal thirds, supports a wide variety of crops because it drains yet holds water and nutrients.

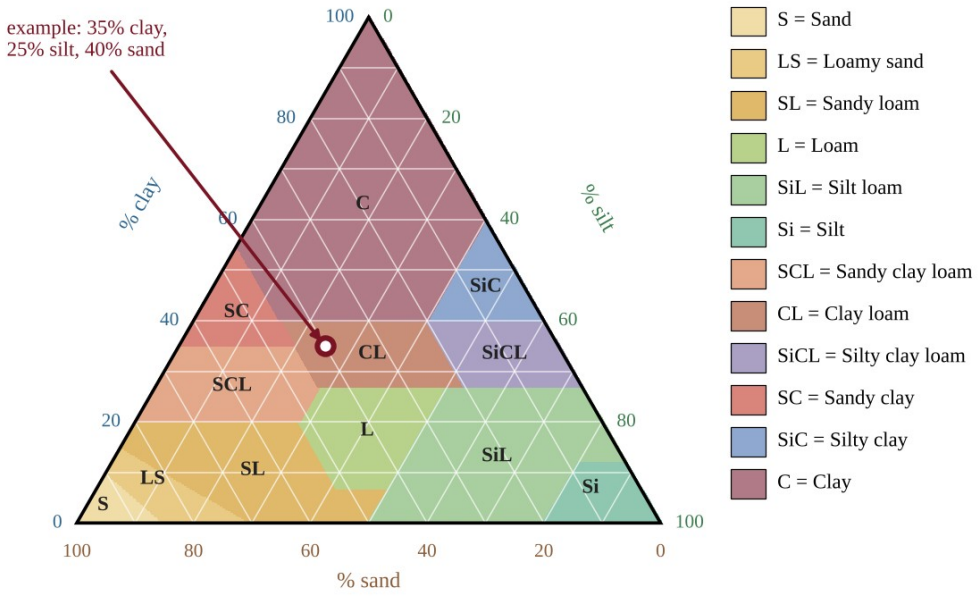


Figure 1.11 — The soil texture triangle; the circled point is 35% clay, 25% silt, 40% sand (clay loam)

Group	Drainage	Water and nutrient holding
Sandy (sand, loamy sand, sandy loam)	Fast	Low to moderate
Loamy (loam, silt loam, silt)	Moderate	Moderate to high
Clay-loamy (clay loam, sandy clay loam, silty clay loam)	Moderately slow	High
Clayey (sandy clay, silty clay, clay)	Very slow	Highest

A **landfill liner** needs clay-rich soil, whose tiny pores block leachate; compost helps a sandy garden hold water. For worked examples we also use a simplified key:

Class	Rule (percent by mass)
Sand	sand ≥ 85
Clay	clay ≥ 40
Silt	silt ≥ 80
Loam	sand 23–52, silt 28–50, clay 7–27

Any mixture fitting no row is "intermediate texture."

Testing soil

- **Physical:** the **jar test** (sand settles in about a minute, silt in hours, clay in a day or more; readings are cumulative, so layer thickness is the difference between readings; percent = layer ÷ total sediment × 100), sieving, and a **percolation test** timing drainage.
- **Chemical:** pH, N–P–K kits, salinity (electrical conductivity).
- **Biological:** organic matter, earthworm counts, soil respiration (CO_2 release).

Results guide decisions: lighter, more frequent irrigation for sand; targeted phosphorus rather than blanket fertilizer; lime for low pH; drainage or raised beds for waterlogged clay. **Infiltration rate** = depth of water ÷ time (mm/h).

FORMULA & MATH SKILL

Percent by mass = (component mass ÷ total mass) × 100. **Water retained** = poured − drained; **percent retained** = retained ÷ poured × 100. **Runoff rate = rainfall rate − infiltration rate** (zero if rain is slower than infiltration).

CLASSROOM EXAMPLE SG-U4-E014

A ring is filled with 30 mm of water on Field A and on Field B. The water soaks in completely in 45 minutes on A and in 6 hours on B. (a) Calculate the infiltration rate of each field in mm/h. (b) A storm drops 12 mm of rain per hour for 3 hours. Calculate the runoff depth from each field during the storm, assuming the rates stay constant. (c) Identify which field is more likely to be compacted clay and state one effect on a nearby stream.

For an investigation, include a **control** (the untreated or bare comparison, such as bare soil beside grass) and name the **independent variable** (what you change, e.g. soil type), the **dependent variable** (what you measure, e.g. drained volume after 10 minutes), the **controlled variables** (same soil mass, water volume, container, time) and **replicates** (at least three, averaged), and write an if–then hypothesis.

COMMON STUDENT MISTAKES

- Confusing weathering (breaks rock in place) with erosion (moves material).
- Calling the whole soil "topsoil." Topsoil is the A horizon (with the O above it).
- Mixing up porosity and permeability, or reading the triangle with one percentage.

AP EXAM STRATEGY

For a protection method give **how it works** (slows water, binds soil, covers surface) and **what it protects** (topsoil, nutrients, water quality). In data questions quote numbers to justify which soil holds the most water or drains fastest.

CLASSROOM EXAMPLE SG-U4-E003

A technician oven-dries and sieves a 50.0 g soil sample and obtains 21.0 g of sand, 18.0 g of silt and 11.0 g of clay. (a) Calculate the percent by mass of sand, silt and clay. (b) Using the simplified key in this guide, name the texture class. (c) State which of the three components is most responsible for holding water against gravity, and explain why in terms of pore size.

CLASSROOM EXAMPLE SG-U4-E004

A student tests three soils (X, Y, Z). For each, she places 400 g of dry soil in an identical funnel lined with filter paper, pours in 200 mL of water at once, and records the volume that has drained into a cylinder after 10 minutes. Results: soil X, 170 mL; soil Y, 110 mL; soil Z, 40 mL. (a) Calculate the percent of the poured water retained by each soil. (b) Identify the independent variable, the dependent variable and two controlled variables. (c) One soil is a sand, one a loam and one a clay. Assign each, and justify using the drained volumes. (d) Propose one improvement that would make her conclusion more reliable.

QUICK CONCEPT SUMMARY

- Soil forms as parent material is weathered, transported and deposited; horizons (O, A, E, B, C, R) differ in composition and organic content.
- Wind and water erode soil; conservation keeps cover and slows flow; healthy soil filters water.
- Texture (sand–silt–clay, read on the triangle) controls porosity, permeability and water-holding; loam is the balanced blend.
- Physical, chemical and biological tests guide irrigation and fertilizer decisions.

1.9 The Rock Cycle

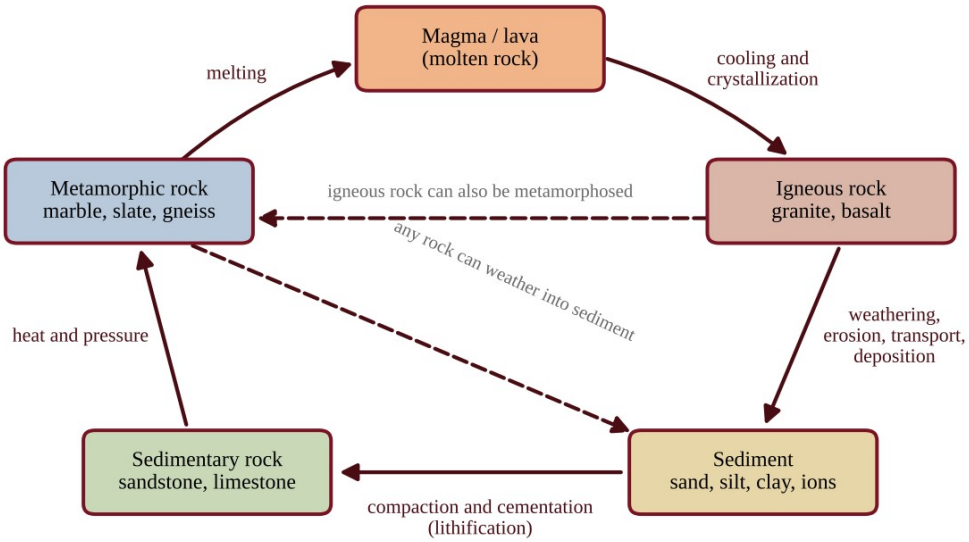
CONCEPT MADE SIMPLE

Rocks are slowly recycled. A mountain wears down to sand, sand is buried and cemented into stone, stone is baked or squeezed into a new rock, and deep enough it melts and may erupt again. Earth's heat and the Sun's energy keep the cycle turning.

Rocks are made of **minerals**, natural solids with a definite chemistry and crystal structure.

Family	How it forms	Clues	Examples
Igneous	Magma or lava cools and crystallizes: *intrusive* (slow, underground, large crystals) or *extrusive* (fast, at the surface, tiny crystals or glass)	Interlocking crystals, no layers	Granite; basalt, obsidian

Family	How it forms	Clues	Examples
Sedimentary	Sediment is compacted and cemented (**lithification**); some form from minerals precipitated from water or from shell and plant remains	Layers, rounded grains, fossils	Sandstone, shale; limestone, coal
Metamorphic	Existing rock changes in the solid state under high heat and pressure; minerals recrystallize or align in bands (*foliation*)	Banded, hard, shiny	Slate, gneiss; marble, quartzite



Uplift by plate tectonics exposes buried rock to weathering; subduction carries rock down to melt.

Figure 1.12 — The rock cycle

Processes: **weathering and erosion** break rock into **sediment** (the parent material of soil) and **deposition** lays it in layers; **compaction and cementation** make sedimentary rock; **metamorphism** (heat and pressure, no melting) makes metamorphic rock; **melting** makes magma; **cooling and crystallization** make igneous rock. The cycle has many routes: igneous rock can be metamorphosed without weathering, and any rock can be weathered again. Plate tectonics drives it: subduction drags rock down to melt, collisions metamorphose rock, volcanoes make new igneous rock and uplift exposes rock to weathering, over thousands to hundreds of millions of years.

REAL-WORLD EXAMPLE

Granite and sandstone are building stone; limestone is crushed for cement and lime; coal is a fuel; many metal ores form where hot fluids move through igneous rock; marble and slate are metamorphic building stones.

COMMON STUDENT MISTAKES

- Believing the cycle runs in one fixed order. Any rock can become any type, and steps can be skipped.
- Saying metamorphic rock melted. It changed in the solid state; if it melts it becomes magma.

CLASSROOM EXAMPLE SG-C1-E007

Sediment accumulates on a shallow sea floor at an average of 0.40 mm per year. When it becomes sedimentary rock, compaction reduces its thickness by 40%. (a) Calculate the thickness of sediment, in meters, deposited in 5.0 million years. (b) Calculate the thickness of rock formed after compaction. (c) Name the process that turns loose sediment into rock. (d) The rock is limestone, which is later buried deep and heated and squeezed without melting. Name the rock family and one example of the rock that forms.

QUICK CONCEPT SUMMARY

- Igneous rock cools from melt; sedimentary rock forms from cemented sediment; metamorphic rock is changed by heat and pressure.
- Weathering, deposition, lithification, metamorphism, melting and crystallization link them; plate tectonics drives the cycle.

1.10 Practice Questions and Answers

Test what you have learned in this chapter with the exam-style questions that follow.

PRACTICE QUESTIONS AND ANSWERS

Classroom examples in this chapter (unsolved; space is provided in each box): SG-C1-E001, SG-U4-E011, SG-U4-E001, SG-C1-E002, SG-C1-E003, SG-C1-E004, SG-C1-E005, SG-U4-E016, SG-U4-E008, SG-C1-E006, SG-U4-E012, SG-U4-E002, SG-U4-E014, SG-U4-E003, SG-U4-E004, SG-C1-E007.

Classified practice: Book Two, Chapter 1 — 40 multiple-choice questions (Section A) followed by 2 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 1. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

CHAPTER
2

UNIT I — EARTH SYSTEMS AND RESOURCES
The Atmosphere

2.1 · Composition
2.2 · Structure
2.3 · Weather and Climate
2.4 · Atmospheric Circulation—Pressure
2.5 · Hurricanes, Tornadoes, and Cyclones
2.6 · El Niño—Southern Oscillation (ENSO)
2.7 · Practice Questions and Answers

LEARNING OBJECTIVES

- Name the major and variable gases of the atmosphere, give their abundances, and explain why trace gases and aerosols matter.
- Describe the five atmospheric layers by their temperature gradients and read a temperature–altitude profile.
- Distinguish weather from climate, and explain how fronts, mountains, oceans and elevation shape both.
- Explain how uneven solar heating, density differences and the Coriolis effect produce the global circulation cells and wind belts.
- Describe how hurricanes and tornadoes form, how they are rated, and why storm surge is so dangerous.
- Describe El Niño and La Niña, their causes and their effects in different regions.

2.1 Composition

CONCEPT MADE SIMPLE

The atmosphere is a thin blanket of gas held to Earth by gravity. Almost all of it is nitrogen and oxygen; the small amounts of other gases and tiny floating particles control warming, ozone protection and air quality.

The major gases

The atmosphere is made up of major gases, each with its own relative abundance. Dry air is about **78% nitrogen**, **21% oxygen** and **0.9% argon**. Everything else is a **trace gas**: present at under 0.1% and usually measured in **parts per million (ppm)**.

Gas	Share of dry air	Role
Nitrogen, N_2	78.08%	Unreactive; enters the nitrogen cycle after fixation
Oxygen, O_2	20.95%	Respiration, combustion; made by photosynthesis
Argon, Ar	0.93%	Inert; no biological or climate role
Carbon dioxide, CO_2	≈0.042% (≈420 ppm)	Greenhouse gas; used in photosynthesis
Neon, helium, methane, N_2O, ozone	ppm or less	Methane and N_2O warm the planet; ozone absorbs UV

Variable gases

- **Water vapor** ranges from almost 0% (cold deserts, polar air) to about 4% (warm tropical air). It is the most abundant greenhouse gas and is the source of clouds and precipitation. Warm air can hold much more water vapor than cold air.

- **Ozone (O_3)** is concentrated in the stratosphere, where it absorbs ultraviolet radiation. Near the ground it is a pollutant.
- **Carbon dioxide** has climbed from about 280 ppm before industrialization to about 420 ppm today, mainly from burning fossil fuels.

Aerosols

An **aerosol** is a tiny solid or liquid particle suspended in air. Natural sources: dust, sea salt, volcanic ash, pollen, wildfire smoke. Human sources: soot from engines and fires, sulfates from burning coal.

- Light-colored aerosols (sulfates, volcanic particles) **reflect sunlight** and cool the surface; dark soot **absorbs sunlight** and warms the air.
- Aerosols are **condensation nuclei**: water vapor condenses onto them to form cloud droplets.
- Fine particles (PM2.5) lodge deep in the lungs and harm health.

KEY DEFINITIONS

Trace gas — a gas present in very small amounts, yet often with a large effect. **ppm** — parts per million; 1% = 10,000 ppm. **Aerosol** — a solid or liquid particle suspended in air.

FORMULA & MATH SKILL

Volume of a gas = total air volume × (percent ÷ 100). **ppm** = percent × 10,000. **Percent change** = (new − old) ÷ old × 100.

COMMON STUDENT MISTAKES

- Saying air is mostly oxygen. It is nearly four-fifths nitrogen.
- Treating water vapor as a fixed gas. It varies from place to place and hour to hour.

CLASSROOM EXAMPLE SG-C2-E001

A sealed classroom measures 240 m³. Its air is 78.08% N_2 and 0.0420% CO_2 by volume. Before industrialization, CO_2 made up 280 ppm of air. (a) Express the present CO_2 level in ppm. (b) Calculate the volume of CO_2 in the room in m³, to the nearest 0.001 m³. (c) Calculate the volume of N_2 in the room in m³, to the nearest 0.1 m³. (d) Calculate the percent increase in the CO_2 level since industrialization. (e) Identify which of the gases N_2, CO_2 and water vapor changes most from place to place.

QUICK CONCEPT SUMMARY

- Dry air is ≈78% N_2, ≈21% O_2, ≈0.9% Ar, with trace gases (CO_2 ≈420 ppm).
- Water vapor and ozone vary; trace gases and aerosols have large effects on climate and health.

2.2 Structure

CONCEPT MADE SIMPLE

Scientists divide the atmosphere into five layers. The layers are not different mixtures of gas. They are separated by whether the air gets colder or warmer as you climb.

Layers defined by temperature

The layers of the atmosphere are based on **temperature gradients**. Where the temperature trend changes direction, one layer ends and the next begins.

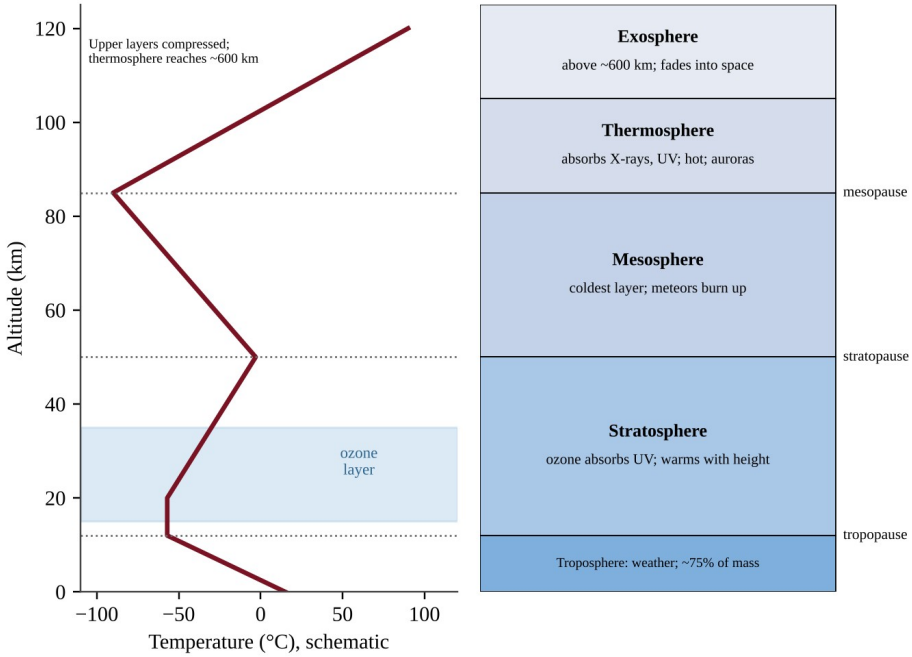


Figure 2.1 — Atmospheric layers with a schematic temperature profile (layer heights are approximate and the upper layers are compressed)

- **Troposphere** (surface to about 12 km; thinner at the poles, thicker at the equator): temperature **falls** with height, by about 6.5°C per km on average. It holds about 75% of the atmosphere's mass and nearly all water vapor, so it is where weather happens.
- **Stratosphere** (about 12–50 km): temperature **rises** with height because the ozone layer (roughly 15–35 km) absorbs ultraviolet radiation. This warm-over-cold arrangement is stable, so there is little vertical mixing and little weather.
- **Mesosphere** (about 50–85 km): temperature **falls** again, reaching the coldest values in the atmosphere (about −90°C). Most meteors burn up here.
- **Thermosphere** (about 85–600 km): temperature **rises** sharply as sparse molecules absorb high-energy X-rays and short-wave UV. Auroras occur here, and the International Space Station orbits in this layer.
- **Exosphere:** the outermost region, where molecules are so few that they drift off into space.

The boundary between layers is named with the suffix *-pause*: the tropopause, stratopause and mesopause.

FORMULA & MATH SKILL

Average lapse rate = (T at the lower altitude − T at the higher altitude) ÷ (altitude difference in km) Units: °C per km. In the troposphere the typical value is about 6.5°C/km.

Reading the profile

On a temperature–altitude graph, the turning points of the line are the tropopause, stratopause and mesopause. A line sloping left going up means air cools with altitude. The stratosphere warms because ozone absorbs UV; the troposphere cools because it is heated from the ground below.

Pressure. Air pressure and density fall steadily with altitude in every layer.

Why particles linger in the stratosphere. Warm air lies above cooler air, so the layer is stable, with little mixing and no rain to wash particles out. Volcanic aerosols there can stay a year or more and, by reflecting sunlight (raising albedo), cool the surface; tropospheric particles last days to weeks. A **thermal**

inversion is different: a local warm layer in the lower troposphere that traps cooler air and pollutants beneath it.

COMMON STUDENT MISTAKES

- Thinking the layers are separated by gas composition. They are separated by temperature trends.
- Thinking the ozone layer is in the troposphere. It is in the stratosphere; ozone near the ground is a pollutant.
- Thinking it is always colder higher up. The stratosphere and thermosphere get warmer with altitude.

AP EXAM STRATEGY

On a temperature–altitude graph, mark the turning points first, then name each section by its trend (cool, warm, cool, warm). Link each warm layer to its absorber: ozone (stratosphere), high-energy radiation (thermosphere).

CLASSROOM EXAMPLE SG-U4-E005

A weather balloon records these temperatures: 0 km, 18°C; 4 km, −8°C; 8 km, −34°C; 12 km, −60°C; 16 km, −60°C; 25 km, −47°C; 50 km, −3°C. (a) Calculate the average rate of temperature change (°C per km) from 0 to 12 km. (b) Identify the layer from 0 to 12 km and the layer from 16 to 50 km. (c) Explain, using a specific gas, why the temperature trend reverses at about 12–16 km. (d) State the altitude at which the tropopause is located in these data.

QUICK CONCEPT SUMMARY

- Troposphere (cools, weather), stratosphere (warms, ozone), mesosphere (cools), thermosphere (warms), exosphere.

2.3 Weather and Climate

CONCEPT MADE SIMPLE

Weather is what the air is doing today. Climate is what the weather is usually like in a place over many years. The sun sets the basic climate for each latitude, but mountains, oceans and elevation change it.

Weather versus climate

Weather is the state of the atmosphere at a place over a short time (minutes to days): temperature, humidity, wind, clouds and precipitation. **Climate** is the long-term pattern of weather in a region, usually averaged over at least 30 years. A cold week in a desert is weather; its low rainfall year after year is climate.

Temperature and precipitation

Temperature depends mainly on **latitude** (sun angle), then on elevation, nearness to oceans and ocean currents. Precipitation forms when air rises, expands and cools until its water vapor condenses into droplets large enough to fall. Air is lifted by ground heating (convection), by a mountain (orographic lift) or at a front.

Air masses and fronts

An **air mass** is a huge body of air with fairly uniform temperature and humidity. A **front** is the boundary between two air masses. Warm air is less dense, so it is always lifted over colder, denser air at a front, producing clouds and precipitation.

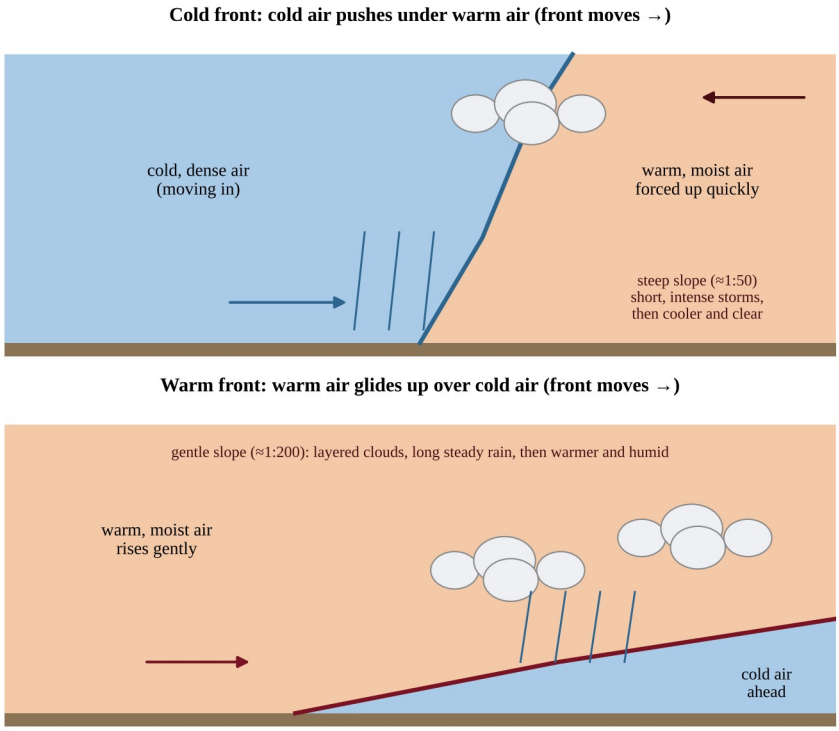


Figure 2.2 — Cold front and warm front in cross-section

At a **cold front** cold air wedges under warm air and lifts it steeply: a narrow band of heavy showers and thunderstorms, then cooler, clearer air. At a **warm front** warm air slides gently up over cold air: a wide area of layered clouds and steady rain, then warmer, humid air. Fronts move west to east in the westerlies, which is why mid-latitude weather changes every few days.

Geologic and geographic factors

Weather and climate are affected not only by the sun's energy but by geologic and geographic factors, such as mountains and ocean temperature.

- **Elevation:** air temperature drops about 6.5°C per km of elevation, so mountain tops are cold even in the tropics.
- **Proximity to oceans:** water heats and cools more slowly than land (high specific heat), so coastal climates are milder, with smaller temperature ranges, than inland ones at the same latitude.
- **Ocean currents:** surface currents move heat across the globe. Warm currents (the Gulf Stream) warm coasts such as western Europe; cold currents (the California Current) cool coasts and favor fog. Air over cold water holds little moisture, one reason for some coastal deserts.
- **Mountains:** force air upward and shape where precipitation falls.

Rain shadows

A **rain shadow** is a region of land that has become drier because a higher elevation area blocks precipitation from reaching the land.

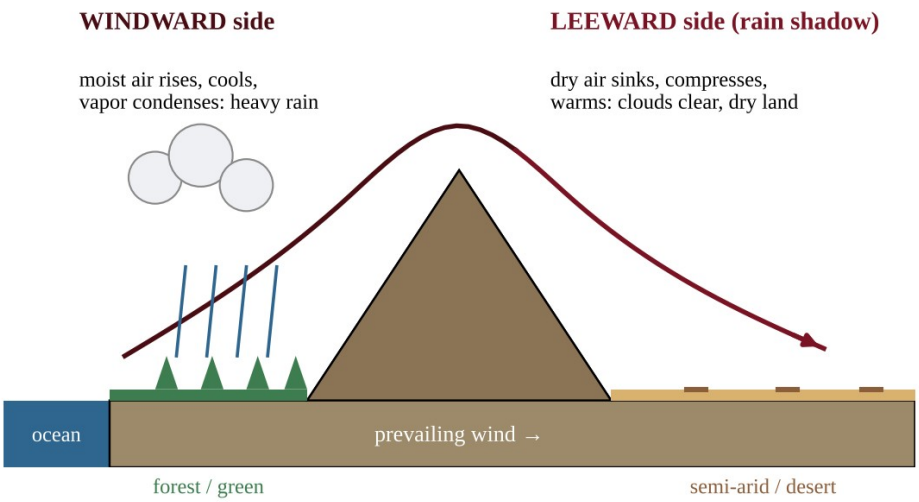


Figure 2.3 — A rain shadow: wet windward side, dry leeward side

PROCESS & CAUSE-AND-EFFECT

Moist air blows toward the mountains (windward side) → forced to rise → expands and cools (about 10°C per km when dry, about 5–6°C per km once saturated) → water vapor condenses into clouds → heavy rain or snow on the windward slope → air crosses the crest, now dry → descends on the leeward side → compresses and warms → clouds evaporate, little rain → dry leeward desert (rain shadow).

Examples. Pacific air rises over the Cascades and Sierra Nevada: western slopes are lush rain forest, while eastern Washington and the Great Basin of Nevada are semi-arid or desert. The Atacama Desert lies in the shadow of the Andes. Farmers on the dry side cope with drought-tolerant crops, drip irrigation or stored runoff.

On a map, find the prevailing wind (Section 2.4), then the mountains: wet side upwind, dry side downwind.

West coast versus east coast. In the middle latitudes the westerlies carry ocean air onto west coasts (Pacific Northwest, western Europe), giving mild winters, cool summers and a small temperature range. East coasts receive air that has crossed the continent, so winters are colder and summers warmer. Global climate is set by solar input, atmospheric gases, the layout of land and ocean, and Earth's rotation and orbit; a thermal inversion is local weather, not a driver of climate.

Reading a climatogram

A **climatogram** (climate graph) shows monthly precipitation as bars (mm) and monthly mean temperature as a line (°C) for one place.

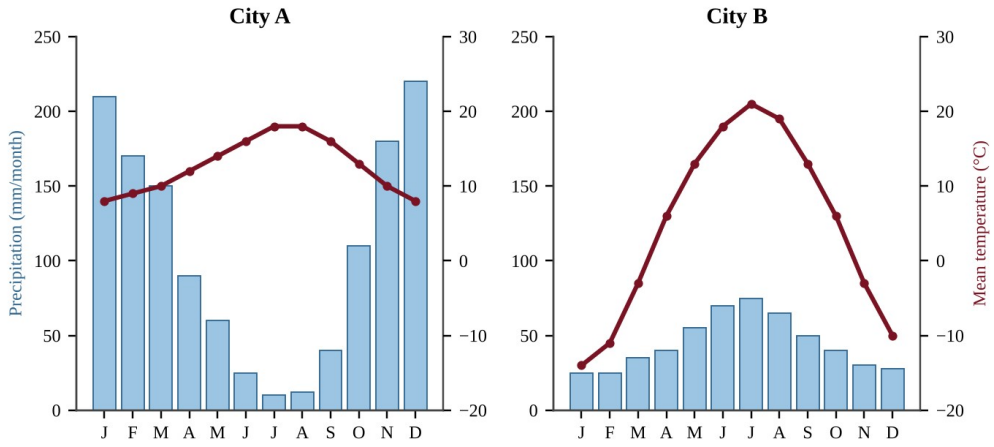


Figure 2.4 — Climatograms of two cities: City A (mild, wet winters, dry summers) and City B (cold winters, warm summers, modest rain)

- **Annual precipitation** = the sum of the 12 bars.
- **Annual temperature range** = warmest month − coldest month. A large range suggests an inland site; a small range a coastal one.
- Look for a **dry season**. City A has a wet winter and dry summer (a west coast in the westerlies); City B's large range points inland.
- For "A is X% less than B," use (B − A) ÷ B × 100, with B (the one after "than") as the base.

COMMON STUDENT MISTAKES

- Calling one hot summer "climate." One season is weather; climate is a 30+ year average.
- Putting the desert on the windward side. The rain shadow is *behind* the mountain (leeward).
- Saying the mountain "makes the air dry by itself." The air loses its moisture as rain on the way up; descending air then warms and cannot easily form clouds.

AP EXAM STRATEGY

Use "up, cool, rain; down, warm, dry," and state the **wind direction** and which side is windward. For a climatogram, show the numbers you used (sum of bars, warmest − coldest month).

CLASSROOM EXAMPLE SG-U4-E009

A mountain range runs north–south in a region where the prevailing winds blow from the west. City M lies on the western foot of the range at 100 m elevation and receives 1,850 mm of precipitation per year. City N lies 60 km away on the eastern foot of the range, also at about 100 m elevation, and receives 420 mm per year. (a) Calculate how many times greater City M's precipitation is than City N's, to the nearest tenth. (b) Identify which city is in the rain shadow. (c) Explain the sequence of events in the air that causes the difference. (d) Predict how the vegetation at City N would differ from City M, and state one reason a coastal city at the same latitude would have a smaller annual temperature range than an inland one.

CLASSROOM EXAMPLE SG-C2-E002

Climate data for two cities at the same latitude and elevation. City P monthly precipitation (mm), January to December: 150, 120, 100, 70, 45, 20, 8, 10, 30, 80, 130, 160. City P monthly mean temperature ranges from 9°C (January) to 19°C (August). City Q monthly precipitation (mm), January to December: 30, 30, 40, 45, 60, 75, 80, 70, 55, 45, 35, 30. City Q monthly mean temperature ranges from −12°C (January) to 22°C (July). (a) Calculate the annual precipitation of each city. (b) Calculate the annual temperature range of each city. (c) Calculate how much less annual precipitation City Q receives than City P, as a percent of City P's, to the nearest percent. (d) Identify which city is more likely coastal and give one reason from the data. (e) Describe City P's dry season.

QUICK CONCEPT SUMMARY

- Weather is short-term; climate is the 30+ year pattern.
- Warm air lifted over cold air at fronts makes clouds and rain; cold fronts bring short, intense storms, warm fronts long, steady rain.
- Mountains, elevation, ocean temperature and currents modify the climate set by the sun; a rain shadow forms on the leeward side of high ground.

2.4 Atmospheric Circulation—Pressure

CONCEPT MADE SIMPLE

The equator gets the most direct sunlight, so air there heats up and rises, spreads toward the poles, cools and sinks. Because Earth spins, the flows are bent sideways, making the steady winds sailors have used for centuries.

Pressure and wind

Air pressure is the weight of the air above a point (about 1,013 hPa at sea level). Warm air is less dense, rises and leaves **low pressure** at the surface; cold air sinks and makes **high pressure**. **Wind is air moving from high toward low pressure.** Lows bring clouds and rain (rising air cools); highs bring clear skies (sinking air warms and dries). In the Northern Hemisphere winds spiral **counterclockwise into a low** and **clockwise out of a high** (reversed in the Southern).

The cause: uneven heating

Global wind patterns primarily result from the most intense solar radiation arriving at the equator, resulting in density differences and the Coriolis effect. Air warmed at the equator expands, becomes less dense and **rises**, creating a belt of low pressure. Rising air cools and its water vapor condenses, so the equator is wet. This belt of converging air is the **intertropical convergence zone (ITCZ)**, which tracks the overhead sun north and south through the year. The now-dry air moves poleward aloft and by about 30° latitude has cooled enough to **sink**, forming high pressure. Sinking air warms and dries, so many great deserts lie near 30° N and S (Sahara, Arabian, Kalahari, Australian interior).

Three cells in each hemisphere

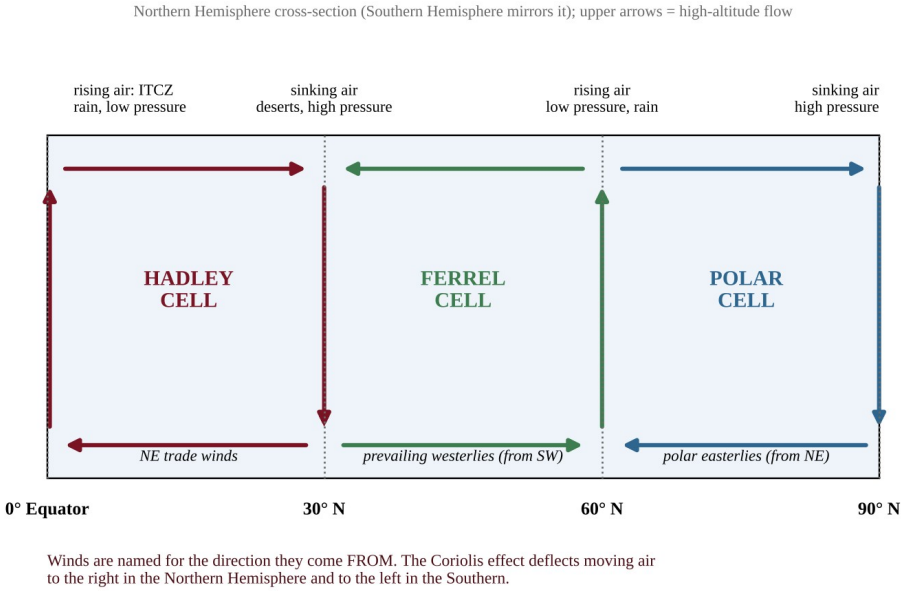


Figure 2.5 — Hadley, Ferrel and Polar cells in the Northern Hemisphere

- The **Hadley cell** (0°–30°): air rises at the equator, flows poleward aloft, sinks near 30° and returns along the surface.
- The **Polar cell** (60°–90°): cold air sinks over the poles, flows equatorward at the surface, rises near 60° and returns aloft.
- The **Ferrel cell** (30°–60°) is driven by the other two: surface air flows poleward, rises near 60° and returns aloft toward 30°.

Rising air means low pressure, clouds and rain (near 0° and 60°). Sinking air means high pressure and dry conditions (near 30° and 90°).

The Coriolis effect

Earth rotates eastward, and points near the equator move faster (about 1,670 km/h) than points near the poles (almost zero). Moving air keeps the eastward speed of where it started, so it appears to curve relative to the ground: **to the right in the Northern Hemisphere and to the left in the Southern Hemisphere**. This apparent deflection is the **Coriolis effect**. It turns the surface flows as follows.

Latitude	Wind belt	Blows from	Hemisphere detail
0°–30°	Trade winds	East (NE in north, SE in south)	Air moving toward equator curves westward
30°–60°	Prevailing westerlies	West (SW in north, NW in south)	Air moving poleward curves eastward
60°–90°	Polar easterlies	East (NE in north, SE in south)	Air moving equatorward curves westward

Winds are named for the direction they come **from**. Storms in middle latitudes generally move west to east, carried by the westerlies.

If Earth did not rotate, air would rise at the equator, sink at the pole and return in one cell per hemisphere, with surface winds blowing straight from the poles to the equator. Rotation breaks the flow into three cells and bends the winds.

COMMON STUDENT MISTAKES

- Saying winds blow from the *direction they go to*. A westerly comes from the west.
- Saying the Coriolis effect *causes* wind. Pressure differences cause wind; Coriolis only turns it.
- Putting deserts at the equator. They are mainly near 30° where air sinks.

AP EXAM STRATEGY
Start at the equator and work poleward: rising (low, wet), sinking at 30° (high, dry), then add the deflection (right in the north, left in the south).

CLASSROOM EXAMPLE SG-U4-E006
A sailing ship leaves the coast of West Africa at about 15° N, bound for the Caribbean. (a) State the name and the direction of origin (compass direction the wind comes from) of the prevailing wind that helps it. (b) Explain, using the Coriolis effect, why that wind does not blow directly from the north. (c) Later, a storm system forms near 45° N over the North Atlantic. State the direction in which it is most likely to be carried and the global wind belt responsible. (d) Explain why a large desert is expected near 30° N on the African continent.

QUICK CONCEPT SUMMARY
- Strongest solar heating at the equator drives rising air and a global circulation; wind blows from high to low pressure.
- Hadley (0–30°), Ferrel (30–60°) and Polar (60–90°) cells; rising air = low pressure and rain, sinking air = high pressure and dry.
- Coriolis deflects air right in the north, left in the south, producing trades, westerlies and polar easterlies.

2.5 Hurricanes, Tornadoes, and Cyclones

CONCEPT MADE SIMPLE
A hurricane is a huge spinning storm, hundreds of kilometers wide, that feeds on warm ocean water. A tornado is a narrow, violent column of rotating air, usually under a kilometer wide, that comes out of a thunderstorm over land.

How a tropical cyclone forms

A **tropical cyclone** is a large, rotating storm that forms over warm tropical oceans. It needs:

1. **Warm surface water**, at least about 26.5°C to roughly 50 m deep.
2. **A starting disturbance**, such as thunderstorms moving west off Africa.
3. **Low wind shear**, so the storm's cloud tower is not torn apart.
4. **The Coriolis effect**, so storms do not form within about 5° of the equator.

PROCESS & CAUSE-AND-EFFECT
Warm ocean evaporates water → moist air rises and condenses → latent heat is released, warming the air and making it rise faster → surface pressure falls → winds rush in toward the low → Coriolis deflection makes them spiral (counterclockwise in the Northern Hemisphere, clockwise in the Southern) → more evaporation → storm strengthens until it moves over land or cold water.

By sustained wind speed it is a **tropical depression** (under 63 km/h), a **tropical storm** (63–118 km/h) or, from 119 km/h, a hurricane, typhoon or cyclone. It weakens over land (no warm water, more friction) or cooler water. Lower central pressure means stronger winds, so pressure rises as a storm weakens.

Damaging winds and surge arrive hours before the eye: arrival time = distance ÷ speed, minus that lead time.

Structure

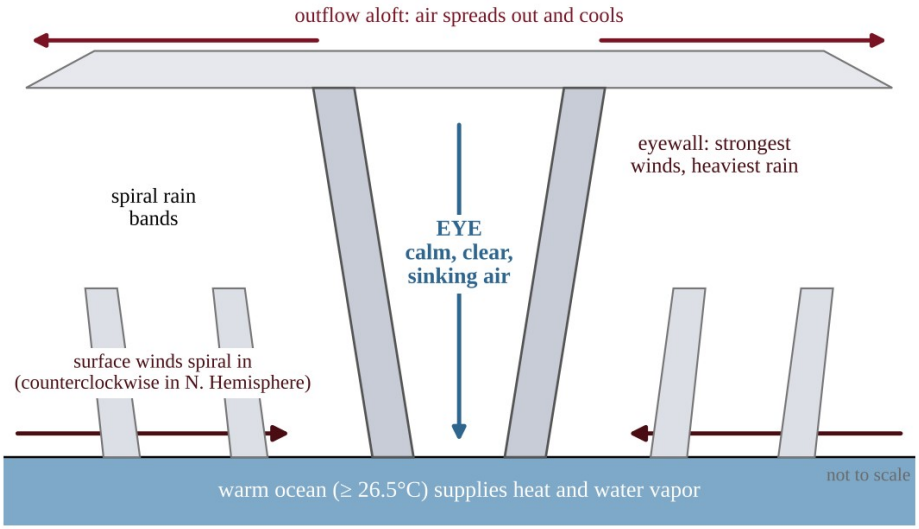


Figure 2.6 — Cross-section of a hurricane

- The **eye** (20–60 km across) is calm and often clear, with sinking air and the lowest pressure.
- The **eyewall** is a ring of towering thunderstorms around the eye; it has the strongest winds and heaviest rain.
- **Spiral rain bands** extend outward for hundreds of kilometers; air spirals in near the surface and flows out at the top.

Regional names

The same kind of storm has different names depending on where it forms.

Name	Where it forms
Hurricane	North Atlantic and the eastern and central North Pacific
Typhoon	Northwestern Pacific (near Japan, the Philippines, China)
Cyclone	South Pacific and Indian Ocean (near Australia, India, Madagascar)

Saffir–Simpson hurricane wind scale

The **Saffir–Simpson scale** rates hurricanes from Category 1 to 5 by **sustained wind speed**. It does not measure rain or storm surge.

Category	Sustained wind (mph)	Sustained wind (km/h)	Typical damage
1	74–95	119–153	Some roof and tree damage
2	96–110	154–177	Major roof damage
3	111–129	178–208	Devastating; roofs and walls lost
4	130–156	209–251	Catastrophic damage
5	157 or more	252 or more	Most buildings destroyed

Storm surge and other hazards

Storm surge is the rise in sea level above the normal tide, pushed onto land mainly by the storm's winds (low pressure adds a little). It is the greatest killer in hurricanes. It is worst on coasts with a shallow, gently sloping seafloor and, in the Northern Hemisphere, on the right-front side of the storm, where wind

and forward motion add together. Other hazards are wind, heavy rain with inland flooding, and tornadoes. Higher sea level lets the same surge reach farther inland. (In 2005, Hurricane Katrina's surge overtopped the levees at New Orleans.)

Tornadoes

A **tornado** is a violently rotating column of air touching the ground, spawned by a thunderstorm, usually a **supercell** (a thunderstorm with a rotating updraft). Rotation develops when warm, moist surface air meets cold, dry air aloft and winds change speed and direction with height (wind shear). A rotating cloud not touching the ground is a **funnel cloud**. The central United States ("Tornado Alley") has the most, mainly in spring.

The **Enhanced Fujita (EF) scale** rates a tornado from EF0 to EF5 by the **damage** it does, which is used to estimate wind speed.

Rating	Estimated wind (mph)	Typical damage
EF0	65–85	Light: branches broken
EF1	86–110	Moderate: roofs stripped
EF2	111–135	Considerable: roofs torn off houses
EF3	136–165	Severe: walls torn off houses
EF4	166–200	Devastating: houses leveled
EF5	over 200	Incredible: strong houses swept away

COMMON STUDENT MISTAKES

- Thinking the calm eye means the storm is over. The second half of the storm, with winds from the opposite direction, follows.
- Rating a tornado by wind speed alone. The EF scale uses damage.

CLASSROOM EXAMPLE SG-C2-E003

A hurricane in the Atlantic has sustained winds of 185 km/h. Its eye is 420 km from the coast and is moving toward it at a steady 14 km/h. Use 1 mph = 1.609 km/h. (a) Convert the wind speed to mph, to the nearest whole number. (b) Use the Saffir–Simpson table to identify the hurricane's category. (c) Calculate how many hours until the eye reaches the coast. (d) Explain why the storm will weaken soon after it moves inland.

QUICK CONCEPT SUMMARY

- Tropical cyclones form over warm (≥ 26.5°C) water with low wind shear and spin because of the Coriolis effect; hurricane, typhoon and cyclone are the same storm in different oceans.
- Hurricanes are rated by wind (Saffir–Simpson, 1–5); storm surge is the greatest hazard.
- Tornadoes are narrow and short-lived, form in supercells, and are rated by damage (EF0–EF5).

2.6 El Niño—Southern Oscillation (ENSO)

CONCEPT MADE SIMPLE

Every few years the Pacific's surface temperature pattern flips for a while. In an El Niño a big pool of warm water slides east and the rain follows it; in a La Niña the cold water along the eastern Pacific gets even colder. Because the Pacific is so large, weather changes worldwide.

Normal (neutral) conditions

Normally the **trade winds** blow east to west across the tropical Pacific, piling a deep warm pool near Indonesia and Australia. Off South America, surface water moves away and cold, nutrient-rich water **upwells**, feeding plankton and fisheries. Warm, moist air rises over the western Pacific (heavy rain) and sinks in the east, which stays drier.

El Niño

El Niño and La Niña are phenomena associated with changing ocean surface temperatures in the Pacific Ocean, and together with the linked atmospheric pressure shift (the Southern Oscillation) make up **ENSO**. In an **El Niño**, the trade winds weaken or even reverse. The warm pool spreads east, the boundary between warm and cold water (the **thermocline**) flattens, and upwelling off South America weakens or stops. Evaporation and rising air shift toward the central and eastern Pacific.

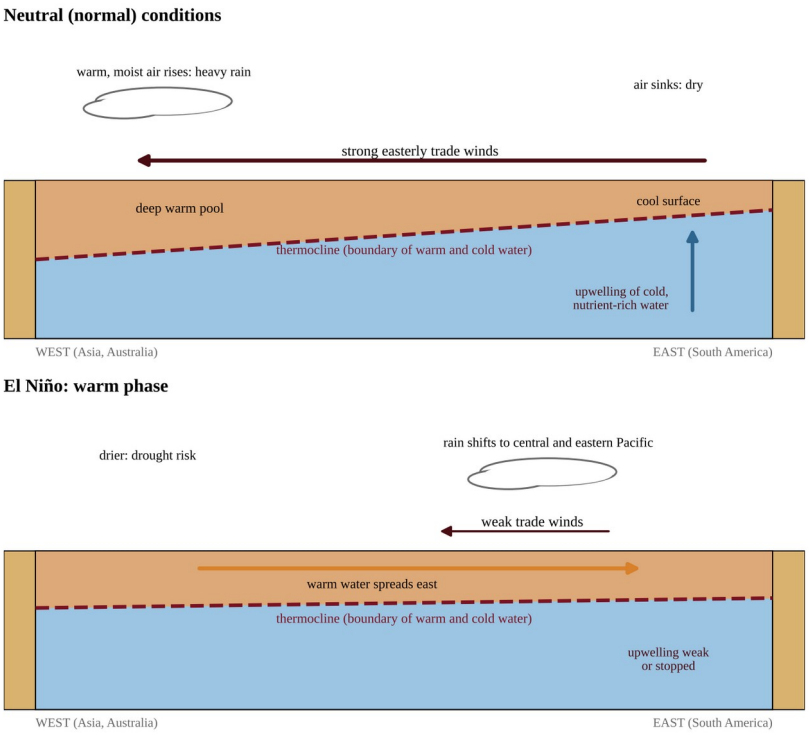


Figure 2.7 — Pacific cross-section: neutral conditions and El Niño

La Niña

In a **La Niña**, the trade winds strengthen, more warm water is pushed west, upwelling off South America intensifies, and the eastern Pacific becomes cooler than normal.

Effects

These phenomena can cause global changes to rainfall, wind and ocean circulation patterns, but effects differ by place. The typical tendencies are:

Region	El Niño	La Niña
Coastal Peru and Ecuador	Heavy rain, floods; fish catch collapses (upwelling lost)	Drier; strong upwelling, rich fishing

Region	El Niño	La Niña
Indonesia and Australia	Drought, wildfire risk	Heavier rain, flooding
Southern United States	Wetter, cooler winter	Drier, warmer winter, drought risk
Northwestern United States	Warmer, often drier winter	Cooler, wetter winter
Atlantic hurricanes	Often fewer (more wind shear)	Often more
Global average temperature	Tends to rise	Tends to dip

Other impacts include coral bleaching and crop failures. ENSO events recur every two to seven years. ENSO reaches North America by shifting the **jet stream**, the fast high-altitude river of air that steers storms.

ENSO is influenced by geological and geographic factors: the size and shape of the Pacific basin, the position of continents and islands, the sloping thermocline, and the trade winds and currents shaped by them. Because each region sits differently relative to ocean and mountains, the same event can bring flood to one place and drought to another.

PROCESS & CAUSE-AND-EFFECT

El Niño: weaker trade winds → warm water flows east → thermocline deepens in the east → less upwelling → warm surface water in the eastern Pacific → rain shifts east → drought in the west Pacific, flooding on the Peru coast, altered weather worldwide. **La Niña:** stronger trades → more upwelling in the east → cooler eastern Pacific → rain concentrated in the west → opposite regional effects.

Peru's anchovy fishery depends on cold upwelling; in an El Niño the lost nutrients lower phytoplankton, then anchovies, then seabirds.

FORMULA & MATH SKILL

Anomaly = observed value − long-term average. Positive = warmer than normal; a larger anomaly means a stronger event. **Mean** = sum ÷ number of values. Classify with the rule given (for example, ≥ +0.5°C El Niño, ≤ −0.5°C La Niña, otherwise neutral), then connect it to the expected effects.

COMMON STUDENT MISTAKES

- Mixing up the phases: El Niño is **warm**, La Niña **cool**.
- Assuming every region gets the same effect.
- Reading an anomaly as the actual temperature. It is the difference from average.
- On exam answers, give the cause chain (trade winds → warm water → upwelling → rainfall) and a located effect.

CLASSROOM EXAMPLE SG-U4-E010

A monitoring program reports the sea-surface temperature anomaly (difference from the long-term average) in the eastern tropical Pacific for four years, along with the anchovy catch off Peru. Year 1: +2.1°C, catch 4 million tonnes. Year 2: −0.3°C, catch 9 million tonnes. Year 3: −1.6°C, catch 11 million tonnes. Year 4: +0.2°C, catch 8 million tonnes. Use this rule: an anomaly of +0.5°C or more is El Niño, −0.5°C or less is La Niña, anything between is neutral. (a) Classify each year. (b) Calculate the percent change in catch from Year 2 to Year 1, to the nearest whole percent. (c) Explain, in terms of trade winds and upwelling, why the catch was lowest in the year you identified as El Niño. (d) Predict the rainfall effect in Indonesia in Year 1 and in Year 3.

QUICK CONCEPT SUMMARY

- ENSO is a Pacific ocean–atmosphere cycle: El Niño (warm, weak trades, less upwelling) and La Niña (cool, strong trades, more upwelling).
- Effects on rainfall, wind and ocean circulation spread globally but differ by place.
- Geology and geography (basin shape, thermocline, trade winds) influence ENSO.

2.7 Practice Questions and Answers

Now put what you have learned to work with the practice set for this chapter.

PRACTICE QUESTIONS AND ANSWERS

Classroom examples in this chapter (unsolved; space is provided in each box): SG-C2-E001, SG-U4-E005, SG-U4-E009, SG-C2-E002, SG-U4-E006, SG-C2-E003, SG-U4-E010.

Classified practice: Book Two, Chapter 2 — 23 multiple-choice questions (Section A) followed by 2 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 2. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

CHAPTER
3

UNIT I — EARTH SYSTEMS AND RESOURCES
Global Water Resources and Use

3.1 · Water Distribution and Properties
3.2 · Freshwater
3.3 · Aquifers
3.4 · Types of Ice
3.5 · Oceans
3.6 · Agricultural, Industrial, and Municipal Use
3.7 · Global Problems
3.8 · Practice Questions and Answers

LEARNING OBJECTIVES

- Describe where Earth's water is stored and explain why so little of it is usable fresh water.
- Explain how the structure of the water molecule gives water its special properties.
- Describe watersheds, aquifers, ice, and the ocean, and explain how water moves through each.
- Compare irrigation methods, and explain how agriculture, industry, and cities use water.
- Analyze global water problems and evaluate solutions such as conservation, dams, and desalination.

3.1 Water Distribution and Properties

CONCEPT MADE SIMPLE

Earth is a "water planet," yet almost all its water is salty and most fresh water is frozen. Water also behaves in unusual ways, and those oddities make life possible.

Where Earth's water is

About 71% of Earth's surface is ocean. The planet holds roughly 1.4×10^9 km³ of water, and about **97.5% of it is salt water**, mostly in the oceans. Only about **2.5% is fresh water**. Of that fresh water:

- about 69% is frozen in glaciers, ice caps, and ice sheets;
- about 30% is groundwater;
- only about 1% is everything else (permafrost, lakes, rivers, wetlands, soil moisture, air, and living things). Lakes and rivers together hold roughly 0.3% of the fresh water.

So the liquid fresh water that people reach easily, in lakes and rivers, is under 0.01% of all the water on Earth. Fresh water is also **unevenly spread**: some regions are water-rich, others (the Middle East, North Africa) have very little.

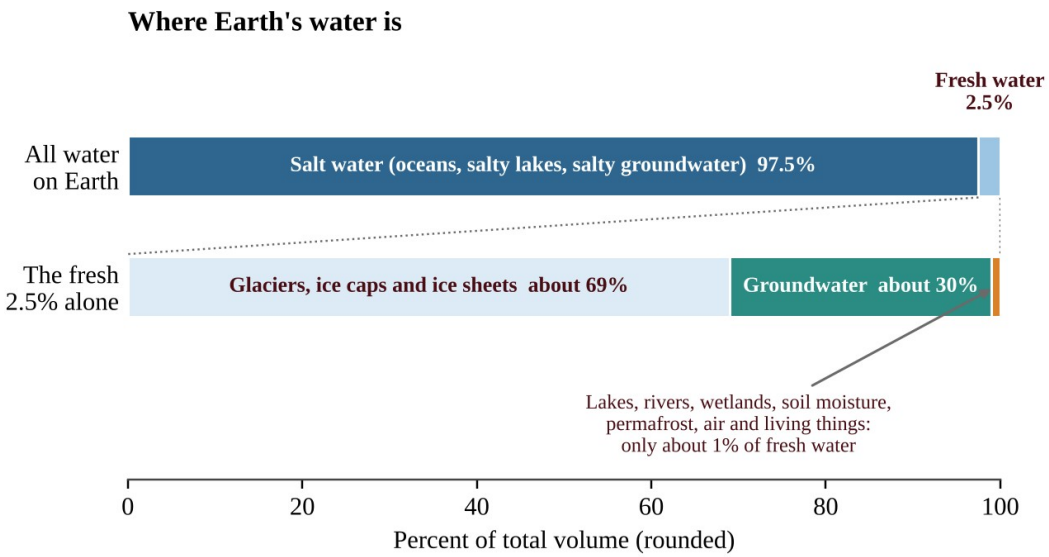


Figure 3.1 — Share of Earth's water that is salty or fresh, and where the fresh water is stored. Values are rounded.

Special properties of water

Water is made of two hydrogen atoms bonded to one oxygen atom in a bent shape. Oxygen pulls the shared electrons harder, so the oxygen end is slightly negative (δ−) and the hydrogen ends are slightly positive (δ+). A molecule with uneven charge is **polar**. The positive end of one molecule is attracted to the negative end of its neighbor, forming a weak **hydrogen bond**. Billions of these bonds explain nearly everything unusual about water.

Property	Cause	Why it matters
Excellent solvent	Polar molecules surround and pull apart ions and other polar substances	Carries nutrients in blood and sap; also carries pollutants and salts
Cohesion and adhesion	Hydrogen bonds hold water molecules to each other and to other surfaces	High surface tension; capillary action lifts water in plants and soil
High specific heat	Energy goes into breaking hydrogen bonds before molecules speed up	Oceans and lakes warm and cool slowly, moderating coastal climates and body temperature
High heat of vaporization	Many hydrogen bonds must break for water to evaporate	Evaporation (sweating, transpiration) cools; moving water vapor carries heat around the planet
Ice is less dense than liquid water	In ice, hydrogen bonds lock molecules into an open, six-sided lattice	Ice floats; lakes freeze from the top down, and the ice insulates the life below

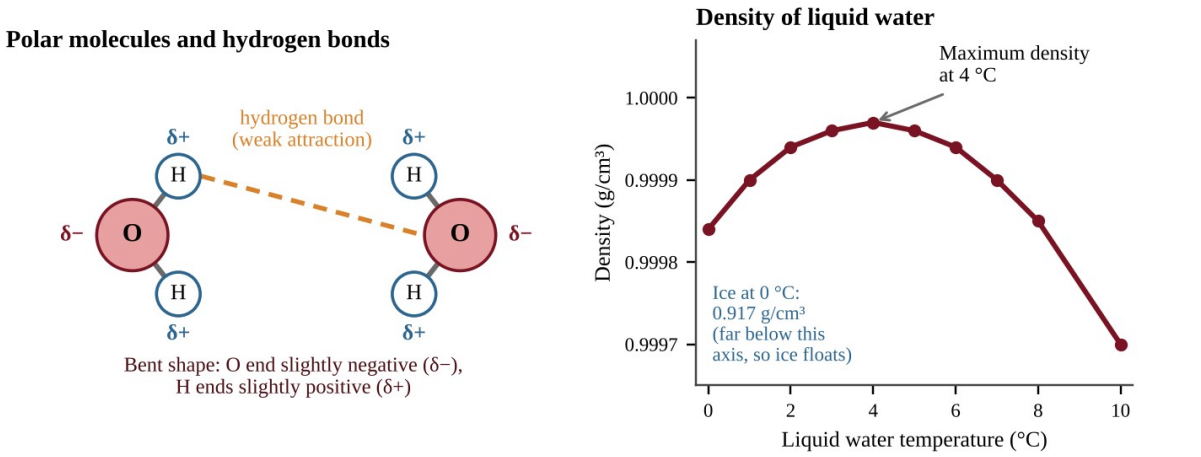


Figure 3.2 — Hydrogen bonding between polar water molecules (left) and the density of liquid water between 0 °C and 10 °C (right).

FORMULA & MATH SKILL

Heat energy Q = m × c × ΔT Q = heat in joules (J); m = mass in grams; c = specific heat in J/(g·°C); ΔT = temperature change in °C. For liquid water, c = 4.18 J/(g·°C). Dry sand is about 0.84 J/(g·°C). Rearrange: ΔT = Q ÷ (m × c).

COMMON STUDENT MISTAKES

- Saying water is densest as ice. Ice is *less* dense than liquid water; liquid water is densest at about 4 °C.
- Calling a hydrogen bond a bond *inside* a water molecule. Hydrogen bonds act *between* molecules; the bonds inside are covalent.

CLASSROOM EXAMPLE SG-C3-E001

Earth holds about 1.4 × 10⁹ km³ of water, of which 2.5% is fresh. Of the fresh water, about 69% is ice and about 30% is groundwater. Lakes and rivers together hold about 0.3% of the fresh water. (a) Calculate the volume of fresh water in km³. (b) Calculate the volume of groundwater, in km³, in scientific notation. (c) Calculate the volume of water in lakes and rivers, in km³, and express it as a percentage of all the water on Earth.

CLASSROOM EXAMPLE SG-C3-E002

A beaker contains 500 g of liquid water and a tray holds 500 g of dry sand, both starting at 20 °C. Each absorbs 20,000 J of heat. The specific heat of water is 4.18 J/(g·°C) and that of sand is 0.84 J/(g·°C). (a) Calculate the temperature change of the water. (b) Calculate the temperature change of the sand. (c) Use your answers to explain why a beach is much hotter than the nearby sea on a sunny afternoon.

QUICK CONCEPT SUMMARY

- 97.5% of Earth's water is salty; of the 2.5% that is fresh, about 69% is ice and about 30% is groundwater.
- Polarity and hydrogen bonds give water cohesion, high specific heat, high heat of vaporization, and a solid form (ice) that floats.
- These properties moderate climate, carry heat, dissolve substances, and protect aquatic life.

3.2 Freshwater

CONCEPT MADE SIMPLE

A watershed is the whole area of land that drains to one river, lake, or sea, like a giant bowl or funnel. Every raindrop that lands inside the rim ends up at the same outlet, carrying along whatever it picked up on the way.

Sources of fresh water

Surface water (rivers, lakes, reservoirs, wetlands) is easy to reach and is refilled by precipitation and runoff, but is exposed to pollution and evaporation. **Groundwater** is stored in soil and rock (Section 3.3). Using fresh water faster than it is renewed makes it behave like a nonrenewable resource.

What a watershed is

A **watershed** (drainage basin) is all the land that drains into a common water body. Its boundary, a ridge or high ground, is the **divide**, which separates it from adjoining watersheds. Small watersheds nest inside larger ones: a creek's watershed lies inside a river's, which lies inside a major basin such as the Mississippi.

Characteristics that control flow

Characteristic	Effect on water movement
Area	Larger area collects more rain and produces more total runoff and flow
Length	A long, narrow watershed delivers flow to the outlet more slowly and evenly than a short, round one
Slope	Steeper slopes speed runoff, increase erosion and give a quick, high peak flow
Soil	Permeable soils soak up water and recharge groundwater; clay or compacted soils increase runoff
Vegetation	Roots and leaf litter slow water, increase infiltration, reduce erosion; bare or paved ground does the opposite

Characteristic	Effect on water movement
Divides	Define which land contributes water and pollutants to the outlet

Natural cover and permeable soil mean most rain infiltrates and moves slowly as groundwater, giving a steady stream. Impervious surfaces (roofs, roads) force water into **runoff**, which reaches the stream quickly, raising peak flows, causing flash floods and carrying oil, fertilizer, sediment and trash. Pollutants flow downstream from wherever they enter, so only points below the entry are affected, and a watershed is the natural unit for managing water quality.

FORMULA & MATH SKILL

Runoff volume = precipitation depth × area × runoff coefficient (C) Precipitation depth in m, area in m^2 (1 $km^2 = 10^6$ m^2), result in m^3. The runoff coefficient C is the fraction of rain that leaves as surface runoff (0 to 1); forest has a low C, pavement a high C. On pavement or a roof, where nearly all rain runs off, volume ≈ depth × area (for example, 0.02 m of rain on a 30 m × 40 m roof gives 24 m^3). For mixed land cover, add the parts: depth × ($A_1C_1 + A_2C_2 + \ldots$). **Discharge (m^3/s) = volume (m^3) ÷ time (s)**; one year = 3.15×10^7 s. **Mass of material carried = concentration × volume**; 1 m^3 = 1,000 L, and 1 metric ton = 10^9 mg. Overflow = inflow − capacity. Percent change = (new − old) ÷ old × 100.

Reading hydrographs and land-use graphs

A **hydrograph** plots a stream's discharge (m^3/s) against time after a storm begins. Compare three features: the **peak** (highest discharge), the **lag time** (how long the peak takes to arrive) and the total area under the curve. A mostly paved watershed gives a taller, earlier, steeper peak than a forested one, because rain runs off instead of soaking in and being released slowly.

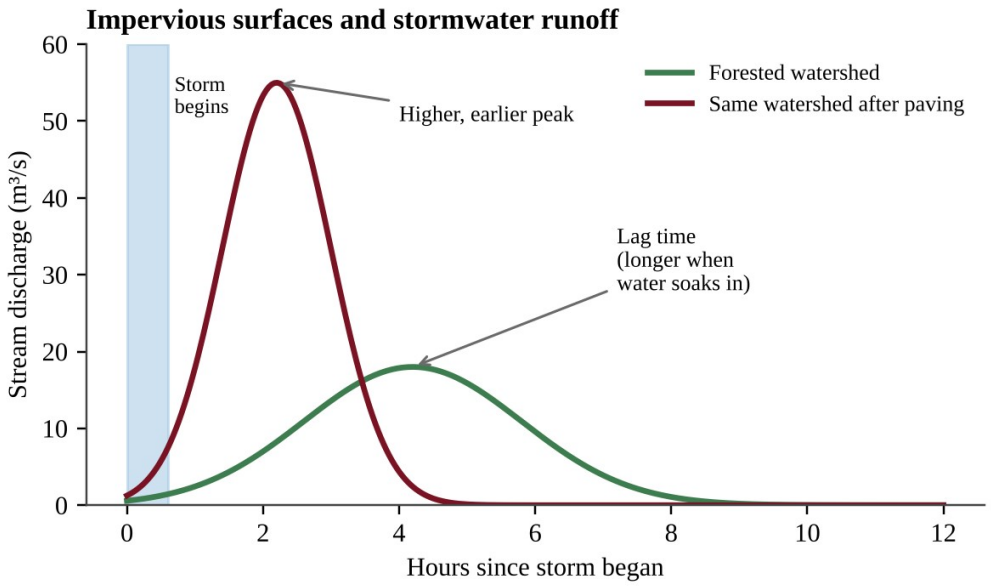


Figure 3.3 — Stream discharge during the same storm before and after a watershed is paved (invented data). Impervious surfaces create a higher, earlier flood peak.

CLASSROOM EXAMPLE SG-U4-E015

Birch Creek drains a 90 km² watershed that receives 1,200 mm of precipitation a year, with an average runoff coefficient of 0.35. Its water carries 80 mg/L of suspended sediment, and 15% of the watershed is cropland. (a) Calculate the annual runoff volume in m³. (b) Calculate the average discharge in m³/s. (c) Calculate the mass of sediment carried each year, in metric tons. (d) Assuming all sediment comes from the cropland and the soil's bulk density is 1.2 t/m³, calculate the average depth of soil lost from the cropland per year, in mm.

REAL-WORLD EXAMPLE

A town upstream converts forest to a shopping center and parking lots. The stream downstream floods more often and carries more sediment and road pollutants, because the larger impervious area increased runoff. Retention ponds, permeable pavement and restored wetlands reduce the effect.

COMMON STUDENT MISTAKES

- Treating a watershed boundary as a political border. It follows ridges, not county or state lines.
- Saying a larger watershed always means faster flooding. Steepness, soil and cover matter as much as size.

AP EXAM STRATEGY

For watershed questions, ask of each characteristic: does it speed up or slow down water? Clearing, paving and compaction raise runoff, flood peaks and pollution loads.

CLASSROOM EXAMPLE SG-U4-E007

A watershed covers 120 km² and receives 0.90 m of precipitation in a year. Before development its runoff coefficient is 0.30 everywhere. (a) Calculate the annual runoff volume in m³ before development, in scientific notation. (b) Later, 20 km² of the watershed is covered with pavement and rooftops (C = 0.85) while the other 100 km² keeps C = 0.30. Calculate the new annual runoff volume in m³. (c) Calculate the percent increase in runoff caused by development. (d) Explain, in terms of infiltration, why runoff increased, and name one watershed characteristic from this topic that changed.

QUICK CONCEPT SUMMARY

- Fresh water is renewable only as fast as the water cycle replaces it; surface water and groundwater are the two main sources.
- A watershed is all land draining to one outlet; divides separate adjoining watersheds.
- Area, length, slope, soil, vegetation and divides shape how fast and how cleanly water moves.
- More impervious area means more runoff, faster peaks and more pollutants.

3.3 Aquifers

CONCEPT MADE SIMPLE

An aquifer is not an underground lake. It is a soaked sponge of sand, gravel, or fractured rock, with water filling the spaces between grains. A well is a straw pushed into it.

What an aquifer is

Rain and snowmelt soak into the ground. Near the surface, pores hold air and water (the **unsaturated zone**); deeper down, every pore is full (the **saturated zone**). The top of the saturated zone is the **water table**. An **aquifer** is a body of rock or sediment that holds enough water, and lets it flow easily enough, to supply wells and springs. Two properties decide this:

- **Porosity** is the fraction of the rock that is empty space. Clay can have high porosity but its pores are tiny.
- **Permeability** is how easily water moves through the connected pores. Sand, gravel, sandstone, and fractured limestone are permeable; clay and unfractured granite are not.

A layer that water cannot easily pass through, such as clay, is an **aquitard** (or confining layer). Groundwater moves slowly (centimeters to a few meters per day). Some deep groundwater is thousands of years old (**fossil water**), so refilling it takes far longer than a human lifetime.

Unconfined and confined aquifers

Feature	Unconfined aquifer	Confined aquifer
Boundaries	Open to the surface above; sits on an aquitard	Sandwiched between two aquitards
Top of the water	The water table, which rises and falls	No water table; water is under pressure
Recharge	Rain soaks in directly above it	Only where the aquifer reaches the surface (the **recharge zone**), often far away
Well behavior	Water stands at the water table	Water rises above the aquifer, and may flow out as an **artesian** well
Vulnerability	Easily polluted from the surface; responds quickly to drought	Better protected, but recharges very slowly

Recharge is the water that moves down into an aquifer. It is greatest where soil is permeable and vegetation is intact, and least where land is paved or covered by clay.

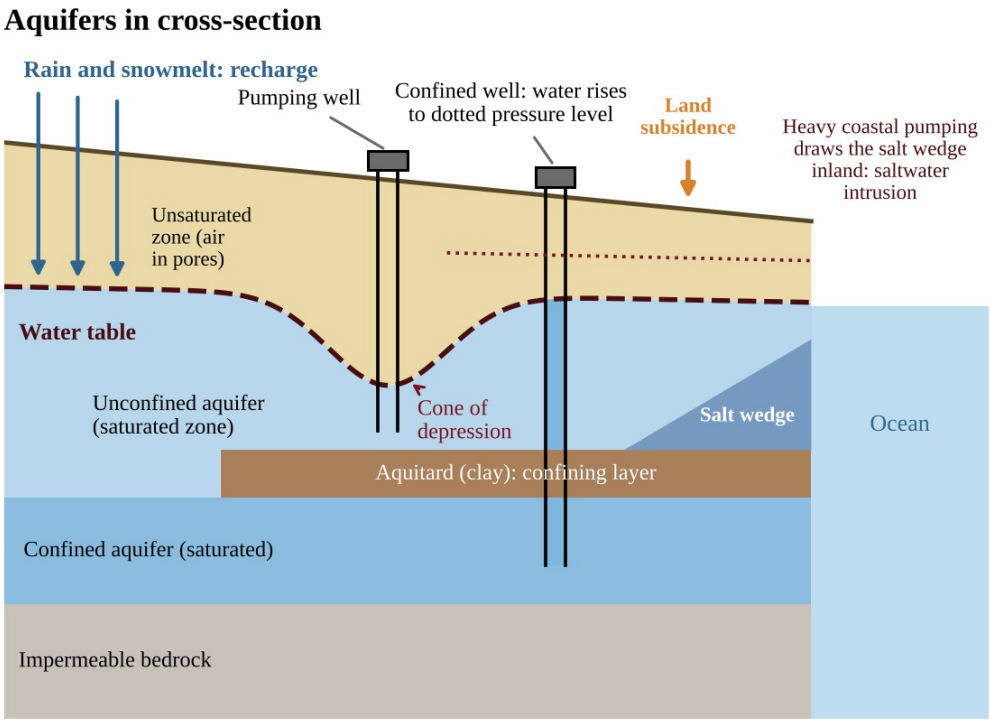


Figure 3.4 — Cross-section of an unconfined aquifer over a confined aquifer, with recharge, two wells, a cone of depression, a saltwater wedge at the coast, and land subsidence.

The Ogallala Aquifer

The **Ogallala (High Plains) Aquifer** lies beneath parts of eight central states, from South Dakota to Texas. It irrigates one of the world's most productive farming regions. Its recharge is slow, only a few millimeters to a few centimeters of water a year in the dry south, while farmers pump it much faster. In parts of Texas and Kansas the water table has dropped more than 30 m, and wells have run dry. Pumping a resource faster than it refills is **groundwater mining**.

Problems caused by pumping

- **Cone of depression.** Pumping draws water toward the well faster than it flows in, so the water table forms a cone-shaped dip around the well. Nearby shallow wells can run dry.
- **Saltwater intrusion.** Fresh water is less dense than salt water, so along a coast a lens of fresh water floats on top of a wedge of salt water. Pumping lowers the fresh water level, and the salt wedge moves inland and upward, so wells begin to produce salty water.
- **Land subsidence.** Water in the pores helps hold up the layers above. When water is pumped from a confined aquifer, the clay layers compress and the land sinks. This is mostly *permanent*, because compacted clay cannot rebound and the aquifer loses storage. Parts of California's Central Valley have sunk by more than 8 m, and Mexico City and Jakarta have sunk several meters.

FORMULA & MATH SKILL

Water stored = area × saturated thickness × porosity (drainable fraction) Area in m² (1 km² = 10^6 m²), thickness in m, porosity as a decimal; result in m³. **Net change = recharge − withdrawal;** years until depleted = water stored ÷ net annual loss (same units). **Change in storage = area × change in water-table height × specific yield.** Specific yield (a little less than porosity) is the fraction of the volume that actually drains. **Recharge = precipitation volume × infiltration fraction.** If pumping is the only outflow, pumping = recharge + loss from storage. A *depth to* the water table that increases means the table is falling.

CLASSROOM EXAMPLE SG-C3-E003

An unconfined aquifer covers 5,000 km² and has an average saturated thickness of 30 m. Its porosity (the fraction of the volume that can be drained) is 0.20. Farmers withdraw 1.5×10^9 m³ of water per year and natural recharge adds 3×10^8 m³ per year. (a) Calculate the volume of water stored, in m³. (b) Calculate the net change in storage per year. (c) Calculate the number of years until the aquifer is empty if both rates stay constant.

QUICK CONCEPT SUMMARY

- An aquifer is porous, permeable material saturated with water; unconfined aquifers have a water table, confined ones are under pressure beneath an aquitard.
- Recharge refills aquifers slowly; the Ogallala is being mined faster than it recharges.
- Overpumping causes a cone of depression, saltwater intrusion, and land subsidence.

3.4 Types of Ice

CONCEPT MADE SIMPLE

The key question about ice is whether it sits on land (melting adds water to the ocean) or already floats (melting changes almost nothing).

Land ice

- A **glacier** is a large, long-lasting mass of ice that flows slowly under its own weight. **Valley (alpine) glaciers** flow down mountain valleys; **ice caps** are smaller domes covering highlands.
- **Ice sheets** are continental-scale masses of ice, over 50,000 km². Only two exist today: **Antarctica** (about 90% of the world's ice) and **Greenland**.
- **Ice shelves** are thick floating extensions of an ice sheet, still attached to land. Where the shelf breaks off, the pieces are **icebergs**; this breaking is **calving**. Shelves act as a buttress that holds back glaciers on land, so a collapse adds little water at once but lets the grounded ice flow faster.

A glacier grows where more snow accumulates than melts. Its **mass balance** is **accumulation − ablation** (ablation is loss by melting, sublimation, and calving). A negative balance means the glacier is retreating. Glaciers store fresh water and release it slowly to rivers in summer. Many people rely on mountain meltwater, so retreating glaciers eventually mean less summer flow.

Floating ice

- **Sea ice** is frozen *sea water* floating on the ocean, mainly in the Arctic Ocean and around Antarctica. It is only a few meters thick. Arctic summer sea ice has shrunk since satellite records began in 1979. When sea water freezes, most of the salt is pushed into the water below, making it saltier and denser; this dense water helps drive deep ocean currents (Section 3.5).
- **Icebergs** are chunks of freshwater land ice. Because ice is about 92% as dense as liquid water (917 kg/m³ versus about 1,025 kg/m³ for sea water), about 90% of an iceberg lies below the surface.

Permafrost

Permafrost is ground (soil, rock, and ice) that stays frozen for at least two consecutive years. It covers about a quarter of Northern Hemisphere land. Only the top **active layer** thaws each summer. It stores huge

amounts of frozen organic carbon; as it thaws, microbes decompose it and release **carbon dioxide and methane**, which adds to warming (a positive feedback). Thaw also collapses buildings and roads.

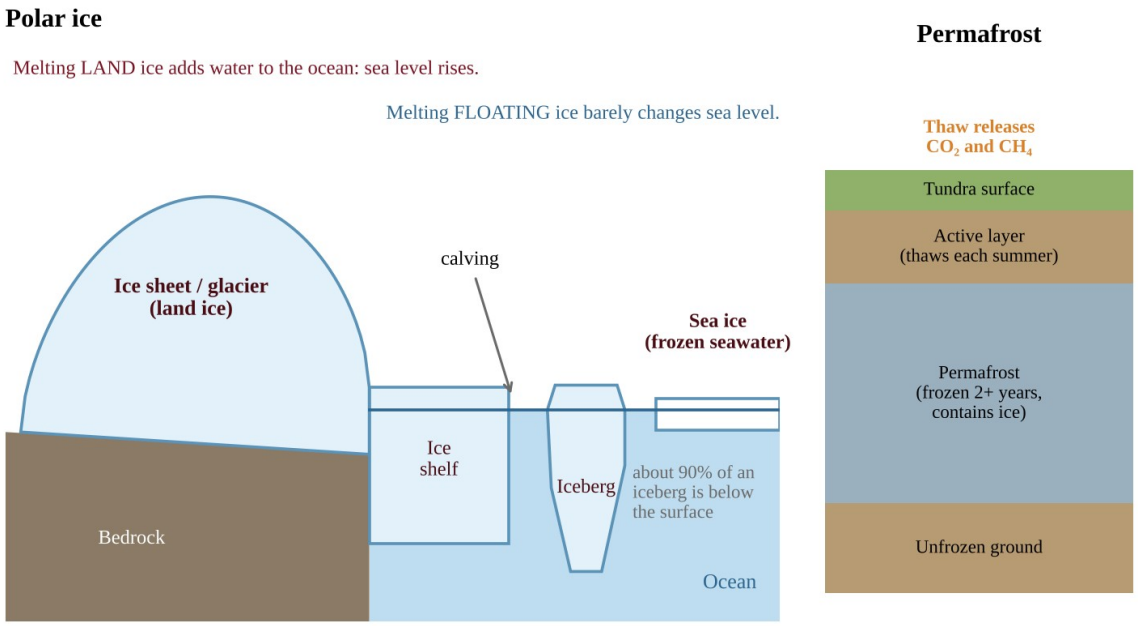


Figure 3.5 — Types of polar ice (left) and a permafrost profile (right).

Why ice matters: sea level and albedo

Sea level. Sea level rises for two reasons: melting **land ice** adds new water to the ocean, and warming makes sea water **expand** (thermal expansion). Melting **floating ice** (sea ice, ice shelves, and icebergs) changes sea level very little, because the floating ice already displaces about its own weight of water. If all of Greenland's ice melted, the oceans would rise about 7 m; Antarctica's ice holds about 58 m of sea-level rise, and all other glaciers about 0.4 m. Sea level now rises about 3 to 4 mm per year.

Albedo. Albedo is the fraction of sunlight a surface reflects. Fresh snow and ice reflect about 80–90% of sunlight, whereas open ocean reflects only about 6%. When ice melts, dark water or land is exposed, more energy is absorbed, and more ice melts. This **ice-albedo feedback** is a positive feedback that makes the poles warm faster than the rest of the planet (**polar amplification**).

FORMULA & MATH SKILL

Sea-level rise = (ice volume × 0.917) ÷ ocean surface area Convert ice to water by multiplying by 0.917 (ice density ÷ water density). Use consistent units: km³ ÷ km² gives km; multiply by 1,000 for m. **Fraction of an iceberg below water = ρ(ice) ÷ ρ(sea water) = 917 ÷ 1,025 ≈ 0.89. Mass balance = accumulation − ablation.**

CLASSROOM EXAMPLE SG-C3-E004

The Greenland ice sheet contains about 3.0×10^6 km³ of ice. Ice has a density 0.917 times that of liquid water. The ocean covers 3.6×10^8 km², and you may assume this area stays constant. (a) Calculate the rise in sea level, in meters, if all of this ice melted and ran into the ocean. (b) An iceberg has a density of 917 kg/m³ and floats in sea water of density 1,025 kg/m³. Calculate the percentage of its volume that is below the water surface. (c) Explain why melting of floating sea ice raises sea level much less than melting of an equal volume of land ice.

QUICK CONCEPT SUMMARY

- Land ice (glaciers, ice caps, ice sheets) raises sea level when it melts; floating ice (sea ice, ice shelves, icebergs) does not.
- Permafrost is frozen ground; thawing releases CO_2 and methane.
- Ice reflects sunlight, so ice loss lowers albedo and speeds warming.

3.5 Oceans

CONCEPT MADE SIMPLE

The ocean is a giant heat reservoir. Winds push its surface, and differences in temperature and salt make water sink or rise, spreading heat and nutrients around the planet.

Ocean zones

The ocean is divided by **depth and light** (vertical zones):

Zone	Depth	Features
Sunlight (euphotic)	0–200 m	Enough light for photosynthesis; phytoplankton and most marine life
Twilight (dysphotic)	200–1,000 m	Dim light; no net photosynthesis
Midnight (aphotic)	1,000–4,000 m	No sunlight; cold, high pressure
Abyssal and hadal	4,000 m to about 11,000 m	Near freezing; life relies on food sinking from above

Temperature falls with depth. A warm **surface mixed layer** sits on top, then a **thermocline** where temperature drops steeply, then cold deep water (2–4 °C). Horizontally, the **intertidal zone** lies between high and low tide, the **neritic zone** is the shallow water over the continental shelf, and the **oceanic zone** is the open sea beyond.

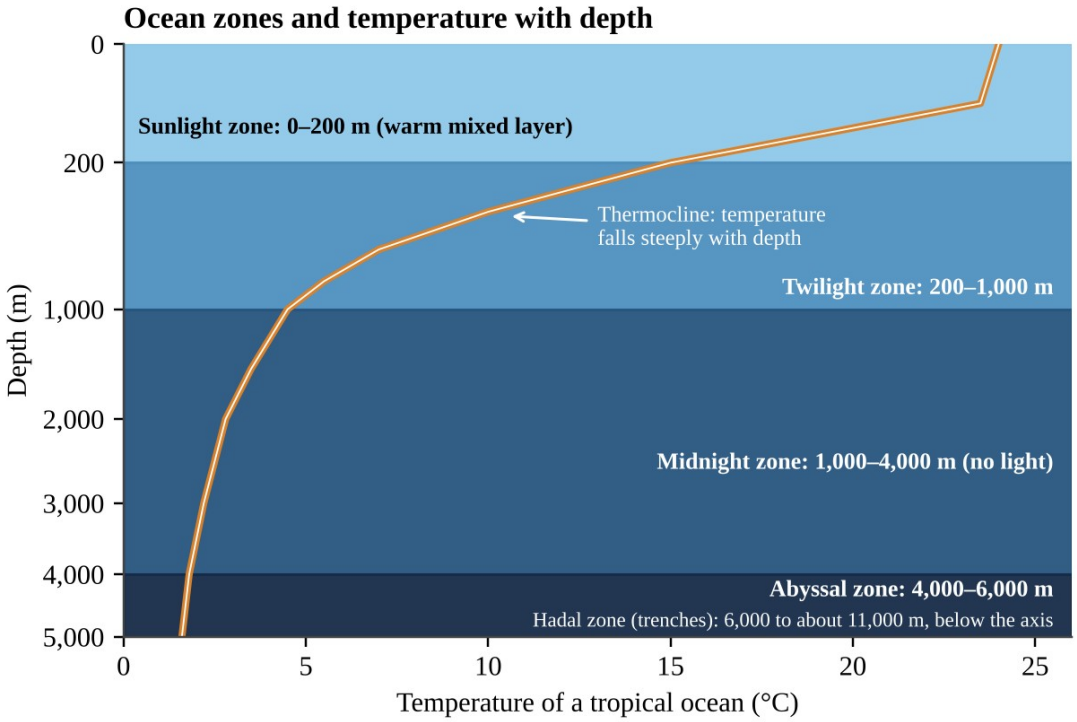


Figure 3.6 — Ocean depth zones with a typical temperature profile for a tropical ocean. The depth axis is compressed with depth so that the thin upper zones are visible.

Salinity

Salinity is the mass of dissolved salts per kilogram of water, in parts per thousand (ppt). The ocean average is about **35 ppt (3.5%)**, mostly sodium and chloride ions. Salts come from rock weathering and undersea vents and stay because evaporation removes only water. Salinity varies:

- **Higher** where evaporation exceeds precipitation, such as the dry subtropics around 25° latitude, and in enclosed seas.
- **Lower** near the equator (heavy rain), near river mouths and estuaries, and in polar waters where ice melts or sea ice thaws.

Density rises as water gets **colder** and **saltier**. This is the key to deep circulation.

The ocean also absorbs CO_2, which lowers its pH (Chapter 11). pH is logarithmic: $[H^+]$ changes by a factor of 10^(pH change), so a drop of 0.3 units roughly doubles $[H^+]$.

Surface currents

Surface currents are driven by **prevailing winds** (the trade winds and westerlies), deflected by the **Coriolis effect** (to the right in the Northern Hemisphere, left in the Southern) and turned by continents. The result is a large circular flow in each ocean basin called a **gyre**: clockwise in the Northern Hemisphere, counterclockwise in the Southern. On the **western** side of each basin, **warm** currents carry tropical heat poleward (Gulf Stream, Kuroshio). On the **eastern** side, **cold** currents flow toward the equator (California, Canary, Humboldt). The Gulf Stream keeps northwestern Europe far milder than other places at the same latitude.

Deep currents (thermohaline circulation)

In the North Atlantic and around Antarctica, surface water becomes cold and salty (partly because forming sea ice leaves salt behind) and **sinks**. It spreads along the ocean floor, slowly crosses the basins, and eventually returns to the surface. This global **thermohaline circulation**, sometimes called the "conveyor belt," takes about a thousand years to make one circuit.

Upwelling

Upwelling is the rise of cold, deep water to the surface. Along many west coasts (California, Peru, Chile, northwest Africa), winds blow toward the equator parallel to the coast. The Coriolis effect turns the surface water offshore, and deep water rises to replace it. Deep water is rich in nutrients from decayed organisms, so upwelling zones feed plankton blooms and some of the world's richest fisheries. During El Niño, upwelling off Peru weakens, and fisheries and seabirds suffer (Chapter 2).

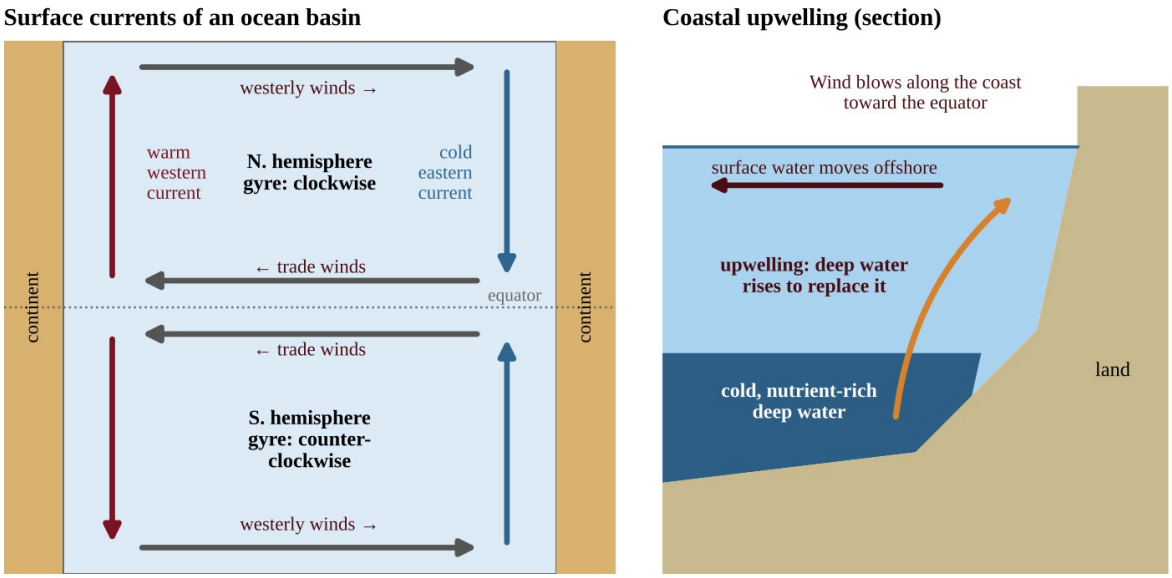


Figure 3.7 — Idealized gyres and surface currents in an ocean basin (left) and coastal upwelling (right).

CLASSROOM EXAMPLE SG-C3-E005

A 2.0 kg sample of sea water has a salinity of 35 ppt (35 g of salt per kg of sea water). The sample is left in the sun and pure water evaporates, but no salt is lost. (a) Calculate the mass of salt in the sample, in g. (b) Calculate the mass of water that must evaporate for the salinity to reach 50 ppt. (c) Explain why the surface water of the dry subtropics is saltier than the surface water at the equator.

QUICK CONCEPT SUMMARY

- Ocean zones are set by depth and light; temperature falls steeply across the thermocline.
- Average salinity is about 35 ppt; it is higher where evaporation dominates and lower where rain, rivers, or ice melt add fresh water.
- Winds drive surface gyres, density drives deep currents, and wind plus Coriolis drives coastal upwelling.

3.6 Agricultural, Industrial, and Municipal Use

CONCEPT MADE SIMPLE

Irrigation is like watering a houseplant. You could flood the pot, dribble water along the edge, spray it overhead, or give it a few drops at the roots. The more precisely you water, the less is wasted, but precision costs more.

Who uses the water

Human water use falls into three sectors:

Sector	Approximate share of global use	Typical uses
Agricultural	about 70%	Irrigating crops, watering livestock, aquaculture
Industrial	about 20%	Cooling in power plants, manufacturing, mining, processing
Municipal (domestic)	about 10%	Drinking, cooking, washing, toilets, lawns, public services

Wealthy countries use more for industry; dry developing countries use most for farming. Two terms matter. **Withdrawal** is water taken from a source, and **consumption** is the part that does not return to the source, because it evaporates, is incorporated into products, or is transpired by crops. Power plants withdraw huge amounts for cooling but return most of it (warmer), while irrigation consumes a large share because much of it is evaporated and transpired. Leaking pipes can lose 15–30% of a city's water. Because agriculture is the biggest user, even a modest gain in irrigation efficiency saves large volumes of water. **Irrigation efficiency** is the percentage of the water applied that actually reaches the crop roots.

Irrigation methods

Method	How it works	Water lost to evaporation and runoff	Cost and energy
Furrow	Channels are cut between crop rows and filled with water	about 1/3	Inexpensive
Flood	The whole field is covered with water	about 20%	Inexpensive; can cause waterlogging
Spray	Groundwater is pumped to spray nozzles over the field	1/4 or less	More expensive than flood and furrow; requires energy
Drip	Perforated hoses release small amounts of water at the roots	about 5%	Most efficient but expensive, so used less often

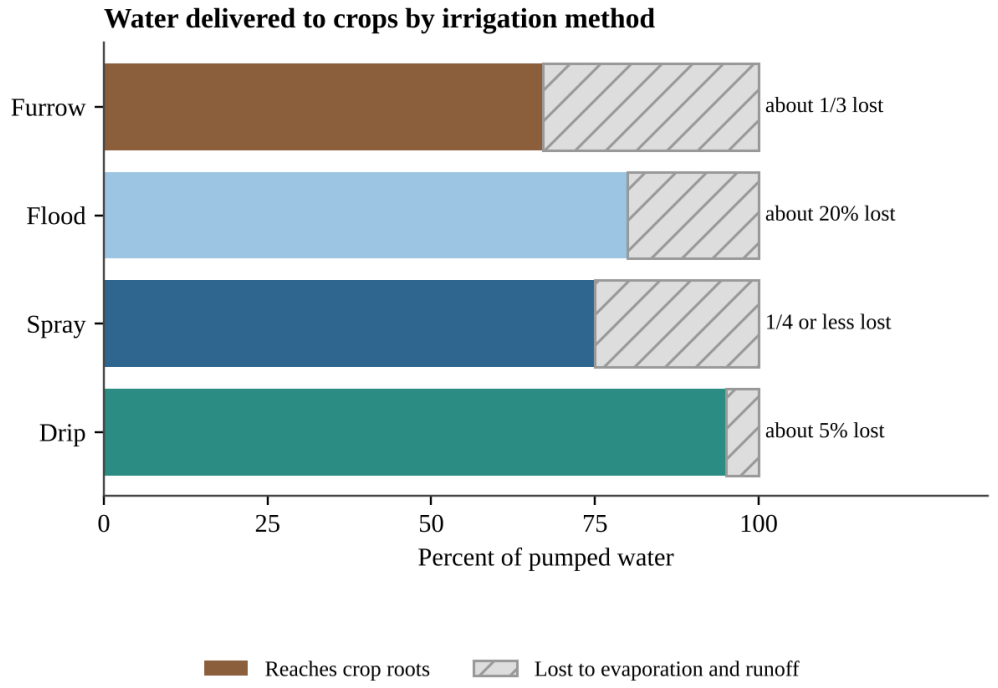


Figure 3.8 — Share of pumped water that reaches the crop roots for four irrigation methods, using the loss values given in the course framework.

Problems caused by irrigation

Waterlogging happens when too much water sits in the soil. It raises the groundwater table until the root zone is saturated, so roots cannot absorb the oxygen they need and plants suffocate. Flood irrigation is especially prone to it.

Salinization occurs when irrigation water, which contains dissolved salts, evaporates and leaves the salts behind. Over time the soil becomes salty enough to be toxic to many plants, reducing yields or forcing farmers to abandon the field. It is worst in hot, dry climates and in poorly drained soils. A white salt crust on the surface is the classic sign. In a flood-irrigated arid field both problems often appear together.

Reclaimed wastewater saves fresh water for irrigation, but it usually carries more dissolved salts (including sodium) than river water, so soil irrigated with it can salinize faster.

Aquifer depletion occurs when water is pumped out faster than rain recharges the aquifer, as in the Ogallala (Section 3.3). As the water table falls, wells must be deepened and pumping costs rise until farming becomes uneconomical. This is also a tragedy of the commons (Chapter 7).

Water productivity. Crops and livestock differ in water use and in income per liter (value ÷ water used). Vegetables earn far more per liter than livestock or pasture, which use the most water; compare bars of the same quantity.

Aquifer budgets and rates of change

An aquifer works like a bank account. The water table falls whenever withdrawal exceeds recharge, *even if pumping is shrinking*. If pumping falls from 1,800 to 1,300 million m³ per year but recharge is only 200 million m³ per year, the aquifer still loses 1,100 million m³ every year. Sustainable use means withdrawal ≤ recharge.

FORMULA & MATH SKILL

Net rate = recharge − withdrawal (negative = decline) **Average rate of change = (final value − initial value) ÷ (final year − initial year) Years until depletion = amount remaining ÷ net annual loss** Units: m/yr or m³/yr; put both rates in the same units first. A negative sign means decline; quote it, or write "declined by".

KEY DEFINITIONS

Irrigation efficiency — (water that reaches the crop ÷ water applied) × 100%. **Waterlogging** — saturation of soil by excess water, raising the water table and depriving roots of oxygen. **Salinization** — build-up of salts in soil as irrigation water evaporates. **Withdrawal / consumption** — water taken from a source / the part of it that does not return.

FORMULA & MATH SKILL

Water to apply = water needed by crop ÷ efficiency (as a decimal) Water lost = water applied × fraction lost Efficiency = 1 − fraction lost (drip loses 5%, so efficiency = 0.95). Dividing by 0.67 is not the same as multiplying by 1.33. **Annual use = people × use per person per day × 365**; 1 m³ = 1,000 L. If a system loses a fraction f in pipes, water produced = water delivered ÷ (1 − f).

COMMON STUDENT MISTAKES

- Treating "efficiency" as the percent *lost*. Efficiency is the percent *delivered*.
- Saying drip irrigation is "free of problems." It is expensive and can clog; and because water is applied only at the roots, salts can still collect at the edges of the wetted zone.
- Confusing withdrawal with consumption. A power plant can withdraw a lot of water yet consume little.

AP EXAM STRATEGY

In "compare irrigation methods" questions give a *cost* point and an *efficiency* point for each method. Many students supply only efficiency. Check whether a graph shows percent *lost* or percent *delivered* before reading it.

CLASSROOM EXAMPLE SG-U5-E004

A 40-hectare farm needs 5,700 m³ of water per hectare to reach its crop roots during the growing season. Assume the loss percentages are exactly 1/3 for furrow irrigation and 5% for drip irrigation. (a) Calculate the volume of water, in m³, that must be pumped for the whole farm using furrow irrigation. (b) Calculate the volume that must be pumped using drip irrigation. (c) Calculate how many cubic meters of water the farm would save per year by switching, and give one reason a farmer might still keep the furrow system.

CLASSROOM EXAMPLE SG-U5-E013

An aquifer holds 36 m of saturated thickness. Pumping lowers the water table by 1.1 m per year and recharge raises it by 0.2 m per year; both rates stay constant. (a) Calculate the net change per year and the number of years until the aquifer is dry. (b) Pumping is then cut by 40% while recharge is unchanged. Calculate the new net change and state whether use is now sustainable. (c) Name one change in irrigation that could cut pumping and one drawback of it.

CLASSROOM EXAMPLE SG-C3-E006

A city of 250,000 people uses an average of 380 L of water per person per day. The city's pipes lose 15% of all the water put into the system through leaks, so customers receive the other 85%. (a) Calculate the volume of water, in m³, that customers use in one year. (b) Calculate the volume of water the city must produce in one year, in m³. (c) Calculate the volume lost to leaks in one year, in m³.

QUICK CONCEPT SUMMARY

- About 70% of human freshwater consumption goes to agriculture; industry uses about 20% and municipal use about 10%.
- Water lost: drip about 5%, spray 1/4 or less, flood about 20%, furrow about 1/3. Cost and energy generally rise with efficiency.
- Irrigation can cause waterlogging, salinization, and aquifer depletion (the Ogallala).

3.7 Global Problems

CONCEPT MADE SIMPLE

Water problems come from too little water where people live, water too dirty to use, and disputes over who gets it.

Water scarcity

Water scarcity exists when demand for water exceeds the supply that is available and safe. Two kinds exist. **Physical scarcity** means there is not enough water in the region (the Middle East, North Africa, parts of India and Central Asia). **Economic scarcity** means water exists but people lack the money or infrastructure to reach it (much of sub-Saharan Africa). A common indicator is renewable water per person per year: below about 1,700 m^3 is called *water stress*, below 1,000 m^3 *scarcity*, and below 500 m^3 *absolute scarcity*. **Water per person = renewable supply ÷ population** (1 $km^3 = 10^9$ m^3). Climate change, population growth, and rising incomes all increase pressure on supplies.

Groundwater depletion

Wells in many farming regions (northwest India, the North China Plain, California's Central Valley) draw water faster than rain recharges it, causing falling water tables, dry wells, subsidence, and saltwater intrusion. To show that pumping, not drought, is the cause, compare a pumped area with an unpumped area of the same climate. Remedies: pricing, pumping limits, efficient irrigation, and artificial recharge.

Dams and reservoirs

A **dam** blocks a river to create a **reservoir**. Large dams store water, generate hydroelectricity, control floods, and supply irrigation, but have costs.

Benefits	Costs
Reliable water for cities and farms	Flooding of land, homes, farms, and habitat; people displaced
Hydroelectric power without burning fuel	Blocks fish migration and traps sediment, so downstream deltas and floodplains lose nutrients and erode
Flood control; recreation	Changes river temperature and flow; evaporation losses; methane from decaying plants in the reservoir
	Rivers may no longer reach the sea (the Colorado River), and dam failure is a risk

Diverting rivers for irrigation can shrink whole lakes: the Aral Sea lost most of its volume after its rivers were diverted to cotton fields.

Desalination

Desalination removes salt from sea water or salty groundwater. In **reverse osmosis**, high pressure forces water through a membrane that blocks salt, and a stream of concentrated brine remains. In **distillation**, water is evaporated and the vapor is condensed, leaving salts behind. It gives coastal regions such as the Arabian Peninsula and Israel a drought-proof supply. Its drawbacks are **high energy use** (about 3–4 kWh per m^3 for reverse osmosis, often from fossil fuels), **high cost**, **brine** with about twice the salinity of sea

water that harms marine life if dumped carelessly, and intake systems that kill small organisms. Because it is costly, it supplies cities, not large-scale irrigation.

Transboundary conflicts

Rivers and aquifers cross borders, so an upstream country can reduce flow and quality for downstream countries. Examples: the **Nile** (Egypt, Sudan, Ethiopia), the **Tigris–Euphrates** (Turkey, Syria, Iraq), the **Indus** (India, Pakistan), and the **Colorado** (United States, Mexico). Treaties and agreed allocations lower the risk of conflict.

Access to clean water

Roughly **2 billion people** lack safely managed drinking water, and about 3.5 billion lack safely managed sanitation. Contaminated water spreads cholera, typhoid, dysentery, and other diarrheal diseases, which kill many young children each year. Solutions include protected wells, filtration and chlorination, sanitation (Chapter 9), and rainwater harvesting.

Solutions

Conserve (fix leaks, price water, use drip irrigation, replace lawns with native drought-tolerant plants, which also need less fertilizer and pesticide); **reuse** graywater and treated wastewater; **protect** watershed wetlands and forests; **recharge** aquifers and harvest rainwater; **cooperate** through treaties.

FORMULA & MATH SKILL

Recovery = product water ÷ feed water × 100% Brine salinity = salt in feed ÷ brine volume (if the product is salt-free, so salinity ≈ feed salinity ÷ (1 − recovery)). **Energy = volume × kWh per m³;** cost = energy × price per kWh.

AP EXAM STRATEGY

"Propose a solution" requires a specific action plus its drawback: for example, "install drip irrigation (saves water but costs more to install)". Match the solution to the problem: pricing and efficiency for depletion, desalination only for coastal cities with money, treaties for transboundary rivers.

CLASSROOM EXAMPLE SG-C3-E007

A coastal reverse-osmosis plant produces 200,000 m³ of fresh water per day with a recovery of 45% (45% of the feed water becomes fresh water; the rest leaves as brine). The feed is sea water at 35 ppt, and the product is salt-free. The plant uses 3.5 kWh of electricity per m³ of fresh water, at a price of $0.10 per kWh. (a) Calculate the volume of sea water taken in per day, in m³. (b) Calculate the volume of brine discharged per day, in m³. (c) Calculate the salinity of the brine, in ppt. (d) Calculate the cost of the electricity per day, in dollars.

QUICK CONCEPT SUMMARY

- Scarcity can be physical or economic; groundwater depletion, dams, and rising demand make it worse.
- Dams store water and power, but displace people and disrupt rivers; desalination provides new water at high energy cost and with brine.
- Shared rivers need treaties, and about 2 billion people still lack safe drinking water.

3.8 Practice Questions and Answers

Practice makes these ideas stick, so use the classroom examples and exam-style questions listed below.

PRACTICE QUESTIONS AND ANSWERS

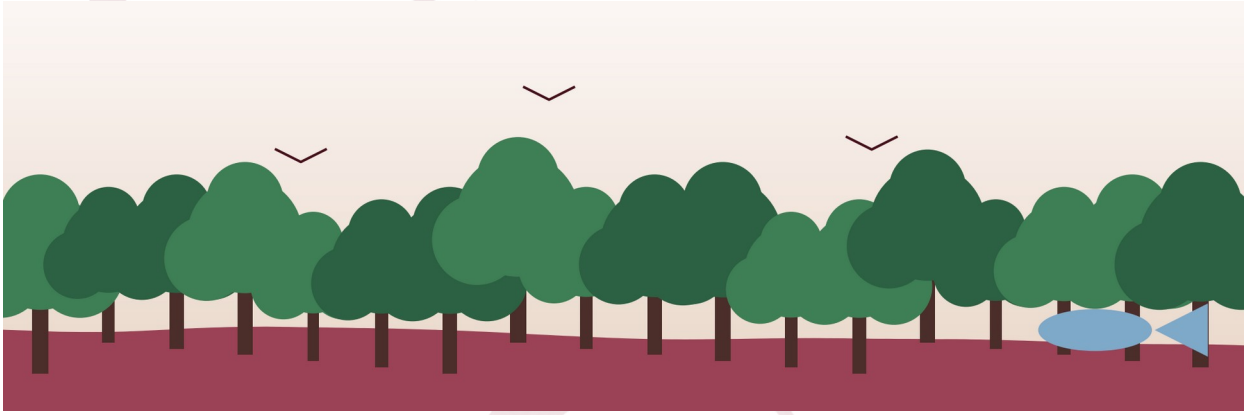
Classroom examples in this chapter (unsolved; space is provided in each box): SG-C3-E001, SG-C3-E002, SG-U4-E015, SG-U4-E007, SG-C3-E003, SG-C3-E004, SG-C3-E005, SG-U5-E004, SG-U5-E013, SG-C3-E006, SG-C3-E007.

Classified practice: Book Two, Chapter 3 — 25 multiple-choice questions (Section A) followed by 3 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 3. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

UNIT II
THE LIVING WORLD



IN THIS UNIT

Chapter 4 **Ecosystems**
Chapter 5 **Natural Biogeochemical Cycles**

CHAPTER
4

UNIT II — THE LIVING WORLD
Ecosystems

4.1 · Ecology
4.2 · Law of Tolerance
4.3 · Energy Flow
4.4 · Biomes
4.5 · Biodiversity
4.6 · Practice Questions and Answers

LEARNING OBJECTIVES

- Explain how limited resources shape interactions among species (predation, symbiosis, competition).
- Describe tolerance ranges and use them to predict where species can live.
- Trace energy through ecosystems: GPP, NPP, trophic levels, the 10% rule, food chains and food webs.
- Identify the major terrestrial and aquatic biomes and explain how climate, salinity, depth and nutrients shape them.
- Explain biodiversity, ecosystem services, island biogeography, natural disruptions, adaptation and succession.

4.1 Ecology

Ecology is the study of how organisms interact with each other and with their environment. It works at several scales: an **organism**, a **population** (one species in one area), a **community** (all the populations living together), an **ecosystem** (a community plus its non-living surroundings), a **biome** (a large region with a characteristic climate and community) and the **biosphere** (all life on Earth).

An ecosystem includes the **biotic** (living) parts and the **abiotic** (non-living) parts of an area and the interactions among them. Abiotic factors include sunlight, temperature, water, soil minerals and oxygen. Biotic factors include every organism and the relationships between them. The central idea is simple: *resources are limited, so organisms interact over them.* Food, water, light, nesting space and mates are all resources. When a resource is scarce, the interactions around it become stronger and more important for survival.

Predator–prey relationships

A **predator** is an organism that kills and eats another organism, the **prey**. The two populations affect each other. When prey is plentiful, predators have more food and may become more numerous. As predators multiply, more prey are eaten and the prey population declines. Predators then face a food shortage, and their numbers fall. This cause-and-effect loop often produces linked rises and falls in both populations.

Symbiosis

Symbiosis is a close, long-term interaction between two species. To identify the type, ask who benefits and who is harmed.

Type	Species A	Species B	Example
Mutualism	benefits	benefits	Bees collect nectar while pollinating flowers; reef-building coral and the algae living inside them
Commensalism	benefits	not affected	Orchids growing on a tree branch for light; barnacles riding on a whale
Parasitism	benefits	harmed	Ticks feeding on a deer; tapeworms in an intestine

Competition and resource partitioning

Competition occurs when organisms need the same limited resource. It can be **intraspecific** (within one species, for example two oak seedlings competing for light) or **interspecific** (between different species). Competition lowers the survival and reproduction of the competitors. Species that use a resource in the same way cannot easily coexist. Coexistence becomes possible through **resource partitioning**: using the resource in a different way, in a different place, or at a different time.

Examples: several warbler species feed in the same spruce tree, each in a different part of it; owls and hawks both hunt rodents, at night and by day.

COMMON STUDENT MISTAKES

- Parasitism is not predation: a parasite lives on or in its host and does not usually kill it quickly; a predator kills its prey.
- Commensalism needs the second species to be truly unaffected; if both benefit it is mutualism.

AP EXAM STRATEGY

When a question describes an interaction, write the effect on each species as a plus, minus or zero. (+,+) is mutualism, (+,0) is commensalism, (+,−) is parasitism or predation, and (−,−) is competition. Then use words from the scenario to justify your answer.

CLASSROOM EXAMPLE SG-U1-E001

Four species of small seed-eating birds live in the same grassland. Observers record the plant height at which each species feeds: species P feeds at 0–10 cm, species Q at 10–30 cm, species R at 30–60 cm, and species S at 60–100 cm. (a) Name the interaction pattern that allows these four species to share the same seed supply. (b) Explain how this pattern reduces the negative impact on each species' survival. (c) If a drought reduces total seed production by half, state whether you expect competition to increase, decrease or stay the same, and give a reason.

QUICK CONCEPT SUMMARY

- Ecosystems = biotic + abiotic parts interacting; limited resources drive species interactions.
- Predation, mutualism, commensalism, parasitism and competition are distinguished by who gains and who loses.
- Resource partitioning (different ways, places or times) lets similar species coexist.

4.2 Law of Tolerance

CONCEPT MADE SIMPLE

Every organism has a comfort zone, and beyond its edges it gets hurt or dies. A polar bear is comfortable in cold; it overheats in a desert. A cactus handles drought; it rots in a swamp. The width of the zone is the organism's tolerance.

The **law of tolerance** says that each species can live only within a range of each environmental factor, and the factor closest to a species' limit decides whether it survives. **Ecological tolerance** is the range of conditions, such as temperature, salinity, flow rate, and sunlight, that an organism can endure before injury

or death results. Each factor has its own range. A trout may tolerate a narrow temperature range but a wide range of sunlight; an estuary crab may tolerate wide swings in salinity.

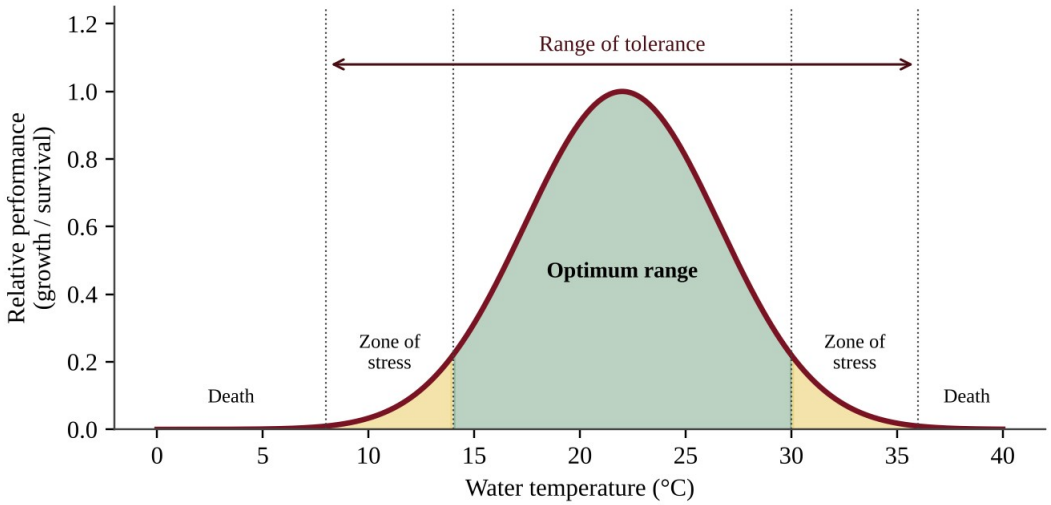


Figure 4.1 — A tolerance curve: performance is highest in the optimum range, falls in the zones of stress, and reaches zero beyond the limits of tolerance.

Reading the graph: the x-axis is the environmental factor, the y-axis is how well the organism does. The peak is the optimum. The two outer limits mark the range of tolerance; outside them survival is not possible. A species with a *wide* curve is a generalist for that factor, and one with a *narrow* curve is a specialist. The **optimum range** is where the organism grows and reproduces best; in the **zone of stress** it survives but performs poorly.

Individuals and species. Tolerance applies at two levels. An *individual* has a tolerance that can shift a little with age, health, and acclimation (a runner adapts to altitude over weeks). A *species* has a tolerance range for the whole group, because individuals vary. A species' range is wider than any single individual's because it includes the hardiest and most sensitive members. A limiting factor that goes beyond tolerance determines where a species can live, so tolerance helps explain geographic range, and why a small shift in temperature, salinity, or flow can remove sensitive species from a stream.

Reading tolerance graphs and tables

- The **range** is where performance is above zero (or the interval in a table); the **optimum** is the peak. Height shows performance at the best; width shows tolerance.
- Test the new condition against *each* species' range. Inside: it persists, perhaps with reduced growth. Outside: it is lost, however wide its range is elsewhere.
- Exposure time matters: a temperature survived for a week may kill within a month.

Designing a tolerance experiment

A field survey shows only an association: distance from a road also changes shade, depth, and predators (**confounding variables**). Only an experiment that changes one factor shows cause. A good design has:

- a **hypothesis** in the form "If X increases, then Y will decrease, because the organism's tolerance limit is exceeded";
- an **independent variable** (the one factor you change, at several graded levels) and a **dependent variable** (what you measure, such as percent survival);
- a **control** (the natural or zero-stress level) and **replicates** (several containers per level, each with the same number of organisms);
- **constants** (temperature, volume, food, light, age and source of organisms, duration).

Graph response against level; the tolerance limit is where survival drops sharply.

COMMON STUDENT MISTAKES

- Mixing up tolerance with adaptation. Tolerance is the range an organism can already survive; adaptation is a heritable change across generations (Section 4.5).
- Thinking the optimum and the limit are the same thing. Organisms can survive in the zone of stress but do poorly.
- Stating that tolerance is one number. It is a range with an upper and lower limit, for each factor.

CLASSROOM EXAMPLE SG-U2-E004

Laboratory trials recorded the survival of two freshwater snail species after 7 days at different water temperatures.

Temperature (°C)	Species A survival (%)	Species B survival (%)
6	0	0
10	0	55
14	80	90
18	100	95
22	95	85
26	70	40
30	0	0
34	0	0

(a) Using only the temperatures tested, identify the lowest and highest tested temperatures at which each species survived at all. (b) Which species is better described as having the wider tolerance for temperature? Justify with the data. (c) A power plant will raise stream temperature from 20 °C to 28 °C. Predict which species is affected more and explain.

CLASSROOM EXAMPLE SG-U2-E010

Acid rain may be lowering the pH of ponds where a salamander breeds. Design an experiment to find the range of pH its larvae can tolerate. (a) State a testable hypothesis. (b) Identify the independent and dependent variables. (c) Describe the treatments, control, and replication. (d) Name two constants. (e) Explain how you would find the tolerance limit.

QUICK CONCEPT SUMMARY

- Ecological tolerance is the range of conditions an organism can endure before injury or death.
- Factors include temperature, salinity, flow rate, and sunlight.
- It applies to individuals and to species.
- Narrow tolerance means a specialist for that factor; wide means a generalist.

4.3 Energy Flow

Energy enters ecosystems as sunlight, is captured by producers, and then passes upward through feeding levels, shrinking at every step. This section follows that energy from the sun to the top predator.

Primary productivity

Primary productivity is the rate at which solar energy is converted into organic compounds through photosynthesis over a unit of time. **Gross primary productivity (GPP)** is the total rate of photosynthesis in an area. Producers use part of that energy for their own cellular respiration (R). **Net primary productivity (NPP)** is the rate of energy storage by photosynthesizers in an area after subtracting the energy lost to respiration. NPP is the energy available to consumers.

FORMULA & MATH SKILL

NPP = GPP − R GPP — gross primary productivity; R — energy used in the producers' respiration; NPP — net primary productivity. Units: energy per unit area per unit time, for example kcal/m²/yr. Carbon units such as g C/m²/yr are also used. Percent of GPP lost to respiration = R ÷ GPP × 100. If NPP and GPP are given, R = GPP − NPP. Total NPP = NPP per unit area × area (1 km² = 10^6 m²).

Productivity is a **rate**, so its units include time. Biomass (total mass of organisms) is an amount at one moment. A forest can have very large biomass but moderate productivity, while an algal bloom can have small biomass but fast turnover.

What controls productivity. Light, temperature, water and nutrients set the rate of photosynthesis. Tropical rainforests, estuaries, marshes and coral reefs have high NPP per unit area; deserts, tundra and the open ocean have low NPP per unit area. The open ocean has low NPP per m², but its enormous area gives it the largest total NPP of any ecosystem type.

Seasonal data. Find NPP = GPP − R for each month. It is negative when respiration exceeds photosynthesis, as in a temperate forest's winter. Annual NPP = annual GPP − annual R. If insects strip leaves, GPP falls while respiration continues, so NPP falls.

Light in water. Photosynthesis stores energy as glucose; oxygen is a by-product. Water absorbs light. Most red light is absorbed in the upper 1 m, and blue light penetrates deeper than 100 m only in the clearest water. Photosynthesis therefore concentrates near the surface, and productivity falls with depth. Floating phytoplankton stay in the sunlit layer, and some algae have accessory pigments that capture the wavelengths that penetrate deeper. Turbid water makes the problem worse.

CLASSROOM EXAMPLE SG-U1-E008

Two ecosystems are studied. Ecosystem A: GPP = 6,000 kcal/m²/yr; producer respiration = 4,200 kcal/m²/yr. Ecosystem B: GPP = 2,400 kcal/m²/yr; producer respiration = 600 kcal/m²/yr. (a) Calculate the NPP of each ecosystem. (b) Calculate NPP as a percentage of GPP for each ecosystem. (c) A student claims that Ecosystem A supports more consumers because its GPP is higher. Evaluate this claim using your results.

CLASSROOM EXAMPLE SG-C4-E001

A lake covers 25 km² and its producers have a net primary productivity of 800 kcal/m²/yr. Assume 10% of energy is passed on at each transfer. (a) Calculate the total NPP of the lake in kcal/yr. (b) Calculate the energy per year available to the primary consumers. (c) Calculate the energy per year available to the secondary consumers.

Trophic levels

A **trophic level** is an organism's feeding position in a food chain. **Producers (autotrophs)** make their own organic compounds, most by photosynthesis. **Consumers (heterotrophs)** get energy from other organisms. **Primary consumers** are herbivores that eat producers, **secondary consumers** eat primary consumers, and **tertiary consumers** eat secondary consumers. Omnivores eat both plants and animals, so they can occupy more than one level. **Detritivores** (for example earthworms and dung beetles) eat dead matter, and **decomposers** (bacteria and fungi) break down dead matter chemically, returning nutrients to the soil.

All ecosystems depend on a **continuous inflow of high-quality energy**, usually sunlight, to maintain their structure and function. In terrestrial and near-surface marine communities, energy flows from the sun to producers in the lowest trophic levels and then upward to higher levels. At each step some energy becomes low-quality heat that radiates away and cannot be recycled, so the flow is one-way.

Matter behaves differently. Atoms of carbon, nitrogen, phosphorus, oxygen and hydrogen move between the environment and organisms through **biogeochemical cycles** (Chapter 5). Each cycle demonstrates the **conservation of matter**: atoms are not created or destroyed, only moved and rearranged.

Energy flow and the 10% rule

Of all the energy a trophic level takes in, only a fraction is stored in its tissues and available to the next level. The **10% rule** approximates that in a transfer of energy from one trophic level to the next, only about 10% of the energy is passed on. The rest is lost mostly as heat from respiration, and also in wastes, uneaten parts (bones, roots, fur) and dead matter that goes to decomposers.

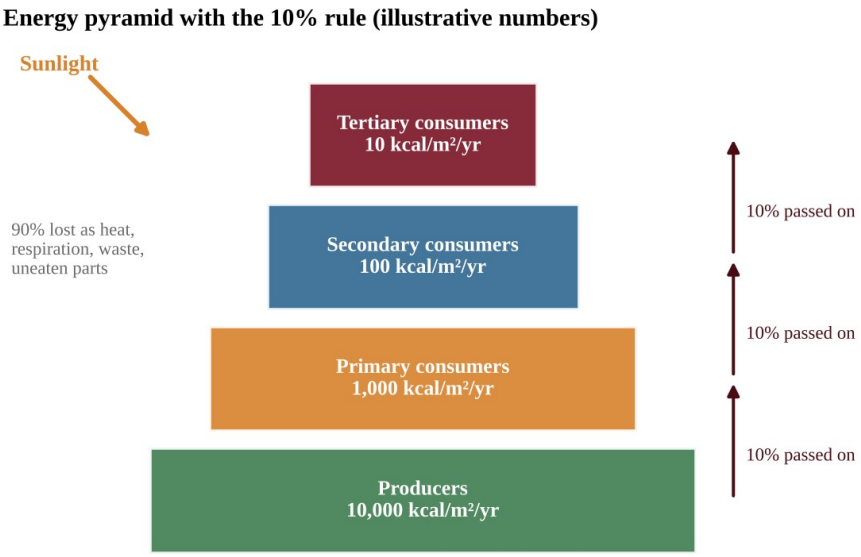


Figure 4.2 — An energy pyramid. Each step passes on about 10% of the energy of the level below.

Why the loss happens: thermodynamics.

- The **first law of thermodynamics** states that energy cannot be created or destroyed, only converted from one form to another. The sunlight captured by producers becomes chemical energy in sugar, then movement and heat.
- The **second law of thermodynamics** states that every energy conversion increases disorder (entropy), and some energy is always converted to low-quality heat. No conversion is 100% efficient. Every organism spends energy to move, grow, stay warm and reproduce, and that energy ends up as heat.

Consequences: food chains are short (usually 3 to 5 levels), top predators are rare, and a large base of producers is needed to support few top consumers.

FORMULA & MATH SKILL

Energy at level n+1 = energy at level n × 0.10 Energy after k transfers = starting energy × $(0.10)^k$ Ecological efficiency (%) = (energy at the higher level ÷ energy at the lower level) × 100 Going down (energy needed at the lower level): divide by 0.10 for each step. Units: the same as the data, such as kcal/m²/yr or joules.

The 10% figure is approximate (real efficiencies range from about 5% to 20%); use 10% unless told otherwise. Count transfers, not levels: an organism at level 4 is three transfers above the producers.

- **Efficiency from data:** for each step, higher level ÷ lower level × 100. Similar values near 10% support the rule; the missing energy is lost mostly as heat (second law), not destroyed.
- **From harvest to area:** harvest mass × energy per kg = energy needed at the top level; divide by 0.10 per transfer down to the producers; divide by NPP (kcal/m²/yr) for area in m²; convert (1 ha = 10^4 m²). Carry units on every line.
- **Diet and land:** each step up multiplies the energy and land needed by about 10, or by the factor given (if 1 kcal of beef needs 20 kcal of corn, 5,000 kcal of beef needs 100,000 kcal of corn). Eating lower on the food chain wastes less energy, so the same land feeds more people. People fed = energy per year ÷ (kcal per person per day × 365).

REAL-WORLD EXAMPLE

Raising beef needs many times more plant energy and land than growing grain for people, because about 90% of the feed energy is lost as heat and waste before the meat is eaten.

AP EXAM STRATEGY

Write the trophic level labels next to the numbers, then draw an arrow for each transfer and multiply by 0.10. Show units, state the answer and check that each lower level has about 10 times the energy of the one above.

CLASSROOM EXAMPLE SG-U1-E009

In an ocean food chain, phytoplankton fix 25,000 kcal/m²/yr. The chain is phytoplankton → zooplankton → small fish → tuna. Assume 10% of the energy is passed on at each transfer. (a) Calculate the energy available to the tuna in kcal/m²/yr. (b) Calculate how much phytoplankton energy would be needed to support 400 kcal/m²/yr in the small fish. (c) Explain, using the second law of thermodynamics, why the tuna receive less energy than the small fish.

Food chains and food webs

A **food chain** shows a single pathway of energy and matter from producers (autotrophs) to primary consumers (herbivores), secondary and tertiary consumers (carnivores and omnivores). The arrows point from the organism being eaten to the organism that eats it, which is the direction of energy flow. Detritivores and decomposers play an essential role: they return nutrients to the soil, where producers use them again.

A **food web** is a model of the interlocking food chains in an ecosystem. It is more realistic than a single chain because most consumers eat several things and are eaten by several predators, so one organism can sit at several trophic levels. A hawk that eats a mouse (a primary consumer) is a secondary consumer, but a hawk that eats a snake that ate a mouse is a tertiary consumer.

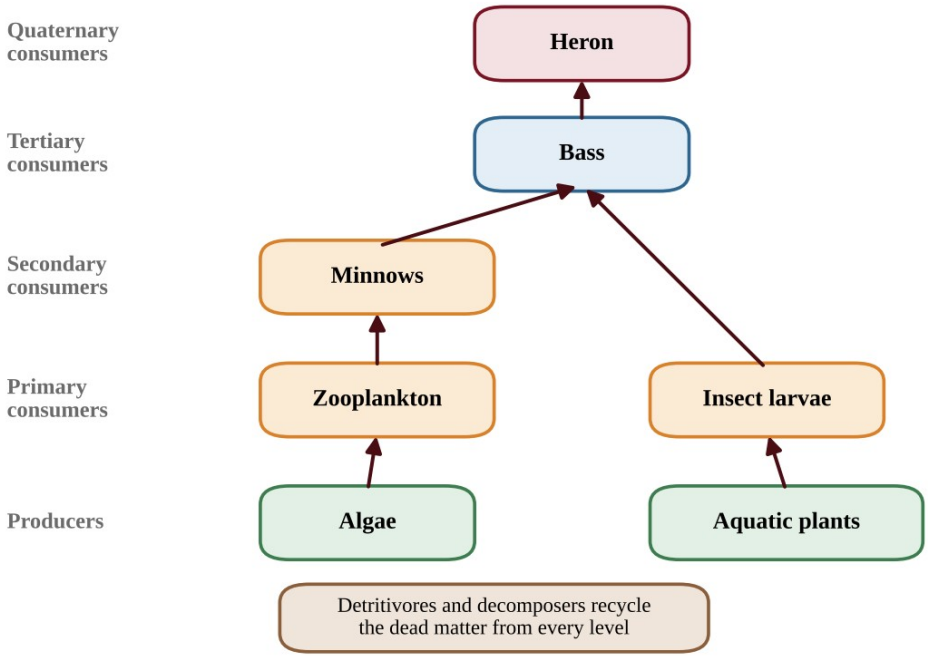


Figure 4.3 — A lake food web. Arrows point from the food to the eater.

To read a food web, find the producers (arrows only leave them), count steps from a producer to give each consumer's trophic level (list every level if there are several paths), and, to predict the effect of removing a species, follow every arrow that enters or leaves it.

Feedback loops and changing the web. When one species is added to or removed from a food web, the rest of the web can change. Feedback loops often drive these changes.

- A **negative (balancing) feedback loop** counteracts a change and stabilizes the system. When rabbits increase, foxes find more food and multiply, so rabbits are reduced.
- A **positive (reinforcing) feedback loop** amplifies a change. In a lake, a large algal bloom dies, decomposers use up the dissolved oxygen as they consume it, fish die, and their decay uses still more oxygen.

A **trophic cascade** is a chain of effects across levels. When sea otters (which eat sea urchins) are removed, urchins increase, they graze down the kelp, and the kelp forest that sheltered many other species disappears.

Tracing effects and judging evidence. To predict a change, start with the first change and flip the direction at each link. Removing bass lets minnows increase; more minnows eat more zooplankton; fewer zooplankton graze less, so algae rise. Predators are far fewer than their prey because only about 10% of energy passes up a level.

Two sites that differ (a no-fishing reserve and a fished reef) show an association, not proof of cause, because depth, temperature or nutrients may also differ. Stronger evidence comes from several matched sites, before-and-after data at one site, or an experiment that removes or adds the predator with untreated controls.

COMMON STUDENT MISTAKES

- GPP is always greater than NPP (unless R = 0); productivity is a rate, biomass an amount.
- Energy flows one way; only matter cycles. An omnivore can belong to more than one trophic level.
- Arrows point from the food to the eater. A web with alternative prey is more resilient.

CLASSROOM EXAMPLE SG-U1-E010

A food web in a dry grassland has these feeding relationships (prey → predator): grass → grasshoppers; grass → rabbits; grasshoppers → lizards; lizards → hawks; rabbits → foxes; rabbits → hawks. (a) Write the longest food chain in this web and identify the trophic level of the hawk in that chain. (b) A disease eliminates nearly all lizards. Predict the direction of change in the grasshopper population, assuming lizards are the only predator of grasshoppers, and explain. (c) As rabbits increase, foxes increase, and then rabbits decline. Identify the type of feedback and justify your answer.

QUICK CONCEPT SUMMARY

- Primary productivity is the rate at which producers convert sunlight into organic compounds; NPP = GPP − R, in energy per area per time (for example kcal/m²/yr). Water absorbs light, so aquatic photosynthesis is concentrated near the surface.
- Producers capture energy; consumers pass it up; detritivores and decomposers recycle dead matter. Ecosystems need a continuous input of high-quality energy but recycle matter (conservation of matter).
- About 10% of energy passes to the next trophic level and about 90% is lost, mainly as heat (first and second laws of thermodynamics), so food chains are short and top predators few.
- Food chains show one pathway; food webs combine many; adding or removing a species affects the web through positive and negative feedback loops.

4.4 Biomes

A **biome** is a large region with a characteristic community of plants and animals that result from, and are adapted to, its climate. Temperature and precipitation are the two main drivers. Because both change with latitude (distance from the equator) and altitude (height above sea level), biomes form recognizable bands and patterns around the globe.

Terrestrial biomes

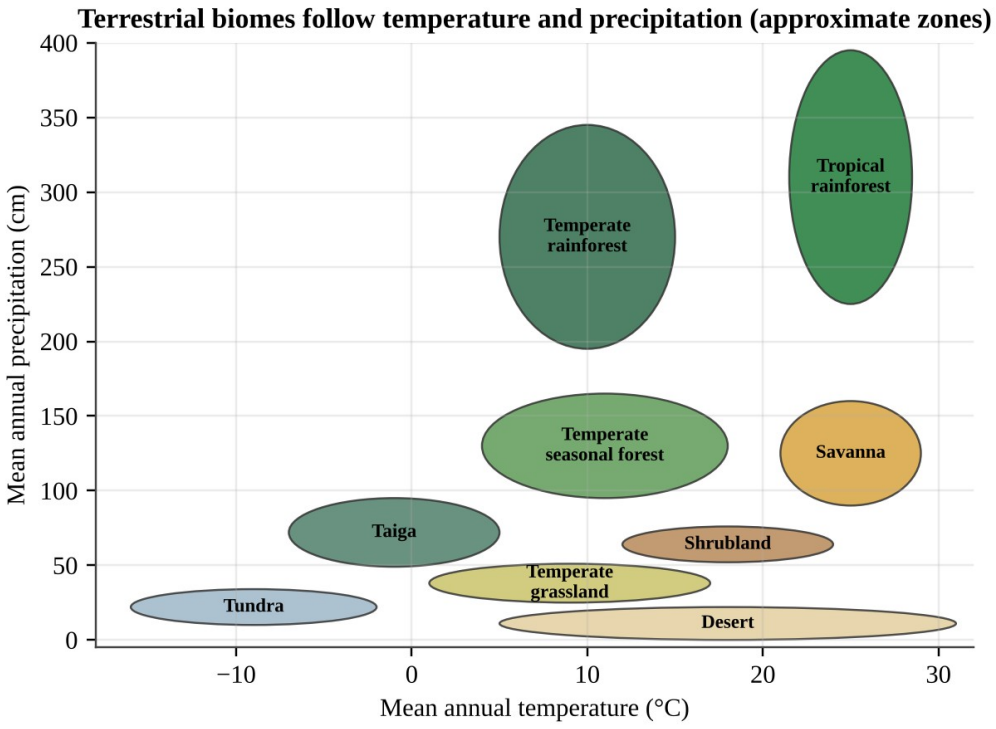


Figure 4.4 — Terrestrial biomes sorted by mean annual temperature and precipitation. Zones are approximate and overlap in nature.

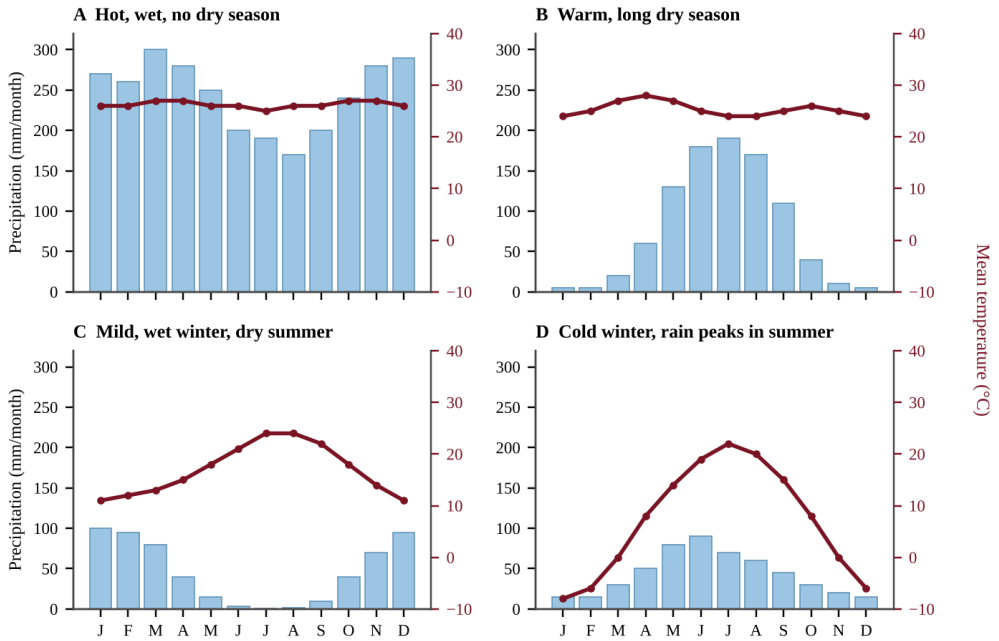
Reading a biome climate graph. Find the site's temperature (x-axis) and precipitation (y-axis) and read which zone contains the point; where zones overlap, other clues (seasonality, soil, vegetation) decide. Cold biomes lie on the left and wet biomes at the top. At a given temperature, more precipitation shifts desert to grassland to forest.

Biome	Climate	Typical features
Tundra	Very cold, low precipitation, short growing season	Permafrost; low shrubs, mosses, lichens; slow decomposition
Taiga (boreal forest)	Long, cold winters; short, mild summers	Evergreen conifers; acidic, nutrient-poor soil; large expanses

Biome	Climate	Typical features
Temperate rainforest	Cool, very wet, mild temperatures	Tall conifers, ferns and mosses; moist, rich in biomass
Temperate seasonal forest	Four distinct seasons; moderate precipitation	Deciduous trees that drop leaves in winter; fertile soil
Tropical rainforest	Hot all year, very wet	Greatest biodiversity; nutrients locked in biomass; thin, poor soil
Shrubland	Hot dry summers, mild wet winters	Drought-tolerant, fire-adapted shrubs
Temperate grassland	Hot summers, cold winters, moderate-to-low rain	Grasses; deep fertile soil; fire and grazing maintain it
Savanna	Warm year-round, distinct wet and dry seasons	Grasses with scattered trees; large grazing herds
Desert	Very low precipitation (typically under about 30 cm/yr)	Drought-adapted plants; animals that avoid daytime heat

Tropical rainforest soil is poor even though the forest is lush: warmth and moisture make decomposition fast and plants absorb nutrients at once, so little is stored in the soil. In cold biomes, decomposition is slow and organic matter accumulates.

Reading a climatograph. Monthly graphs show precipitation as bars and temperature as a line. Ask: how warm is the line, and does it change? Is rain spread evenly, or is there a dry season? Does rain fall in the warm or the cool months? In Figure 4.5, a flat warm line with tall bars every month is tropical rainforest (A); warmth with a long dry season is savanna (B); a mild wet winter and nearly rainless summer is shrubland (C); a freezing winter with rain peaking in summer is temperate grassland (D). Very short bars in every month mean desert.



Blue bars = precipitation (left axis); red line = temperature (right axis). Same scales in every panel.

Figure 4.5 — Four climatographs. Compare the temperature line (right axis) and the rainfall bars (left axis) to identify the biome.

Comparing biomes. Species diversity and net primary productivity (NPP) per m² peak where warmth and water are abundant all year. Diversity ranks tropical rainforest, temperate seasonal forest, temperate grassland, tundra (high to low); NPP is highest in tropical rainforest and lowest in desert and tundra. On a world map, rainforest lies near the equator, temperate seasonal forest at mid-latitudes, then taiga and tundra toward the pole.

Why resources and biomes are not spread evenly. Non-mineral natural resources such as fresh water and timber are unevenly distributed because of **climate, geography, latitude and altitude, nutrient availability and soil**. Timber is concentrated in forest biomes; water is abundant in rainforests and scarce in deserts. Biome distribution is **dynamic**: it changed in the past, for example as ice sheets advanced and retreated, and may shift again with climate change. Warming can push forest boundaries toward the poles and up mountainsides.

COMMON STUDENT MISTAKES

- Deserts are defined by low precipitation, not heat; some are cold.

CLASSROOM EXAMPLE SG-U1-E002

Use Figure 4.4. Three field stations report the following climate data. Station A: mean annual temperature 25 °C, precipitation 300 cm/yr. Station B: mean annual temperature −9 °C, precipitation 22 cm/yr. Station C: mean annual temperature 18 °C, precipitation 12 cm/yr. (a) Identify the most likely biome at each station. (b) Explain why decomposition and nutrient turnover would differ between the biomes at Stations A and B.

CLASSROOM EXAMPLE SG-U1-E011

Station M: temperature 24–28 °C every month; precipitation 150–230 mm per month from November to April and under 15 mm per month from May to October. Station N: temperature −6 °C in January and 20 °C in July; precipitation 15–25 mm per month in winter and 60–90 mm per month from May to August. (a) Identify the biome at each station and cite two pieces of evidence for each. (b) Explain how the dry season at Station M shapes its plant community. (c) Predict how Station N's biome would change if the region warmed and became drier, and justify it.

Aquatic biomes

Freshwater biomes include streams, rivers, ponds and lakes. They contain little dissolved salt (typically under about 0.5 parts per thousand) and are a vital source of drinking water, irrigation and food. **Marine biomes** are salty (the open ocean averages about 35 parts per thousand).

- **Oceans** cover most of Earth's surface. Surface algae (phytoplankton) supply a large portion of Earth's oxygen and take in carbon dioxide from the atmosphere.
- **Coral reefs** occur in warm, clear, shallow water. They host great biodiversity and rely on a mutualism between coral animals and algae. They are sensitive to warming and pollution.
- **Marshland** (including salt marshes and mangroves) is shallow and nutrient-rich, with plants rooted in saturated soil; it filters runoff and protects coasts.
- **Estuaries** form where a river meets the sea, creating water of intermediate salinity that varies with tides and river flow. They are nutrient-rich, turbid and serve as nurseries for many fish and shellfish.

Why aquatic resources vary. The distribution of non-mineral marine resources such as different fish varies with **salinity, depth, turbidity, nutrient availability and temperature**. Fish that need cold, nutrient-rich water gather in areas where deep water rises to the surface, bringing nutrients with it. Turbid (cloudy) water blocks light and limits photosynthesis to a thin surface layer. Where nutrients are scarce, as in much of the open ocean, phytoplankton stay sparse even in clear, warm, sunlit water.

Depth zones. In a lake, the shallow **littoral zone** near shore is lit to the bottom, holds rooted emergent plants such as cattails, and has the richest variety of life; the open sunlit water is the **limnetic zone** and the dark bottom layers are the **profundal zone**. In the ocean, the sunlit surface layer (the **photic zone**) holds nearly all photosynthesis (Section 4.3), and shallow coastal waters hold the most species. The deep ocean is dark, cold and under great pressure (about 1 atmosphere more for every 10 m of depth), so no producers live there and only specially adapted animals do.

COMMON STUDENT MISTAKES

- Algae and phytoplankton, not large plants, produce most marine oxygen.
- Do not call an estuary "freshwater." Its salinity is intermediate and changes with the tides.

CLASSROOM EXAMPLE SG-U1-E003

A student records data for two aquatic sites. Site X: salinity 34 parts per thousand, temperature 27 °C, depth 8 m, very clear water, with corals and many fish species. Site Y: salinity 12 parts per thousand where a river meets a bay, depth 2 m, cloudy water full of sediment and rich in nutrients, with young fish and oysters. (a) Identify the aquatic biome at each site. (b) Explain how low turbidity at Site X helps its primary producers, and state one reason Site Y supports so many young fish.

QUICK CONCEPT SUMMARY

- Terrestrial biomes follow temperature and precipitation, which vary with latitude and altitude. Know the nine biomes, their climate and signature adaptations; biome boundaries are dynamic and may shift with climate change.
- Freshwater biomes: streams, rivers, ponds, lakes. Marine biomes: oceans, coral reefs, marshland, estuaries.
- Climate, soil and geography control non-mineral land resources; salinity, depth, turbidity, nutrients and temperature control marine resources.
- Marine algae produce a large share of Earth's oxygen and absorb atmospheric CO_2.

4.5 Biodiversity

Levels of biodiversity

CONCEPT MADE SIMPLE

Biodiversity is variety, counted at three scales. Think of a school. The students differ from one another (genetic variety within one species). The school has many kinds of clubs (variety of species). The campus has a gym, library, garden, and cafeteria (variety of habitats). A healthy ecosystem has lots of variety at every scale: when something goes wrong, someone can always step in.

Genetic diversity is the variety of genes (alleles) among the individuals of one species. **Species diversity** is the variety of different species in an ecosystem. **Habitat diversity** (also called ecosystem diversity) is

the variety of physical environments, such as forest, meadow, stream, and wetland, in an area. Each habitat supports its own set of species, so more habitats usually means more species.

KEY DEFINITIONS
Species richness — the number of different species found in an ecosystem. It counts kinds, not individuals. **Population bottleneck** — a sharp drop in population size that leaves the survivors with only a small sample of the original genes. **Specialist** — a species with narrow requirements for food, habitat, or conditions. **Generalist** — a species that can use many foods and habitats and tolerate a wide range of conditions.

Why genetic diversity matters. A genetically diverse population contains individuals with many different traits. When a stressor arrives, such as a new disease, a hot summer, or a pesticide, a few individuals are likely to carry traits that let them survive and reproduce. Those survivors rebuild the population. A population with little genetic variety may have no such individuals, so the same stressor can wipe out nearly everyone. A field planted with one identical crop variety can be lost to a single fungus, while a mixed wild meadow rarely is. Crops and livestock bred from a few uniform lines carry fewer alleles than wild populations, and breeding among close relatives in small populations makes harmful alleles show up more often.

A **bottleneck** causes loss of genetic diversity. When a population collapses, whether from hunting, fire, or disease, only the survivors pass genes forward. Rare alleles are often lost by chance, and even if numbers recover, the population stays genetically uniform.

Species richness and recovery. If a meadow plot contains 14 plant species, its richness is 14, whether it holds 20 plants or 2,000. Ecosystems with more species are more likely to recover from disruptions. More species means more different ways of doing the same job (several kinds of plants fix carbon, several kinds of insects pollinate). If a drought or pest eliminates one species, another with a different tolerance fills in. A low-diversity system depends on a few species, so losing one leaves a gap nothing can fill.

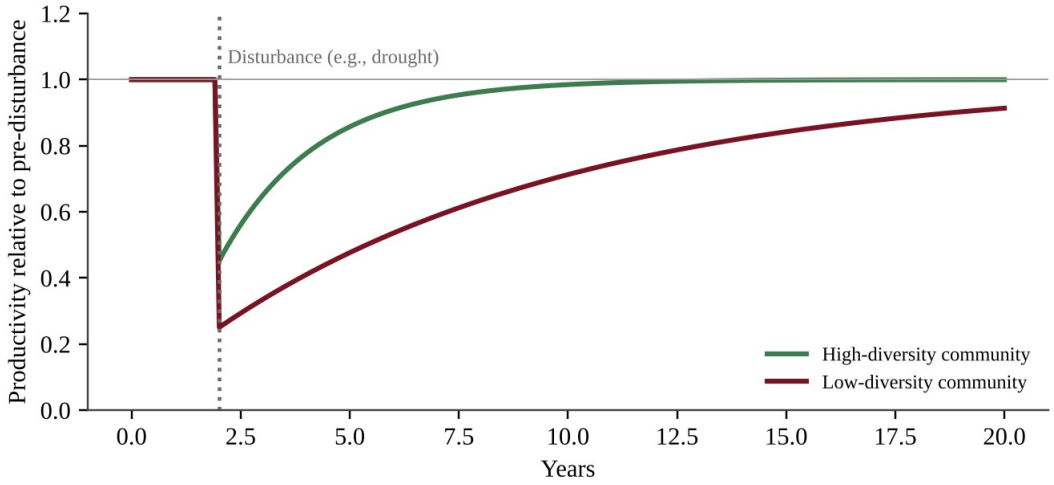


Figure 4.6 — A high-diversity community loses less productivity after a disturbance and recovers faster than a low-diversity community.

Reading the graph: the drop at the dotted line shows the size of the damage; the climb back to 1.0 shows resilience. The green line drops less and returns sooner.

Habitat loss. When habitat is destroyed or fragmented, species do not vanish all at once. **Specialists** go first because they cannot use any other habitat or food. Generalists follow later if the loss continues. Species with **large territorial requirements** (wolves, big cats, eagles) decline in number early, because a shrunken or chopped-up patch cannot hold enough territories. Habitat loss is the leading cause of biodiversity loss worldwide (Chapter 11). A **biodiversity hotspot** is a region with many endemic species (found nowhere else) that has lost much of its original habitat. Roads, fields, and clear-cuts split one large habitat into small patches. A small patch is mostly edge, where sun, wind, and outside predators change conditions, so forest-interior specialists lose space first. Corridors (strips of habitat joining patches) let

animals move between them. To decide who vanishes first, compare patch size with territory size, then ask how narrow the species' needs are.

COMMON STUDENT MISTAKES

- Confusing richness with total number of individuals. A plot with 500 ants of one species has richness 1.
- Saying a recovered population has "regained genetic diversity." A bottleneck loss of alleles is not undone by regrowth in numbers.
- Believing generalists are the first to disappear from habitat loss. Specialists go first.
- Treating "biodiversity" as only species. Genetic and habitat diversity are part of it too.

AP EXAM STRATEGY

When a question asks you to *explain* why diversity helps, give a chain: more genetic or species variety → some individuals or species tolerate the stressor → they survive and reproduce or take over the lost function → the ecosystem keeps functioning or recovers sooner. A bare statement such as "diversity is good" earns no credit.

CLASSROOM EXAMPLE SG-U2-E001

For each situation, name the level of biodiversity (genetic, species, or habitat) that is being described, and say in one sentence whether it is high or low in that situation. (a) A cheetah population descended from a few dozen survivors shows almost no variation in the genes that control immune response. (b) A 40-hectare reserve contains pine forest, a lake, a peat bog, and open grassland. (c) A 100 m² plot of coral reef holds 63 fish species.

CLASSROOM EXAMPLE SG-C4-E002

A forest plot was surveyed before and after selective logging. Before: oak 12 trees, maple 7, birch 3, hickory 5, beech 2. After: oak 9 trees, maple 4, birch 0, hickory 3, beech 0. (a) State the species richness of the plot before and after logging. (b) Calculate the percent change in species richness. (c) Calculate the percent change in the total number of trees, to the nearest percent. (d) Explain why richness and the number of individuals are different measures of biodiversity.

Ecosystem services

KEY DEFINITIONS

Ecosystem services — the benefits that people obtain from functioning ecosystems. **Provisioning services** — the material goods ecosystems supply: food, fresh water, timber, fiber, fuel, medicines. **Regulating services** — the ways ecosystems control conditions: climate regulation (carbon storage), flood control, water purification, pollination, disease and pest control, erosion control. **Cultural services** — the non-material benefits: recreation, tourism, spiritual value, education, and beauty. **Supporting services** — the basic processes on which all the others depend: nutrient cycling, soil formation, photosynthesis (primary production), and habitat for species.

Ecosystem services are the free benefits nature provides. Memory trick: provisioning = *products*; regulating = *rules* of the environment; cultural = *culture and enjoyment*; supporting = *the foundation*. One ecosystem usually gives all four. A mangrove forest supplies fish and wood (provisioning), buffers storm surge and traps sediment (regulating), draws ecotourists (cultural), and acts as a nursery for young fish while recycling nutrients (supporting). To classify a benefit, ask what it does: supplies a good, moderates a process or hazard, gives enjoyment, or keeps the ecosystem running.

Service category	Wetland example	Forest example
Provisioning	Fish, rice, drinking water	Timber, mushrooms, medicinal plants
Regulating	Absorbs floodwater, filters pollutants	Stores carbon, reduces erosion on slopes
Cultural	Bird watching, education	Hiking, spiritual significance
Supporting	Nutrient cycling, nursery habitat	Soil formation, wildlife habitat

Human disruption of ecosystem services. Anthropogenic (human-caused) activities can damage or remove these services, and the results are both **ecological** and **economic**. Cutting a mountain forest for timber gains short-term income (provisioning) but can destroy flood and erosion control (regulating): the next rains send mud into the river, the reservoir silts up, and the downstream town pays for dredging and flood repair. Pollinator decline from pesticide overuse lowers crop yields and forces farmers to pay for hand pollination or rented hives. Draining wetlands removes natural water purification, so a city must build an expensive treatment plant.

PROCESS & CAUSE-AND-EFFECT

Human activity (clearing, draining, polluting) → ecosystem function damaged → service lost → ecological consequence (flooding, soil loss, fewer species) → economic consequence (repair costs, lower yields, lost tourism)

Putting a dollar value on services. Two common methods compare costs and benefits. **Market value** is quantity × price (a harvest). **Replacement cost** is what a machine or plant would cost to do the same job (a treatment plant replacing a wetland's filtering).

FORMULA & MATH SKILL

Value of a service ($/yr) = area (ha) × rate (units/ha/yr) × price ($/unit) Cost over t years = one-time cost + (annual cost × t) Net benefit = total benefits − total costs Convert every service to $ per year before adding; compare options over the same number of years. Example: 40 ha × 150 kg/ha/yr × $5/kg = $30,000/yr.

When an ecosystem is converted, compare the developer's profit with the *total* value of all services lost; the loss usually falls on the public. Solutions: build on already-degraded land, restore habitat elsewhere, require new building to stay back from the shore, or make the developer pay for lost services. Each has a cost, which is its tradeoff.

CLASSROOM EXAMPLE SG-U2-E009

A 120-ha tidal marsh filters nutrients from a bay. A treatment plant doing the same job would cost $9 million to build and $0.5 million per year to run. Protecting the marsh costs $120 per hectare per year. The marsh also supports 90 kg/ha/yr of shellfish worth $8/kg. (a) Calculate the yearly value of the shellfish harvest. (b) Calculate the 15-year cost of the plant and of protecting the marsh, and the difference. (c) Classify the filtering and the shellfish as ecosystem services. (d) State one tradeoff of relying on the marsh.

CLASSROOM EXAMPLE SG-U2-E002

A river valley has a floodplain forest. Classify each of the following benefits as provisioning, regulating, cultural, or supporting: (i) the forest soaks up spring floodwater; (ii) families harvest wild nuts; (iii) leaf litter decomposes and returns nitrogen to the soil; (iv) a canoe-rental company runs tours. Then, if the forest were cleared for farmland, describe one ecological and one economic consequence for the town downstream.

Island biogeography

Big, near islands hold the most species; small, remote ones the fewest. **Island biogeography** is the study of the ecological relationships and distribution of organisms on islands, and of the community structure of those organisms. The ideas apply to any isolated habitat: a mountaintop, a lake, a forest fragment surrounded by farmland, or a nature reserve in a city.

Islands were **colonized** in the past by species that arrived from elsewhere: birds flying, seeds blown by wind or carried in feathers, insects rafting on floating logs, and sometimes human introductions. The number of species on an island reflects a balance between new species arriving (**colonization**) and species being lost (**extinction**).

PROCESS & CAUSE-AND-EFFECT

Distance: island near mainland → more colonists arrive → more species. Island far away → fewer colonists → fewer species. **Size:** large island → more habitats and resources → larger populations → lower extinction rate → more species. Small island → small populations → more likely to disappear → fewer species.

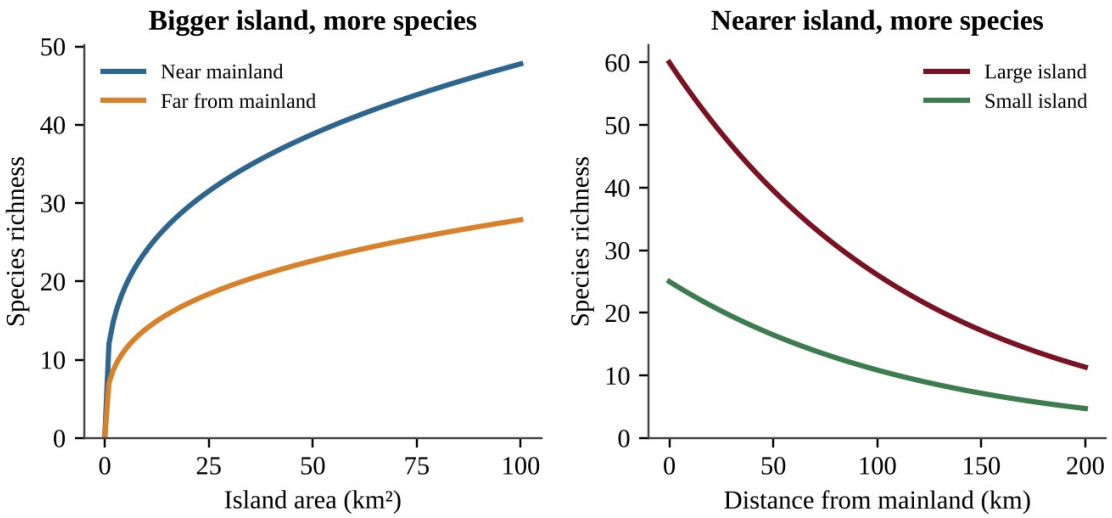


Figure 4.7 — Species richness rises with island area (left) and falls with distance from the mainland (right).

Reading the graphs: on the left, both lines rise with area, and the near-mainland line sits higher at every size. On the right, richness drops as distance grows, and the large-island line sits above the small-island line. Describe both variables: "richness increases with island area and decreases with distance."

Equilibrium: arrivals balance losses. As an island fills, fewer new species remain to arrive, so the **immigration rate falls**. More species means smaller populations and more competition, so the **extinction rate rises**. Where the curves cross is the **equilibrium**: arrivals equal losses and richness stays steady. Read the number of species straight down from the crossing.

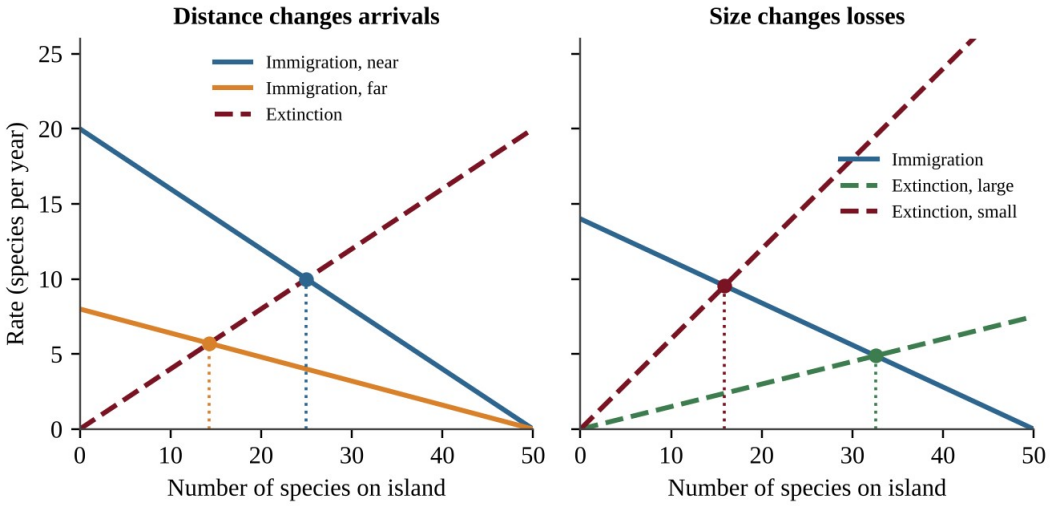


Figure 4.8 — Distance lowers the immigration curve (left); small size raises the extinction curve (right). Each dot marks an equilibrium.

Reading the graphs: a far island has a lower immigration curve and a small island a steeper extinction curve; both push the crossing left. Equilibrium is a *number*, not a fixed list: species keep arriving and disappearing (turnover), so two surveys can show the same richness but different species. **Species–area graphs** rise quickly on small areas and slowly on large ones. On a logarithmic area axis each equal step multiplies area by 10, so 300 ha sits about halfway between the 100 and 1,000 ha marks. To predict an island's richness, bracket its area between islands at a similar distance.

Island biogeography and evolution. Island resources such as food and territory are limited. Over time many island species evolve to be **specialists** that use one narrow food or habitat, which reduces competition among the few species present. Isolation also lets populations diverge from mainland relatives, so islands are well known for unusual species found nowhere else. This specialization carries a risk. Invasive species that arrive on islands are typically **generalists**: they eat many foods, tolerate many

conditions, and reproduce quickly. A generalist can outcompete a specialist for food or space, or prey on animals that evolved without defenses against such predators. The long-term survival of island specialists can therefore be jeopardized when invasive species are introduced.

COMMON STUDENT MISTAKES

- Saying a bigger island has more species "because it is closer." Size and distance are two separate variables; keep them separate.
- Claiming islands have more species than the mainland. Islands generally have *fewer* species than a comparable mainland area.
- Assuming invasive species are specialists. They are typically generalists.

CLASSROOM EXAMPLE SG-U2-E003

Four islands were surveyed for land-bird species.

Island	Area (km²)	Distance from mainland (km)	Bird species
P	4	10	18
Q	4	120	9
R	90	10	41
S	90	120	24

(a) Describe the effect of distance on bird richness, using a pair of islands that differ only in distance, and give the percent decrease for that pair. (b) Describe the effect of area on richness for islands at the 120 km distance. (c) Predict which island is most likely to lose a specialist species through chance extinction, and justify your choice.

Natural disruptions to ecosystems

Nature disturbs ecosystems all the time, sometimes harder than humans do. Natural disruptions include fires, floods, hurricanes, volcanic eruptions, earthquakes, drought, disease outbreaks, and slow shifts in climate. **Short-term** disruptions (a storm, a fire) act over hours to years. **Long-term** disruptions (an ice age, a change in sea level) act over thousands to millions of years. The consequences of a natural disruption, for a given occurrence, may be as great as or greater than those of many human-made disruptions. A major eruption can bury a whole landscape in ash, and a hurricane can flatten a forest.

Earth system processes operate over a range of time scales, and they differ in how predictable they are:

KEY DEFINITIONS

Periodic — happens at regular, predictable intervals (seasons, annual monsoon flooding, daily tides). **Episodic** — happens in occasional bursts, with some pattern in how often but not on a fixed schedule (hurricane seasons with active and quiet years; large wildfires every few decades). **Random** — has no predictable pattern in timing (a meteorite impact, a single earthquake or volcanic eruption).

Climate and sea level. Earth's climate has changed over geological time for many reasons, including changes in Earth's orbit and tilt, variations in solar output, volcanic activity, shifting continents, and changes in greenhouse gas levels. During cold (glacial) periods large amounts of water are locked in

continental ice, so **sea level falls**; during warm periods ice melts and **sea level rises**. Falling sea level can expose land bridges and shallow shelves; rising sea level floods coastal lowlands. Continental-shelf islands often share mammals with the mainland because lower glacial sea level once joined them by land.

Habitat change and migration. Major environmental change or upheaval commonly results in **large swaths of habitat change**: grassland may turn to desert, forest may retreat or advance, shorelines move inland. Wildlife responds. **Migration** may be short-term (a seasonal trip to avoid winter, or fleeing a fire and returning) or long-term (a species' range shifting poleward over centuries as climate warms). Animals migrate for many reasons, including food, breeding, temperature, and natural disruptions. A wet winter in a fire-prone region grows extra grass and shrubs; once dry, that fuel makes the next fire larger. As climate warms, species track their temperature range poleward or upslope, but a narrow-tolerance species stranded on a mountaintop or island may have nowhere to go.

COMMON STUDENT MISTAKES

- Believing that natural disruptions are always smaller than human ones. A single natural event can exceed many human-caused disruptions.
- Mixing up "episodic" and "periodic." Periodic = regular clock-like timing; episodic = occasional bursts; random = unpredictable.
- Saying sea level falls when ice melts. Melting land ice *raises* sea level.

CLASSROOM EXAMPLE SG-U2-E005

A valley forest has a net primary productivity of 1,200 g/m²/yr before a hurricane. The measured values afterward are: Year 1 = 300, Year 3 = 540, Year 5 = 780, Year 8 = 1,020. (a) Calculate the percent decrease in productivity in Year 1 compared with the pre-hurricane value. (b) Calculate the average increase in productivity per year between Year 1 and Year 5. (c) Classify hurricanes as periodic, episodic, or random for a coastal region with several active years and several quiet years, and justify. (d) Describe one way wildlife might respond in the short term.

Adaptations

Adaptation is a population change across generations, not something an individual does. Individuals in a population vary, and many differences are **heritable** (passed from parents to offspring through genes). Some variants help an organism survive and reproduce in its particular environment. Those individuals leave more offspring, so the helpful trait becomes more common in the next generation. Over many generations these incremental changes at the genetic level make the population **adapted** to its environment.

Example: a beetle population contains green and brown beetles. Birds in a brown, dry landscape eat green beetles more often. Brown beetles survive longer, breed more, and the next generation has a higher percentage of brown beetles. No individual beetle changed color. The *population's* gene frequencies shifted. **Genetic diversity supplies the variation** that natural selection needs; a population with no variation cannot adapt. Species that interact over long periods adapt to each other (**coevolution**); native pollinators and native plants fit each other closely, so introduced pollinators often serve native plants less well.

KEY DEFINITIONS

Natural selection — the process in which individuals with heritable traits that improve survival and reproduction leave more offspring, so those traits become more common. **Adaptation** — a heritable trait that increases an organism's fitness in its environment; also the process of a population becoming better suited over generations. **Heritable trait** — a trait controlled by genes that can be passed to offspring.

When the environment changes. Environmental change may be sudden (a volcanic eruption, an oil spill) or gradual (slow warming, a rising sea). Either may threaten a species' survival. Individuals have three possible outcomes: **alter behavior** (hunt at night when days get too hot), **move** (migrate to a better area), or **perish**. Behavioral change and movement happen within an individual's lifetime and are not genetic adaptation. Whether the *species* survives depends on whether enough individuals can do one of these or whether the population has heritable variation that selection can act on quickly enough. Fast change, few individuals, long generation times, and low genetic diversity all work against survival.

Kinds of adaptation and selection in action. Adaptations may be **structural** (large heat-shedding ears), **physiological** (concentrated urine that saves water), or **behavioral** (nocturnal activity). A behavior counts as an adaptation when it is inherited; a one-time response to a situation is not. **Mutations** are random changes in DNA that supply new alleles. The environment does not cause helpful mutations on demand; it only decides which existing variants survive.

Resistance is natural selection. Each spraying of one pesticide or herbicide kills susceptible individuals and leaves resistant ones to breed, so the resistant fraction rises every generation, fastest in species with many offspring and short generations. To slow it, reduce selection pressure: leave untreated refuges so susceptible individuals keep mating with survivors, rotate products with different modes of action, and spray only when needed. Higher doses or more frequent spraying strengthen selection.

COMMON STUDENT MISTAKES

- Saying "the organism adapted because it needed to." Selection acts on existing variation; needs do not create traits.
- Claiming individuals evolve during their lives. Populations change across generations.
- Confusing learned behavior or acclimation (not inherited) with adaptation (heritable change).

CLASSROOM EXAMPLE SG-U2-E011

A weed species was sprayed with the same herbicide for six seasons. The share of resistant plants rose: 3% (season 1), 9%, 26%, 58%, 81%, 93% (season 6). (a) Describe the trend. (b) Explain it as natural selection, naming the variation, the selective agent, and the outcome. (c) Propose one change that would slow resistance and justify it.

CLASSROOM EXAMPLE SG-U2-E006

A population of 2,000 field mice is reduced to 40 individuals by a disease outbreak, then recovers to 2,000 over six years. Before the outbreak, 5 different alleles of an immune-response gene were present; after recovery only 2 of them are found. (a) Name the process that explains the loss of alleles and state what happened to genetic diversity even though the population size returned to normal. (b) A new, more virulent disease then appears. Explain why this population is at greater risk than the original population. (c) Distinguish between a mouse changing its behavior to avoid the disease and the population adapting to it.

Ecological succession

KEY DEFINITIONS

Primary succession — community development in an area with no soil, such as bare rock after a lava flow, a retreating glacier, or a new sand dune. It starts from nothing and takes hundreds to thousands of years. **Secondary succession** — community development in an area where soil remains after a disturbance, such as an abandoned field, a burned forest, or a cleared lot. It is faster because the soil already holds nutrients, organic matter, and seeds. **Pioneer species** — the first species to colonize unoccupied or newly disturbed habitat, usually hardy, fast-growing, and good at dispersing. **Keystone species** — a species whose activities play a particularly significant role in determining community structure, much more than its abundance would suggest. **Indicator species** — a plant or animal whose presence, abundance, scarcity, or chemical composition shows that some distinctive aspect of the character or quality of an ecosystem is present.

Succession is an ecosystem rebuilding itself in stages: hardy species move in, change the place, and let new species follow until the community is stable. **Primary succession** typically proceeds: bare rock → lichens and mosses (pioneers) that slowly break down rock and add organic matter → thin soil → grasses and herbs → shrubs → trees → mature community. **Secondary succession** skips the soil-building stage: the field or burned forest starts with weeds and grasses, then shrubs, fast-growing trees, and finally a mature forest. A storm or fire that leaves the soil restarts *secondary* succession.

Pioneers and the origin of new species. Pioneer members of an early successional species commonly move into unoccupied habitat. Over time they adapt to its particular conditions, which may result in the origin of new species: a founding population in a new, isolated place faces new selective pressures and can diverge from its parent population. Pioneers also change the environment: lichens create soil, and early plants add organic matter and shade, making the habitat suitable for later species.

What changes during succession. Succession in a disturbed ecosystem affects **total biomass** (the mass of living organisms), **species richness**, and **net productivity** (the rate at which producers store energy as new biomass, after their own respiration) over time.

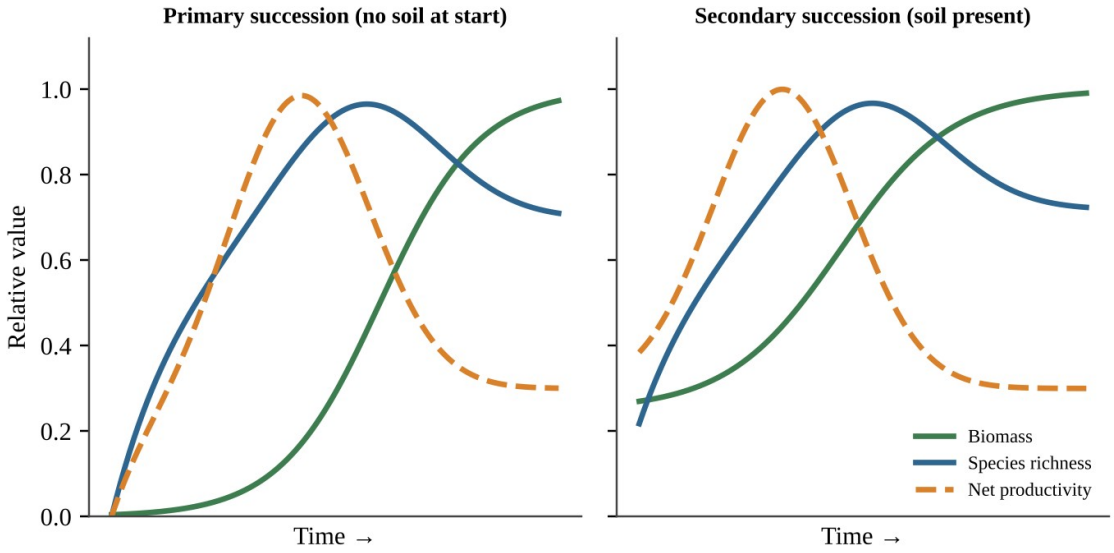


Figure 4.9 — Typical trends during primary (left) and secondary (right) succession. Secondary succession starts with higher values because soil is already present.

- **Biomass** increases steadily and levels off in a mature community, where large long-lived plants store much material.
- **Net productivity** is highest during the fast-growing middle stages and declines as the community matures, because a large mature community uses much of what it produces for its own respiration and maintenance.
- **Species richness** generally increases through the early and middle stages as habitats become more varied, and may level off or decline slightly late in succession when a few dominant, shade-casting species crowd others out.

Reading the graph: identify the type of succession from where the lines begin (near zero means no soil, so primary; already substantial means soil present, so secondary). Then describe each line using words such as "increases," "peaks," and "levels off."

As the canopy closes, light at the ground falls, so sun-loving pioneers (such as sweet gum or pine) decline and shade-tolerant, late-successional trees (oak, hickory, beech) take over, leading to a stable mature (climax) community. After clear-cutting, such a forest takes many decades to centuries to return.

FORMULA & MATH SKILL

Productivity per biomass (yr^{-1}) = NPP ÷ total biomass Young stands give high ratios; old stands give low ratios, because a mature stand holds much woody tissue that adds no growth but needs energy for respiration. Example: NPP 0.9 $kg/m^2/yr$ with biomass 6.0 kg/m^2 → 0.9 ÷ 6.0 = 0.15 yr^{-1}.

Keystone and indicator species. A **keystone species** has an influence far larger than its numbers. Sea otters eat sea urchins; without otters, urchins overgraze kelp, and the kelp forest community collapses. Beavers build dams that create wetlands for dozens of other species. An **indicator species** signals something about ecosystem condition. Certain stonefly larvae need clean, oxygen-rich water, so their presence indicates good water quality and their disappearance suggests pollution. Lichens that are sensitive to air pollution work the same way. A keystone species *shapes* the community; an indicator species *reports* on it.

COMMON STUDENT MISTAKES

- Saying primary succession occurs after a forest fire. Burned forest keeps its soil, so that is secondary.
- Assuming a keystone species must be abundant or large. Its importance comes from its effect, not its numbers.
- Treating indicator and keystone species as the same. One reports conditions; the other controls community structure.
- Stating that net productivity keeps rising until the end of succession. It peaks in the middle and falls in mature stages, while biomass keeps climbing and then levels off.

CLASSROOM EXAMPLE SG-U2-E007

Researchers measured two plots after a disturbance. Plot X began as bare, scoured rock after a glacier retreated. Plot Y was a farm field abandoned after plowing.

Year	Plot X species	Plot Y species	Plot Y biomass (kg/m²)
1	0	12	0.3
10	3	26	2.1
30	9	31	6.5
80	15	24	14.0

(a) Identify the type of succession in each plot and justify using the data. (b) Calculate the percent increase in Plot Y biomass between Year 10 and Year 30. (c) Describe the trend in Plot Y species richness over the 80 years and give one reason the trend changes after Year 30. (d) A lichen species grows only on Plot X's rock in Year 1–10; explain how lichens make later colonists possible.

CLASSROOM EXAMPLE SG-U2-E008

A coastal island 80 km from the mainland has an area of 6 km². Its only ground-nesting bird, species G, eats only the seeds of one native grass and has a population of about 300. A generalist invasive rat arrives. In 15 years the rat population reaches 40,000 and species G declines to about 20 birds. (a) Describe two reasons a species like G is vulnerable on this island, using island biogeography and specialist-versus-generalist concepts. (b) Calculate the percent decrease in species G. (c) Propose one action to protect the remaining G population and justify it. (d) Explain why even a successful rat removal would leave species G with reduced ability to respond to future stressors, if its numbers stay near 20 for several generations.

Math & Skills Toolkit

- **Species richness:** count the number of different species and ignore how many individuals each has (oak 12, maple 3, birch 1 → richness 3).

- **Percent change** = (new value − old value) ÷ old value × 100. Divide by the *original* value; a negative result is a decrease (richness falls from 40 to 30 → (30 − 40) ÷ 40 × 100 = −25%). **Percent of whole** = part ÷ whole × 100.
- **Average rate of change** = (end value − start value) ÷ time elapsed, with units (biomass 2 to 8 kg/m² in 12 years → 0.5 kg/m² per year).
- **Conversions:** multiply by a fraction equal to 1 so units cancel (25 km² = 25 × 10^6 m²). **10% rule:** × 0.10 per upward transfer. **NPP = GPP − R.**
- **Graphs:** for tolerance curves, range = upper limit − lower limit. For recovery and succession graphs, note the starting value, the peak, where each line levels off, quote two data points and give a cause.

QUICK CONCEPT SUMMARY

- Biodiversity (genetic, species, habitat) builds resilience; bottlenecks reduce genetic diversity and habitat loss removes specialists first. Ecosystem services (provisioning, regulating, cultural, supporting) can be disrupted with ecological and economic costs.
- Island richness rises with area and falls with distance; invasive generalists threaten island specialists.
- Natural disruptions are periodic, episodic or random; populations adapt by natural selection, while individuals alter behavior, move or perish.
- Succession: primary from bare substrate, secondary from soil; biomass rises and levels off, net productivity peaks mid-succession. Keystone species shape communities; indicator species report conditions.

4.6 Practice Questions and Answers

Test what you have learned in this chapter with the classroom examples and exam-style questions below.

PRACTICE QUESTIONS AND ANSWERS

Classroom examples in this chapter (unsolved; space is provided in each box): SG-U1-E001, SG-U2-E004, SG-U2-E010, SG-U1-E008, SG-C4-E001, SG-U1-E009, SG-U1-E010, SG-U1-E002, SG-U1-E011, SG-U1-E003, SG-U2-E001, SG-C4-E002, SG-U2-E009, SG-U2-E002, SG-U2-E003, SG-U2-E005, SG-U2-E011, SG-U2-E006, SG-U2-E007, SG-U2-E008.

Classified practice: Book Two, Chapter 4 — 48 multiple-choice questions (Section A) followed by 7 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 4. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

CHAPTER

5

UNIT II — THE LIVING WORLD

Natural Biogeochemical Cycles

5.1 · Carbon Cycle
5.2 · Nitrogen Cycle
5.3 · Phosphorus Cycle
5.4 · Sulfur Cycle
5.5 · Water Cycle
5.6 · Role of H_2O, C, N, P, and S in the Human Body
5.7 · Practice Questions and Answers

LEARNING OBJECTIVES

- Trace carbon, nitrogen, phosphorus, sulfur and water through their reservoirs, naming every process and every arrow of the cycle.
- Distinguish short-term from long-term reservoirs and explain what residence time tells us.
- Explain how humans speed up each cycle (fossil fuels, fertilizer, sewage, SO_2, land use) and predict the result.
- Calculate carbon budgets, residence times, water budgets and mass conversions.
- Explain why water, carbon, nitrogen, phosphorus and sulfur are essential to the human body.

Matter is not made or destroyed in an ecosystem; it is used again and again. A **biogeochemical cycle** is the path an element takes between living things (*bio*), the air, water and rocks (*geo*) and chemical forms (*chemical*). Each cycle has **reservoirs** (pools that hold the element) and **fluxes** (the movements between them). A reservoir that releases more than it takes in is a **source**; one that takes in more than it releases is a **sink**. A reservoir can be a source in one process and a sink in another: a growing forest is a carbon sink, a burning forest is a carbon source.

5.1 Carbon Cycle

CONCEPT MADE SIMPLE

Carbon is the backbone of all living things. Plants pull CO_2 from the air and build sugar. Eating, breathing and rotting put CO_2 back. Burning fossil fuels adds carbon that was locked underground for millions of years.

The **carbon cycle** is the movement of atoms and molecules containing carbon between sources and sinks.

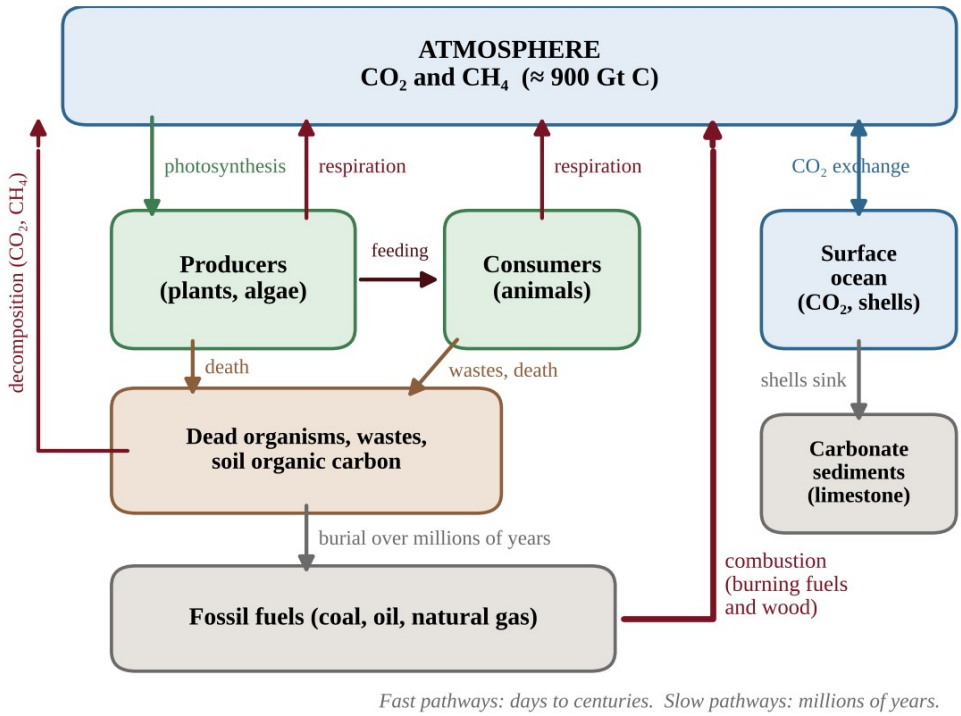


Figure 5.1 — The carbon cycle. Red arrows return carbon to the atmosphere; grey arrows show the slow geologic route.

How carbon moves

- **Photosynthesis** moves carbon from atmospheric CO_2 (or CO_2 dissolved in water) into organic compounds in producers.
- **Cellular respiration** returns carbon to the air as CO_2 when producers, consumers and decomposers break down organic compounds for energy.
- **Feeding** passes carbon through food chains.
- **Decomposition** releases CO_2 (and, where oxygen is scarce, as in wetlands and landfills, methane, CH_4) from dead matter and waste.
- **Gas exchange with the ocean:** CO_2 dissolves in and escapes from surface water, where it forms carbonic acid ($CO_2 + H_2O \rightleftharpoons H_2CO_3$), so the ocean takes up CO_2 but becomes more acidic (Chapter 11). Phytoplankton fix carbon, and some marine organisms build calcium carbonate shells; sinking remains become sediment.
- **Burial and fossilization:** plant and animal remains buried and compressed over millions of years form coal, oil and natural gas; carbonate sediments form limestone.
- **Combustion** of fossil fuels and wood moves carbon quickly into the atmosphere as CO_2.

Short-term and long-term reservoirs

Some reservoirs hold carbon for a short time and others for very long periods. The sizes below are rounded, approximate values.

Reservoir	Carbon stored (Gt C)	How long carbon stays
Atmosphere (CO_2, CH_4)	≈ 900	years
Living vegetation	≈ 450–650	years to centuries
Soil organic matter	≈ 1,500–2,400	decades to centuries
Surface ocean	≈ 1,000	years to decades
Deep ocean	≈ 38,000	centuries to a thousand years
Fossil fuels	several thousand	millions of years
Sedimentary rock (limestone)	tens of millions	tens to hundreds of millions of years

Burning fossil fuels returns carbon stored for millions of years to the atmosphere within decades, about 10 Gt C per year, which is why atmospheric CO_2 is accumulating (Chapter 11). Other human sources are deforestation, cement production, and methane from livestock, rice paddies and landfills.

Carbon budgets and seasonal data

To find the net change in a reservoir from a flux diagram, add every arrow into it and subtract every arrow out, keeping the units (for example Gt C/yr). A positive result means the reservoir is growing.

An ecosystem's carbon balance is carbon taken up by photosynthesis minus carbon released by *all* respiration (plants, animals, soil microbes). Positive is a net **sink**; negative is a net **source**. Add all seasons before judging: a tundra can release carbon in winter yet absorb more over the year.

Atmospheric CO_2 at a Northern Hemisphere station zigzags each year: land plants photosynthesize from late spring to early autumn, so CO_2 falls; in cold months respiration and decay win, so it rises. The rising baseline is fossil fuel burning adding carbon faster than sinks remove it.

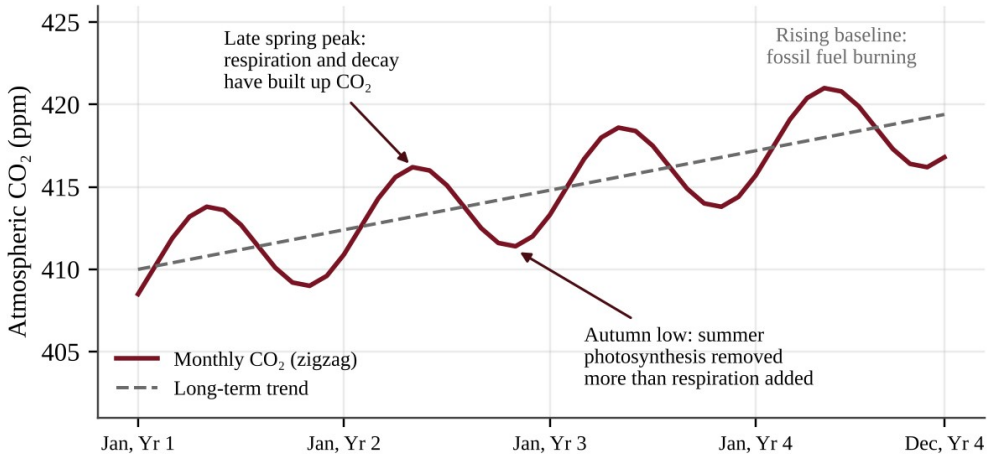


Figure 5.2 — Seasonal rise and fall of atmospheric CO_2 on top of a long-term upward trend.

Stored soil carbon becomes a source when disturbed. In cold or waterlogged soils (peat, mangrove, permafrost) decomposition is slow for lack of warmth or oxygen; draining, burning and warming speed it up and release CO_2. Plowing does the same: it aerates the soil and breaks up the aggregates that protect organic matter, so microbes respire faster.

KEY DEFINITIONS

Flux — the amount moving between reservoirs per unit time (for example Gt C/yr). **Fossil fuel** — coal, oil or natural gas formed from buried organisms over millions of years.

FORMULA & MATH SKILL

Net change in reservoir = Σ inputs − Σ outputs Ecosystem carbon balance = GPP − total respiration (positive = net sink) Mass of CO_2 = mass of C × 44 ÷ 12 ≈ mass of C × 3.67 (molar masses: CO_2 44, C 12) Photosynthesis: 6 CO_2 + 6 H_2O + light energy → $C_6H_{12}O_6$ + 6 O_2 Cellular respiration: $C_6H_{12}O_6$ + 6 O_2 → 6 CO_2 + 6 H_2O + usable energy Combustion of a hydrocarbon: fuel + O_2 → CO_2 + H_2O + heat

COMMON STUDENT MISTAKES

- Plants also respire, releasing CO_2; they are not only sinks.
- Fossil fuel carbon was out of the fast cycle for millions of years; burning it adds *new* carbon to the fast cycle.

AP EXAM STRATEGY

Label each arrow with its process and say whether it moves carbon into or out of the atmosphere. For a budget, show "inputs − outputs" with units.

CLASSROOM EXAMPLE SG-U1-E004

A carbon atom is part of a coal seam. A power plant burns the coal, and the carbon enters the air. Later that atom is taken up by a wheat plant, the wheat is eaten by a cow, and the cow exhales it. (a) List, in order, the four named processes that move the atom from the coal seam to the exhaled breath, and give the form of the carbon after each step. (b) State whether the coal seam acted as a source or a sink in this sequence, and explain why this carbon was not part of the fast cycle before the plant was built.

CLASSROOM EXAMPLE SG-U1-E012

A 50-ha peatland stores 600 Mg of carbon per hectare in its top 1 m of soil. After it is drained and plowed, 12% of that carbon is oxidized to CO_2 over ten years. (a) Calculate the mass of CO_2 released, in Mg. (b) Explain why draining speeds up the release. (c) State whether the peatland is now a source or a sink.

CLASSROOM EXAMPLE SG-C5-E001

A simplified yearly budget for atmospheric carbon (Gt C per year) lists these inputs: fossil fuel burning 9.5, land-use change 1.1, respiration and decay on land 120, release from the ocean 90. The outputs are: photosynthesis on land 123 and absorption by the ocean 93. (a) Calculate the net change in atmospheric carbon in Gt C per year. (b) If 2.1 Gt C added to the atmosphere raises CO_2 by 1 ppm, calculate the yearly rise in CO_2 in ppm. (c) Convert the 9.5 Gt C from fossil fuels into Gt of CO_2.

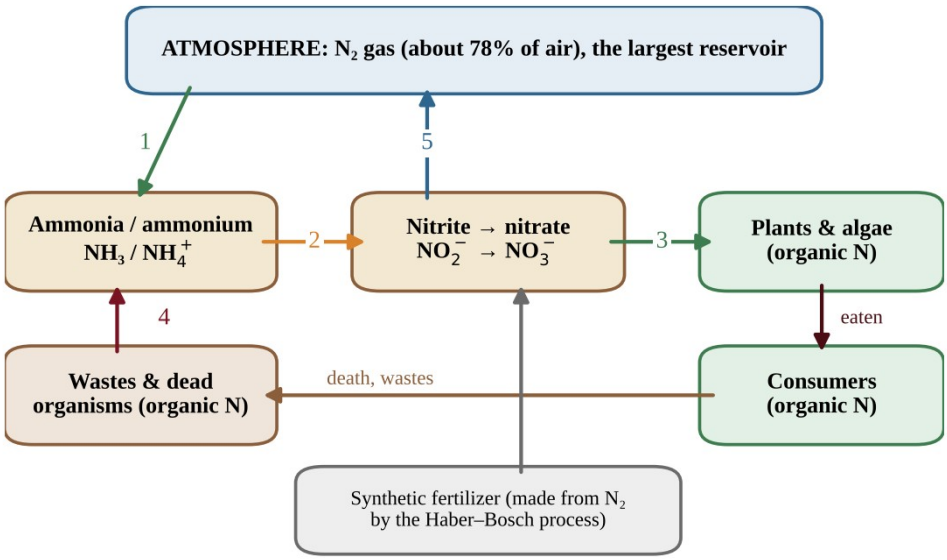
QUICK CONCEPT SUMMARY

- Carbon moves between sources and sinks; photosynthesis and respiration link living things to the atmosphere.
- Short-term reservoirs: atmosphere, organisms, soil, surface ocean. Long-term: fossil fuels, rocks, deep-ocean sediments.
- Burning fossil fuels moves carbon stored for millions of years into the atmosphere within decades.

5.2 Nitrogen Cycle

CONCEPT MADE SIMPLE
Nitrogen is in every protein. Air is almost 80% nitrogen, but plants cannot use the gas. Special soil bacteria must first change it into forms roots can absorb.

Nitrogen is a key part of proteins, DNA and chlorophyll. Air is about 78% nitrogen gas (N_2), yet most organisms cannot use it, because the two atoms in N_2 are held by a very strong triple bond. The **nitrogen cycle** is the movement of nitrogen-containing atoms and molecules between sources and sinks. It involves several steps, and microorganisms in the soil do most of the work.



1 Nitrogen fixation: $N_2 \rightarrow NH_3$ (soil bacteria; also lightning)
2 Nitrification: $NH_4^+ \rightarrow NO_2^- \rightarrow NO_3^-$ (soil bacteria, needs O_2)
3 Assimilation: roots take up NH_4^+ / NO_3^- and build organic N
4 Ammonification: decomposers turn organic N back into NH_4^+
5 Denitrification: $NO_3^- \rightarrow N_2$ (bacteria in oxygen-poor soil)

Figure 5.3 — The nitrogen cycle. Numbers match the five steps described in the text.

The five steps

1. **Nitrogen fixation** converts atmospheric N_2 into ammonia (NH_3) by certain soil bacteria, including bacteria in root nodules of legumes such as beans and clover. In soil, ammonia quickly becomes **ammonium (NH_4^+)**, which is available for biological uptake. Lightning also fixes a smaller amount.
2. **Nitrification** is the conversion of ammonium to nitrite (NO_2^-) and then nitrate (NO_3^-) by soil bacteria that need oxygen. Nitrate is the main form that plant roots absorb. It dissolves easily, so it is also easily lost by leaching.
3. **Assimilation** is the uptake of ammonium and nitrate by plants and algae and their conversion into proteins and nucleic acids. Animals assimilate nitrogen by eating plants or other animals.
4. **Ammonification** (decay) is the conversion of nitrogen in wastes and dead organisms back to ammonium by decomposers.
5. **Denitrification** is the conversion of nitrate to nitrogen gas by bacteria in oxygen-poor (anaerobic) conditions such as waterlogged soils, returning nitrogen to the atmosphere.

Reservoirs and limits

The **atmosphere is the largest reservoir** of nitrogen, but it is not directly available. Most of the other reservoirs (soil nitrate and ammonium, living organisms, wastes) hold nitrogen for relatively short periods. The availability of nitrogen compounds in the soil is limited by the rate of nitrogen fixation. Because of

this, nitrogen often **limits primary production** by plants and other producers: when more nitrogen is added, plants grow faster.

Human effects

Humans have greatly increased fixation. The Haber–Bosch process makes ammonia from N_2 for fertilizer, legume crops fix nitrogen on farms, and high-temperature burning makes nitrogen oxides. Excess nitrate runs off into waterways and causes eutrophication (Chapter 9). Nitrate leaches easily because it is negatively charged and does not cling to soil particles, which are also mostly negative.

COMMON STUDENT MISTAKES

- Plants cannot take up N_2 from the air; only certain bacteria (and lightning) fix it.
- Nitrification needs oxygen; denitrification occurs when oxygen is scarce.
- Do not confuse nitrification (makes nitrate) with denitrification (removes nitrate).

AP EXAM STRATEGY

Name the step, the organism and the nitrogen form before and after. Flooded or compacted soil means oxygen-poor, so think denitrification.

CLASSROOM EXAMPLE SG-U1-E005

A farmer notices that, after a week of flooding, an oxygen-poor, waterlogged field has much less nitrate and the corn is yellowing and stunted. (a) Name the process that increased in the waterlogged soil and state the product that leaves the soil. (b) Explain why the loss of nitrate reduces plant growth. (c) Name the process by which dead corn leaves, left on the field, would become a new supply of ammonium.

CLASSROOM EXAMPLE SG-C5-E002

Urea, $CO(NH_2)_2$, is a common nitrogen fertilizer. Atomic masses: C = 12, O = 16, N = 14, H = 1. (a) Calculate the percent of nitrogen by mass in urea. (b) A farmer wants to apply 150 kg of nitrogen per hectare to 40 ha. Calculate the mass of urea needed, in kg. (c) Suppose 20% of the applied nitrogen is lost as nitrate by leaching. Calculate the mass of nitrogen lost, in kg, and explain why nitrate leaches more easily than ammonium.

QUICK CONCEPT SUMMARY

- Nitrogen cycle steps: fixation, nitrification, assimilation, ammonification, denitrification; soil microorganisms do most of the work.
- The atmosphere is the largest reservoir, but N_2 is unusable until fixed. Most other reservoirs turn over quickly.
- Nitrogen frequently limits plant growth; humans add large amounts through fertilizer and combustion.

5.3 Phosphorus Cycle

CONCEPT MADE SIMPLE

Phosphorus is in DNA, in the cell's energy molecule and in bones. It never becomes a gas, so it moves slowly through rock, soil and water. Because it is scarce, a little extra can turn a lake green with algae.

Phosphorus is part of DNA, ATP (the cell's energy molecule) and cell membranes. The **phosphorus cycle** is the movement of phosphorus-containing atoms and molecules between sources and sinks. It differs from the carbon and nitrogen cycles in one important way: **it has no significant atmospheric component.**

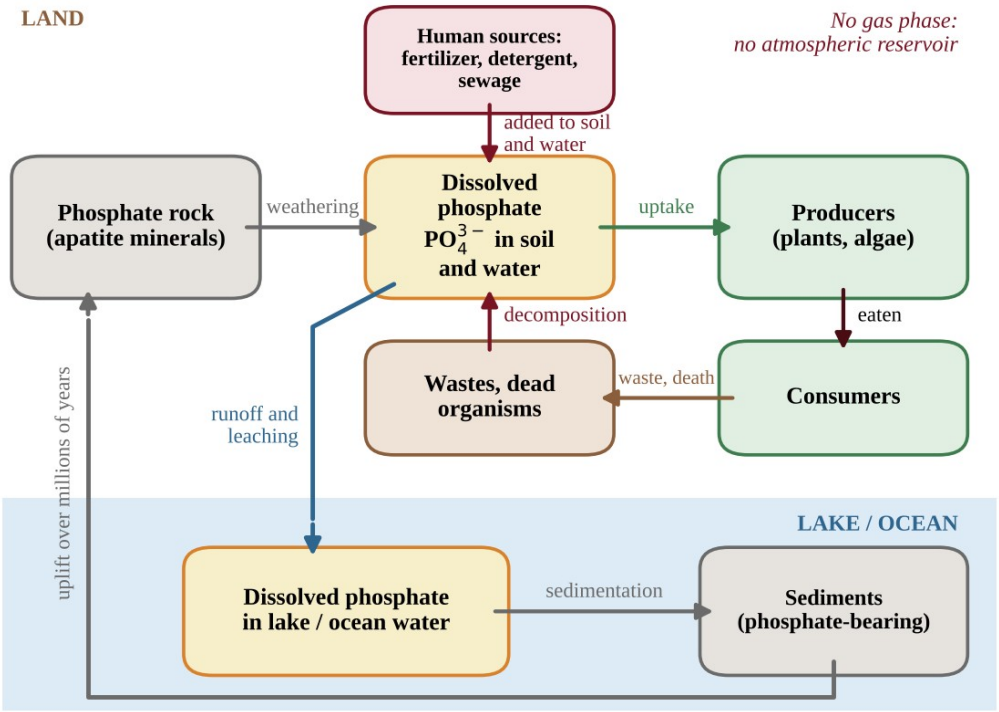


Figure 5.4 — The phosphorus cycle. Phosphorus has no gas phase; the slow return from sediments to rock takes millions of years.

The steps

1. **Weathering** slowly breaks down phosphorus-bearing rock and releases phosphate (PO_4^{3-}) into soil and water.
2. **Uptake** by plants and algae takes dissolved phosphate through roots or cell surfaces.
3. **Transfer** moves phosphorus through the food chain as consumers eat producers. (Phosphate-rich seabird droppings, guano, are mined as fertilizer.)
4. **Return** occurs through wastes and decomposition, which release phosphate back to the soil or water for reuse.
5. **Sedimentation** carries phosphate to ocean and lake bottoms, where it forms sediments. Over millions of years, uplift can bring sedimentary rock back to land, restarting the cycle.

The **major reservoirs** are rock and ocean sediments that contain phosphorus-bearing minerals. Because rocks weather slowly, phosphorus is relatively scarce in ecosystems. It is therefore often a **limiting nutrient** for producers, particularly in freshwater and some terrestrial ecosystems. This is why a small addition of phosphorus (from fertilizer, sewage or some detergents) can cause an algal bloom in a lake. Cutting the supply of the limiting nutrient helps: match fertilizer and manure to crop needs, keep vegetated buffer strips and wetlands that trap runoff, and remove phosphate at sewage plants.

Designing and judging a nutrient experiment

A limiting nutrient is found by adding it and watching the response.

- **Hypothesis:** an "if … then …" prediction naming both variables: if phosphorus limits growth, bottles with added phosphorus will hold more algae than bottles without it.
- **Independent variable:** nutrient added. **Dependent variable:** chlorophyll *a* or algal biomass.
- **Control:** identical water with nothing added.
- **Treatments:** N alone, P alone, both, and the control.
- **Constants:** light, temperature, volume, water source, duration.
- **Replication:** several bottles per treatment, averaged.

If nitrogen alone changes little, phosphorus alone raises growth a lot, and both together raise it most, phosphorus is the main limiting nutrient and nitrogen limits next. Growth in the control does not show that nothing limits it. Change one factor at a time: fertilizing plots in a different field, adding several things at once, or comparing only "before and after" with no untreated plots cannot isolate the cause.

FORMULA & MATH SKILL

Nutrient ratio = mass of N ÷ mass of P in the water. Algae use N and P at about 7 g N per 1 g P (about 16 N : 1 P by atoms). A water N:P ratio above about 7 suggests phosphorus is limiting; below 7 suggests nitrogen is limiting. Percent increase = (new − control) ÷ control × 100

COMMON STUDENT MISTAKES

- Do not include an atmospheric reservoir or a gas phase in the phosphorus cycle.
- Phosphorus is limiting in many freshwater lakes; nitrogen is more often limiting in many marine and terrestrial systems. Both can be limiting, so use the data in the question.

AP EXAM STRATEGY

Name the control, the independent and dependent variables and one constant. The limiting nutrient is the one whose addition gives the biggest change from the control.

CLASSROOM EXAMPLE SG-U1-E006

Researchers add nutrients to four identical experimental ponds for six weeks and measure algae concentration as chlorophyll in micrograms per liter (µg/L). Control (nothing added): 10 µg/L. Nitrogen only: 11 µg/L. Phosphorus only: 38 µg/L. Nitrogen and phosphorus together: 40 µg/L. (a) Identify the limiting nutrient in these ponds and justify your answer using the data. (b) Calculate the percent increase in algae concentration in the phosphorus-only pond compared with the control. (c) Explain why phosphorus is naturally scarce in freshwater ecosystems.

CLASSROOM EXAMPLE SG-C5-E003

Algae use nitrogen and phosphorus at a mass ratio of about 7 g N to 1 g P. Lake A has 1.4 mg/L of nitrogen and 0.10 mg/L of phosphorus. Lake B has 0.35 mg/L of nitrogen and 0.08 mg/L of phosphorus. (a) Calculate the N:P mass ratio of each lake. (b) State which lake is limited by phosphorus and which by nitrogen, and justify. (c) A treatment plant adds 0.12 mg/L of phosphorus to Lake A without changing its nitrogen. Calculate the new N:P ratio and state which nutrient now limits growth.

QUICK CONCEPT SUMMARY

- Phosphorus moves through rock, soil, water, organisms and sediments, with no significant atmospheric reservoir.
- Slow weathering makes it scarce, so it often limits producers, especially in fresh water.
- Added phosphorus can trigger rapid algal growth.

5.4 Sulfur Cycle

CONCEPT MADE SIMPLE

Sulfur gives rotten eggs and burning matches their smell. Living things need a little for proteins. It moves between rocks, ocean, air and organisms. Burning coal and oil adds sulfur gas to the air, where it makes acid rain.

The **sulfur cycle** is the movement of sulfur-containing atoms and molecules between rocks, water, air and living things. Like phosphorus, sulfur is stored mostly in rocks. Like carbon and nitrogen, it has a gas phase, so it can travel through the air.

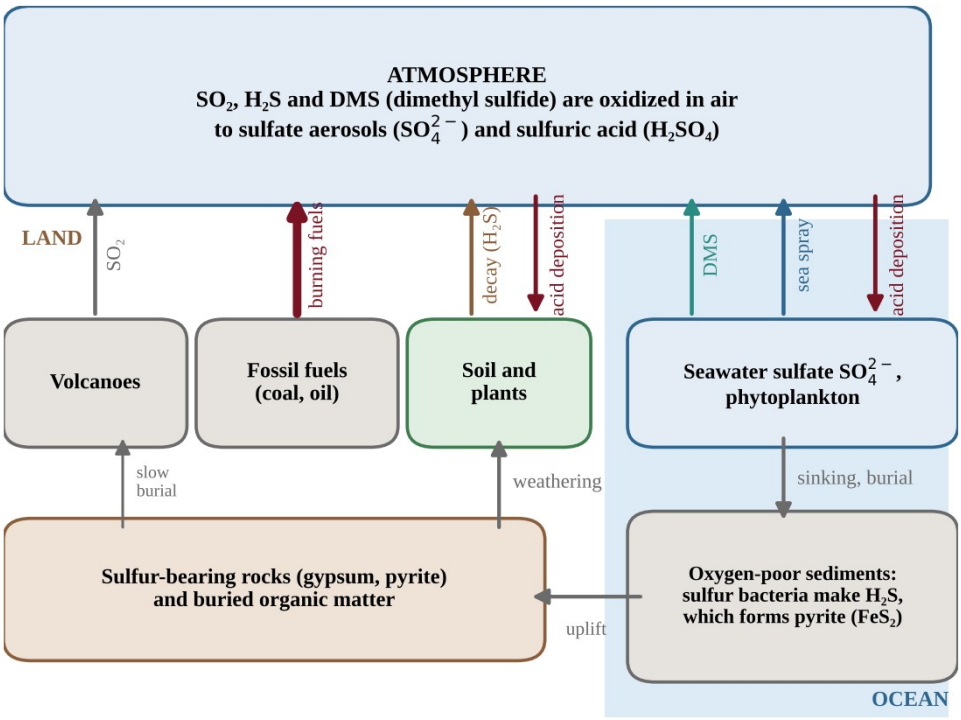


Figure 5.5 — The sulfur cycle. The thick red arrow is human burning of sulfur-bearing fuels, which now rivals all natural gas sources together.

Reservoirs

- **Rocks and sediments** are the largest reservoir: gypsum (calcium sulfate, $CaSO_4 \cdot 2H_2O$), pyrite (iron sulfide, FeS_2) and sulfur bound in shale. These release sulfate (SO_4^{2-}) slowly by weathering.
- **The ocean** holds huge amounts of dissolved sulfate and is the largest *surface* reservoir.
- **Fossil fuels** (coal, oil, natural gas) contain sulfur from the organisms they formed from. Coal commonly holds 0.5–5% sulfur by mass.
- **Living things and soil** hold sulfur in proteins.
- **The atmosphere** holds little (SO_2, H_2S, DMS, sulfate aerosols) but moves sulfur fast and far.

Steps and fluxes

1. **Weathering and uptake.** Weathering releases sulfate to soil and water. Plants and algae absorb it and build amino acids (assimilation); animals get sulfur by eating.
2. **Decomposition.** Decomposers break down dead organisms and release sulfur as hydrogen sulfide (H_2S), the "rotten egg" gas, or back to sulfate if oxygen is present.
3. **Volcanoes** release SO_2 and H_2S. A large eruption can lift SO_2 into the stratosphere, where sulfate aerosols reflect sunlight and cool the planet for a year or two.
4. **Ocean emissions.** Phytoplankton release dimethyl sulfide (DMS), which is oxidized in air to sulfate aerosols that seed clouds. Sea spray adds sulfate too.
5. **Fossil fuel burning and smelting.** Burning coal or oil, and smelting metal ores such as copper sulfide, oxidizes sulfur to SO_2: $S + O_2 \rightarrow SO_2$. Human emissions now rival or exceed all natural gas sources combined.
6. **Atmospheric chemistry.** In the air SO_2 is oxidized and reacts with water to form sulfuric acid (H_2SO_4) and sulfate aerosols: $2\ SO_2 + O_2 \rightarrow 2\ SO_3$, then $SO_3 + H_2O \rightarrow H_2SO_4$.
7. **Deposition.** Sulfuric acid and sulfate return to Earth as acid rain and snow (wet deposition) or as dry particles and gas (dry deposition).
8. **Burial.** In oxygen-poor mud and swamps, sulfate is used by **sulfate-reducing bacteria**, which make H_2S. The H_2S combines with iron to form pyrite, and the sulfur is buried in sediment. Over millions of years uplift returns it to land.

Sulfur bacteria

Sulfate-reducing bacteria live without oxygen and use sulfate in place of oxygen, producing H_2S. **Sulfur-oxidizing bacteria** convert H_2S and sulfur back into sulfate. Some live at deep-sea vents, where the energy of H_2S builds food (chemosynthesis) and supports whole vent communities. Purple and green sulfur bacteria use H_2S in a form of photosynthesis.

Acid deposition and aerosols

Sulfuric acid lowers the pH of rain and lakes. It frees aluminum and washes calcium and magnesium from soil, which stresses trees and kills fish and their eggs, and it corrodes marble and limestone. Sulfate aerosols also scatter sunlight and brighten clouds, so they cool the climate. Cutting SO_2 (scrubbers, low-sulfur fuel, cap-and-trade rules) reduces acid rain but also removes this slight cooling.

REAL-WORLD EXAMPLE

Mount Pinatubo, 1991. The eruption injected about 20 million tonnes of SO_2 into the stratosphere. The sulfate aerosols spread worldwide and cooled Earth's surface by about 0.5 °C for roughly two years. This natural event shows how fast the sulfur cycle can move sulfur through the air.

FORMULA & MATH SKILL

$S + O_2 \rightarrow SO_2$ (molar masses: S 32, SO_2 64) Mass of SO_2 = mass of S × 64 ÷ 32 = mass of S × 2 Mass of S in fuel = mass of fuel × fraction sulfur

COMMON STUDENT MISTAKES

- Most sulfur is in rocks, sediments and the ocean, not the air.
- SO_2 is a gas; sulfate (SO_4^{2-}) is what it becomes after oxidation.
- Burning fuel does not create sulfur; it moves it from a long-term reservoir into the air.

AP EXAM STRATEGY

To link a cause to acid rain, write the chain: burning sulfur-bearing fuel → SO_2 → sulfuric acid → lower pH of rain.

CLASSROOM EXAMPLE SG-C5-E004

A power plant burns 2.0×10^6 metric tons of coal per year. The coal is 1.8% sulfur by mass, and all of the sulfur is converted to SO_2 (atomic masses: S = 32, O = 16). (a) Calculate the mass of sulfur burned per year, in metric tons. (b) Calculate the mass of SO_2 produced per year. (c) A scrubber removes 90% of the SO_2. Calculate the mass of SO_2 still released per year. (d) Explain how the SO_2 that escapes can lower the pH of rain far from the plant.

QUICK CONCEPT SUMMARY

- Sulfur is stored mostly in rocks, sediments and ocean sulfate; the atmosphere holds little but moves sulfur quickly.
- Natural sources: weathering, volcanoes, decomposition and marine DMS. Human source: burning sulfur-bearing fuels and smelting, which makes SO_2.
- SO_2 becomes sulfuric acid and sulfate aerosols (acid deposition and cooling); sulfur bacteria drive reduction and oxidation steps.

5.5 Water Cycle

CONCEPT MADE SIMPLE
Water evaporates, forms clouds, falls as rain or snow and flows back to the sea. The sun supplies the energy; gravity brings the water down.

The **hydrologic cycle**, powered by the sun, is the movement of water in its solid, liquid and gaseous phases between sources and sinks. Solar energy evaporates water, and gravity returns it to Earth.

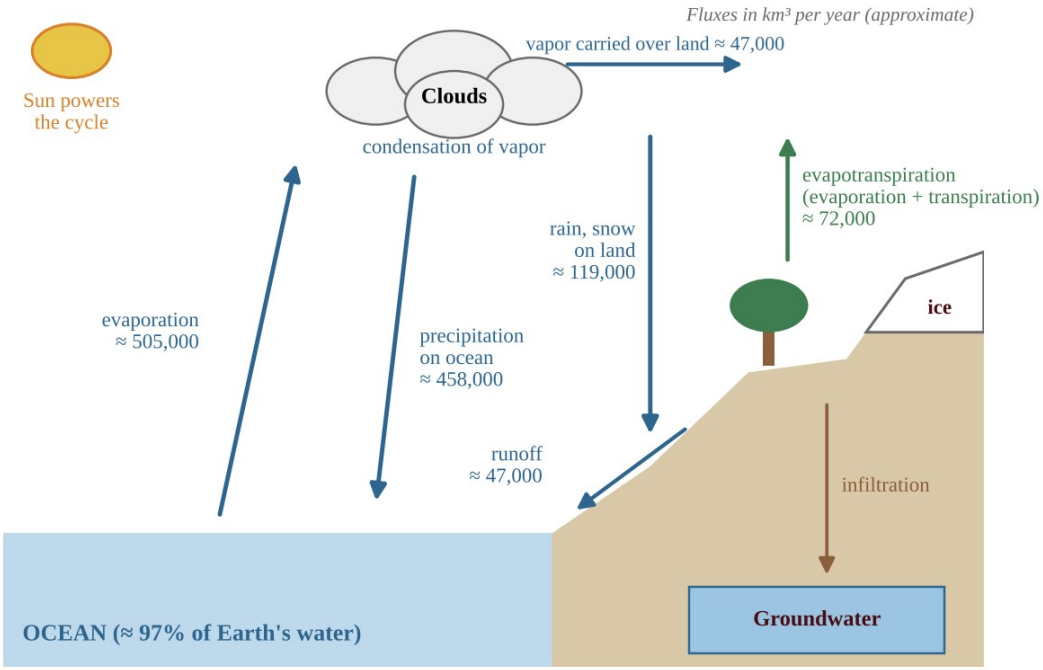


Figure 5.6 — The water cycle with approximate global fluxes in km³ per year. Land precipitation (119,000) equals evapotranspiration (72,000) plus runoff (47,000).

Processes

- **Evaporation** changes liquid water to vapor from oceans, lakes and soil.
- **Transpiration** is the release of water vapor from plant leaves. Evaporation plus transpiration together are called **evapotranspiration**.
- **Condensation** changes vapor to liquid droplets, forming clouds.
- **Precipitation** returns water as rain, snow, sleet or hail.
- **Infiltration and percolation** move water into and through soil to **groundwater**. Water stored in porous rock is an **aquifer**.
- **Runoff** is water flowing over land surfaces into streams, lakes and the ocean. The land that drains to one stream or river is its **watershed**.
- **Sublimation, freezing and melting** move water between ice and the other phases.

Reservoirs

The **oceans are the primary reservoir** of water at Earth's surface, holding roughly 97% of it. Ice caps and groundwater act as much smaller reservoirs; together they hold nearly all fresh water.

Reservoir	Share of all water	Typical residence time
Oceans	≈ 97%	thousands of years
Ice caps and glaciers	≈ 1.7%	thousands to tens of thousands of years
Groundwater	≈ 1.7%	weeks to over 10,000 years

Reservoir	Share of all water	Typical residence time
Lakes, soil, rivers, atmosphere, organisms	≈ 0.1% in total	days (air, rivers) to years (lakes)

Water spends only about nine days in the atmosphere but thousands of years in a deep aquifer or an ice cap, so turnover differs greatly among reservoirs.

Residence time, water budgets and land use

Residence time is the average time a water molecule stays in a reservoir. A watershed's **water budget** balances what arrives against where it goes.

FORMULA & MATH SKILL

Residence time = volume stored ÷ rate of inflow (or outflow) Precipitation (P) = evapotranspiration (ET) + runoff + change in storage Volume = depth × area (convert units first: 1 km² = 10^6 m²; 1,000 mm = 1 m)

Pavement, roofs and compacted soil block **infiltration**, so more precipitation leaves quickly as **runoff** and less recharges groundwater. Clearing vegetation lowers transpiration, so more of the same precipitation is left to run off. Forest and wetland soils slow runoff and feed groundwater.

Math and skills toolkit

- **Percent change** = (new − original) ÷ original × 100 (divide by the *original*). **Percent of whole** = part ÷ whole × 100.
- **Conversions:** multiply by a fraction equal to 1 so units cancel (80 cm × 1 m ÷ 100 cm = 0.80 m; 25 km² = 25 × 10^6 m²).
- **Volume = area × depth** (m² × m = m³). Show formula, substitution with units, answer with units.

COMMON STUDENT MISTAKES

- Transpiration is water loss from plants; evaporation is from surfaces.
- Most fresh water is in ice and groundwater, not in lakes and rivers.
- Residence time uses the stored volume divided by the *rate* of flow, not the rate times the volume.

AP EXAM STRATEGY

Write P = ET + runoff first, then convert every quantity to the same units.

CLASSROOM EXAMPLE SG-U1-E007

A watershed has an area of 25 km² and receives 80 cm of precipitation in a year. Of the precipitation, 60% returns to the atmosphere by evapotranspiration, and the remaining water leaves the watershed as stream flow (ignore groundwater storage and change). (a) Calculate the volume of stream flow in cubic meters per year. Use 1 km² = 10^6 m². (b) Name the process by which the largest portion of the water leaves the watershed, and give one biological contributor to it.

CLASSROOM EXAMPLE SG-U1-E013

A groundwater aquifer stores 3.0 × 10³ km³ of water. Recharge and natural discharge are each 12 km³ per year. (a) Calculate the average residence time of water in the aquifer. (b) A city covers part of the recharge area with pavement, so recharge falls to 9 km³ per year while natural discharge stays at 12 km³ per year. Calculate the annual change in stored water and state whether the aquifer is growing or shrinking. (c) Explain how pavement changes infiltration.

CLASSROOM EXAMPLE SG-C5-E005

Global fluxes in thousand km³ per year: evaporation from the ocean 505, precipitation on the ocean 458, precipitation on land 119, evapotranspiration from land 72. The atmosphere holds about 12,900 km³ of water vapor. (a) Calculate the net amount of vapor carried from the ocean to the land. (b) Calculate the global runoff from land, assuming no change in storage. (c) Calculate the percent of all evaporation and evapotranspiration that comes from the ocean. (d) Calculate the average residence time of water in the atmosphere, in days, using total evaporation plus evapotranspiration as the flux.

QUICK CONCEPT SUMMARY

- The sun-powered water cycle moves water as solid, liquid and gas among oceans, ice, groundwater, surface water, soil, organisms and the air.
- Oceans are the largest surface reservoir; ice caps and groundwater are much smaller.
- Key processes: evaporation, transpiration, condensation, precipitation, infiltration, runoff. Residence time = volume ÷ flow rate.

5.6 Role of H_2O, C, N, P, and S in the Human Body

CONCEPT MADE SIMPLE

Your body is part of every cycle in this chapter. The water you drink, the carbon in your food, the nitrogen in your proteins, the phosphorus in your bones and the sulfur in your hair all came from soil, water and air, and will return there.

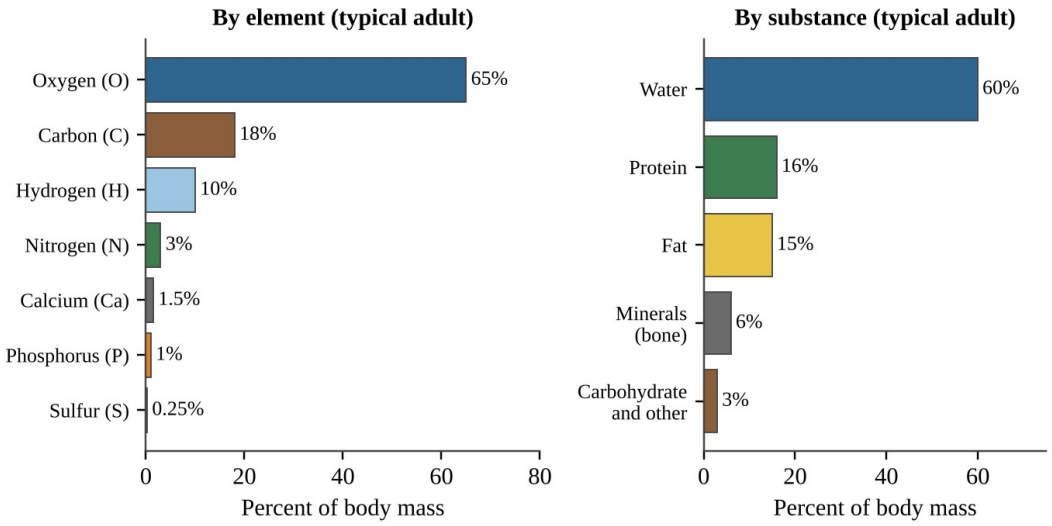


Figure 5.7 — Typical composition of the human body by element and by substance (percent of body mass; values vary with age, sex and body fat).

Water (H_2O): about 60% of body mass

Water makes up roughly 50–65% of an adult's mass and is the **solvent** for cell chemistry. It carries nutrients, oxygen, hormones and wastes in blood, takes part in reactions such as digestion, and cools the body through sweat because its high heat of vaporization absorbs much heat. An adult must replace about 2–3 liters a day lost in urine, sweat, breath and feces, so **clean drinking water** links the body directly to the water cycle (Section 5.5).

Carbon (C): the backbone of life

Carbon atoms form chains and rings and bond with H, O, N, P and S. All **organic molecules** are built on carbon: carbohydrates, lipids, proteins and nucleic acids. Carbon is about 18% of body mass, second only to oxygen (water is mostly oxygen by mass). Plants take carbon from CO_2; we eat it and release CO_2 in respiration, closing the carbon cycle (Section 5.1).

Nitrogen (N): proteins and DNA

Nitrogen is in the **amino group** of every amino acid, so it is in all proteins (enzymes, muscle, hemoglobin, antibodies). It is also in the **bases** of DNA and RNA and in ATP. We cannot use the N_2 we breathe; our nitrogen comes from dietary protein, which began as nitrate or ammonium taken up by plant roots and fixed by soil bacteria (Section 5.2). About 16% of protein by mass is nitrogen. Too little protein stunts growth; too much nitrate in well water can interfere with oxygen transport in infants ("blue-baby syndrome").

Sulfur (S): protein shape

Sulfur is in two amino acids, **cysteine** and **methionine**. Pairs of cysteine form **disulfide bridges** that hold proteins in shape, giving strength to hair, nails and skin (keratin). Sulfur is also in some vitamins and in coenzyme A, which helps release energy from food. We get it from protein foods that trace back to soil sulfate absorbed by plants (Section 5.4). Body sulfur is about 0.25% of mass.

Phosphorus (P): genes, energy and bones

Phosphate groups form the **sugar–phosphate backbone of DNA and RNA**, carry the energy in **ATP**, and make up the **phospholipids** of every cell membrane. About 85% of the body's phosphorus (about 1% of body mass) is in **bones and teeth** as calcium phosphate (hydroxyapatite). It comes from food grown on soil phosphate (Section 5.3).

Why the cycles matter for health and food

- **Food supply.** Crops need N, P and S from soil. Haber–Bosch nitrogen and mined phosphate help feed about half of the world's people, but over-application pollutes water (Chapter 9).
- **Nutrient loss.** Erosion, leaching and harvest remove N, P and S from fields; manure, compost, crop rotation or fertilizer must replace them. Phosphate rock is limited, so recovering phosphorus from sewage and manure also reduces algal blooms.
- **Health.** Too little protein or water causes illness; polluted water and acid rain harm health directly and through food.

KEY DEFINITIONS

Amino acid — building block of proteins; contains N, and cysteine and methionine also contain S. **ATP** — adenosine triphosphate, the cell's energy-carrying molecule, containing P. **Disulfide bridge** — a bond between two sulfur atoms that holds a protein in its folded shape. **Organic molecule** — a carbon-based molecule made by living things (carbohydrate, lipid, protein, nucleic acid). **Hydroxyapatite** — the calcium phosphate mineral of bones and teeth.

COMMON STUDENT MISTAKES

- The human body is mostly *oxygen* by mass, because water is 89% oxygen by mass, not mostly carbon.
- Breathing in nitrogen does not supply the body's nitrogen; it comes from dietary protein.
- Phosphorus is not only in bones; it is also in DNA, ATP and membranes.

AP EXAM STRATEGY

When asked "why is element X essential?", name the molecule that contains it (for example amino acids for N and S, ATP and DNA for P), say what that molecule does, and link X to its cycle and source in food.

CLASSROOM EXAMPLE SG-C5-E006

An adult has a body mass of 70 kg. Water is 60% of body mass. Protein is 16% of body mass, and protein is 16% nitrogen by mass. Phosphorus is 1.0% of body mass, and 85% of that phosphorus is in bones and teeth. (a) Calculate the mass of water in the body, in kg. (b) Calculate the mass of nitrogen contained in the body's protein, in kg. (c) Calculate the mass of phosphorus in bones and teeth, in kg. (d) Name one molecule other than bone mineral that contains phosphorus and state its function.

QUICK CONCEPT SUMMARY

- Water (about 60% of body mass) is the solvent for cell chemistry, transport and cooling.
- Carbon is the backbone of all organic molecules; nitrogen is in amino acids and nucleic acid bases; sulfur is in cysteine and methionine (protein shape).
- Phosphorus is in DNA, RNA, ATP, membranes and bones. Body elements come from the cycles through food and water.

5.7 Practice Questions and Answers

Now put the cycles to work: the classroom examples below, together with the extra question sets, will turn these ideas into exam skills.

PRACTICE QUESTIONS AND ANSWERS

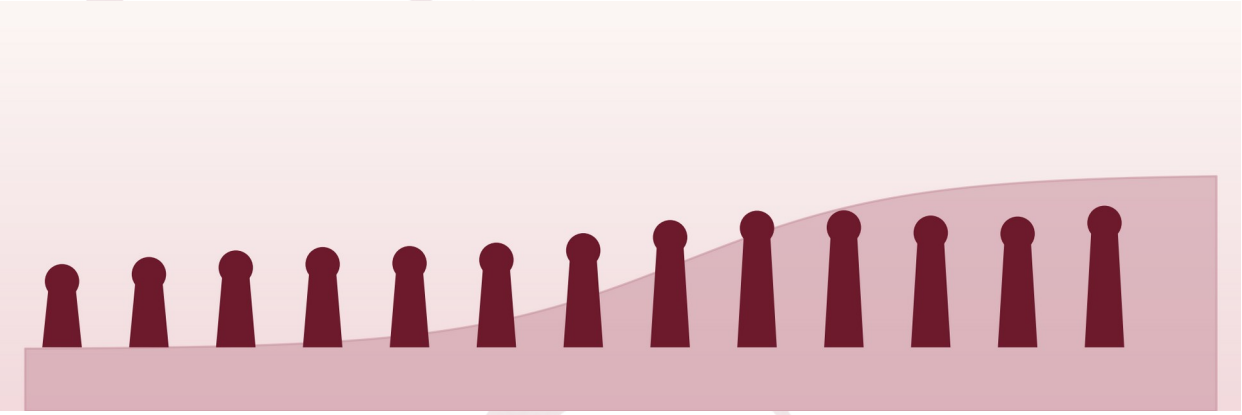
Classroom examples in this chapter (unsolved; space is provided in each box): SG-U1-E004, SG-U1-E012, SG-C5-E001, SG-U1-E005, SG-C5-E002, SG-U1-E006, SG-C5-E003, SG-C5-E004, SG-U1-E007, SG-U1-E013, SG-C5-E005, SG-C5-E006.

Classified practice: Book Two, Chapter 5 — 18 multiple-choice questions (Section A) followed by 1 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 5. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

UNIT III
POPULATION



IN THIS UNIT

Chapter 6 Populations

CHAPTER
6

UNIT III — POPULATION
Populations

6.1 · Population Ecology
6.2 · Survivorship
6.3 · Human Population Dynamics
6.4 · Demographic Transition
6.5 · Age-Structure Diagrams
6.6 · Population Size
6.7 · Impacts of Population Growth
6.8 · Practice Questions and Answers

LEARNING OBJECTIVES

- Compare generalist and specialist species, and r-selected and K-selected species, and predict who is favored when habitats stay constant or change.
- Explain carrying capacity, overshoot and dieback, and read survivorship curves.
- Apply the growth mathematics: percent change, logistic growth, RNI, doubling time, TFR and resource budgets.
- Describe human population dynamics: limiting factors, fertility, infant mortality, the demographic transition and age structure.
- Explain how population growth and consumption create environmental impacts, and read ecological footprint data.

A population is a group of individuals of one species living in the same place at the same time. This chapter asks two big questions. Why do populations of different species grow, shrink and survive in such different ways? And how do the same ideas explain the growth of the human population and the pressure it puts on Earth? It is the most math-heavy chapter in the course, so work every calculation with a pencil as you read.

6.1 Population Ecology

CONCEPT MADE SIMPLE

Every species makes a living in its own way. Some are flexible, some are fussy. Some have many cheap babies, some have a few precious ones. Whatever the strategy, every population eventually meets a ceiling set by its environment, called the carrying capacity.

Generalist and specialist species

A **niche** is a species' job and address: what it eats, where it lives, and the conditions it tolerates. A **generalist species** has a wide niche. It eats many foods, lives in many habitats and tolerates wide ranges of temperature and moisture. Raccoons, cockroaches, coyotes, house mice and white-tailed deer are generalists. A **specialist species** has a narrow diet, a narrow habitat or narrow tolerance limits. A koala eats almost only eucalyptus leaves, and a giant panda depends overwhelmingly on certain bamboos. Think of a Swiss army knife (generalist: does everything adequately) and a surgeon's scalpel (specialist: one job done superbly, useless for anything else).

In a **constant habitat**, such as a stable rain forest, a specialist fine-tunes its body and behavior to exact conditions and out-competes the generalist for its one resource. In a **changing habitat** the story flips. A fire, drought, new road or climate shift removes the specialist's one food or habitat, so its population crashes. The generalist simply switches food or moves, so it persists or spreads.

PROCESS & CAUSE-AND-EFFECT

Stable habitat → specialist uses its one resource very efficiently → specialist is favored Habitat changes or is lost → specialist's only resource disappears → specialist declines or goes extinct Habitat changes → generalist switches food or habitat → generalist persists or spreads

REAL-WORLD EXAMPLE

A coastal wetland is drained for farmland and crop fields replace the reed beds. A rare marsh bird that nests only in tall reeds disappears within two seasons. Crows and starlings, generalists that eat grain, insects and scraps and nest almost anywhere, increase in the same area. The same change produced opposite results because of niche width.

KEY DEFINITIONS

Niche — the role of a species in its ecosystem, including its food, habitat and tolerance range. **Generalist species** — broad niche; advantaged in changing habitats. **Specialist species** — narrow niche; advantaged in constant habitats and vulnerable to change.

COMMON STUDENT MISTAKES

- A generalist is not "better." It wins only when conditions change; in a stable habitat the specialist wins.
- Living in a small area does not make a species a specialist. Niche width (diet and tolerance) is what counts.

CLASSROOM EXAMPLE SG-U3-E001

Two beetle species live in a mixed forest. Beetle A lays its eggs only on the bark of one tree species and its larvae eat only that bark. Beetle B eats wood of more than twenty tree species and can survive in forests, parks and wood piles. A fungal disease kills nearly all trees of the species Beetle A depends on. (a) Classify each beetle as a generalist or a specialist. (b) Predict what happens to each population over the next five years and explain why. (c) State which beetle would likely be favored if the forest stayed completely unchanged for a century, and give one reason.

To test whether a disturbance favors generalists, compare **paired plots**: several similar disturbed plots (for example, partly logged) and several undisturbed ones, with equal trapping effort in every plot and counts of at least one generalist and one specialist species. Replicates separate the disturbance from ordinary differences between sites. One plot, one species, or a published number from another forest cannot answer the question. Design rules are in Section 6.6.

K-selected and r-selected species

Every species has a limited energy budget. It can spend it on **many cheap offspring** (the r strategy) or **a few expensive ones** (the K strategy). The letters come from population math: **r** is the growth rate of a population and **K** is carrying capacity. **r-selected species** put energy into rapid reproduction, so they grow fast when conditions are good. **K-selected species** stay near the carrying capacity of their environment and put their energy into surviving and competing.

Trait	r-selected	K-selected
Body size	Small	Large

Trait	r-selected	K-selected
Offspring per birth event	Many	Few
Energy invested per offspring	Minimal	Large; long parental care
Age at maturity	Early	Late (long youth)
Life span	Short	Long
Reproduction	Often once, or in a burst	Several times in a lifetime
Habitat and competition	Disturbed or unpredictable; competition usually low	Stable; competition usually high
Examples	Bacteria, weeds, mosquitoes, mice	Elephants, whales, humans, oaks

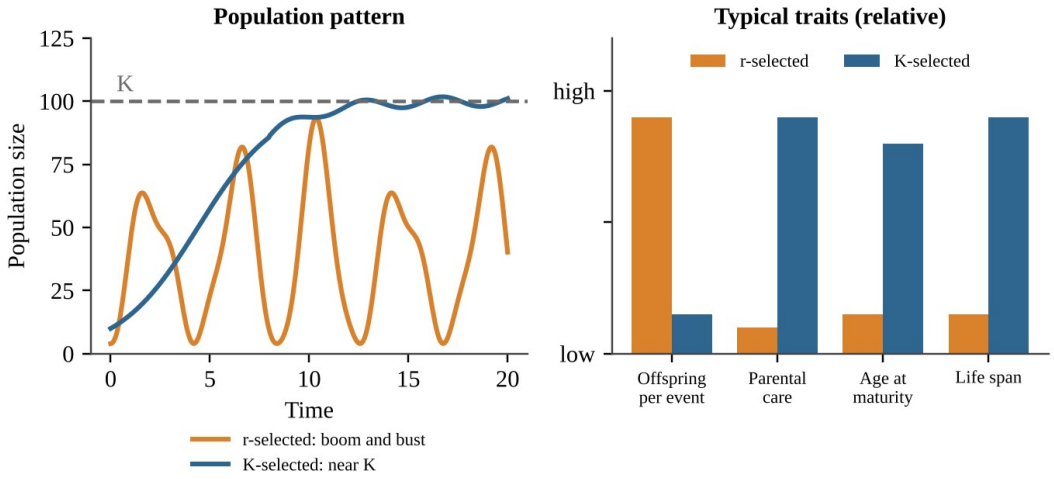


Figure 6.1 — r-selected populations tend to boom and bust, while K-selected populations settle near carrying capacity. The traits on the right are relative, not exact.

Biotic potential is the maximum reproductive rate of a population under ideal conditions: unlimited food, no predators, no disease. A bacterium has an enormous biotic potential; an elephant has a tiny one. To rank species, weigh three traits together: more offspring per reproductive event, an **earlier** age at first reproduction, and a **shorter** time between generations. Body size and long life add nothing to reproductive speed. Biotic potential is almost never reached in nature, because environmental resistance (limited resources, predators, disease) holds populations back.

Most real species fall between the extremes, and many change strategy with conditions. A sea turtle lives a long time (K-like) but lays more than a hundred eggs (r-like). Describe a species by its **tendencies**, not as a perfect r or K.

Invasive species. Most invasive species are r-selected: they reproduce quickly, spread widely and tolerate disturbance. They hurt K-selected natives most, because slow breeders cannot replace their losses quickly. An island's slow-breeding native bird suffers far more from introduced rats than a fast-breeding native does. r-selected natives rebound quickly and are minimally affected. The same logic predicts recovery: after hunting removes 30% of a population, an r-selected species rebounds within a few seasons, but a K-selected species needs many years.

KEY DEFINITIONS

r-selected species — small, many offspring, little investment per offspring, early maturity, short life, often reproduce once; usually low competition. **K-selected species** — large, few offspring, high investment, late maturity, long life, reproduce repeatedly; usually high competition. **Biotic potential** — the maximum reproductive rate of a population in ideal conditions.

COMMON STUDENT MISTAKES

- r and K are tendencies on a spectrum. Do not claim "all large animals are K and all small ones are r" without checking the traits.
- K the strategy is not K the carrying capacity. They are connected (K-selected species live near K) but they are different things.

AP EXAM STRATEGY

Answer "compare" prompts with a trait pair: "r-selected species produce many offspring while K-selected species produce few." A list of traits with no comparison earns less credit than one clear contrast per trait.

CLASSROOM EXAMPLE SG-U3-E002

A biologist records the following about three species. Species P: adult mass 2 g, 400 eggs per season, matures at 3 months, lives about 1 year. Species Q: adult mass 4,500 kg, one calf every 3 years, matures at 12 years, lives 60 years. Species R: adult mass 25 kg, 2 pups per year, matures at 2 years, lives 12 years. (a) Classify P and Q as r-selected or K-selected, citing two traits each. (b) State which of P and Q would be more severely harmed by an aggressive invasive predator and explain why. (c) Explain why R is difficult to classify as purely r or purely K.

Carrying capacity

A lake can feed only so many ducks. **Carrying capacity (K)** is the largest population an environment can support **indefinitely** with its food, water, space and other resources. When a few individuals enter a resource-rich area, the population first grows slowly, then rapidly, then levels off near K. The graph is an **S-shaped (logistic) curve**. With no limits, growth would follow a **J-shaped (exponential) curve**, climbing faster and faster. Real populations may follow a J-curve briefly, but limiting factors bend it into an S or into a boom-and-bust pattern.

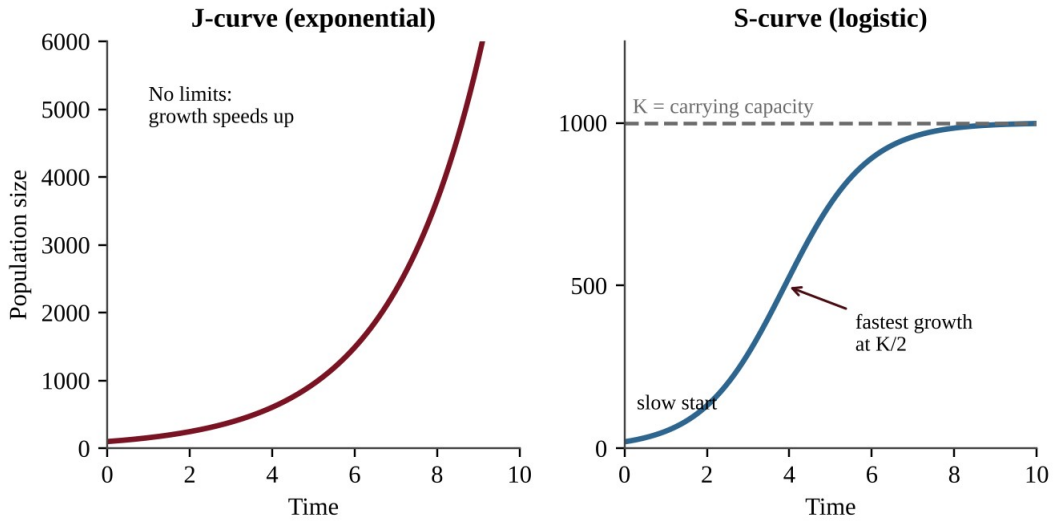


Figure 6.2 — Left: unlimited exponential growth (J-curve). Right: growth limited by resources (S-curve) leveling at carrying capacity K.

Carrying capacity is **not fixed**. A wet year increases plant growth and raises K for deer; a drought or a pollutant lowers it; a new competitor can reduce it for the original species.

Overshoot and dieback. Overshoot happens when a population exceeds K. The population often keeps growing for a time after resources start to run short because of a **time lag**: births already underway, long-lived adults and slow feedback. During overshoot, consumption exceeds renewal, so the habitat is damaged: vegetation is overgrazed, soil erodes, water and prey are depleted. The result is **dieback** (a population crash), a sharp, often severe decline caused by famine, disease or conflict. Because the habitat was damaged, the new carrying capacity may be lower than the old one.

PROCESS & CAUSE-AND-EFFECT

Population is below K → resources plentiful → population grows (S-curve rising) Population exceeds K (overshoot) → resource use > resource renewal → resource depletion and habitat damage Resource shortage → famine, disease, conflict → dieback → population falls to or below K (K may be lower)

REAL-WORLD EXAMPLE

A small number of goats is released on a rocky island with sparse grass. The herd grows quickly for years, eats nearly all the ground cover, and then starves in a dry summer. The bare hillside erodes in winter storms, so a smaller herd can be supported afterward. Overshoot, dieback and a lowered K occurred in one story.

KEY DEFINITIONS

Carrying capacity (K) — the maximum population size an environment can sustain over the long term. **Overshoot** — a population exceeding K. **Dieback** — a severe decline after overshoot because of resource shortage.

COMMON STUDENT MISTAKES

- K is not "the number of births." It is a ceiling set by resources, and it can change.
- Overshoot does not cause instant dieback; the lag lets a population overshoot before the crash.

CLASSROOM EXAMPLE SG-U3-E004

Deer were introduced to a 25 km² island. Population counts: year 0: 30; year 2: 80; year 4: 190; year 6: 330; year 8: 430; year 10: 480; year 12: 495; year 14: 500; year 16: 500. (a) Estimate the carrying capacity and calculate the population density at that size in deer per km². (b) Identify which two-year interval had the greatest increase in number of deer, and show the calculation. (c) Explain, in terms of resources, why growth slowed after year 8.

CLASSROOM EXAMPLE SG-U3-E005

On a small island, the forage supports a carrying capacity of 300 goats. Goats are released and the herd grows to 560 before a severe drought. Afterward the herd falls to 140. (a) Calculate by what percent the herd exceeded carrying capacity at its peak. (b) Name the term for the decline from 560 to 140. (c) Explain two environmental impacts of the overshoot and one reason the new carrying capacity might be below 300.

What holds a population at K

At low density each individual finds plenty to eat, so the **per-capita birth rate** (births per individual per year) is high and the **per-capita death rate** is low. As density rises, competition, disease and stress push births down and deaths up. The population grows while births per individual exceed deaths per individual, stops growing where they are equal, and shrinks beyond that. That density is K.

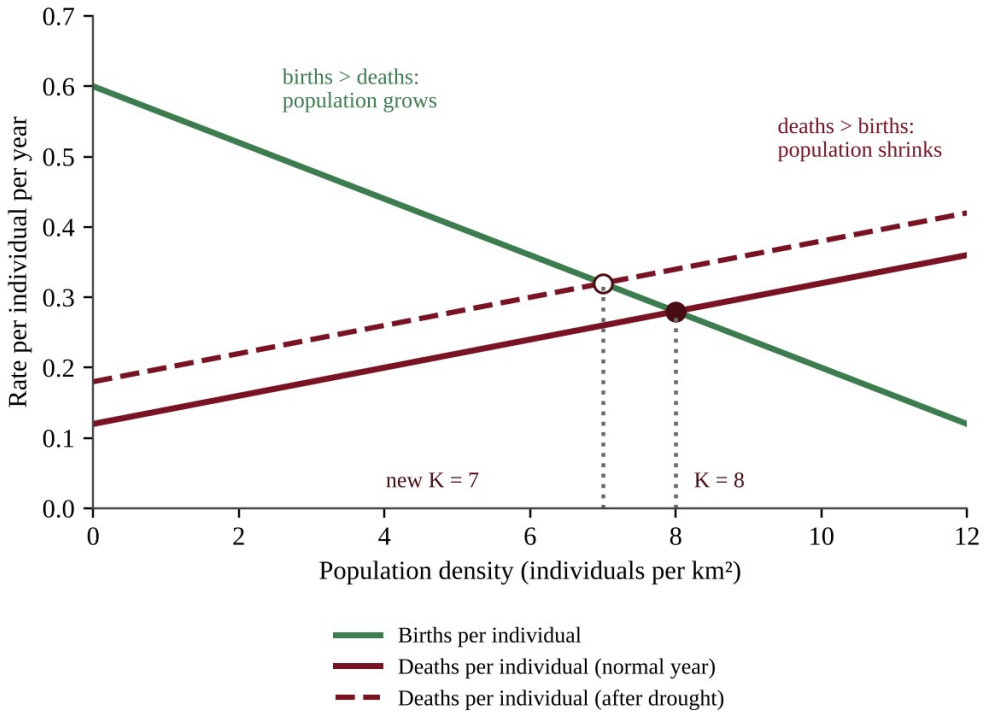


Figure 6.3 — Per-capita birth and death rates change with density. K lies where the lines cross. Extra mortality (dashed) moves the crossing to a lower density, so K falls.

Finding K from a table. Look for the density where birth and death values are equal. If no row matches, find the two rows where births switch from above deaths to below; with straight-line trends, K lies between them, in proportion to the gaps (a gap of +0.10 at density 6 and −0.10 at density 8 crosses at 7). A stress that raises deaths or lowers births at every density moves the crossing left, so K falls; more food or shelter moves it right, so K rises.

Predators and prey

Predation is a density-dependent limit and a **negative feedback loop**. When prey are plentiful, predators eat well and increase; more predators remove more prey; prey decline; predators then go hungry and

decline; prey recover. This produces linked cycles in which the predator peak comes **after** the prey peak, because predators need time to breed.

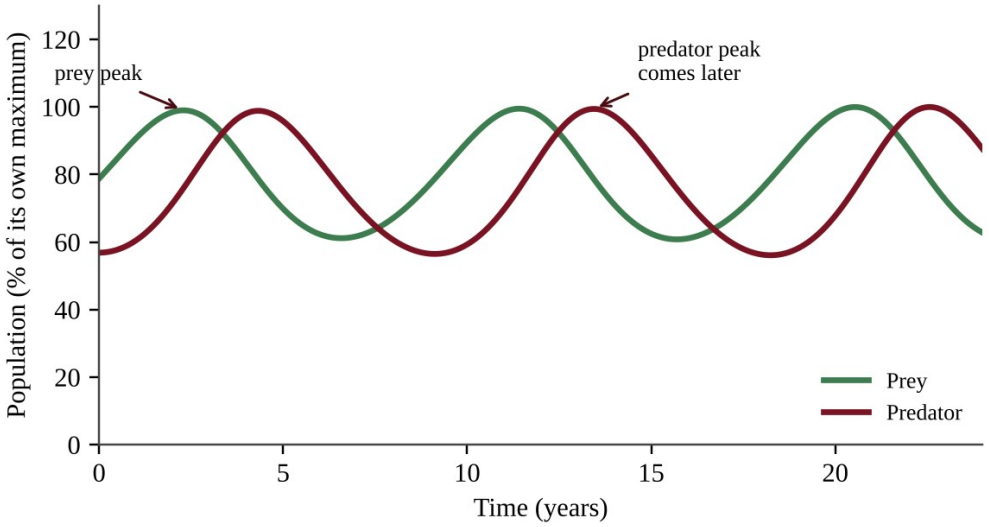


Figure 6.4 — Predator and prey cycles. The predator peak lags behind the prey peak.

In a census table, find the prey peak first, then the predator peak; the predator column trails. A prey decline that follows a predator rise is predation, not necessarily overshoot of the prey's own K.

Reading growth graphs and tables

- On a labeled graph: the steep early climb is the exponential phase; the dashed line is K; a peak above it is overshoot; the low point after the peak is the dieback; a later plateau is the new K.
- In a census table, the **greatest increase** is the largest difference between consecutive counts. If the gaps differ, divide by the years to compare yearly change. Percentage growth is highest very early, when N is small, even though the largest numerical gain comes near K/2.
- Estimate K as the plateau.

CLASSROOM EXAMPLE SG-U3-E013

A vole population was sampled at five densities (voles per hectare): 2, 4, 6, 8, 10. Births per vole per year: 0.70, 0.58, 0.46, 0.34, 0.22. Deaths per vole per year: 0.30, 0.38, 0.46, 0.54, 0.62. (a) Identify the carrying capacity and explain your reasoning. (b) Calculate the net change per vole per year at 4 voles per hectare and state whether the population is growing or shrinking there. (c) A disease raises deaths by 0.05 per vole per year at every density. Estimate the new carrying capacity by straight-line interpolation and state the direction of the change.

CLASSROOM EXAMPLE SG-U3-E014

Census counts on a prairie. Year: 0, 2, 4, 6, 8, 10. Rabbits: 900, 1,500, 1,100, 600, 800, 1,400. Foxes: 40, 60, 90, 70, 40, 55. (a) Identify the years of the rabbit peak and the fox peak and state the lag. (b) Explain the lag in terms of fox reproduction and food supply. (c) Calculate the percent decrease in rabbits from year 2 to year 6. (d) Name the type of feedback loop and state whether predation is density-dependent or density-independent.

QUICK CONCEPT SUMMARY

- Specialists win in constant habitats; generalists win in changing ones.
- r-selected: small, many offspring, early maturity, short life. K-selected: large, few offspring, late maturity, long life. Most invasive species are r-selected; K-selected natives suffer most.
- K is the long-term maximum and can change. Overshoot → resource depletion → dieback from famine, disease or conflict.

6.2 Survivorship

CONCEPT MADE SIMPLE

Follow 1,000 newborns of one species. Graph how many are still alive at each age. If most live long and die old, you get Type I. If death strikes evenly at every age, Type II. If most die very young, Type III.

A **survivorship curve** displays the relative survival of a **cohort** (a group of individuals born at the same time) from birth to the greatest age reached by any member. The x-axis is age, often as a percent of maximum life span so that species can be compared. The y-axis is the number (or percent) of the cohort still alive, commonly on a log scale so that a drop from 1,000 to 10 survivors is as visible as a drop from 100 to 1.

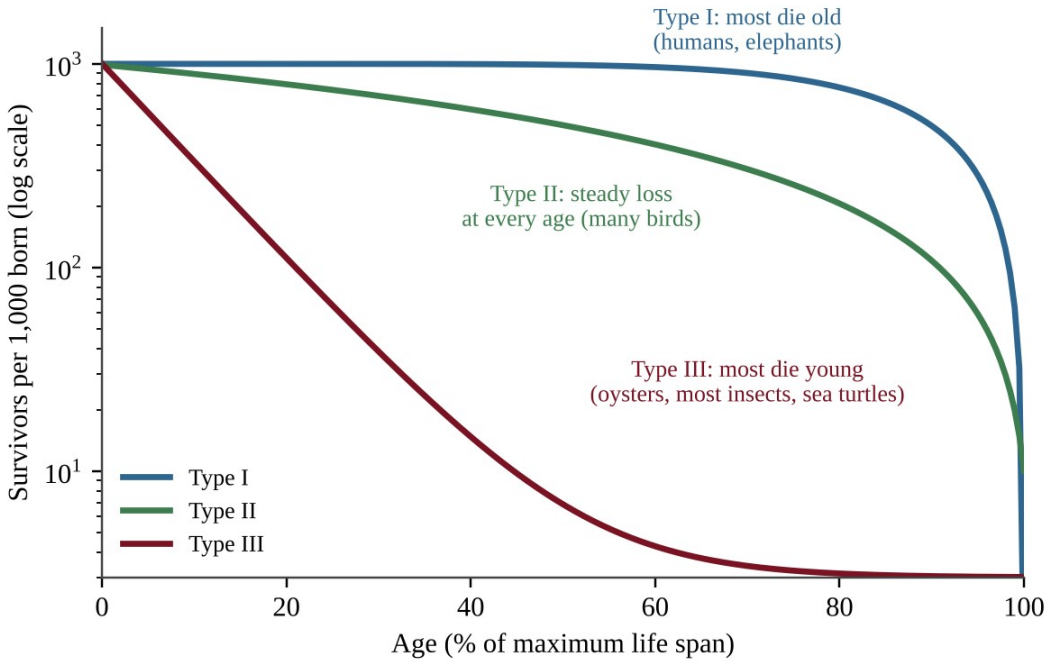


Figure 6.5 — The three survivorship curves. The y-axis is logarithmic, so equal vertical distances mean equal percentage losses.

Type	Shape	Pattern of death	Typical species and strategy
I	Flat early, then drops steeply late	Low death at young and middle ages; most die in old age	Humans, elephants, large mammals (K-selected); heavy parental care
II	Straight diagonal line	Roughly constant chance of dying at every age	Many songbirds, some lizards and rodents (intermediate)
III	Steep early drop, then flat	Very high death of the young; survivors often live long	Oysters, most fish, insects, sea turtles, weeds (r-selected); many offspring, little care

Reading the curve

Steepness tells you where deaths happen: a steep section means many individuals die in that age range, a flat one means few. Type III species offset massive early losses by producing huge numbers of offspring, so a few survive. Type I species produce few offspring but protect each one. The cause-and-effect link: high parental investment → high juvenile survival → Type I; low investment → high juvenile mortality → Type III. K-selected species typically follow Type I or II; r-selected species follow Type III.

FORMULA & MATH SKILL

Percent surviving to age x = (number alive at age x ÷ number in starting cohort) × 100 **Number dying in an interval** = survivors at start of interval − survivors at end of interval Counts are individuals; the first result is a percent.

Working from a survivorship table. Convert each count to a **percent of the starting cohort**; the percent that died before an age is 100 minus that. Then compare the **percent lost** in each interval, not the raw number. If ages are unevenly spaced (0, 1, 2, 4, 8 years), divide each loss by the length of its interval before comparing yearly rates. Decision rule: most of the cohort gone in the first interval, then slow loss means Type III; almost no loss until late, then rapid loss means Type I; about the same percent lost in every equal interval means Type II.

COMMON STUDENT MISTAKES

- The curve shows **survival**, not population size. A Type III curve does not mean the population is shrinking.
- Do not say Type III species "live short lives." Many survivors live long; the point is high mortality when young.
- On a log scale, a straight line means a constant **percent** loss per time step, not a constant number lost.

AP EXAM STRATEGY

Locate where the curve is steepest, name the life stage (young, middle, old), then name the type, and support it with a number read from the graph: "Only 8% survive to 10% of maximum age, so most deaths occur in the young."

CLASSROOM EXAMPLE SG-U3-E003

A researcher follows a cohort of 800 tree-frog eggs. Survivors at the end of each age: age 0: 800; age 1 year: 160; age 2 years: 80; age 4 years: 40; age 8 years: 20. (a) Calculate the percent of the cohort that survives to age 1 year. (b) Calculate how many individuals die between age 1 and age 2. (c) Identify the survivorship type and state whether the species is more likely r-selected or K-selected, giving a reason based on the data.

CLASSROOM EXAMPLE SG-C6-E001

A researcher follows a cohort of 1,000 individuals of a large mammal. Survivors at each age: age 0: 1,000; 10 years: 950; 20 years: 900; 30 years: 840; 40 years: 700; 50 years: 400; 60 years: 100. (a) Calculate the percent of the cohort that survives to age 40. (b) Calculate how many individuals die between ages 40 and 60. (c) Calculate the percent of the starting cohort that dies between ages 0 and 20, and between ages 40 and 60, and state which interval has the greater loss. (d) Identify the survivorship type and state whether the species is more likely r-selected or K-selected, giving a reason.

QUICK CONCEPT SUMMARY

- A survivorship curve tracks a cohort from birth to maximum age; steepness shows when most deaths occur.
- Type I (K-selected, much parental care), Type II (steady mortality), Type III (r-selected, high juvenile mortality).

6.3 Human Population Dynamics

CONCEPT MADE SIMPLE

A population is like a bathtub. Births and immigrants are water coming in; deaths and emigrants are water going down the drain. If more comes in than leaves, the level rises. The only questions are how fast and for how long.

The four inputs

Human population growth or decline is controlled by four rates: **birth, death, immigration and emigration**. Birth and death rates are strongly affected by access to **education, family planning, healthcare and nutrition**.

FORMULA & MATH SKILL

Rate of natural increase (RNI, %) = (crude birth rate − crude death rate) ÷ 10 CBR and CDR are births and deaths per 1,000 people per year; dividing by 10 converts per 1,000 to percent. **Population growth rate (%)** = [(births + immigrants) − (deaths + emigrants)] ÷ starting population × 100 or = (CBR − CDR + immigration rate − emigration rate) ÷ 10, with every rate per 1,000 per year. **Doubling time (years)** ≈ 70 ÷ annual growth rate (%) (the Rule of 70)

The Rule of 70 is an estimate: 2% growth doubles a population in about 35 years, 1% in about 70 years, 3.5% in about 20 years. After one doubling the population is 2× the original; after two, 4×; after three, 8×. For a declining population, 70 ÷ percent decline gives the time to halve.

KEY DEFINITIONS

Crude birth rate (CBR) — births per 1,000 people per year. **Crude death rate (CDR)** — deaths per 1,000 people per year. **Rate of natural increase (RNI)** — CBR minus CDR, usually expressed as a percent. **Doubling time** — years for a population to double at a constant growth rate.

COMMON STUDENT MISTAKES

- Forgetting to convert per 1,000 to percent: a difference of 20 per 1,000 is 2.0%, not 20%.
- Using the decimal in 70 ÷ r. Use the percent number (2, not 0.02).
- Mixing RNI with total growth: RNI ignores migration.

AP EXAM STRATEGY

In a calculation FRQ, label each step with its unit (people per year, %, years). Write units in the setup so they cancel to the unit asked for.

CLASSROOM EXAMPLE SG-U3-E009

A coastal town has 25,000 residents at the start of a year. During the year there are 400 births, 250 deaths, 350 people moving in and 150 people moving out. (a) Calculate the town's population growth rate as a percent for the year. (b) Use the Rule of 70 to estimate the doubling time. (c) Identify whether the town's growth in this year came mostly from natural increase or from net migration, and show the two numbers you compared.

CLASSROOM EXAMPLE SG-U3-E010

Country M has a crude birth rate of 28 per 1,000, a crude death rate of 8 per 1,000 and an emigration rate of 1.5 per 1,000 per year. Immigration is negligible. (a) Calculate the rate of natural increase as a percent. (b) Calculate the overall population growth rate as a percent. (c) Calculate the doubling time using the overall growth rate and state how many times larger the population would be after three doubling times. (d) Classify an earthquake and an increase in disease transmission as density-independent or density-dependent factors.

Total fertility rate and infant mortality

Total fertility rate (TFR) is the average number of children a woman would have in her lifetime if she lived through all her childbearing years at today's age-specific birth rates. **Replacement-level fertility** is about 2.1 children per woman in countries with low infant mortality. The extra 0.1 covers children who die young and the slightly higher number of boys born. Where infant mortality is high, replacement is higher, because more births are needed to ensure two children survive to adulthood. At replacement level a population is considered relatively stable (ignoring migration, and with a lag because of momentum, Section 6.5). Above it the population grows; below it the population eventually shrinks.

FORMULA & MATH SKILL

TFR = (sum of age-specific fertility rates for all age groups) × (width of age group in years) ÷ 1,000 Age-specific fertility rate = births per 1,000 women in that age group per year; for 5-year groups, width = 5. TFR is in children per woman. **Infant mortality rate (IMR)** = (deaths of infants under 1 year ÷ live births) × 1,000, in deaths per 1,000 live births.

Factors that lower TFR:

- **Age at first child.** Later first births leave fewer childbearing years and tend to mean fewer children overall.
- **Educational opportunities for females.** Educated women gain job options and delay marriage and childbirth.
- **Access to family planning.** Contraception and information let families choose the number and timing of children.
- **Government acts and policies.** Examples are incentives or penalties, family-planning programs, and laws on marriage age or women's rights.

Infant mortality falls when mothers have access to good **healthcare** (prenatal care, safe delivery, vaccines) and good **nutrition**, and when families have clean water and sanitation. When infant mortality falls, families no longer need many births to ensure survivors, and fertility tends to fall after a lag.

TFR can fall far below 2.1 where housing and child-rearing are costly, women have wide education and career options, and late marriage is the norm. These countries face a shrinking workforce and a rising share of older people. Governments may adopt **pro-natalist** policies that encourage births (child allowances, paid parental leave, tax breaks) or **anti-natalist** policies that discourage them (family-planning programs, schooling for girls, penalties for larger families).

COMMON STUDENT MISTAKES

- TFR is not babies per 1,000 people; that is the crude birth rate.
- TFR of 2.1 does not instantly stop growth, because of momentum.
- A graph showing TFR falling as girls' schooling rises is a **negative association**. It does not prove schooling alone caused the fall, because income, contraceptive access, urbanization and infant survival change at the same time. Write "associated with" or "tends to," not "always" or "directly causes."

CLASSROOM EXAMPLE SG-U3-E008

In Country Z, age-specific fertility rates (births per 1,000 women per year) are: ages 15–19: 10; 20–24: 70; 25–29: 105; 30–34: 85; 35–39: 40; 40–44: 8; 45–49: 0. (a) Calculate the total fertility rate, using 5-year age groups. (b) Compare it with the replacement level of 2.1 and state whether, ignoring migration, the long-term population is likely to grow, stay stable or decline. (c) Describe two factors that could change Country Z's TFR and the direction each would push it.

CLASSROOM EXAMPLE SG-C6-E002

In one year Country H records 250,000 live births and 11,500 infant deaths. After a program of prenatal clinics and vaccination, a later year records 240,000 live births and 7,500 infant deaths. (a) Calculate the infant mortality rate for each year. (b) Calculate the percent change in the infant mortality rate. (c) Explain why a lower infant mortality rate tends to lower a country's total fertility rate after a lag.

Limits to human population

Earth's carrying capacity. Earth's resources (food, fresh water, energy, space, a stable climate) are finite, so the human population ultimately has a limit, though technology, diet and consumption level change where that limit lies.

Malthusian theory. Thomas Malthus argued that population grows **geometrically** (2, 4, 8, 16) while food production grows only **arithmetically** (2, 4, 6, 8). Eventually population outruns food and is held back by famine, disease and war. Technology has delayed these limits for humans, but the basic idea, that growth in numbers cannot continue forever on finite resources, remains central to the debate.

Global trend. World population is still rising, but its growth rate has fallen since the 1960s as fertility has dropped (girls' education, family planning, urbanization, lower infant mortality). Most projections show the total leveling off later this century. As fertility falls and life spans lengthen, older people become a larger share of the population.

Density-dependent and density-independent factors

Kind	Effect depends on population density?	Examples
Density-dependent	Yes; stronger when crowded	Competition for food, clean water and air, disease transmission, territory size, predation
Density-independent	No; hits regardless of size	Major storms, fires, heat waves, droughts, floods

A crowded city is more likely to see a fast-moving epidemic (density-dependent), but a hurricane harms a village and a city equally per person (density-independent). Density-dependent factors tend to regulate populations near K, while density-independent events can cause sudden crashes at any size. A factor is density-dependent if its **rate** changes with density, so compare rates, not totals: percent infected, deaths per individual, eggs per female. A flood that kills about the same percent at every density is density-independent.

Environmental impact depends on how many people there are **and** how much each uses: total use = population × use per person.

KEY DEFINITIONS

Density-dependent factor — a limit whose strength changes with population density. **Density-independent factor** — a limit that acts regardless of density.

CLASSROOM EXAMPLE SG-U3-E018

Hares in four meadows. Density (hares per hectare): 5, 10, 20, 40. Percent infected with a parasite: 4, 9, 18, 37. Percent killed by a spring flood: 12, 11, 13, 12. (a) Identify which factor is density-dependent and which is density-independent, citing the rates. (b) Explain why the total number of infected hares would be a poor test. (c) Country S has 10 million people each using 8 tons of a resource per year; Country T has 40 million people each using 1 ton. Calculate each country's total use and state which has the larger impact.

QUICK CONCEPT SUMMARY

- Growth depends on births, deaths, immigration and emigration, influenced by education, family planning, healthcare and nutrition.
- RNI = (CBR − CDR) ÷ 10; doubling time ≈ 70 ÷ growth rate (%); replacement TFR ≈ 2.1.
- Lower TFR: later first births, female education, family planning, policy. Lower infant mortality: healthcare and nutrition.
- Malthus: population grows geometrically, food arithmetically. Density-dependent limits (disease, food, water, territory) differ from density-independent ones (storms, fires, heat waves, droughts).

6.4 Demographic Transition

CONCEPT MADE SIMPLE

As a country develops, death rates fall first (better food, medicine and sanitation). Birth rates fall later (families adjust). In the gap between, the population surges. The **demographic transition model (DTM)** tells that story in four stages.

The **demographic transition** is the shift from high birth and death rates to low birth and death rates as a country moves from a **pre-industrial** to an **industrialized** economy.

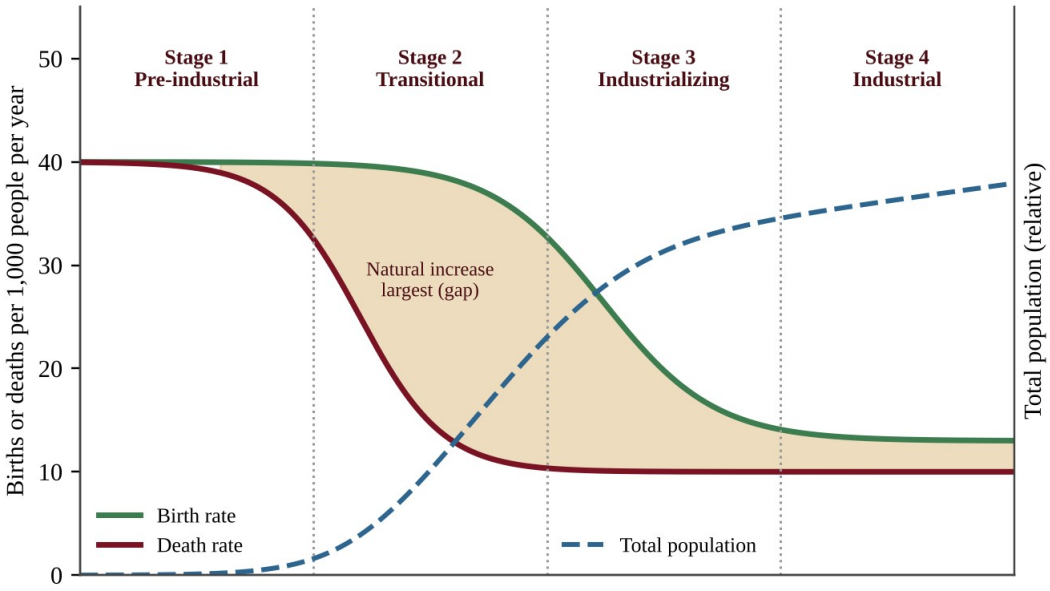


Figure 6.6 — The demographic transition model. The shaded gap between birth and death rates drives population growth; total population (dashed) rises most steeply in Stage 2.

Stage	Birth rate	Death rate	Growth	What is happening
1 Pre-industrial	High	High	Very slow	Famine, disease and poor sanitation; many children die; children help with farm work
2 Transitional	High	Falling fast	Rapid	Better food, clean water, medicine; infant mortality drops but families still large
3 Industrializing	Falling	Low	Slowing	Urbanization, women's education and work, family planning, children cost more
4 Industrial	Low	Low	Very slow, near zero or negative	Stable or aging population; TFR near or below replacement

Why death rates fall before birth rates. Public health and food improvements spread quickly because they are cheap and obvious, but family size depends on social and economic change (education, jobs, urban living, cost of children, expectations) that takes decades. The lag creates Stage 2 growth.

Developing versus developed countries. Developing countries (Stages 2 and early 3) typically have **higher infant mortality**, more **children in the workforce**, lower female education, lower income and a young age structure. Developed countries (Stage 4) have low infant mortality, children in school rather than at work, high education levels and an older age structure. Long life expectancy and low infant mortality signal better health care, nutrition and sanitation.

PROCESS & CAUSE-AND-EFFECT

Stage 1: high births and deaths → Stage 2: death rate falls (medicine, sanitation, food), births stay high → population surge Stage 3: birth rate falls (education, family planning, urbanization, women's work) → growth slows Stage 4: low births and deaths → stable, aging population

REAL-WORLD EXAMPLE

A country in Stage 2 sees its death rate fall from 30 to 12 per 1,000 after a vaccination campaign and clean-water projects, while its birth rate stays at 38. The RNI jumps from nearly zero to 2.6%. Its schools and cities strain to keep up. Twenty years later, as girls' schooling and family planning spread, the birth rate falls to 22 and the country enters Stage 3.

KEY DEFINITIONS

Demographic transition — the move from high to low birth and death rates as a country industrializes.
Demographic transition model (DTM) — the four-stage model describing this shift.

Placing a country in a stage

Stage names vary between sources (Stage 3 may be called "industrial," Stage 4 "post-industrial"). Rely on the rates. Both high = Stage 1; birth far above death with death falling = Stage 2; birth falling but still clearly above death, so growth is positive, = Stage 3 (birth 16 and death 5 per 1,000 gives 1.1% growth, Stage 3, even though both numbers look low); both low and nearly equal (growth near zero or negative) = Stage 4. Then check the age structure for agreement. On a graph, read both curves at the year asked, subtract, divide by 10, then apply the Rule of 70.

Plotting data. Put the independent variable (time) on the x-axis and the measured variable on the y-axis. Label both axes with units, choose an even scale that uses most of the grid, plot every point, and join the points or draw a smooth curve as directed. Plot two data sets on one grid with a key.

COMMON STUDENT MISTAKES

- Stage 4 is not zero growth in every country; some have slight growth or decline. Say "very slow, near zero."
- The population is **largest in Stage 4** but **grows fastest in Stage 2**.
- Death rate falls first, not birth rate.

AP EXAM STRATEGY

Compare the two rates to place a country, check the age structure, then state the cause of the gap (falling death rate from medicine and sanitation, or falling birth rate from education and family planning).

CLASSROOM EXAMPLE SG-U3-E011

A country reports these rates per 1,000 people per year.

Year	Birth rate	Death rate
1900	44	41
1930	43	30
1960	38	14
1990	24	9
2020	12	10

(a) Calculate the rate of natural increase (as a percent) for each year. (b) Assign each year to a stage of the demographic transition model. (c) Identify the year of fastest natural growth and explain the cause of the gap between birth and death rates then. (d) Describe the likely age structure shape in 2020.

CLASSROOM EXAMPLE SG-U3-E012

Country X has 36 million people, a growth rate of 2.8% per year, a total fertility rate of 4.8, an infant mortality rate of 62 per 1,000 live births, and only 28% of girls completing secondary school. (a) Estimate the doubling time and the population after 50 years if the growth rate stayed constant. (b) Identify the stage of the demographic transition model that best fits Country X and justify it with two data values. (c) Propose two policies and explain, for each, how it would lower TFR over time. (d) Explain why the population would probably keep growing for decades even if TFR fell to 2.1 tomorrow.

CLASSROOM EXAMPLE SG-U3-E019

A country's rates per 1,000: 1950, birth 36, death 20; 1970, birth 33, death 10; 1990, birth 21, death 8; 2010, birth 12, death 9. (a) Plot both rates against year on one graph with labeled axes and scales. (b) Calculate the RNI for 1970 and the doubling time, then give the year the population would double if that rate continued. (c) Assign 1970, 1990 and 2010 to a stage and justify each from the rates.

QUICK CONCEPT SUMMARY

- Stage 1 both rates high; Stage 2 death rate falls and population surges; Stage 3 birth rate falls; Stage 4 both low.
- Developing countries have higher infant mortality and more children in the workforce.

6.5 Age-Structure Diagrams

CONCEPT MADE SIMPLE

An **age structure diagram** (population pyramid) stacks bars for each age group, males on one side and females on the other. A wide base means lots of children: the population will grow. A narrow base means few children: the population will shrink.

Each horizontal bar shows the number or percent of people in an age band. The population is often split into three groups: **pre-reproductive** (under 15), **reproductive** (about 15–44) and **post-reproductive** (45 and older). The shape predicts where the population is heading, because today's children become tomorrow's parents.

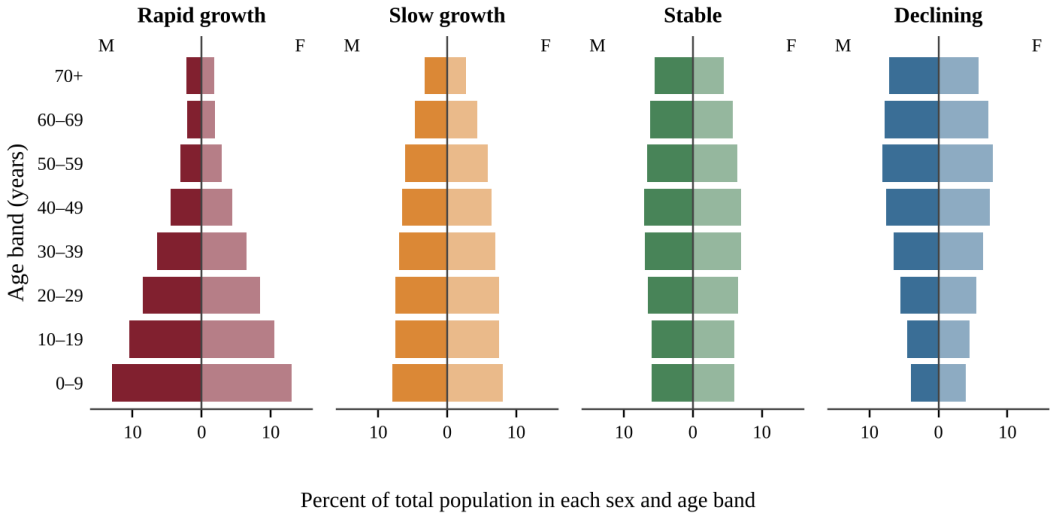


Figure 6.7 — Age structure diagrams for four patterns. Each bar is a 10-year age band; males on the left, females on the right.

Shape	What it means	Proportion of young
Wide base, narrowing quickly (triangle)	Rapid growth: many children will become parents	Highest
Wider at base, gentle taper	Slow growth	Moderate
Nearly vertical sides	Stable, near zero growth	Lower
Narrow base, bulge in middle or top	Declining population	Lowest

A rapidly growing population has a **higher proportion of young people** than a stable or declining one. This produces **population momentum**: even if each family chose to have fewer children, the population keeps growing for decades because so many young people are about to reach reproductive age. A bulge in the middle ages (a "baby boom") moves upward over time; a sudden narrow band shows a past drop in births, a famine, a war or an epidemic. A narrow base and large top predicts a rising share of elderly people and fewer workers.

To find the largest age range, add the bars in each range; several narrow bars can outweigh one wide bar. The **support ratio** is people aged 15–64 ÷ people aged 65 and older. A falling ratio means fewer workers for each older person, which strains pensions and health care. A **stable age distribution** means the proportion in each age group stays constant; the whole population may still grow or shrink if every group changes by the same percent.

KEY DEFINITIONS

Age structure diagram — a graph of the number or percent of a population in each age group, split by sex. **Pre-reproductive / reproductive / post-reproductive** — age groups before, during and after the child-bearing years. **Population momentum** — continued growth caused by a large young group entering reproductive age even after fertility falls.

COMMON STUDENT MISTAKES

- A pyramid shows the **current** structure; you infer the trend from it.
- Read the **base** (young ages) for growth, not the width of the top bars.
- A high proportion of young does not by itself tell you the death rate.

AP EXAM STRATEGY

Describe the shape first ("wide base, sharply tapering"), then state the inference ("rapid growth because many young will reproduce"), then give a consequence (schools, jobs, health care; for a young, growing population, rising demand for resources and waste).

CLASSROOM EXAMPLE SG-U3-E007

The table gives the percent of each country's population in three age groups.

Country	Under 15	15–44	45 and over
A	41	40	19
B	25	41	34
C	13	36	51

(a) Identify which country is growing fastest and which is most likely declining, and justify each using the data. (b) Explain what population momentum means and name the country where it is strongest. (c) Describe one challenge that Country C's age structure is likely to create.

CLASSROOM EXAMPLE SG-U3-E017

Percent of population by age. Country P: under 15, 36; 15–64, 59; 65 and older, 5. Country Q: under 15, 14; 15–64, 66; 65 and older, 20. (a) Calculate the support ratio for each country. (b) State which country faces greater pension and health-care strain and explain why. (c) Describe the shape of each country's diagram and state which has the greater population momentum.

QUICK CONCEPT SUMMARY

- The base of a diagram predicts growth: wide = rapid growth, narrow = decline.
- Rapidly growing populations have a higher proportion of young people, and momentum keeps them growing after fertility drops.

6.6 Population Size

CONCEPT MADE SIMPLE

Food, water and space are a bank account. When the account is full, a population spends freely and grows faster and faster. When the account shrinks, there is not enough to go around, and the population stops growing or shrinks.

The logic of limits

Population growth is limited by environmental factors, above all resources and space, which are **finite at every scale**: a year's rain, a lake's fish, an island's soil. When resources are abundant, growth usually **accelerates**. When the resource base shrinks, some individuals get enough and others do not. The result is higher mortality, lower fecundity (fewer offspring), or both, so growth falls to or below carrying capacity.

Percent change and exponential growth

FORMULA & MATH SKILL

Percent change = [(final value − initial value) ÷ initial value] × 100 **Average annual growth rate (%)** = percent change ÷ number of years Values are numbers of individuals; time in years; the answer is in % per year. Setup only: 400 → 450 in 5 years gives (50 ÷ 400) × 100 = 12.5% total, 2.5% per year. For a decline of p%, final = initial × (1 − p ÷ 100).

Exponential growth means a **constant percentage** increase each period, so the population takes the same time to double again and again (the J-curve). **Linear growth** adds a constant *number* each period. Growth at biotic potential is exponential. To project forward, count doublings: doublings = years ÷ doubling time, and N = N_0 × 2^doublings. A population of 3 million growing 1.4% per year doubles in 70 ÷ 1.4 = 50 years and reaches 12 million after 100 years (two doublings). For a constant annual rate, N after t years = N_0 × (1 + rate)^t, with the rate as a decimal.

Logistic growth

In a limited environment the speed of growth depends on how much room is left. The **logistic model** says the growth rate is the unlimited rate multiplied by the fraction of capacity still open.

FORMULA & MATH SKILL

Population change per time (dN/dt) = r × N × [(K − N) ÷ K] N = current population (individuals); K = carrying capacity (individuals); r = per-capita rate of increase under ideal conditions (per year); dN/dt = individuals added per year. When N is small, (K − N) ÷ K ≈ 1, so growth is nearly exponential. When N = K/2, growth is fastest. When N = K, growth is 0. When N > K, growth is negative.

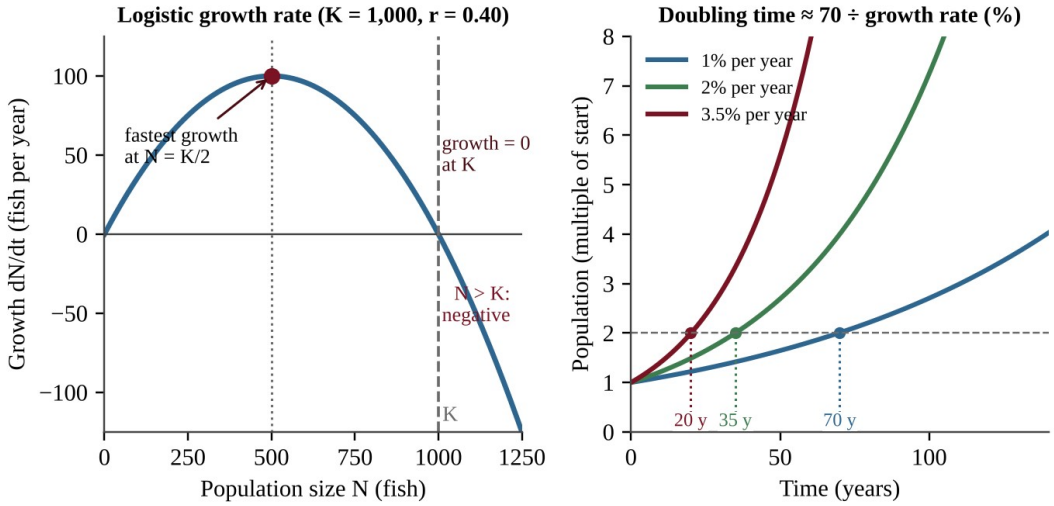


Figure 6.8 — Left: the logistic growth rate is largest at K/2, zero at K and negative above K. Right: the higher the growth rate, the shorter the doubling time (70 ÷ rate).

On an S-curve the steepest part (the fastest addition of individuals) occurs about halfway to K, not at K. Above K the curve falls, a sign of overshoot. If a graph shows the population falling after peaking far above a dashed K line, say "overshoot followed by dieback."

COMMON STUDENT MISTAKES

- The fastest growth in numbers is near K/2, not at K.
- Percent change uses the **initial** value in the denominator, not the final value.
- A population that is still growing may be growing at a slower rate. Rate and size are different things.

AP EXAM STRATEGY

Write the formula, substitute with units, calculate, and put a unit on the answer. Partial credit depends on visible work.

CLASSROOM EXAMPLE SG-U3-E006

A lake's carrying capacity for a fish species is K = 2,000 fish, and the species' per-capita rate of increase is r = 0.40 per year. Use dN/dt = r × N × (K − N) ÷ K. (a) Calculate the yearly population change when N = 500, when N = 1,000, and when N = 1,900. (b) State at which of these sizes growth is fastest and explain why growth is slowest of the three at 1,900. (c) Calculate dN/dt when N = 2,600 and interpret the sign of your answer.

CLASSROOM EXAMPLE SG-C6-E003

A city of 480,000 people grows at a constant 2.5% per year. (a) Use the Rule of 70 to estimate the doubling time. (b) Estimate the population after 56 years by counting doublings. (c) Use $N = N_0 \times (1 + \text{rate})^t$ to calculate the population after 10 years, to the nearest thousand.

Carrying capacity from supply and need

When one resource is the limit, carrying capacity is a division, and human demand is a multiplication.

FORMULA & MATH SKILL

K = total resource supplied per year ÷ amount one individual needs per year **Percent over capacity** = (N − K) ÷ K × 100 **Demand** = population × need per person **Shortfall** = demand − supply (a positive answer is a deficit) Convert units first: per day → per year (× 365), kg → metric tons (÷ 1,000). Supply and demand must share one unit.

K changes in proportion to supply: 25% less forage means 25% fewer animals can be fed.

Designing an investigation

- **Hypothesis**: an if–then statement with a direction, such as "If nitrate is added to tanks, then the algae population will reach a larger plateau."
- **Independent variable**: the one factor you change. **Dependent variable**: what you measure (a count, a birth rate). **Controlled variables**: everything else held equal (light, temperature, container size, starting number).
- **Control group**: an untreated group for comparison.
- **Replicates**: several units per treatment, so chance differences average out (reliability). Assigning treatments at random and changing only one factor improves validity. Count at regular intervals and long enough to see the plateau.
- **Supporting results**: treated groups plateau higher than the control. Small-scale results may not scale to a whole lake.

Math and skills toolkit

- **Unit conversions.** Per 1,000 → percent: divide by 10. Percent → per 1,000: multiply by 10. Write units next to every number.
- **Density** = individuals ÷ area, with one area unit throughout.
- **Scale-ups.** Multiply along a chain so the units cancel (hectares × termites per hectare × kg per termite per year). To rescale a result, multiply by the ratio of the changed factor: a rate one-third as large gives one-third of the total.
- **Exam routine.** Formula → substitution with units → answer with units → one-sentence interpretation ("the population doubles about every 35 years"). Round at the end; keep two or three significant figures.

CLASSROOM EXAMPLE SG-U3-E015

A pasture yields 240,000 kg of usable forage per year, and one sheep needs 600 kg per year. A farmer keeps 520 sheep. (a) Calculate K. (b) Calculate by what percent the flock exceeds K. (c) A drought cuts forage by 25%. Calculate the new K and the number of sheep that must be removed. (d) A town of 80,000 people eats 0.45 kg of rice per person per day and its farms supply 12,000 metric tons per year. Calculate the annual demand in metric tons and the shortfall.

CLASSROOM EXAMPLE SG-U3-E016

A student suspects that phosphorus limits a green alga in a pond. She has 12 identical clear tanks, pond water, a phosphorus solution and a light bank. (a) Write an if–then hypothesis. (b) Identify the independent and dependent variables and two controlled variables. (c) Describe the treatment and control groups, how many tanks each gets, and why. (d) State the results that would support the hypothesis and one reason the result might not apply to the whole pond.

QUICK CONCEPT SUMMARY

- Abundant resources → accelerating growth; shrinking resource base → higher mortality and lower fecundity → growth falls toward or below K.
- Percent change = (final − initial) ÷ initial × 100. Logistic growth: dN/dt = rN(K − N)/K; fastest at K/2, zero at K, negative above K.
- K = supply ÷ need per individual.

6.7 Impacts of Population Growth

CONCEPT MADE SIMPLE

Every person needs food, water, energy and space, and every person makes waste. More people usually means more demand and more waste, unless each person uses less or uses it more cleverly. Your **ecological footprint** is a receipt for your lifestyle, written in hectares of land and water.

From more people to more pressure

Population growth raises the total demand on Earth's resources, and each extra person in a high-consumption society adds more demand than one in a low-consumption society. The impacts follow the resource chains met throughout this book.

Demand	Typical environmental impact
Food	Cropland expands into forests and grasslands (habitat loss); fertilizer and pesticide runoff; heavy irrigation; overfishing

Demand	Typical environmental impact
Fresh water	Aquifer depletion, lower river flows, saltwater intrusion, water shortages
Energy	Fossil-fuel combustion adds CO_2 and air pollutants; mining and drilling damage land
Land for housing, roads and industry	Urban sprawl, impervious surfaces, loss of farmland and wildlife habitat
Materials and goods	Mining, deforestation, solid waste and landfills
Waste	Water and air pollution, greenhouse gas emissions, climate change

When demand exceeds what ecosystems can supply or absorb, the human population is in **overshoot** in the same sense as the goats in Section 6.1: resources are used faster than they regenerate, and the damage may lower future carrying capacity. Technology, such as the Green Revolution, has raised Earth's carrying capacity for humans, but at costs such as water use and pollution. Malthus's warning (Section 6.3) is therefore a question about how much each person uses as well as how many people there are.

FORMULA & MATH SKILL

Impact (I) = Population (P) × Affluence (A) × Technology (T) P = number of people; A = resource use per person (for example energy in GJ per person per year); T = impact per unit of resource (for example tons of CO_2 per GJ). Impact falls when any factor falls: fewer people, less use per person, or cleaner technology.

REAL-WORLD EXAMPLE

City M has 2 million people using 300 liters of water per person per day; City N has 8 million people using 60 liters. N has four times the people, yet M has the larger total demand, because affluence (use per person) counts as much as population. If N's incomes double and its per-person use doubles, its demand doubles even with no population growth.

Ecological footprints

An **ecological footprint** compares the resource demands and waste production of an individual or a society with the Earth's capacity. It adds up the area needed for:

- food: cropland, pasture and fishing grounds;
- wood and fiber: forest products;
- built-up land: housing, roads and industry;
- energy: the forest area needed to absorb the CO_2 released from fossil-fuel use.

Footprints are usually expressed in **global hectares (gha)** per person; a global hectare is a hectare of average biological productivity. **Biocapacity** is the productive area available to supply resources and absorb waste. When the footprint exceeds biocapacity there is an **ecological deficit**: the region is using resources faster than they regenerate or importing them from elsewhere. When biocapacity is larger there is an **ecological reserve**. Footprints are larger where consumption of energy, meat and goods is high. Distinguish **per-person** from **total** values.

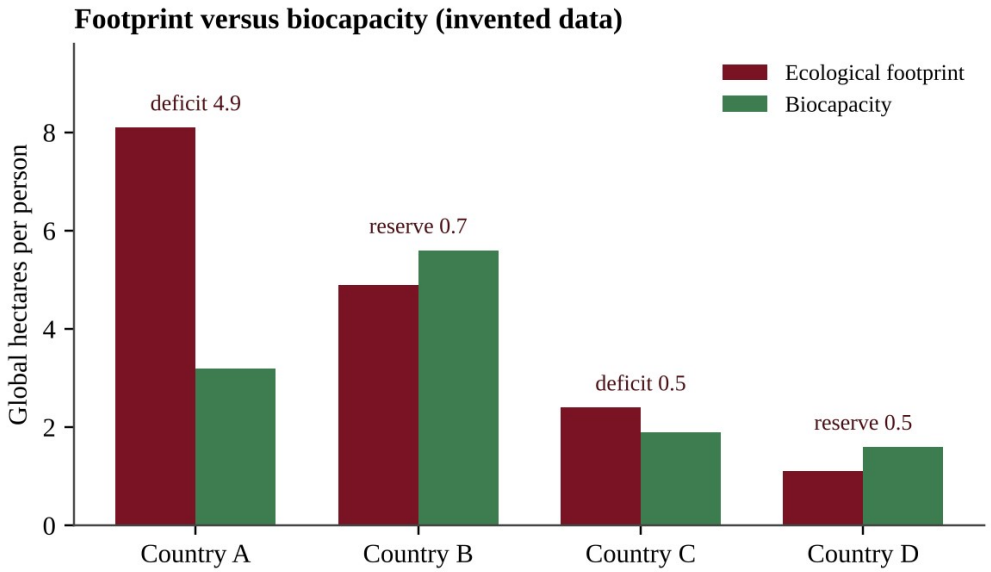


Figure 6.9 — Per-person footprint and biocapacity for four invented countries. A deficit means the footprint exceeds biocapacity.

FORMULA & MATH SKILL

Total footprint = per-person footprint × population Deficit (or reserve) = biocapacity − footprint (negative = deficit); for a population, multiply the per-person deficit by the number of people **Percent cut needed to reach biocapacity = (footprint − biocapacity) ÷ footprint × 100% Earths required = global footprint per person ÷ biocapacity available per person on Earth** Units: global hectares (gha) or gha per person.

Reducing the impact

Slower population growth (Section 6.3), lower per-person use (efficiency, less meat, less waste) and cleaner technology (renewable energy, recycling) all shrink the footprint. A country in deficit can import resources, but that moves the pressure elsewhere.

KEY DEFINITIONS

Ecological footprint — estimate of the land and water area required to supply a person's or society's resources and absorb its wastes. **Biocapacity** — capacity of ecosystems to regenerate resources and absorb waste. **Ecological deficit** — footprint greater than biocapacity.

COMMON STUDENT MISTAKES

- Ecological footprint is not carbon footprint. The carbon footprint is only the greenhouse gases produced; the ecological footprint includes all land uses.
- More people does not always mean a bigger impact than fewer people; compare total use (P × A).

AP EXAM STRATEGY

For footprint data questions, identify the unit, compare footprint with biocapacity, then say what drives the difference (consumption, population or limited land). For "propose a solution," name the factor (P, A or T) your action changes.

CLASSROOM EXAMPLE SG-U5-E010

Country X has 20 million people, a footprint of 6.0 gha per person and a biocapacity of 4.5 gha per person. (a) Calculate Country X's total footprint and total biocapacity in millions of global hectares. (b) Calculate the size of its ecological deficit as a number and as a percent of its biocapacity. (c) Identify two ways the country could reduce its deficit and one way it can meet its demand without lowering its footprint, and state a consequence of the latter.

CLASSROOM EXAMPLE SG-C6-E004

Country K has 50 million people and a footprint of 3.6 gha per person. Biocapacity available per person on Earth is 1.6 gha. (a) Calculate Country K's total footprint in millions of global hectares. (b) Calculate how many Earths would be required if everyone lived like the people of Country K. (c) Calculate the percent by which Country K's per-person footprint would have to be cut to equal Earth's biocapacity per person.

CLASSROOM EXAMPLE SG-C6-E005

Country P has 60 million people, each using 90 GJ of energy per year, and its energy releases 0.07 tons of CO_2 per GJ. Later, its population rises 10% to 66 million, efficiency cuts energy use to 72 GJ per person, and cleaner sources cut emissions to 0.05 tons of CO_2 per GJ. (a) Calculate total CO_2 emissions at the start, in millions of tons. (b) Calculate total emissions later. (c) Calculate the percent change in emissions and name which of P, A and T each change represents.

QUICK CONCEPT SUMMARY

- More people and more consumption increase demand for food, water, energy and land and increase pollution and habitat loss. Impact = P × A × T.
- Footprint compares demand with biocapacity; footprint > biocapacity = deficit.

6.8 Practice Questions and Answers

Now put the chapter to work with the classroom examples and exam questions below.

PRACTICE QUESTIONS AND ANSWERS

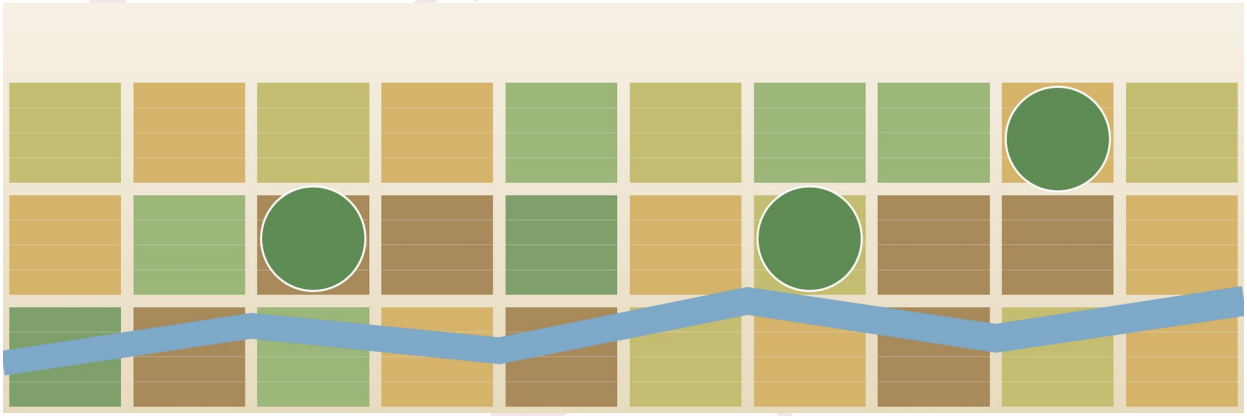
Classroom examples in this chapter (unsolved; space is provided in each box): SG-U3-E001, SG-U3-E002, SG-U3-E004, SG-U3-E005, SG-U3-E013, SG-U3-E014, SG-U3-E003, SG-C6-E001, SG-U3-E009, SG-U3-E010, SG-U3-E008, SG-C6-E002, SG-U3-E018, SG-U3-E011, SG-U3-E012, SG-U3-E019, SG-U3-E007, SG-U3-E017, SG-U3-E006, SG-C6-E003, SG-U3-E015, SG-U3-E016, SG-U5-E010, SG-C6-E004, SG-C6-E005.

Classified practice: Book Two, Chapter 6 — 47 multiple-choice questions (Section A) followed by 4 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 6. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

UNIT IV
LAND AND WATER USE



IN THIS UNIT

Chapter 7 Land and Water Use

CHAPTER
7

UNIT IV — LAND AND WATER USE
Land and Water Use

7.1 · Feeding a Growing Population
7.2 · Types of Agriculture
7.3 · Agricultural Productivity
7.4 · Sustainable Agriculture
7.5 · Controlling Pests
7.6 · Forestry
7.7 · Forest Fires (Wildfires)
7.8 · Wildfire Suppression
7.9 · Deforestation—Tree Harvesting
7.10 · Rangelands
7.11 · Urban Land Development
7.12 · Transportation Infrastructure
7.13 · Public and Federal Lands
7.14 · Mining
7.15 · Global Mineral Reserves
7.16 · Fishing
7.17 · Global Economics
7.18 · Tragedy of the Commons
7.19 · Practice Questions and Answers

LEARNING OBJECTIVES

- Explain how a growing population changes the demand for food, wood, minerals, fish and land, and why meat, farming, forestry, mining and cities each leave an environmental footprint.
- Compare agricultural systems and practices, and propose and justify sustainable alternatives (soil conservation, IPM, rotational grazing, aquaculture, sustainable forestry).
- Describe how wildfire works, why fire suppression can backfire, and how forests, rangelands and public lands are managed.
- Explain the impacts of urban land development, roads, mining and fishing, and the methods that reduce them.
- Use data skills: percent change, yield, ore grade, runoff, carbon-to-CO_2 conversion, CPUE, sustainable yield and carrying capacity.
- Explain the tragedy of the commons and how economics, trade and subsidies shape how people use land and water.

7.1 Feeding a Growing Population

CONCEPT MADE SIMPLE

Every meal, board, battery and parking lot comes from land or water. As people and their appetite for meat grow, so do the demands on farms, forests, mines, oceans and cities. This chapter follows that demand.

About 11% of Earth's ice-free land is cropland, and about twice that area is grazing land. Little good land is left to plow, so farmers have mostly met rising demand by raising **yield**, the harvest per hectare, with the Green Revolution (Section 7.3).

Why diet matters: energy lost at each step

Energy is lost at every step in a food chain; on average only about **10%** passes from one trophic level to the next (Chapter 4). A person who eats grain gets energy one step from the producer. A person who eats beef from grain-fed cattle gets energy two steps away, so about ten times more grain (and land and water) is needed for the same food energy.

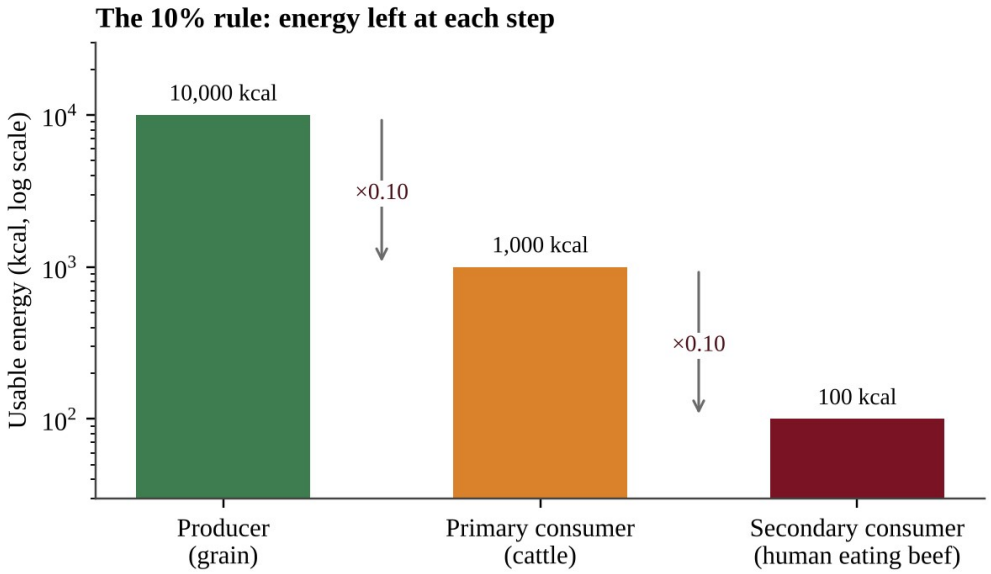


Figure 7.1 — Following 10,000 kcal of grain energy through two trophic steps leaves about 100 kcal for a person who eats beef from grain-fed cattle (vertical axis is logarithmic).

Meat production therefore needs more land, water and energy per gram of protein than plant foods, and it adds nutrient pollution and greenhouse gases such as methane (CH_4); flooded rice paddies also release methane because their waterlogged soil is oxygen-poor. The impact depends on the animal and the system (Section 7.2).

Eating less meat

Less consumption of meat, especially from **ruminants** such as cattle and sheep, can lower emissions of CO_2, CH_4 and N_2O, conserve fresh water and reduce reliance on antibiotics and growth hormones. Better feed quality and precision farming (sensors, GPS and data used to apply feed, water and chemicals only where needed) also reduce impacts. How large the benefit is depends on the production methods used and on what happens to land no longer used for livestock: restored forest stores carbon, but land converted to another intensive use may not.

KEY DEFINITIONS

Yield — the amount of crop harvested per unit of land, for example tonnes per hectare. **Ruminant** — a grazing mammal, such as a cow or sheep, whose gut microbes produce methane.

FORMULA & MATH SKILL

Energy at a later level = energy at the start × 0.10^(number of steps) Working backward, divide by 0.10 for each step. Keep units (kcal) at every step; a person's daily need ÷ energy per hectare gives person-days.

CLASSROOM EXAMPLE SG-U5-E006

A grain farmer produces 900,000 kcal of grain energy per hectare per year. Assume that only 10% of the energy in feed is stored in the bodies of cattle fed that grain, and that a person gets all of the energy stored in the cattle's bodies. (a) Calculate the kilocalories of beef that one hectare could provide if all the grain were fed to cattle. (b) A person needs 2,000 kcal per day. Calculate how many person-days of food one hectare could supply as grain eaten directly and as beef. (c) Explain two other environmental drawbacks of cattle production beyond land use.

CLASSROOM EXAMPLE SG-C7-E001

A country of 40 million people needs an average of 2,200 kcal of food energy per person per day. Its farmland supplies 5.0×10^6 kcal per hectare per year. (a) Calculate the food energy needed per person in one year (365 days). (b) Calculate the total food energy the country needs in one year. (c) Calculate the number of hectares of farmland needed to supply it.

QUICK CONCEPT SUMMARY

- Rising population and diet change push up demand for land, water and energy; most new food comes from higher yield, not more land.
- Each trophic step keeps about 10% of the energy, so meat needs far more resources than plants; less meat, especially from ruminants, lowers emissions and water use.

7.2 Types of Agriculture

CONCEPT MADE SIMPLE

Farming is a trade: we take food and the land pays. Some farms feed one family from a small plot; others are food factories. Each system works, and each has a hidden cost.

System	Typical features	Main environmental concern
Subsistence farming	Small plots, family labor, food mainly eaten at home	Soil exhaustion if land is not rested
Shifting cultivation (slash-and-burn)	Forest cut and burned, farmed a few years, then left to regrow	Deforestation and fire if fields are too many or fallow too short
Industrial (commercial) agriculture	Large monocultures, machines, fertilizer, pesticides, irrigation	Erosion, nutrient runoff, pesticide resistance, fossil-fuel use
Pastoral and ranching	Livestock graze open land	Overgrazing (Section 7.10)

Practices that cause damage

Tilling (plowing) turns over soil to bury weeds and prepare a seedbed. It leaves bare soil exposed, breaks up soil structure, speeds the decomposition of organic matter (releasing CO_2) and lets wind and rain carry away fine particles. Topsoil takes hundreds of years to form, and eroded sediment clouds streams and buries aquatic habitat. Heavy machines also compact soil.

In **slash-and-burn** farming, trees are cut and burned. Ash adds a pulse of nutrients, so the first crops grow well. In tropical forests, however, most nutrients are in living biomass rather than thin soils, so fertility falls in a few years and farmers move on. With few farmers and decades of regrowth the forest can recover; with too many it is lost. Burning releases CO_2 and particulates and destroys habitat.

Fertilizers replace the nutrients crops remove, mainly nitrogen and phosphorus. Plants take up only part of what is applied. The rest runs off or leaches into water. Nutrient pollution fertilizes algae; when they die, bacteria use up dissolved oxygen and create low-oxygen "dead zones" (eutrophication, Chapter 9). Nitrate in drinking water is a health risk for infants, and nitrogen fertilizer also releases nitrous oxide (N_2O), a greenhouse gas.

Meat production systems

A **concentrated animal feeding operation (CAFO)** raises many animals in confined spaces on grain and soy. **Free-range** or **pasture-based** systems let animals eat grass or forage for most of their lives; **rotational grazing** moves them between paddocks so plants can regrow.

Feature	Feedlots / CAFOs	Pasture-based (rotational or free-range)
Housing and feed	Confined; grain and soy	Outdoors; grass and forage
Land needed	Less	More
Cost to consumers	Lower (more economically efficient)	Higher
Waste	Large, concentrated manure; can contaminate waterways if not managed	Manure fertilizes soil; runoff and erosion if animal density is high
Antibiotics	Routine use can add to global antibiotic resistance	Not always antibiotic-free

CAFO manure also causes odors and releases methane and ammonia.

Feed conversion

The **feed conversion ratio (FCR)** is the mass of feed an animal eats per mass of weight it gains. An FCR of 7 means 7 kg of feed per 1 kg of body mass; a lower ratio means less feed, and less land, per kilogram of meat. Chains of conversions are multiplied in order with units kept: if a feed is 25% fishmeal and each kilogram of fishmeal needs 4 kg of wild fish, then 1 kg of farmed fish that ate 1.5 kg of feed used 1.5 × 0.25 × 4 = 1.5 kg of wild fish. Footprint tables per 100 g of protein scale with the protein served: value × (protein served ÷ 100 g).

KEY DEFINITIONS

Tilling — mechanical turning of soil before planting. **Slash-and-burn** — clearing land by cutting vegetation and burning it. **CAFO** — a facility that confines large numbers of animals and feeds them grain and soy. **Feed conversion ratio (FCR)** — kilograms of feed eaten per kilogram of body mass gained.

FORMULA & MATH SKILL

Feed needed = FCR × body mass gained Usable forage = forage per hectare × hectares × fraction allowed to be eaten Animals supported = usable forage ÷ forage eaten per animal per year Units: kg of feed or forage; keep "per year" and "per hectare" consistent.

COMMON STUDENT MISTAKES

- Believing free-range always means antibiotic-free or low impact.

CLASSROOM EXAMPLE SG-U5-E014

A ranch has 600 hectares that produce 1,800 kg of dry forage per hectare per year; managers allow cattle to eat 40% of it. Each cow eats 3,000 kg per year. (a) Calculate the largest number of cows the ranch can support sustainably. (b) The owner keeps 400 cows. Calculate by what percent the herd exceeds the sustainable number. (c) A feedlot with an FCR of 7 adds 25,000 kg of beef. Calculate the feed it needs, and state one environmental problem of the feedlot system.

CLASSROOM EXAMPLE SG-C7-E002

A CAFO holds 4,000 cattle. Each animal produces 30 kg of manure per day, and the manure is 0.5% nitrogen by mass. (a) Calculate the mass of manure produced per day. (b) Calculate the mass of nitrogen in that manure per day and per year (365 days). (c) If 4% of the yearly nitrogen reaches a nearby stream, calculate the mass of nitrogen entering the stream per year.

QUICK CONCEPT SUMMARY

- Damaging practices: tilling (erosion, CO_2 release), slash-and-burn (quick fertility, then decline, habitat and CO_2 loss) and fertilizer (nutrient runoff, dead zones, N_2O).
- CAFOs are cheap and efficient but concentrate manure and antibiotics; pasture systems need more land and cost more but return manure to soil.

7.3 Agricultural Productivity

CONCEPT MADE SIMPLE

Before the Green Revolution a field was like a family garden. Afterward it was more like a factory: improved seeds, fertilizer, pesticides, irrigation and machines made the same land yield far more food.

Beginning in the mid-1900s, plant breeders developed high-yield varieties of staple crops such as wheat and rice that responded strongly to fertilizer and water. Combined with other technologies, they greatly increased food production and helped feed a rapidly growing population. Together these changes are the **Green Revolution**, a shift to new agricultural strategies with both positive and negative results.

Strategy	What it does	Benefit	Cost or risk
Mechanization	Tractors, harvesters, planters	Higher efficiency and profit; less labor	Fossil-fuel use, equipment cost, fewer farm jobs
High-yield and genetically modified crops (GMOs)	Selected or engineered traits	Higher yield, pest or drought tolerance	Seed cost; loss of genetic diversity
Synthetic fertilizer	Adds nitrogen, phosphorus	Higher yield per hectare	Nutrient runoff, eutrophication

Strategy	What it does	Benefit	Cost or risk
Irrigation (Chapter 3)	Adds water	Farming in dry regions; extra harvests	Depleted aquifers, salinization
Pesticides	Kill pests	Less crop damage	Resistance, toxicity to wildlife

Machines, fertilizers and pesticides all use fossil energy, so farms depend on oil prices; the tillage and fertilizer damage in Section 7.2 is the price of higher yield.

Reading for perspective (skill 3.B): if an author praises the Green Revolution only in tonnes of grain, ask what the author values, what evidence is left out (such as water pollution) and who benefits from that view.

KEY DEFINITIONS

Green Revolution — the shift, beginning in the mid-1900s, to high-yield crops plus fertilizer, irrigation, pesticides and machinery to increase food production. **Genetically modified organism (GMO)** — an organism whose DNA was altered by genetic engineering to give it a desired trait. **Mechanization** — the use of machinery in place of human or animal labor.

FORMULA & MATH SKILL

Yield = harvest ÷ area (so harvest = yield × area); **percent change = (new − old) ÷ old × 100%** Convert units first (1 ha = 10,000 m²). Divide by the *old* value: a yield rising from 2.0 to 3.0 t/ha is a 50% increase.

CLASSROOM EXAMPLE SG-U5-E003

A district grows wheat on 12,000 hectares. Before adopting Green Revolution methods the district averaged 1.8 t/ha; after adopting them it averaged 3.6 t/ha, while nitrogen fertilizer use rose from 10 kg/ha to 90 kg/ha and diesel use for machinery rose from 20 L/ha to 55 L/ha. (a) Calculate the percent increase in yield and the total additional grain, in tonnes, from the whole district. (b) Calculate the additional kilograms of nitrogen fertilizer applied to the district each year. (c) Describe one environmental consequence of the increased fertilizer use and one of the increased diesel use.

QUICK CONCEPT SUMMARY

- Green Revolution = high-yield crops, fertilizer, irrigation, pesticides, GMOs and mechanization.
- It raised food production but increased reliance on fossil fuels and created pollution and genetic-diversity problems.

7.4 Sustainable Agriculture

CONCEPT MADE SIMPLE

Good soil is a bank that took centuries to fill. Sustainable farming protects the vault (stops erosion) and makes regular deposits (restores fertility).

Soil conservation: preventing erosion

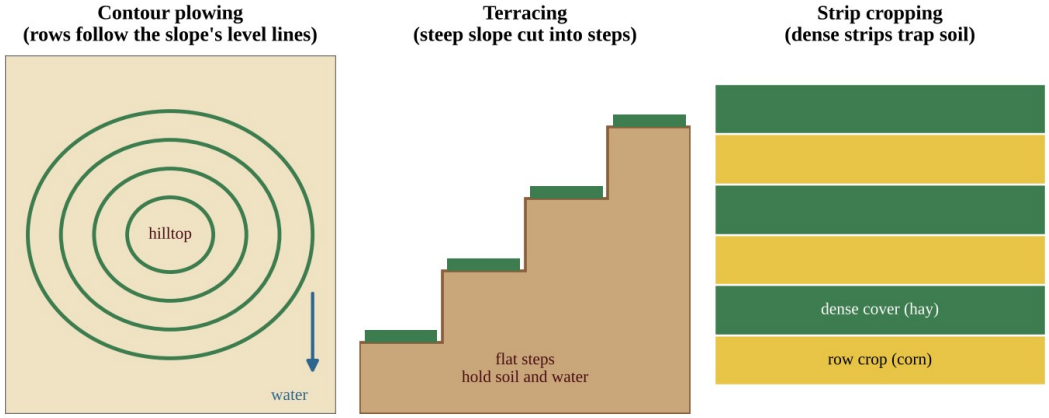


Figure 7.2 — Three ways to slow erosion: plowing along the contours, building terraces, and alternating row crops with dense cover strips.

Practice	What the farmer does	How it works	Trade-off
Contour plowing	Plow across the slope rather than up and down	Furrows slow runoff and trap soil	Best on gentle slopes; needs careful layout
Terracing	Build step-like flat platforms on steep slopes	Breaks long slopes into short flat sections that hold water	Expensive, labor-intensive
Strip cropping	Alternate rows of row crops and dense cover crops along contours	Dense strips catch soil washed from row crops	More complex planting and harvesting
Windbreaks	Rows of trees or shrubs along field edges	Slow wind and reduce wind erosion	Take land and years to grow
Perennial crops	Plant crops that live many years	Roots hold soil year-round; fewer plowings	May take years to produce income
No-till agriculture	Plant seeds directly into last year's residue without plowing	Residue protects soil; stores carbon; saves fuel	May need more herbicide; special seeding equipment

Soil fertility strategies

- **Crop rotation:** alternate crops, for example a nitrogen-hungry grain with a legume. Legumes host bacteria that convert atmospheric nitrogen to forms plants can use, so soil nitrogen is replenished, and pests are disrupted.
- **Green manure:** growing cover crops and plowing them in (or leaving them as mulch) adds organic matter and nutrients. Compost does the same and, like mulch, reduces runoff and sediment loss.
- **Limestone:** raises the pH of acidic soils, improving nutrient availability for crops.

Rotational grazing

Rotational grazing is the regular rotation of livestock between pastures to avoid overgrazing in any one area. Plants have time to regrow, soil stays covered and manure is spread across the land (Sections 7.2 and 7.10).

COMMON STUDENT MISTAKES

- Mixing up contour plowing (along the slope) and terracing (building steps).

CLASSROOM EXAMPLE SG-C7-E003

A 40-hectare field loses 15 tonnes of soil per hectare per year under conventional tillage. Switching to no-till reduces soil loss by 80%. Topsoil forms at about 1 tonne per hectare per year. (a) Calculate the total soil lost per year under conventional tillage. (b) Calculate the soil lost per year under no-till and the tonnes saved on the whole field. (c) Calculate how many times greater the conventional loss rate per hectare is than the topsoil formation rate.

QUICK CONCEPT SUMMARY

- Soil conservation: contour plowing, windbreaks, perennial crops, terracing, no-till, strip cropping.
- Soil fertility: crop rotation, green manure, limestone; rotational grazing protects pasture.

7.5 Controlling Pests

CONCEPT MADE SIMPLE

Spraying a pesticide is like using one antibiotic over and over. It kills most of the target, but the few survivors pass on their toughness, and soon the treatment stops working. A smarter approach is a toolbox, not a hammer.

A **pest** is any organism that harms crops, livestock or people. Farmers use **pesticides**: insecticides (insects), herbicides (weeds), fungicides (fungi) and rodenticides (rodents). Pest control decreases crop damage, raises yield and limits insects that spread disease. The drawbacks are toxicity to non-target species such as pollinators and fish, contamination of water and residues on food. People are exposed by breathing spray, drinking contaminated water, eating residues or absorbing the chemical through skin.

Resistance by artificial selection

Within a pest population, individuals vary genetically. A few may survive a chemical because of a gene that makes them resistant. With repeated spraying, susceptible individuals die and resistant survivors reproduce, so the share of resistant pests rises each generation. Because humans apply the selective pressure, this is **artificial selection**. Pests with short generation times and many offspring evolve resistance fastest, and farmers respond with stronger or more frequent doses, a cycle called the "pesticide treadmill."

For example, suppose 1 insect in 500 (0.2%) is resistant and a spray kills 95% of susceptible insects and none of the resistant ones. Of 500,000 insects, 1,000 are resistant; after the spray 1,000 resistant and about 24,950 susceptible survive, so resistant insects rise to 1,000 ÷ 25,950 ≈ 3.9% of the survivors. Rotating pesticides, leaving untreated refuges and spraying only above a pest threshold weaken this selection.

Genetically engineered crops

Crops can be **genetically engineered** to increase resistance to pests and diseases, which can cut pesticide use and raise yield. Examples are a built-in insect toxin (Bt crops) and herbicide tolerance (repeated use of one herbicide selects for resistant weeds). The risk is a **loss of genetic diversity**: if farmers plant one variety, a single new disease or pest that overcomes its resistance can destroy the entire crop.

Integrated pest management (IPM)

Integrated pest management (IPM) combines biological, physical and limited chemical methods to control pests effectively while minimizing disruption to the environment.

Tool	Example	Why it works
Biocontrol	Release or protect natural predators, parasites or pathogens of the pest	Pest numbers fall without chemicals
Crop rotation	Change the crop each season	Pests of one crop starve when the next crop is not their host
Intercropping	Plant two or more crops together	Confuses pests and supports natural enemies
Monitoring and thresholds	Scout fields; spray only when pest numbers pass an economic threshold	Avoids unnecessary spraying
Physical methods	Traps (including pheromone traps that lure insects), barriers, hand removal	Reduces pests with no toxin
Limited chemicals	Targeted, low-toxicity pesticides applied precisely	Reduces non-target harm and slows resistance

IPM reduces the risk pesticides pose to wildlife, water supplies and human health, and it slows resistance because selection pressure is weaker. A biocontrol organism brought in from elsewhere is itself a risk: if not specific to the pest it may attack native species or become invasive, so it must be tested first. IPM can be complex and expensive: farmers need training, monitoring and sometimes more labor, and results are slower than a quick spray. **Organic farming** (USDA standards) bans synthetic pesticides and fertilizers and genetically engineered crops, but allows compost, manure, crop rotation, biocontrol and pheromone-based mating disruption.

KEY DEFINITIONS

Artificial selection — selection by humans, intentional or not, of which individuals survive and reproduce. **Resistance** — heritable ability to survive a chemical that once killed an organism's ancestors. **Genetic diversity** — variation in genes within a species or crop. **Biocontrol** — use of a pest's natural enemies to control it. **Intercropping** — growing two or more crops in the same field.

COMMON STUDENT MISTAKES

- Writing that pests "adapt" or "learn" to resist. Resistance is *selection* acting on existing genetic variation, not an individual changing.
- Believing IPM bans all pesticides. It allows *limited* chemical use.
- Saying IPM is cheaper. It can be complex and expensive.

AP EXAM STRATEGY

For "propose a solution" questions about pests, name two IPM tools matched to the scenario, a benefit and a drawback (cost, training, slower results).

CLASSROOM EXAMPLE SG-U5-E005

A cabbage grower sprays an insecticide each generation. In generation 1, 2% of the moths carry a resistance allele. Assume the resistant share of the population doubles in every generation, and that the population is rebuilt to its original size each time. (a) Calculate the percentage of resistant moths in generations 2, 3 and 4. (b) Explain why this is an example of artificial selection. (c) Propose one change in pest control that would slow this process, and describe one disadvantage of your proposal.

CLASSROOM EXAMPLE SG-U5-E011

A farm with 50 hectares compares four soil and pest-management plans on separate trial plots. Data per hectare per year: Plan 1 (conventional tillage, weekly insecticide) loses 18 t of soil and uses 12 sprayings; Plan 2 (no-till, weekly insecticide) loses 4 t of soil and uses 12 sprayings; Plan 3 (no-till plus IPM) loses 4 t of soil and uses 3 sprayings; Plan 4 (terraced plus IPM) loses 3 t of soil and uses 3 sprayings. (a) Calculate the percent reduction in soil loss from Plan 1 to Plan 2, and the total tonnes of soil saved per year on the whole farm by Plan 4 compared with Plan 1. (b) Identify which plan best reduces both soil loss and pesticide use, with data, and describe the pest-control tools it likely uses. (c) Justify why a farmer might still choose Plan 2 despite its environmental disadvantage.

QUICK CONCEPT SUMMARY

- Pesticides cut crop damage and raise yield but harm non-target organisms and select for resistance; genetically engineered crops can resist pests but may reduce genetic diversity.
- IPM = biocontrol, crop rotation, intercropping, physical methods, monitoring and limited chemicals: lower risk to wildlife, water and health, but complex and often expensive.

7.6 Forestry

CONCEPT MADE SIMPLE

A sustainable forest is managed like a garden: harvest some, plant more, keep the soil and the stream safe, and clear the dry brush before it catches fire.

Forests provide **timber** and fuelwood, but they also deliver services that markets rarely price: they store carbon, filter air and water, hold soil, moderate floods and shelter most of the world's land species. **Forestry** is the management of forests for these goods and services. Plantations of one species grow wood fast but support less biodiversity and resist pests and fire less well than diverse forests.

Reducing deforestation

- **Reforestation:** replanting trees after harvest or on degraded land, restoring habitat, soil protection and carbon storage.

- **Sustainably harvested wood:** choosing and buying wood from ecologically sustainable forestry, often identified by third-party certification. Methods include **selective cutting** (removing only certain mature trees), **strip cutting** (cutting narrow bands and leaving the rest to reseed), and harvesting no more than regrowth. (Figure 7.5 in Section 7.9 compares these with clearcutting.)
- **Reusing wood:** reclaimed lumber, recycled paper and reuse of furniture reduce demand for new harvest.

Selective cutting costs more per tonne and needs more skill than clearcutting, but it keeps a canopy, so there is less erosion, less stream warming and more habitat.

Protecting forests from pests and fire

- **Pathogens and insects:** use IPM (Section 7.5) and **remove affected trees** to stop the spread. Removing infested trees too late is less effective; healthy, diverse forests resist outbreaks better than single-species plantations.
- **Prescribed burns:** forests are deliberately set on fire under controlled conditions (specific weather, trained crews) to reduce the occurrence of damaging natural fires. They burn off dry underbrush and litter, so a wildfire is less likely to become an intense crown fire (Sections 7.7 and 7.8).

KEY DEFINITIONS

Reforestation — replanting trees on cut or degraded land. **Selective cutting** — removing only some mature trees, leaving the canopy mostly intact. **Certification** — independent verification that wood comes from sustainably managed forest.

FORMULA & MATH SKILL

Sustainable harvest ≈ annual growth per hectare × hectares Harvest above this rate shrinks the standing forest. Percent over = (harvest − sustainable) ÷ sustainable × 100%.

CLASSROOM EXAMPLE SG-C7-E004

A forest company manages 500 hectares of forest in which trees add 6 cubic meters of wood per hectare per year. (a) Calculate the largest sustainable harvest per year in cubic meters. (b) The company actually removes 4,200 cubic meters per year. Calculate by what percent the harvest exceeds the sustainable harvest. (c) Describe one forestry method that would let the company keep earning income while protecting the forest, and one drawback of it.

QUICK CONCEPT SUMMARY

- Reduce deforestation with reforestation, sustainably harvested (certified) wood and reuse of wood.
- Protect forests from pests with IPM and removal of affected trees, and from catastrophic fire with prescribed burns.

7.7 Forest Fires (Wildfires)

CONCEPT MADE SIMPLE

A wildfire needs something to burn, heat to start it and air to feed it. Take away one and it stops. Fire is also a natural, if rough, housecleaning: it clears dead material, recycles nutrients and gives new plants room.

The fire triangle and fire behavior

Burning needs **fuel**, **heat** and **oxygen**. In a forest, fuel is litter, branches, logs, grass, shrubs and trees; heat comes from lightning, human sparks, power lines or another fire; oxygen is always in the air. Fire fighters remove one side: clearing fuel with firebreaks, cooling with water or retardant, or smothering flames.

Fire behavior depends on the **fuel** (amount, moisture, continuity), the **weather** (temperature, humidity, wind, drought) and the **topography** (fire runs faster uphill because flames preheat the slope above). Dry fuel, low humidity and strong wind are the dangerous combination.

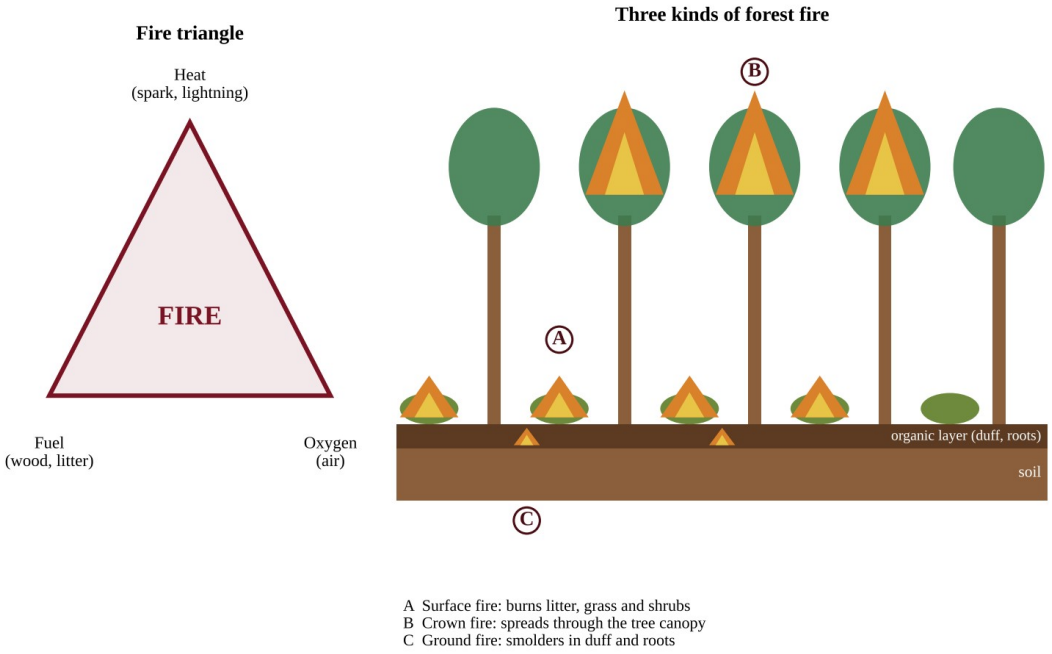


Figure 7.3 — The fire triangle, and the three kinds of forest fire that can occur in the same forest.

Surface, crown and ground fires

Type	What burns	Typical effect
Surface fire	Litter, grass, small shrubs and fallen branches on the forest floor	Often low intensity; most mature trees survive
Crown fire	Leaves and branches in the tree canopy	Very hot and fast; spreads on wind and often kills the whole stand
Ground fire	Organic layer (duff, peat) and roots below the surface	Slow, smoldering, can last for days; kills roots and is hard to see or extinguish

Ladder fuels (small trees and shrubs under taller trees) let a surface fire climb into the canopy and become a crown fire.

The natural role of fire

Many ecosystems evolved with fire and depend on it. Fire clears dead material, returns nutrients as ash, opens the canopy so sunlight reaches seedlings, controls pests and disease, and creates a mosaic of young and old patches that supports many species. **Fire adaptations** include thick bark (ponderosa pine), heat-opened cones (lodgepole and jack pine) and resprouting from roots (oaks, aspen, prairie grasses).

A **fire regime** describes the usual pattern of fire in a region: how often it burns (frequency), how hot (intensity), how much it kills (severity), the season and the size. Fire scars in tree rings date past fires; the **mean fire-return interval** = years between the first and last fire ÷ the number of intervals (one fewer than the number of fires).

Ecosystem	Typical fire regime
Grassland and savanna	Frequent (every 1–5 years); surface fires; grasses regrow quickly
Open ponderosa pine forest	Frequent (about every 5–30 years); low-severity surface fires
Chaparral and lodgepole pine	Infrequent (decades to a century or more); high-intensity, high-severity
Boreal forest	Infrequent (every 50–200 years); stand-replacing crown fires

Harm from wildfire

Fire becomes a problem when it is more intense than the ecosystem is adapted to or when it reaches people. Wildfire kills people and animals, destroys homes, and releases CO_2 and smoke (fine particulates, PM2.5) that harm health far downwind. Burned slopes lose their plant cover, and some soils turn water-repellent after severe fire, so storms cause floods, erosion and mudslides and carry ash into reservoirs. Climate change lengthens fire seasons and dries fuels, and burning forests release stored carbon, a feedback that adds to warming (Chapter 11).

KEY DEFINITIONS

Fuel load — the mass of burnable material per unit area. **Fire regime** — the typical frequency, intensity, severity, season and size of fires in an ecosystem. **Fire adaptation** — a trait (thick bark, heat-opened cones, resprouting) that lets a species survive or use fire.

COMMON STUDENT MISTAKES

- Believing all fire is destructive. Many ecosystems need fire; the problem is fires that are unnaturally intense.
- Mixing up intensity (how hot) with severity (how much damage).

CLASSROOM EXAMPLE SG-C7-E005

A fire spreads outward in every direction at 1.5 kilometers per hour for 6 hours. Assume the burned area is a circle. (a) Calculate the radius of the burned area in kilometers. (b) Calculate the burned area in square kilometers and in hectares (1 km² = 100 ha). (c) Identify which part of the fire triangle a firebreak removes.

QUICK CONCEPT SUMMARY

- Fire needs fuel, heat and oxygen; its behavior depends on fuel, weather and topography.
- Surface, crown and ground fires differ in what burns; ecosystems have adapted to characteristic fire regimes.
- Fire clears, recycles and renews, but unusually intense fires kill, pollute and trigger erosion.

7.8 Wildfire Suppression

CONCEPT MADE SIMPLE

Putting out every small fire is like never cleaning your garage: it feels safe for years, then one spark finds a mountain of clutter. Modern managers try to manage fire, not just stop it.

A history of suppression

The "Big Burn" of 1910 in the northwestern United States (about 3 million acres) convinced the young US Forest Service that fire was an enemy. For most of the 20th century the policy was to put out every fire quickly (the "10 a.m. policy" aimed to control each fire by the next morning), backed by the Smokey Bear campaign. It worked in the short term, but it also removed the frequent low-intensity fires that had kept many forests open.

The unintended result: fuel build-up

Without fire, litter and fallen branches accumulate, small trees grow up between the big trees as ladder fuels, and forests become dense. Crowded trees are stressed and open to insects and disease, which adds dead fuel. When a fire finally escapes, it is far more intense, burns in the canopy and kills mature trees that a surface fire would have spared.

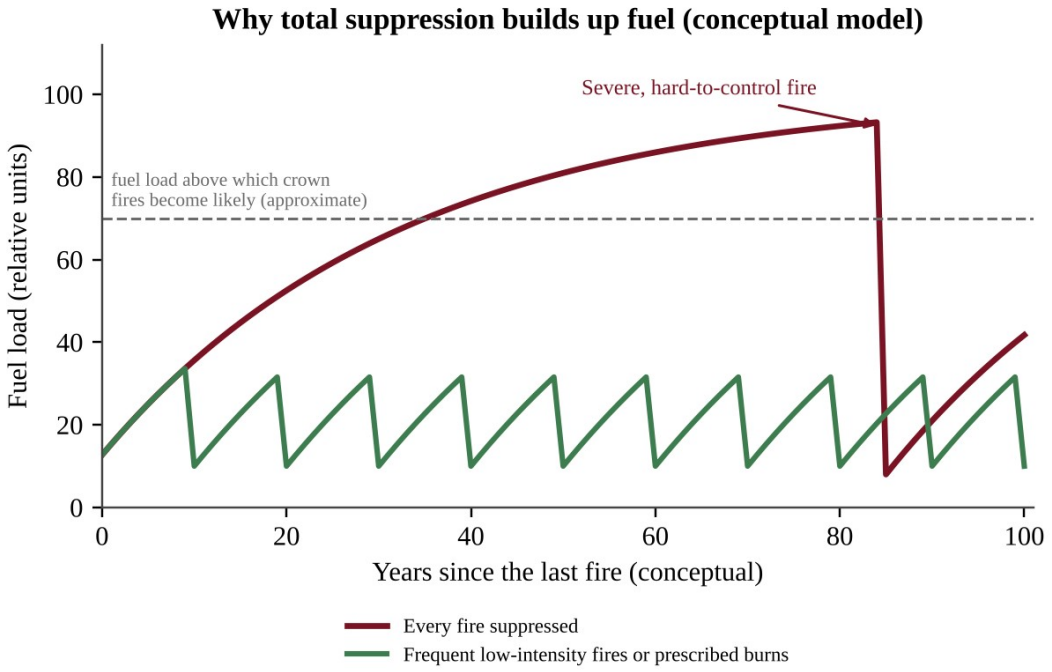


Figure 7.4 — A conceptual comparison of fuel load with total suppression and with frequent low-intensity fire. Numbers are illustrative.

Modern approaches

Method	What it does	Benefit	Limitation
Prescribed burns	Planned, controlled fires under set weather	Remove surface fuel; restore natural regimes; cheap per hectare	Smoke, risk of escape; narrow safe weather windows; needs trained crews
Mechanical thinning	Cut small trees and remove ladder fuels	Reduces crown-fire risk; yields wood	Costly; leaves slash; no ecological "reset"
Managed natural fires	Allow lightning fires to burn under monitoring when risk is low	Cheap; natural process	Public fear; can escape in dry, windy weather
Firebreaks and fire fighting	Clear lines; aircraft and crews	Protect homes and lives	Does not remove fuel; very expensive

The wildland–urban interface (WUI)

The **wildland–urban interface** is the zone where houses meet or mix with flammable vegetation. As more people build there, more homes are at risk and fighting fires to protect property is costly and

dangerous. Homeowners can create **defensible space** (cleared vegetation near the house), use ember-resistant vents and fire-resistant roofs, and communities can plan evacuation routes and limit building in high-risk areas.

KEY DEFINITIONS

Prescribed burn — a fire set deliberately under controlled conditions to reduce fuel. **Thinning** — mechanical removal of small trees and brush to lower fuel density. **Wildland–urban interface (WUI)** — the area where development meets or mixes with wildland vegetation. **Defensible space** — a cleared zone around a building that slows fire.

CLASSROOM EXAMPLE SG-C7-E006

A forest agency plans a prescribed burn on 2,000 hectares at a cost of $55 per hectare. A wildfire on the same land would cost $1,100 per hectare to fight and repair. Without the burn the chance of a wildfire on this land in the next 10 years is 35%; with the burn it is 10%. (a) Calculate the cost of the prescribed burn. (b) Calculate the cost of a wildfire on the 2,000 hectares. (c) Calculate the expected reduction in wildfire cost over 10 years (chance reduction × cost of a wildfire) and the net benefit after subtracting the cost of the burn.

QUICK CONCEPT SUMMARY

- Total suppression let fuel build up, making later fires larger and more intense.
- Prescribed burns, thinning and managed fire reduce fuel; the WUI and defensible space link wildfire to housing decisions.

7.9 Deforestation—Tree Harvesting

CONCEPT MADE SIMPLE

A forest is like a giant sponge wearing an umbrella. The canopy softens rain, roots hold soil in place, and shade keeps streams cool. Clearcutting removes the umbrella and the roots all at once.

Deforestation is the permanent conversion of forest to another land use. Its main causes are clearing for crops, pasture and buildings, timber, fuelwood and charcoal, and urban growth. **Clearcutting** is the harvest of all or nearly all trees in an area in one operation. It is the cheapest and fastest method: heavy machines reach every tree, logging roads are simple, and the sunlit ground can be replanted with one fast-growing species. That economic advantage is real, so acknowledge it in a balanced answer. But the ecological costs are large.

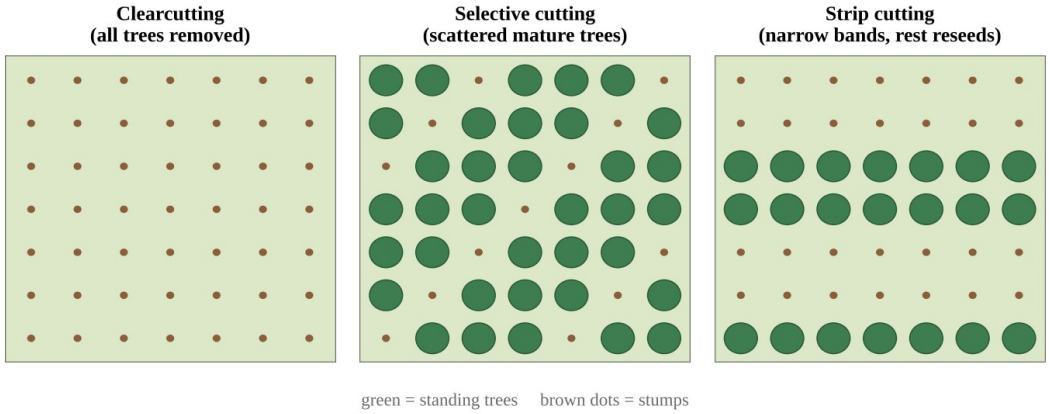


Figure 7.5 — Three harvest methods. Clearcutting removes every tree; selective cutting and strip cutting leave trees standing to hold soil and seed the ground.

PROCESS & CAUSE-AND-EFFECT

Trees removed → roots die and no longer hold soil, and rain strikes bare ground → **soil erosion** increases → sediment washes into streams and clouds the water → the loss of shade raises **soil and stream temperatures** → warmer water holds less dissolved oxygen, stressing cold-water fish → with no canopy interception and less water uptake by roots, runoff rises → **flooding** downstream increases.

Climate and pollution effects

Forests are carbon sinks. Trees absorb CO_2 during photosynthesis and store the carbon in wood, roots and soil, and their leaves capture air pollutants such as particulates and ozone. When trees are cut and burned, or the wood and debris decay, the stored carbon is released as CO_2, which adds to the greenhouse effect and contributes to climate change. Less forest also means less future CO_2 uptake, so the loss is felt twice. Clearcutting also destroys habitat, reducing biodiversity, and exposed soil loses nutrients and organic matter, slowing regrowth.

From stored carbon to CO_2

Forest data are usually given as the mass of *carbon* in the wood, but the gas released is CO_2, which also contains oxygen. One carbon atom (12 g/mol) becomes one CO_2 molecule (44 g/mol), so each tonne of carbon released as CO_2 is $44 \div 12 \approx 3.67$ tonnes of CO_2.

FORMULA & MATH SKILL

Carbon released = stored carbon per hectare × hectares × fraction released Mass of CO_2 = mass of carbon × 44 ÷ 12 Units: tonnes of C, then tonnes of CO_2. Apply the fraction released first and the 44/12 conversion last; the 44/12 step is only needed when the question asks for CO_2 rather than carbon.

KEY DEFINITIONS

Deforestation — the permanent conversion of forest to another land use. **Carbon sink** — a reservoir that absorbs more carbon than it releases.

COMMON STUDENT MISTAKES

- Forgetting the 44/12 conversion, or applying it before the fraction released.

CLASSROOM EXAMPLE SG-U5-E002

A 100-hectare watershed is clearcut. In the following year, monitors record that the stream's maximum summer temperature rises from 14 °C to 21 °C, that suspended sediment in the stream increases, and that peak flow during a storm is much higher than before the cut. (a) Explain how removing the trees causes each of these three changes. (b) Describe one effect of this clearcut on the carbon cycle. (c) State one economic advantage of clearcutting that explains why it is still used.

CLASSROOM EXAMPLE SG-U5-E012

A 30-hectare stand holds 140 tonnes of carbon per hectare. It is clearcut, and 65% of the stored carbon is released as CO_2 within a few years (C = 12 g/mol, CO_2 = 44 g/mol). (a) Calculate the tonnes of carbon released. (b) Convert your answer to tonnes of CO_2. (c) Describe one way wood could still be obtained while keeping more of this carbon stored.

QUICK CONCEPT SUMMARY

- Clearcutting is cheap and efficient but increases erosion, stream and soil temperature, and flooding.
- Forests absorb pollutants and store CO_2; cutting and burning releases CO_2 and adds to climate change.

7.10 Rangelands

CONCEPT MADE SIMPLE

Rangeland grass is a savings account: animals may live off the "interest" (new growth) but not the "principal" (roots and cover).

Rangelands are lands dominated by grasses, forbs and shrubs that are grazed by livestock or wild animals: prairies, savannas, desert grasslands and shrublands. They cover roughly a third of the land area of the United States and a large share of Earth's land, much of it too dry or too steep to farm. They provide meat, milk and wool, wildlife habitat, watershed protection and carbon storage in soil and roots.

Grazing and carrying capacity

Grazing is not harmful by itself: many grasses evolved with grazers and regrow from the base. Damage begins when animals eat plants faster than they regrow. The **carrying capacity** of a range is the number of animals it can support year after year without degrading the vegetation. It depends on the forage produced per hectare (which varies with rainfall and soil) and the fraction that may be taken, often "take half, leave half." Use the forage formulas in Section 7.2.

Overgrazing

Overgrazing occurs when the livestock population exceeds the land's capacity to regenerate vegetation. Animals graze the best plants again and again, so palatable perennial grasses decline while unpalatable weeds and woody shrubs spread. Plant cover shrinks, soil is exposed and compacted, less water infiltrates, erosion increases and fertility drops. Biodiversity falls and the land stores less carbon.

PROCESS & CAUSE-AND-EFFECT

Too many animals → plants grazed faster than they regrow → less cover and fewer roots → soil exposed, compacted and eroding → less water infiltrates and fertility falls → even less forage → still more pressure on the remaining plants.

Desertification

In arid and semi-arid regions overgrazing can lead to **desertification**, the conversion of productive land into desert-like land that can no longer support vegetation. Drought, deforestation and over-cultivation also contribute, as in the African Sahel. It is hard to reverse because lost topsoil takes centuries to rebuild.

Rangeland management

- **Stocking rates:** match animal numbers to forage, and cut them in drought years.
- **Rotational grazing:** move animals between paddocks so each gets a rest period to regrow (Section 7.4).
- **Riparian buffers:** fence cattle out of stream banks and supply water elsewhere to cut erosion, turbidity and manure pollution.
- **Restoration:** reseed native grasses, control invasive weeds, use prescribed fire against woody shrubs and apply soil conservation. Restoration, soil conservation and improved grazing can slow or reverse desertification.
- **Permits:** on public rangeland, agencies limit animal numbers (Section 7.13).

KEY DEFINITIONS

Carrying capacity (grazing) — the number of animals a range can support sustainably. **Overgrazing** — livestock numbers exceed the vegetation's ability to regenerate. **Desertification** — degradation of arid or semi-arid land into desert-like condition.

COMMON STUDENT MISTAKES

- Saying grazing is always harmful. Well-managed grazing can keep grassland healthy; overgrazing is the problem.
- Thinking desertification means "deserts expanding." It means land *degrading*, often far from existing deserts.

CLASSROOM EXAMPLE SG-C7-E007

A rangeland of 2,400 hectares produces 900 kg of forage per hectare per year, and managers allow 50% of the forage to be eaten. Each cow eats 4,500 kg of forage per year. (a) Calculate the number of cows the range can support sustainably. (b) The rancher grazes 300 cows. Calculate by what percent the herd exceeds the sustainable number. (c) A drought cuts forage production by 30%. Calculate the new sustainable number of cows.

QUICK CONCEPT SUMMARY

- Rangelands are grazed grasslands and shrublands; carrying capacity depends on forage produced and the share that can be taken.
- Overgrazing reduces cover, compacts and erodes soil, lowers biodiversity and carbon storage, and can cause desertification; stocking limits, rotation and restoration reduce it.

7.11 Urban Land Development

CONCEPT MADE SIMPLE

A city is a giant machine that takes in water, energy, food and land and gives out waste, heat and runoff.

Urbanization is the growth of cities as more people move from rural to urban areas. Cities can be efficient (shared transport and buildings), but they concentrate resource demand and waste.

Effects on resources and cycles

- **Resource depletion and the water cycle:** Large populations draw heavily on rivers and aquifers. In coastal cities, pumping groundwater lowers the water table and lets salt water move into the freshwater aquifer, called **saltwater intrusion**, which makes wells unusable.
- **Carbon cycle:** Burning fossil fuels for vehicles, power and heating, and the decomposition of waste in landfills, add CO_2 (and methane from landfills) to the atmosphere, increasing the greenhouse effect.
- **Impervious surfaces:** Roads, roofs, sidewalks and parking lots do not allow water to reach the soil. Rain that would soak in becomes fast surface runoff, causing flash floods, carrying oil, metals and trash into streams, and giving less groundwater recharge.
- **Heat:** Dark pavement and buildings absorb sunlight and hold heat (the urban heat island effect, Chapter 9).

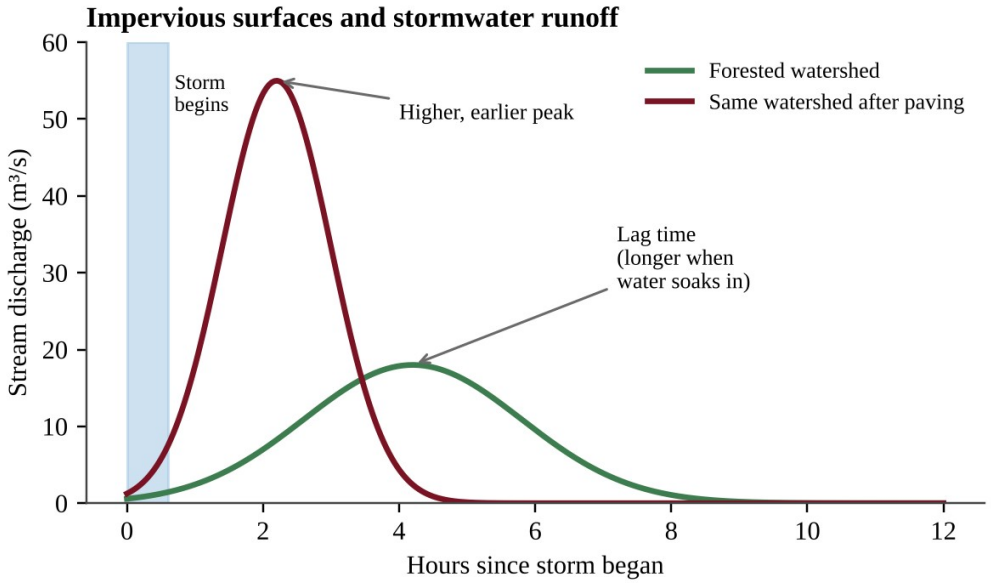


Figure 7.6 — Stream discharge during the same storm before and after a watershed is paved (invented data). Impervious surfaces create a higher, earlier flood peak.

Read a hydrograph by comparing the **peak height** (flood severity), the **lag time** (rainfall to peak) and the **area** under the curve (total runoff). Pavement raises the peak, shortens lag time and increases total runoff. Where old cities send storm runoff and sewage down one pipe, a heavy storm can overwhelm the treatment plant and release untreated mixture into streams.

FORMULA & MATH SKILL

Runoff volume (m^3) = area (m^2) × rainfall depth (m) × runoff coefficient 1 ha = 10,000 m^2; 1 cm = 0.01 m; 1 mm = 0.001 m. Find each surface separately and add.

Urban sprawl

Urban sprawl is the change in population distribution from high-density areas to low-density suburbs that spread into rural lands. Typical consequences: loss of farmland and habitat, more roads and parking (more impervious surface), longer commutes and more vehicle emissions (traffic-related respiratory disease, less walking), higher infrastructure costs per person and greater energy use per household. **Smart growth** counters it with mixed-use zoning, urban growth boundaries, infill and brownfield redevelopment, mass transit and bike paths.

Methods to reduce urban runoff

The goal is to **increase infiltration**, the movement of water into soil, so that less water leaves as runoff:

Method	How it helps	Limitation
Permeable pavement	Replaces traditional pavement so water seeps through to the soil	Can clog with sediment; higher cost; not for all heavy-traffic roads
Planting trees (and green spaces)	Roots and soil absorb water, leaves intercept rain, shade reduces heat	Takes years to grow; needs space and maintenance
Public transportation	Fewer cars means fewer lanes, garages and parking lots, and less pollution	Needs funding and enough population density
Retention basins, bioswales, rain barrels, green roofs	Store, slow and filter runoff; rain barrels and roofs hold water at the source	Need space and upkeep
Building up, not out	Taller buildings on smaller footprints leave more open land	Cost and building regulations; higher local density

Designing an investigation (skill 4.B)

Use this checklist when a question asks you to test a land-use idea (permeable pavement, mulch on a slope, fertilizer rate, rotational grazing, ladybugs against aphids).

- **Hypothesis:** testable and directional ("If slopes are covered with mulch, then less sediment will wash into runoff than from bare slopes").
- **Independent variable:** the one thing you change. **Dependent variable:** what you measure, with units (runoff in L, nitrate in mg/L, percent cover, turbidity).
- **Control group:** gets no treatment or the standard one (bare soil, asphalt). Do not confuse it with **controlled variables**, the conditions kept the same in every plot (slope, soil, area, rainfall, measurement time).
- **Replication and randomization:** several plots per treatment, assigned at random; report the mean and the scatter.
- **Supporting result:** the treatment mean differs from the control in the predicted direction by more than the scatter among replicates.

Weak designs change two variables at once, have no control group, or use one plot.

KEY DEFINITIONS

Impervious surface — human-made surface (road, roof, parking lot) that does not allow water to reach the soil. **Urban sprawl** — the outward spread of low-density development into rural land. **Saltwater intrusion** — movement of salt water into a coastal freshwater aquifer, often because of overpumping.

COMMON STUDENT MISTAKES

- Mixing up the control group (no treatment) with controlled variables (conditions kept equal).

CLASSROOM EXAMPLE SG-U5-E009

A 40-hectare site is a forest. A 25-mm rainstorm falls on it. Use runoff coefficients of 0.15 for pervious ground and 0.90 for impervious surfaces, and runoff volume = area (m^2) × rainfall depth (m) × coefficient. (1 hectare = 10,000 m^2). (a) Calculate the runoff volume from the forested site. (b) The site is developed so that 60% of its area is impervious. Calculate the new runoff volume and how many times greater it is than before. (c) Describe one consequence for the stream below the site and one design change that would reduce the runoff.

CLASSROOM EXAMPLE SG-U5-E017

A 12-hectare school campus has 3 ha of roofs and 4 ha of asphalt (runoff coefficient 0.90 each) and 5 ha of lawn (0.20). A storm drops 3.0 cm of rain. (a) Calculate the runoff volume in m^3. (b) Half of the asphalt is replaced by permeable pavement (coefficient 0.35). Calculate the new runoff volume and the percent reduction. (c) Describe one limitation of permeable pavement.

CLASSROOM EXAMPLE SG-U5-E018

A town wants to know whether a layer of wood-chip mulch on bare construction slopes reduces the sediment that reaches a creek. (a) Write a hypothesis. (b) Identify the independent variable, the dependent variable (with a unit), the control group and two controlled variables. (c) Describe how many plots you would use, how you would assign treatments and what result would support your hypothesis.

QUICK CONCEPT SUMMARY

- Urbanization depletes resources, can cause saltwater intrusion, and adds CO_2 through fossil-fuel use and landfills; impervious surfaces prevent infiltration and cause flooding; sprawl spreads low-density development into rural land.
- Reduce runoff with permeable pavement, trees, public transportation and building up rather than out.

7.12 Transportation Infrastructure

CONCEPT MADE SIMPLE

A road is more than pavement. It cuts a landscape into pieces, brings people and weeds into wild places, and sends oil and salt into every stream it crosses.

Roads, railways, airports, pipelines and ports connect people and goods, but they take land and reshape ecosystems. The United States alone has more than 4 million miles of public roads, and the area affected is much larger than the pavement: the **road-effect zone** extends tens to hundreds of meters on each side.

Habitat fragmentation and edge effects

A road divides a large habitat into smaller patches, called **habitat fragmentation**. Small patches hold fewer individuals, so populations die out more easily, and many animals will not or cannot cross a road; isolated populations lose genetic exchange. Every patch has an **edge**, where light, wind, noise, invasive plants and predators differ from the interior. Small patches may be nearly all edge, so interior species decline.

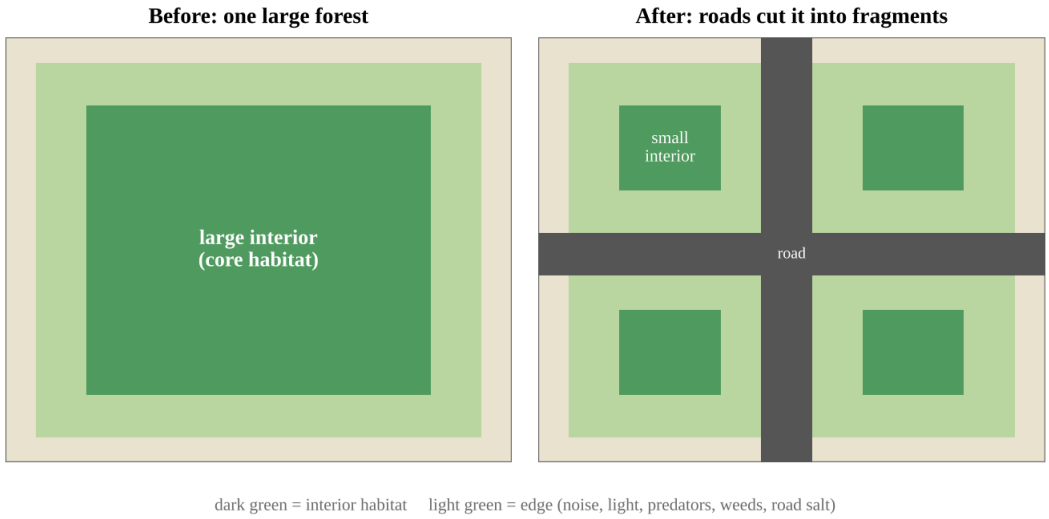


Figure 7.7 — A road network turns one large forest into small fragments, and the edge zone replaces much of the interior.

Vehicles kill wildlife directly, and roads open remote areas to hunting, logging, fires and invasive species. Construction exposes soil, so erosion and stream sediment rise.

Runoff and pollution

Roads are impervious (Section 7.11). Runoff carries oil, heavy metals and tire particles, and winter road salt, into streams and groundwater; salt can harm freshwater organisms. Traffic adds air pollution, noise and CO_2 (Chapters 8 and 9).

Roads and sprawl

New roads make distant land easy to reach, so houses and shops spread outward, creating more traffic and demand for more roads. This feedback is **induced demand**: added capacity fills up with new trips. Sprawl in turn raises the cost of roads and services per household.

Reducing the impacts

- **Wildlife crossings:** overpasses, underpasses and culverts, with fencing that guides animals to them.
- **Fewer and narrower roads:** limit road density in sensitive habitat; close old logging roads.
- **Better runoff control:** permeable pavement, vegetated swales and retention basins, and less road salt.
- **Compact development and transit:** mass transit, walking and cycling, and building up not out, so fewer new roads are needed (Chapter 8).

KEY DEFINITIONS

Habitat fragmentation — division of a large continuous habitat into smaller, isolated patches. **Edge effect** — change in conditions (light, wind, predators, invasive species) along the border of a habitat patch. **Induced demand** — new road capacity encourages new trips and development, so traffic rises again.

COMMON STUDENT MISTAKES

- Measuring a road's impact only by the area of pavement. The affected area includes the edge and road-effect zone.

CLASSROOM EXAMPLE SG-C7-E008

A square forest block measures 4 km on each side. Treat the outer 0.2 km along every boundary as edge habitat; the rest is interior. (a) Calculate the interior area of the block in square kilometers. (b) A straight road (ignore its width) is built through the middle of the block, parallel to one side, splitting it into two equal rectangles of 2 km × 4 km. Calculate the total interior area of the two fragments. (c) Calculate the percent decrease in interior area and the area of the largest single interior fragment.

QUICK CONCEPT SUMMARY

- Roads fragment habitat, create edges, kill wildlife, spread invasive species and carry polluted runoff.
- Roads and sprawl feed each other; crossings, fewer roads, runoff control and compact development reduce the impacts.

7.13 Public and Federal Lands

CONCEPT MADE SIMPLE

Public land belongs to everyone, so everyone wants something different: loggers want trees, ranchers grass, hikers quiet, wildlife habitat. The law gives each agency a different job.

The federal government owns roughly 640 million acres, about 28% of the United States, mostly in the West and Alaska. Four agencies manage most of it. Acreages below are approximate.

Agency	About how much	Mission	Typical uses
Bureau of Land Management (BLM)	245 million acres	**Multiple use** and sustained yield	Grazing, mining, oil and gas, recreation, wildlife
US Forest Service (USFS)	193 million acres (national forests and grasslands)	Multiple use and sustained yield	Timber, grazing, recreation, watershed protection
National Park Service (NPS)	85 million acres	**Preservation** and public enjoyment	Hiking, camping, education; no logging, hunting or mining in most parks
Fish and Wildlife Service (FWS)	89 million acres of refuge land	Conservation of wildlife and habitat	Wildlife viewing, regulated hunting and fishing

Multiple use versus preservation

Multiple use means managing land for many uses at once (timber, grazing, mining, recreation, water, wildlife) without destroying its ability to produce in the future (**sustained yield**). **Preservation** means keeping land natural, with only low-impact recreation and study. The goals often conflict: logging roads

cross wildlife habitat, grazing and mining affect streams, and off-road vehicles disturb quiet recreation. **Conservation**, the wise use of resources so they last, lies between them.

Wilderness and other designations

- **Wilderness areas** (Wilderness Act, 1964) are designated parts of federal land with the highest protection: no roads, no motorized vehicles and no permanent structures. Wilderness can lie within any of the four agencies' lands.
- **National monuments** can be set aside by the president under the Antiquities Act (1906).
- The National Environmental Policy Act (NEPA, 1970) requires environmental impact reviews before major federal projects.
- Public lands are leased or permitted for grazing, timber, oil and gas, and mining (the General Mining Law of 1872 still governs hard-rock claims). Fees are often below market value, a debated subsidy (Section 7.17).

KEY DEFINITIONS

Multiple use — managing public land for several uses at the same time. **Sustained yield** — harvesting no more than the land regrows or renews over time. **Preservation** — protecting land in its natural condition, with little human use. **Wilderness area** — federal land protected from roads, vehicles and permanent development.

COMMON STUDENT MISTAKES

- Mixing up the agencies. USFS and BLM are multiple-use agencies; NPS is preservation and recreation; FWS is wildlife conservation.
- Treating preservation and conservation as identical. Preservation excludes most use; conservation allows wise use.

CLASSROOM EXAMPLE SG-C7-E009

A western state has 62.0 million acres of land. The federal government manages 31.0 million acres of it: BLM 14.5 million, USFS 9.5 million, NPS 3.0 million and FWS 4.0 million acres. (a) Calculate the percent of the state's land that is federally managed. (b) Calculate the percent of the federal land that is managed by the two multiple-use agencies (BLM and USFS). (c) Identify which agency's mission is mainly preservation and recreation.

QUICK CONCEPT SUMMARY

- BLM and USFS manage for multiple use and sustained yield; NPS for preservation and recreation; FWS for wildlife.
- Multiple use versus preservation is a recurring land-use conflict; wilderness areas have the strictest protection.

7.14 Mining

CONCEPT MADE SIMPLE

The rich pieces of the cake go first. In mining, the rich parts are high-grade ore, and as they run out more rock must be moved for each tonne of metal.

Mining extracts minerals, metals and fuels such as coal from the Earth. It supplies low-cost energy and the materials used to make products, from wires to bridges, and it provides jobs. Its drawbacks are habitat destruction, pollution and large volumes of waste.

Ore grade and diminishing returns

An **ore** is rock containing enough valuable mineral to be mined profitably. The **grade** is the concentration of the valuable material in the ore. As the most accessible, highest-grade ores are mined to depletion, operations are forced to use lower-grade ores. This requires more energy, water and machinery, and it produces more waste and pollution per tonne of product.

PROCESS & CAUSE-AND-EFFECT
High-grade ore depleted → miners turn to lower-grade ore → more rock must be excavated per tonne of metal → more energy, water and chemicals are used → more overburden and tailings → more land disturbance and pollution.

Mining methods

Surface mining versus underground mining (cross-section)

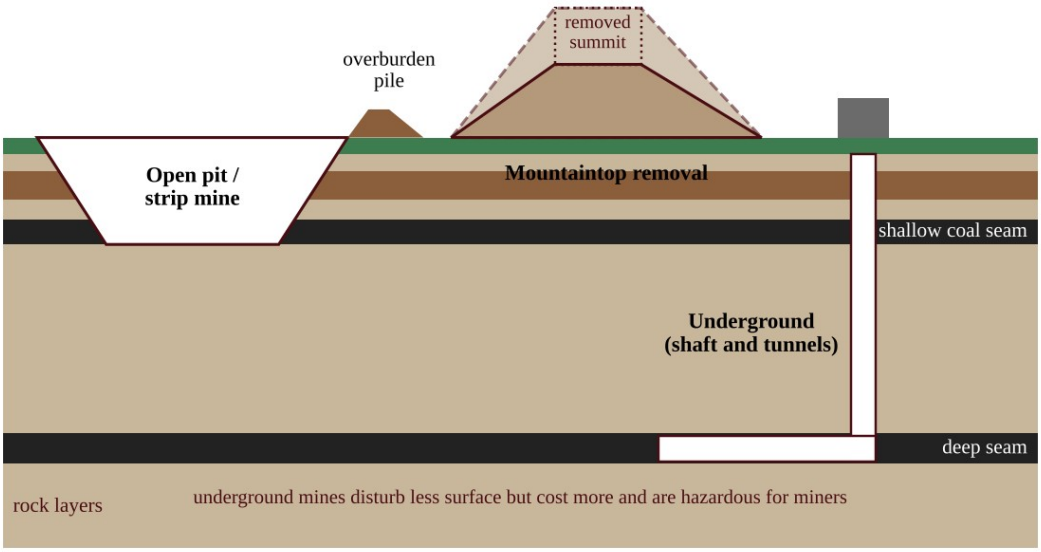


Figure 7.8 — Mining methods in cross-section: surface methods (open pit or strip mine, mountaintop removal) remove the overburden above the ore; underground mines reach deeper seams through shafts and tunnels.

Method	How it works	Example impact
Surface mining	Soil and rock above the ore, called **overburden**, are removed	Habitat loss, erosion, dust
Strip mining	Vegetation and overburden removed in long strips to expose a shallow seam	Vegetation loss makes the area susceptible to erosion; acidic runoff
Open-pit mining	Large pit dug to reach ore close to the surface	Huge pits, tailings piles
Mountaintop removal	Mountain summit blasted off to reach coal seams, debris dumped in valleys	Stream burial, large-scale habitat loss
Subsurface (underground) mining	Shafts and tunnels reach deeper deposits	Very expensive; hazards to miners; subsidence; methane

As shallow coal reserves get smaller, it becomes necessary to use subsurface mining, which is very expensive.

Waste and pollution

Mining wastes include the soil and rock moved to gain access to the ore (overburden or spoil) and the **tailings** and **slag** that remain after minerals are separated or smelted. Water that touches exposed sulfide minerals can become acidic (**acid mine drainage**) and carry heavy metals into streams and groundwater. Pyrite (iron sulfide) exposed to air and water is oxidized to sulfuric acid, so stream pH can fall to about 3–4 and metals dissolve; this continues long after a mine is abandoned. Some mines, such as gold mines, use cyanide solutions to separate the metal, so leaks from tailings ponds can poison water. Coal mining can destroy habitat, contaminate groundwater, and release dust particles and methane, a greenhouse gas. Acid drainage is slowed by keeping waste covered and dry or neutralized with crushed limestone.

Reclamation regrades the land to its original contours, spreads topsoil and organic matter, replants native cover and monitors for years. It is slower in arid regions, where low rainfall and thin soil make plants hard to establish and bare ground erodes.

FORMULA & MATH SKILL

Metal recovered = ore × grade (decimal) × recovery fraction, so **ore needed = metal wanted ÷ (grade × recovery)** (recovery = 1 if not given) **Waste (tailings) mass = ore mined − metal recovered** Units: tonnes (t); 0.5% = 0.005. Halving the grade doubles the ore needed.

Grades can also be given as a fraction of the *volume* (very small values such as 2×10^{-6} are common for gold). Then:

FORMULA & MATH SKILL

Volume of metal = volume of rock × grade (as a fraction) Mass of metal = volume of metal × density (1 $m^3 = 10^6$ cm^3, so convert volume to cm^3 if density is in g/cm^3) **Break-even price = total cost of mining and processing ÷ mass of metal recovered** Method: write each conversion as a fraction equal to 1 (for example 10^6 $cm^3/1$ m^3) so the unwanted units cancel.

KEY DEFINITIONS

Ore — rock containing a valuable mineral in a concentration high enough to mine. **Overburden** — soil and rock removed to reach ore. **Tailings / slag** — waste left after minerals are separated from ore (tailings) or smelted (slag). **Reclamation** — restoring mined land after mining.

COMMON STUDENT MISTAKES

- Saying mining waste is only "overburden." Tailings and slag are separate.
- Thinking mining is purely harmful. Describe both the economic benefit and the ecological cost.

CLASSROOM EXAMPLE SG-U5-E008

A mine must produce 1,000 tonnes of copper per year. Assume all of the copper in the ore is recovered and the rest of the ore becomes tailings. (a) Calculate the tonnes of ore that must be processed per year when the ore grade is 0.8% copper and when it is 0.4% copper. (b) Calculate the tonnes of tailings produced per year at each grade. (c) Explain why the lower-grade ore is likely to increase both the cost and the environmental impact of the mine, and propose one way to reduce the demand for newly mined copper.

CLASSROOM EXAMPLE SG-U5-E016

A block of rock measures 50 m × 40 m × 10 m. Gold makes up 3×10^{-7} of its volume, and gold has a density of 19 g/cm³. Mining and processing the block costs $400,000. (a) Calculate the volume of the rock and the volume of the gold in m³. (b) Calculate the mass of gold in kilograms. (c) Calculate the break-even price per gram, and explain why a fall in the ore grade would raise it.

QUICK CONCEPT SUMMARY

- Surface mining removes overburden (strip mining, open-pit, mountaintop removal); subsurface mining is deeper and costlier.
- Declining ore grade means more waste, energy and pollution per tonne; mining waste includes overburden, tailings and slag, and acid mine drainage.

7.15 Global Mineral Reserves

CONCEPT MADE SIMPLE

Think of the ground as a pantry. "Reserves" are the counted shelves worth opening at today's prices.

Reserves and resources

A mineral **resource** is the total amount of a mineral that exists in Earth's crust in a form that could eventually be extracted, including deposits not yet found. A **reserve** is the part of a resource that has been identified and can be extracted profitably with current technology and at today's prices. Reserves are therefore *smaller* than resources and can change: a higher price, cheaper technology or a new discovery moves material into the reserves, while a falling price or tougher regulation moves it out. Reserve numbers are not a count of how much is left in the ground.

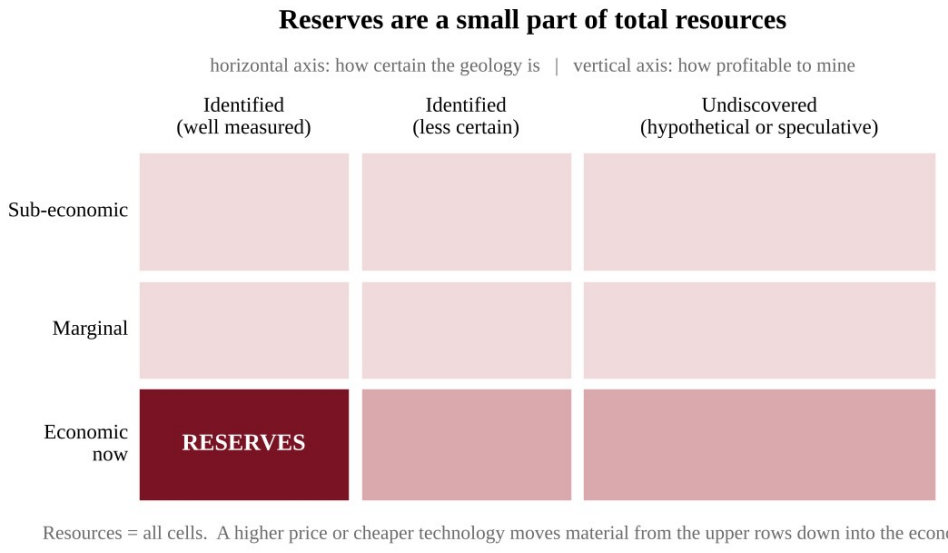


Figure 7.9 — Resources include reserves plus identified but uneconomic and undiscovered deposits. Price, technology and exploration change how much counts as reserve.

Ore grade and how long reserves last

Richer ores are mined first, so average grade falls (Section 7.14) and the energy, water and waste per tonne of metal rise. The **static reserve index** estimates how long reserves last at a constant rate of use:

FORMULA & MATH SKILL

Years remaining (static) = reserves ÷ annual production Units: reserves and production in the same unit (tonnes; tonnes per year). If production grows each year the true lifetime is shorter; recycling, new discoveries and substitution make it longer.

Uneven distribution and critical minerals

Minerals form where geologic processes concentrated them (Chapter 1), so they are unevenly spread. Examples (approximate): much of the world's cobalt comes from the Democratic Republic of the Congo, lithium from Australia, Chile and China, copper from Chile and Peru, platinum-group metals from South Africa and Russia, and most rare-earth processing takes place in China. A **critical mineral** is essential to technologies such as batteries, wind turbines, electronics and solar panels, and its supply can be disrupted because few countries produce or process it. Countries respond by diversifying sources, stockpiling, recycling and finding substitutes.

Recycling and reducing demand

Recycling metals avoids new mining and saves energy: recycling aluminum uses about 95% less energy than making it from bauxite ore. Collecting metal from old electronics ("urban mining") can be profitable because scrap can be richer in metal than natural ore. Substitution, longer-lived products and using less material also reduce demand.

KEY DEFINITIONS

Mineral resource — all of a mineral that exists in or on Earth, including deposits not yet found or not yet profitable to mine. **Reserve** — the part of a resource that is identified and can be mined profitably now. **Critical mineral** — a mineral essential to modern technology whose supply is vulnerable to disruption. **Urban mining** — recovering metals from discarded products and waste.

COMMON STUDENT MISTAKES

- Saying reserves are "all the metal that exists." Resources are larger; reserves depend on price and technology.

CLASSROOM EXAMPLE SG-C7-E010

A metal has world reserves of 120 million tonnes and annual production of 3.0 million tonnes. (a) Calculate the number of years the reserves would last at the current production rate. (b) Recycling supplies enough metal to cut the demand for newly mined metal by 25%. Calculate the new annual mine production and the new number of years. (c) Explain why a rise in the metal's price could increase the reserves without any new deposit being found.

QUICK CONCEPT SUMMARY

- Reserves are the profitable, identified part of the resource; they rise or fall with price and technology.
- Minerals are unevenly distributed, so critical minerals carry supply risks; recycling and substitution reduce demand for new mining.

7.16 Fishing

CONCEPT MADE SIMPLE

A fish population is like a bank account that earns interest. You can spend the interest forever, but if you spend the principal, the interest shrinks and the account empties.

Overfishing is harvesting fish faster than the population can replace itself. It has led to the extreme scarcity of some fish species, which can lessen biodiversity in aquatic systems and harm people who depend on fishing for food and commerce. A fishery is a classic commons, so the tragedy of the commons applies (Section 7.18).

Causes

- Growing demand for seafood as population and incomes increase.
- Technology (sonar, GPS, factory ships), open access in international waters, weak enforcement and subsidies that keep too many boats at sea.

Fishing methods and their impacts

Method	Description	Main problems
Bottom trawling	Weighted net dragged along the seafloor	Destroys bottom habitat; heavy **bycatch**
Purse seining	Net encircles a school and is closed at the bottom	Bycatch (dolphins, sharks)
Drift-net or gill-net	Long walls of netting that hang in the water	Entangles non-target animals, including turtles and marine mammals
Longlining	Line with thousands of baited hooks	Hooks seabirds, turtles and sharks

Bycatch is the unintended capture of non-target species. Removing top predators or a keystone species can cascade through the food web and reduce biodiversity; for example, fishing out the herbivorous fish on a reef lets algae overgrow and smother the coral.

Reading fishery data

Total catch alone can mislead. **Catch per unit effort (CPUE)** is the total catch divided by the fishing effort (boats, boat-days or hours fished). In a healthy stock, adding boats raises the catch and CPUE stays about the same. In a shrinking stock, each boat finds fewer fish even with sonar and bigger nets, so CPUE falls while effort rises. Total catch can stay high for years because more boats and better technology hide the decline, and then collapse. Always work out CPUE before concluding that a stock is healthy.

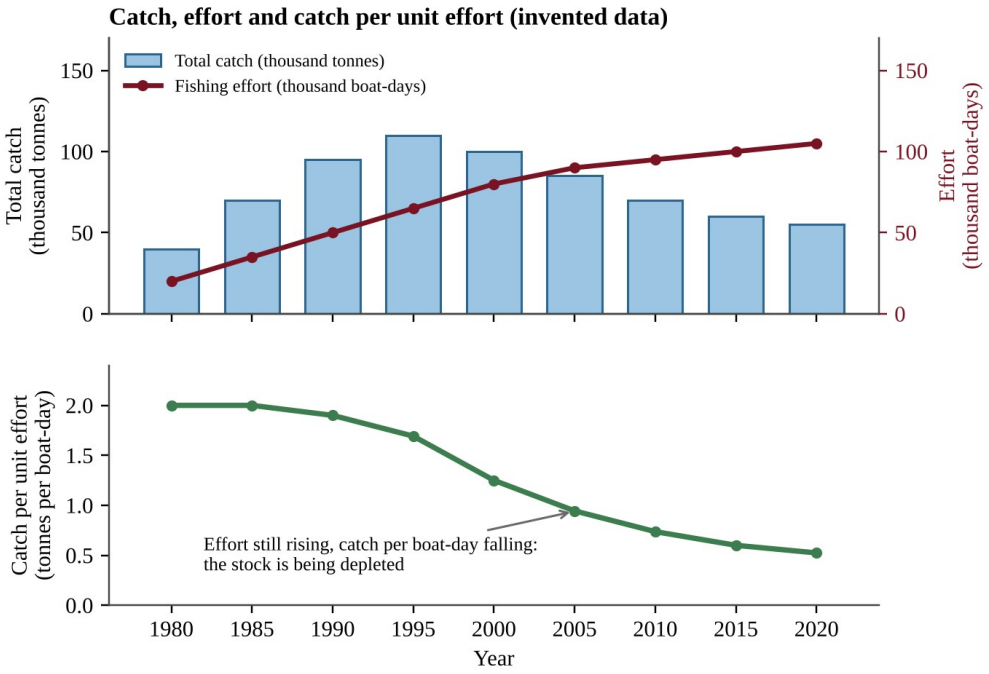


Figure 7.10 — Total catch, fishing effort and catch per unit effort for a fishery (invented data). Catch peaks and falls while effort keeps rising, and CPUE falls throughout.

FORMULA & MATH SKILL

CPUE = total catch ÷ fishing effort (for example, tonnes per boat-day) **Change in stock = natural growth − harvest** (per year)

Sustainable yield

The **sustainable yield** of a fishery is the amount that can be harvested without reducing the supply. The fish stock replaces itself fastest at intermediate stock sizes. A harvest below the peak of the growth curve can continue indefinitely; a harvest above it will shrink the stock every year.

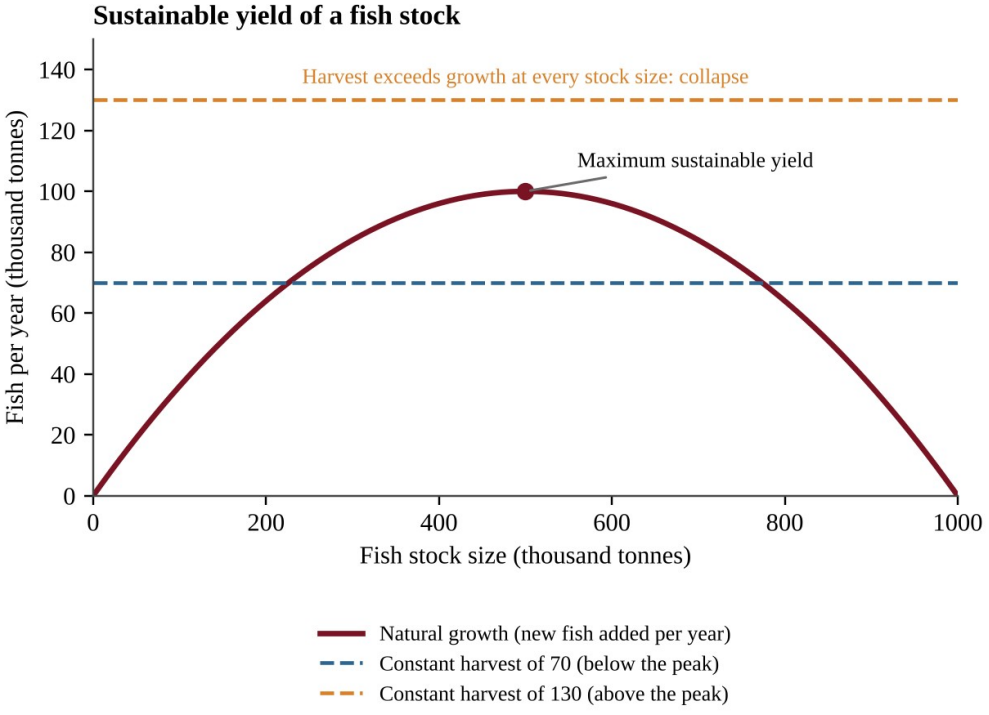


Figure 7.11 — Natural growth of a fish stock at different stock sizes with two constant harvest levels (invented numbers). The peak is the maximum sustainable yield.

Compare the growth curve with the harvest line at the stock size you start from. Where the curve is above the line, the stock rises; where it is below, the stock falls. Where they meet is an equilibrium. The right-hand crossing is stable (a small drop in stock makes growth exceed harvest, so the stock recovers); the left-hand crossing is unstable (below it the stock falls to collapse). A harvest line above the peak never meets the curve, so the stock collapses. Harvesting exactly at the peak leaves no safety margin, so quotas are set below it.

Solutions

- Catch limits and quotas based on stock surveys.
- Closed seasons and marine protected areas where fish can breed.
- Gear restrictions: larger mesh sizes, turtle-excluder devices, bans on destructive methods.
- Certification, international agreements, enforcement, and aquaculture.

Aquaculture

Aquaculture is the farming of fish, shellfish and aquatic plants in ponds, tanks or ocean pens. It has expanded because it is highly efficient, requires only small areas of water and uses little fuel (no long trips to far fishing grounds). It can relieve pressure on wild stocks, provide a reliable supply and create jobs in coastal and rural communities.

Its drawbacks:

- **Wastewater contamination:** uneaten feed, feces and chemicals flow into surrounding water, adding nutrients and causing eutrophication.
- **Escapes:** farmed fish that escape may compete with wild fish for food and habitat, or breed with them and change the wild gene pool.
- **Disease:** crowded fish have high disease and parasite rates, and these can be transmitted to wild fish.
- **Other impacts:** shrimp ponds can replace coastal mangroves, and carnivorous farmed species may be fed wild-caught fish.

KEY DEFINITIONS

Bycatch — unintended capture of non-target species. **Sustainable yield** — the harvest that can be taken year after year without reducing the stock. **Aquaculture** — the farming of fish, shellfish and aquatic plants.

CLASSROOM EXAMPLE SG-U5-E007

A fish stock's natural growth, measured as tonnes of new fish added in a year, depends on stock size as follows: stock 200 → growth 64; stock 400 → growth 96; stock 500 → growth 100; stock 600 → growth 96; stock 800 → growth 64 (all in thousands of tonnes). A fleet takes 90 thousand tonnes every year. (a) State the maximum sustainable yield shown by the data and the stock size at which it occurs. (b) If the stock is 800 thousand tonnes, calculate the change in stock after one year of fishing. (c) Predict what will happen to the stock if the stock falls to 200 thousand tonnes while the fleet still takes 90, and explain how this relates to the tragedy of the commons.

CLASSROOM EXAMPLE SG-U5-E015

A lake fishery records effort of 30,000 boat-days and a catch of 60,000 tonnes in 2005; 50,000 boat-days and 70,000 tonnes in 2015; and 80,000 boat-days and 56,000 tonnes in 2025. (a) Calculate the catch per unit effort for each year. (b) Describe the trend in total catch and in CPUE, and state what each suggests about the fish stock. (c) Explain why the 2015 total catch could mislead a manager, and propose one management action.

QUICK CONCEPT SUMMARY

- Overfishing: harvest exceeds replenishment. It reduces biodiversity and harms fishing communities; destructive gear causes bycatch and habitat damage.
- Falling CPUE signals decline; sustainable yield sets safe catch limits.
- Aquaculture is efficient and uses little space and fuel, but its waste pollutes water, escapees compete and breed with wild fish, and dense populations spread disease.

7.17 Global Economics

CONCEPT MADE SIMPLE

A banana, a T-shirt or a phone ties you to farms, mines and factories abroad. The price rarely includes the damage done to land and water far away.

Trade and globalization

Trade is the exchange of goods and services between regions or countries. **Globalization** is the growing connection of economies through trade, investment, transport and communication. Trade lets countries specialize in what they produce most cheaply, which can lower prices and raise incomes. But it also moves environmental impact away from consumers: forests are cleared for beef, soy and palm oil sold abroad; water is used to grow exported crops (**virtual water**); and long transport burns fossil fuel. A country's ecological footprint (Chapter 6) can exceed the land inside its borders.

Prices, subsidies and externalities

Prices guide choices, but market prices often leave out **externalities**, costs of pollution, habitat loss and depleted water that others pay. When the price is too low, people use too much. **Subsidies** (government payments or tax breaks) can make this worse: payments for fuel to fishing fleets encourage overfishing, subsidized irrigation water encourages waste, and cheap fees for grazing and mining on public land lower costs for the users. Subsidies can also help the environment, for example when they support solar power or soil conservation.

Environmental economics tools (Chapter 10) correct prices: a tax or fee equal to the external cost, tradable permits, and removal of harmful subsidies.

Development and the environment

Poorer countries often export raw materials (timber, crops, minerals, oil) to raise income and may have weak enforcement. When one resource dominates an economy, price swings and environmental damage can leave the country worse off (the "resource curse"). As incomes rise, people can afford cleaner

technology and stricter rules, but consumption per person usually rises too. **GDP per capita** (the value of all goods and services a country produces in a year ÷ its population) is the usual income measure.

KEY DEFINITIONS

Globalization — the increasing connection of the world's economies through trade, investment and communication. **Subsidy** — a government payment or tax break that lowers the cost of producing or buying something. **Externality** — a cost (or benefit) of an activity borne by people who did not choose it. **Virtual water** — the water used to produce a traded product.

FORMULA & MATH SKILL

Full-cost price = market price + external cost per unit Tax to internalize the cost = external cost per unit; revenue = tax × quantity Percent increase = (new − old) ÷ old × 100%

COMMON STUDENT MISTAKES

- Saying trade or subsidies are always harmful. Both can help or harm; evaluate who gains and which costs are left out.

CLASSROOM EXAMPLE SG-C7-E011

A farm product sells for $80 per tonne. Producing it causes $22 per tonne of damage to water and soil that the producer does not pay. A government adds a tax equal to the damage on each tonne. The farms produce 500,000 tonnes per year. (a) Calculate the full-cost price per tonne. (b) Calculate the annual tax revenue. (c) Calculate the percent increase in price caused by the tax, and describe one effect the higher price could have on use of the resource.

QUICK CONCEPT SUMMARY

- Trade and globalization link consumers to distant land and water use, often hiding environmental costs.
- Externalities and subsidies distort prices; taxes, fees and subsidy reform correct them.

7.18 Tragedy of the Commons

CONCEPT MADE SIMPLE

Picture a shared pizza with no rule about slices. Each person thinks, "If I do not take another slice, someone else will," and the pizza is gone in minutes. No one wanted that result, yet everyone's sensible-looking choice caused it.

A **commons** is a resource that belongs to everyone and so to no one in particular: a fishing ground, the air, a groundwater aquifer, a public pasture, the open ocean, the atmosphere's capacity to absorb carbon dioxide. Such a resource is usually *rival* (my use leaves less for you) and *hard to exclude people from*. The tragedy of the commons suggests that individuals will use shared resources in their own self-interest rather than in keeping with the common good, thereby depleting the resources. Each user gets the full benefit of taking more but shares the cost of depletion with everyone, so the rational self-interested choice is to take more. Overfishing (Section 7.16), overgrazing (Section 7.10) and public-land use (Section 7.13) are all examples.

PROCESS & CAUSE-AND-EFFECT

Open access to a shared resource → each user gains the full benefit of extra use → the cost (depletion) is spread across all users → every user has an incentive to take more → total use exceeds the resource's ability to renew itself → the resource declines or collapses → everyone, including the heaviest users, loses.

Why it happens and how it is prevented

The problem is not that people are evil; it is that the *rules* reward overuse. Solutions therefore change the rules:

- **Regulation and quotas:** catch limits for each fishing season, pumping limits on a shared aquifer, permits for grazing on public land, emissions caps.
- **Privatization or assigned shares:** if a person owns the pasture or a transferable share of the fishery, he or she pays the long-term cost of overuse. The drawback is that ownership can favor the wealthy and may ignore the ecosystem services that benefit everyone.
- **Community management:** a village that sets its own rules for a shared forest or irrigation canal, enforced by social pressure, often succeeds when the group is small and can monitor itself.

KEY DEFINITIONS

Tragedy of the commons — the idea that individuals acting in their own self-interest will use a shared resource in ways that do not serve the common good, thereby depleting or degrading it. **Commons (common-pool resource)** — a shared resource open to many users, such as a fishery, aquifer, pasture or the atmosphere.

COMMON STUDENT MISTAKES

- Believing the tragedy happens only to *land*. Air, oceans, groundwater and the climate are commons too.
- Saying people are "greedy" or "ignorant." On the exam, the cause is *self-interest combined with lack of rules or enforcement*.

AP EXAM STRATEGY

When a question gives a shared resource, answer in three moves: (1) identify the commons and who uses it, (2) explain that each user benefits fully from taking more while the cost is shared, and (3) propose a specific rule (quota, permit, fee) and state one limitation of it.

CLASSROOM EXAMPLE SG-U5-E001

Eight ranchers each graze cattle on a public meadow that can feed 80 cattle per year without losing grass cover. Each rancher currently grazes 14 cattle. A rancher who adds one more animal gains the full sale value of that animal, while the loss of grass is shared among all eight ranchers. (a) Calculate the total number of cattle on the meadow and by what percent it exceeds the sustainable number. (b) Explain, using the concept of the tragedy of the commons, why each rancher might add another animal even though all of them know the meadow is overgrazed. (c) Propose one rule that would prevent further damage and identify one drawback of that rule.

QUICK CONCEPT SUMMARY

- The tragedy of the commons: self-interested use of a shared resource depletes it for everyone.
- Cause: full personal benefit, shared cost, no enforceable limits; remedies: quotas, permits, assigned shares, community rules.

Math and Skills Toolkit

Calculators are allowed. Write the formula, the substitution with units, and the answer with units.

- **Percent change** = (new − old) ÷ old × 100%; divide by the *old* value. **Rate of change** = change ÷ time interval. **Percent of total** = part ÷ whole × 100%.
- **Yield** = harvest ÷ area (1 ha = 10,000 m²). **Runoff** = area × depth × coefficient (convert mm to m first).
- **Chained conversions:** multiply the factors in order and cancel units with fractions equal to 1.
- **Ore** = metal ÷ grade; **waste** = ore − metal. **Carbon to CO_2:** × 44 ÷ 12. **CPUE** = catch ÷ effort. **Stocking** = forage × allowed fraction ÷ forage per animal. **Static reserve life** = reserves ÷ annual production.
- **Graphs:** name variables and units first; the peak of a growth curve is the maximum sustainable yield; on a hydrograph compare peak, lag time and area; quote data values.
- **Task verbs:** *Describe* = characteristics; *Explain* = cause and effect; *Calculate* = formula, substitution, answer with units; *Justify* = reason plus evidence; *Propose* = a specific action.

7.19 Practice Questions and Answers

Practice makes the ideas in this chapter stick, so use the classroom examples and question sets below.

PRACTICE QUESTIONS AND ANSWERS

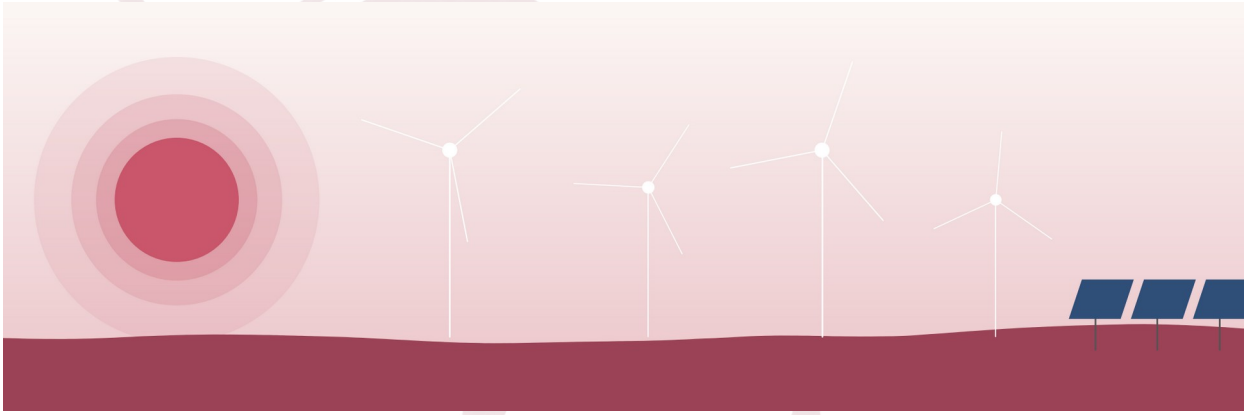
Classroom examples in this chapter (unsolved; space is provided in each box): SG-U5-E006, SG-C7-E001, SG-U5-E014, SG-C7-E002, SG-U5-E003, SG-C7-E003, SG-U5-E005, SG-U5-E011, SG-C7-E004, SG-C7-E005, SG-C7-E006, SG-U5-E002, SG-U5-E012, SG-C7-E007, SG-U5-E009, SG-U5-E017, SG-U5-E018, SG-C7-E008, SG-C7-E009, SG-U5-E008, SG-U5-E016, SG-C7-E010, SG-U5-E007, SG-U5-E015, SG-C7-E011, SG-U5-E001.

Classified practice: Book Two, Chapter 7 — 56 multiple-choice questions (Section A) followed by 4 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 7. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

UNIT V
ENERGY RESOURCES AND CONSUMPTION



IN THIS UNIT

Chapter 8 Energy

CHAPTER

8

UNIT V — ENERGY RESOURCES AND CONSUMPTION

Energy

8.1 · Energy Concepts
8.2 · Energy Consumption (Past)
8.3 · Energy Use (Present)
8.4 · Renewable Energy Sources
8.5 · Energy Conservation
8.6 · Transportation
8.7 · Mass Transit
8.8 · Practice Questions and Answers

LEARNING OBJECTIVES

- Distinguish renewable from nonrenewable sources, and use energy, power, efficiency and the laws of thermodynamics correctly.
- Describe how energy use grew from wood to coal, oil and gas, and what drives demand today.
- Explain how fossil-fuel, nuclear, biomass, solar, hydro, geothermal, hydrogen and wind systems generate power, and what each costs the environment.
- Calculate energy, power, efficiency, capacity factor, payback, EROEI, R/P ratio, half-life, fuel economy and emissions per passenger-kilometer.
- Compare ways to conserve energy and to move people (cars, hybrids, electric vehicles, buses and rail).

Energy is the "currency" of modern life, and every energy choice leaves an environmental bill. Learn how each source works and what it costs, then learn the arithmetic that lets you compare sources fairly.

8.1 Energy Concepts

CONCEPT MADE SIMPLE

Think of a bank account. A **nonrenewable** source is a savings account with a fixed balance: each withdrawal leaves less, and nothing refills it in your lifetime. A **renewable** source is a paycheck: new money arrives continuously, but you can only spend what arrives each period.

Renewable and nonrenewable sources

Nonrenewable energy sources exist in a fixed amount and involve an energy transformation that cannot easily be reversed. Coal, crude oil and natural gas formed from organisms buried for millions of years. Uranium-235, the fuel for nuclear plants, is mined from ore that formed with the Earth. **Renewable energy sources** are replenished naturally at or near the rate we consume them, and they can be reused: sunlight, wind, flowing water, Earth's internal heat and fast-growing biomass. Wood is renewable only if trees regrow as fast as they are cut.

Source	Category	Why
Coal, oil, natural gas	Nonrenewable	Formed over millions of years; used far faster than formed
Uranium-235	Nonrenewable	Fixed supply of ore
Sunlight, wind, moving water	Renewable	Driven continuously by the Sun and gravity
Geothermal heat	Renewable	Heat flows continuously from Earth's interior (if water use is sustainable)
Biomass	Renewable if regrown	Plants regrow within years to decades

KEY DEFINITIONS

Nonrenewable energy source — exists in a fixed amount and cannot be replaced at the rate it is used.

Renewable energy source — replenished naturally at or near the rate of consumption and can be reused.

Fossil fuel — fuel formed from the buried remains of ancient organisms: coal, crude oil and natural gas.

Forms of energy and how they change

Energy is the ability to do work or transfer heat. It comes in several forms, and every energy system is a chain of conversions between them.

Form	Meaning	Example
Kinetic	Energy of motion	Moving air (wind), a car
Potential	Stored by position	Water behind a dam
Chemical	Stored in bonds	Gasoline, coal, food
Thermal	Random motion of particles (heat)	Hot water, a boiler
Radiant	Electromagnetic waves	Sunlight
Electrical	Moving charges	Current in a wire
Nuclear	Stored in atomic nuclei	Uranium-235

A coal plant converts chemical → thermal → kinetic (steam, turbine) → electrical energy. A car engine converts chemical → thermal → kinetic.

The laws of thermodynamics

- **First law (conservation of energy):** energy cannot be created or destroyed, only changed from one form to another. In every device, energy in = useful energy out + waste energy out.
- **Second law:** every conversion turns some energy into low-quality, dispersed heat that cannot do the same work again, so no machine is 100% efficient. This is why a thermal power plant loses about two-thirds of its fuel energy as waste heat, and why using the highest-quality energy (electricity) for low-quality jobs (heating water) is wasteful.

Power, energy, units and efficiency

Do not mix up **power** and **energy**. Power is the *rate* of using or producing energy; energy is power multiplied by time. A 1 kW heater running for 1 hour uses 1 kilowatt-hour (kWh).

FORMULA & MATH SKILL

Energy = Power × Time (kWh = kW × h; J = W × s). If a question gives kW and hours, the answer is kWh. If it gives kWh and hours, divide to get kW.

Efficiency (%) = (useful energy output ÷ total energy input) × 100. Efficiency can never exceed 100%.

Chains multiply: a plant at 35% sending power through a line at 90% delivers 0.35 × 0.90 = 31.5% of the fuel energy.

Electricity cost = energy (kWh) × price ($/kWh)

Conversion	Value
1 kWh	3.6×10^6 J = 3.6 MJ ≈ 3,412 BTU
1 BTU	≈ 1,055 J
1 W	1 J/s; 1 kW = 1,000 W; 1 MW = 10^6 W
1 year	8,760 hours
Prefixes	k = 10^3, M = 10^6, G = 10^9, T = 10^{12}, P = 10^{15}, E = 10^{18}
1 gallon of gasoline	≈ 121 MJ ≈ 33.7 kWh

Write every conversion as a fraction equal to 1 and cancel units.

COMMON STUDENT MISTAKES

- "Renewable means clean." Burning biomass releases particulates and nitrogen oxides, and a large dam floods habitat. Renewable describes supply, not pollution.
- "Nuclear is renewable because it makes no air pollution." Uranium is finite, so nuclear is **nonrenewable**.
- Mixing kW (a rate) with kWh (an amount). A utility bill charges for kWh.

AP EXAM STRATEGY

When a question asks you to "identify" or "distinguish," state the test explicitly: *is it replenished at or near the rate of use?*

CLASSROOM EXAMPLE SG-U6-E001

A student sorts five energy sources into two columns: wind, uranium-235, natural gas, sunlight, and coal. (a) List which belong in the nonrenewable column. (b) Using the definition of nonrenewable, explain in one sentence why natural gas is placed there even though it burns more cleanly than coal. (c) One of the nonrenewable sources releases no air pollutants during operation. Identify it.

CLASSROOM EXAMPLE SG-C8-E001

A 1,500 W electric space heater runs 6.0 hours per day for 30 days. Electricity costs $0.14 per kWh. (a) Calculate the energy used, in kWh. (b) Express that energy in megajoules and calculate the cost. (c) The electricity comes from a power plant that is 35% efficient. Calculate the fuel energy, in kWh, the plant must burn to supply it.

QUICK CONCEPT SUMMARY

- Nonrenewable: fixed amount, not replaced on human timescales. Renewable: replenished at or near the rate of use. Neither word means "clean" or "dirty."
- Energy changes form but is conserved (first law); every conversion loses some energy as waste heat (second law).
- Energy = power × time; efficiency = useful output ÷ input; efficiencies in a chain multiply.

8.2 Energy Consumption (Past)

CONCEPT MADE SIMPLE

Energy use follows wealth and industry. A country with factories, cars and air conditioning uses far more energy per person than a country where most people cook with wood. Over the last two centuries the world moved from burning wood to burning coal, then oil and gas, and total demand exploded.

From wood to coal to oil

For most of history people relied on **muscle power, firewood, and water or wind** for mills and sails. Wood is bulky, and growing cities stripped nearby forests.

- **1700s–1800s, the Industrial Revolution.** Coal packs more energy per kilogram than wood, and it fed the **steam engine**, which powered pumps, factories, locomotives and ships. Coal also smelted iron into steel. Wood gave way to coal first in Britain, then worldwide.
- **Late 1800s.** The first commercial oil wells (1859, Pennsylvania), the internal combustion engine and the electric generator created demand for liquid fuels and for electricity, made mostly from coal.
- **1900s.** Oil overtook coal as cars, trucks and aircraft multiplied; natural gas pipelines carried a cleaner fuel to homes and industry. **Nuclear power** began in the 1950s and hydroelectric dams spread.
- **Since 2000.** Wind and solar have grown fastest, yet fossil fuels still supply most energy.

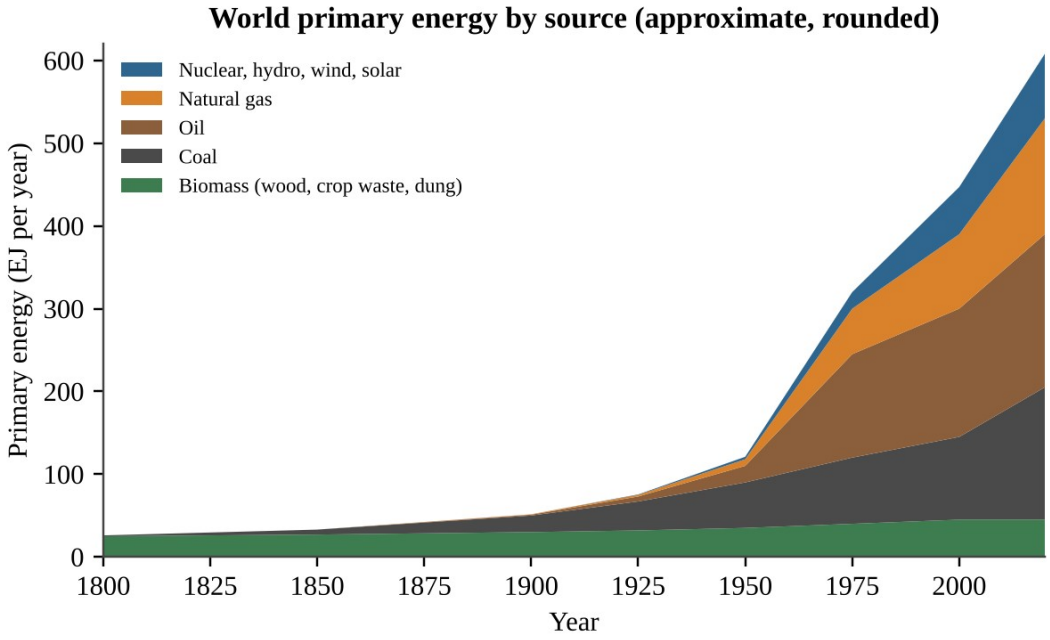


Figure 8.1 — World primary energy by source, 1800–2020 (approximate, rounded values). Total demand grew about twelvefold in 120 years while fossil fuels replaced biomass as the largest supply.

Reading Figure 8.1. Biomass was nearly the only large source in 1800. Each new source *added to* the old ones rather than replacing them: coal use is higher today than in 1950.

Who uses energy, and from what?

Use of energy is **not evenly distributed**. Developed countries have a small share of the world's people but a large share of its energy use, so their use **per capita** is high. Developing countries use less per person, but they hold most of the world's people and their total demand grows fastest. Worldwide, the **most widely used sources are fossil fuels**: oil, coal and natural gas.

Trends and their causes

- **Industrialization raises demand.** Factories, mining, transport, cement and steel need energy, and rising incomes bring more electricity, vehicles and appliances.
- **Development increases fossil-fuel reliance.** Developing countries often start with traditional biomass (firewood, charcoal, dung) because it is nearby and free. As incomes rise they shift to coal, oil and gas, which are concentrated, transportable and often cheap, so fossil fuel use rises before cleaner sources catch up.
- **Availability, price and government regulation shape choices.** A country with abundant coal tends to burn coal. A rise in oil price pushes users toward alternatives. Subsidies, carbon taxes, emission standards and renewable-energy mandates change which source is cheapest or required.

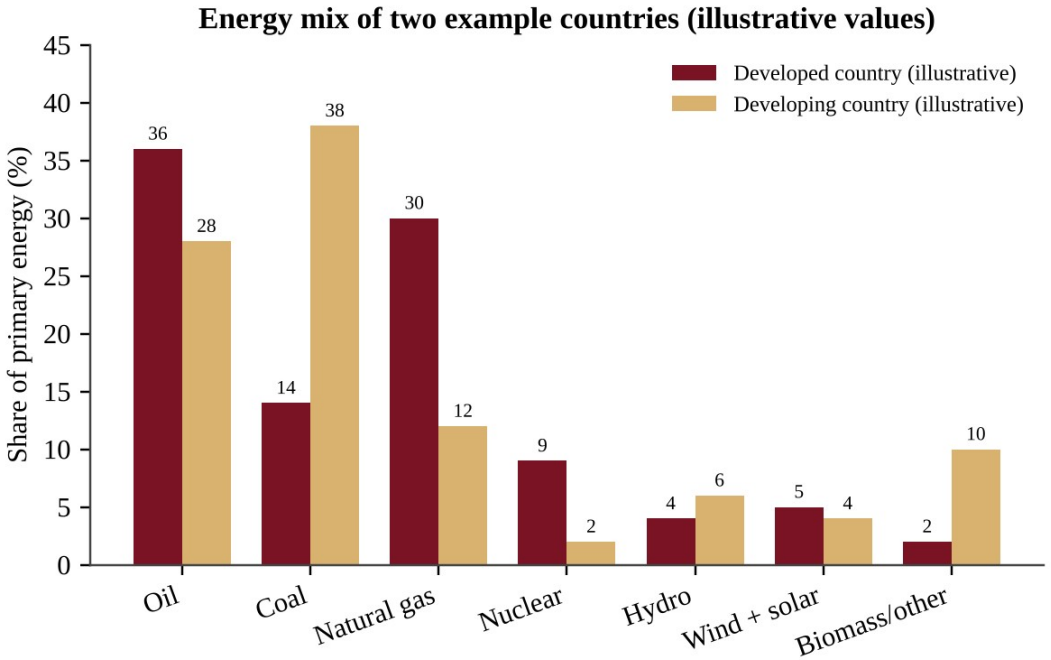


Figure 8.2 — Energy mix of two example countries (illustrative values). Each country's shares total 100%.

Reading Figure 8.2. The developing country gets a much larger share from coal (38% versus 14%) and from biomass (10% versus 2%), while the developed country relies more on oil and natural gas and has more nuclear. Both are dominated by fossil fuels: oil + coal + gas = 80% and 78%. State the pattern, give one numerical comparison, and link it to a cause such as local resources or stage of development.

FORMULA & MATH SKILL

Energy per capita = total energy used ÷ population

Percent share = (energy from one source ÷ total energy) × 100

Percent change = (new − old) ÷ old × 100

Total energy use = energy per capita × population

Doubling time (years) = 70 ÷ annual growth rate (%) (rule of 70)

Percent of capacity used = demand ÷ plant capacity × 100

Demand after t years at a steady rate r (decimal) = present demand × $(1 + r)^t$

Units: J, MJ (10^6 J), GJ (10^9 J), EJ (10^{18} J) or kWh. Per capita is reported as GJ or kWh per person per year.

REAL-WORLD EXAMPLE

Worked example. A country of 50 million people uses 4.0×10^{18} J (4.0 EJ) in a year, and 3.0 EJ comes from fossil fuels.

Per capita: 4.0×10^{18} J ÷ 5.0×10^{7} people = 8.0×10^{10} J = 80 GJ per person. Fossil share: 3.0 ÷ 4.0 × 100 = 75%. If next year's use is 4.4 EJ, percent change = (4.4 − 4.0) ÷ 4.0 × 100 = 10% increase.

CLASSROOM EXAMPLE SG-U6-E002

Country A has 300 million people and uses 90 EJ of energy in a year. Country B has 1,500 million people and uses 45 EJ. (a) Calculate each country's energy use per person in GJ. (b) Calculate how many times greater Country A's per-person use is than Country B's. (c) Country B gets 60% of its energy from coal. Calculate the energy from coal in EJ. (d) Describe one reason Country B's total use is likely to rise over the next 20 years.

Total demand and doubling time

Total demand depends on population and use per person, so a country with low use per person but a huge population can still have very large total use and CO_2 emissions. The rule of 70 works for any steady percentage growth: at 3.5% per year, demand doubles in 70 ÷ 3.5 = 20 years. For a table of towns, find each town's demand (people × kW per person), add them, and compare with plant capacity.

On a graph of **shares**, a source can fall in share while its **amount** still rises, so check the axis. State only causes the data support: a falling coal share beside a rising gas share shows competition, not by itself a cause such as price.

CLASSROOM EXAMPLE SG-U6-E013

A city of 400,000 people uses 3.0 kW per person and a town of 90,000 uses 5.0 kW per person; one 2,000 MW plant serves both. (a) Calculate the percent of capacity used. (b) Demand grows 3.5% per year; use the rule of 70 to estimate the doubling time. (c) Explain why the plant will be overloaded before that time has passed.

CLASSROOM EXAMPLE SG-C8-E002

World primary energy was about 50 EJ in 1900, of which coal supplied 20 EJ. In 2020 the total was about 600 EJ, of which coal supplied 160 EJ. (a) Calculate coal's percent share of the total in each year. (b) Calculate the percent change in coal energy and in total energy between 1900 and 2020. (c) Explain how coal's share can fall while the amount of coal used rises.

COMMON STUDENT MISTAKES

- Comparing countries by total energy instead of per capita. Always use "per capita" or "percent" when countries differ in size.
- Reading a falling *share* as a falling *amount*.

QUICK CONCEPT SUMMARY

- Wood gave way to coal in the Industrial Revolution, then oil and gas in the 1900s; each new source added to, rather than erased, the old ones.
- Energy use is uneven: higher per person in developed countries; demand grows fastest where industrialization is happening.
- Fossil fuels are the most widely used sources; availability, price and regulation influence choices.

8.3 Energy Use (Present)

CONCEPT MADE SIMPLE

Fuels are "energy in a package." The more energy per kilogram (and the less water and ash), the more useful the fuel. Wood is a loose package; gasoline is a compact one. Fossil fuels still supply roughly four-fifths of the world's primary energy.

Fuels you must know

- **Wood.** Used as firewood and as **charcoal** (wood heated with little oxygen so it is mostly carbon). Common in developing countries because it is accessible and often free. Heavy use causes deforestation and indoor smoke.
- **Peat.** Partially decomposed plant material from waterlogged bogs; the early stage on the way to coal. It burns smokily. Peat bogs store large amounts of carbon, so harvesting them releases CO_2.
- **Coal.** Formed from buried plant matter. **Heat, pressure and depth of burial** determine the type. **Lignite** (soft, brown, lowest energy content), **bituminous** (the most commonly burned, medium to high energy) and **anthracite** (hard, shiny, highest carbon and energy content). Higher rank means more carbon, more energy per kilogram and less moisture.
- **Natural gas.** Mostly **methane (CH_4)**. The **cleanest fossil fuel** per unit of energy: less CO_2 and almost no particulates or SO_2. Leaked methane is itself a strong greenhouse gas.
- **Crude oil and tar sands.** Crude oil is pumped from underground reservoirs. **Tar sands** are a mix of clay, sand, water and a thick hydrocarbon called **bitumen**; oil is recovered by mining or by heating the deposit, which uses much energy and water and disturbs land.
- **Refined fuels.** Crude oil is separated by boiling point into specialized fuels: gasoline for cars, diesel for trucks and trains, kerosene for jet fuel, heating oil and propane.

KEY DEFINITIONS

Cogeneration — using one fuel source to produce both useful heat and electricity.

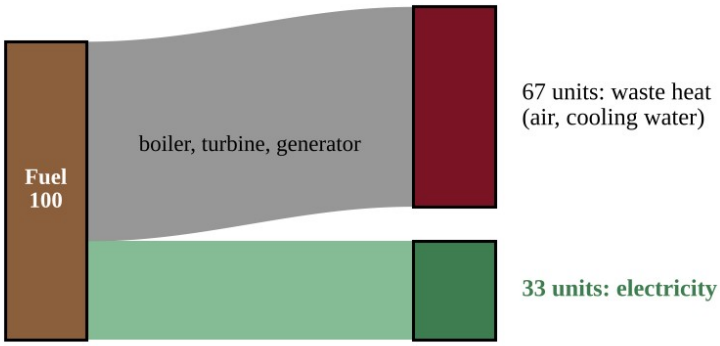
Bitumen — thick, tar-like hydrocarbon in tar sands.

Charcoal — fuel made by heating wood with little oxygen.

Cogeneration

Making electricity in a thermal plant wastes about two-thirds of the fuel's energy as heat. **Cogeneration** (combined heat and power) captures that heat for buildings or industrial steam. Total efficiency rises, so less fuel is burned and less pollution released for the same useful output.

Energy flow in a coal-fired plant (illustrative values)



Energy is conserved: 33 + 67 = 100. Efficiency = 33 ÷ 100 = 33%.

Figure 8.3 — Energy flow in a coal plant (illustrative values). The 67 units of waste heat are the energy cogeneration tries to put to use.

FORMULA & MATH SKILL

Efficiency (%) = (useful energy output ÷ total energy input) × 100

Cogeneration efficiency (%) = (electrical output + useful heat output) ÷ fuel energy input × 100

Input = useful output + waste energy. Use the same unit (J, kWh or BTU) for every term.

CLASSROOM EXAMPLE SG-U6-E003

A cogeneration plant burns fuel containing 200 MJ of energy. It produces 60 MJ of electricity and delivers 90 MJ of useful heat to nearby buildings. (a) Calculate the plant's overall efficiency. (b) Calculate the energy that is wasted, in MJ. (c) Explain why a plant that produced only the 60 MJ of electricity would need more fuel to supply the same electricity plus the same heating.

Where energy resources are found

The global distribution of coal, crude oil, natural gas and ores is **not uniform**. It depends on each region's **geologic history**: what environments existed millions of years ago, how sediments were buried, and how rock layers were folded and sealed.

- **Coal** formed in warm, wet, swampy lowlands where large amounts of plant matter were buried (parts of North America, China, India, Russia, Australia and Europe).
- **Oil and natural gas** formed from marine microorganisms buried in sedimentary basins, then trapped under impermeable rock. The Middle East, parts of Russia, North America and offshore basins are rich in oil and gas; many countries have none.
- **Uranium ores** occur where geological processes concentrated the mineral (Kazakhstan, Canada, Australia).
- **Renewable resources are uneven too.** Geothermal heat is best near plate boundaries and hot spots (Iceland, the Pacific Ring of Fire); sunlight is strongest in low-latitude, cloud-free regions; wind is steadier on coasts, plains and mountain passes.

Countries without resources must import them; resource-rich countries may gain income, but extraction can bring conflict or damage. To compare regions on a graph, divide a region's share of world reserves by its share of world consumption: above 1 marks a likely exporter, below 1 an importer. A useful measure of supply is the **reserves-to-production (R/P) ratio**.

FORMULA & MATH SKILL

R/P ratio (years) = proven reserves ÷ annual production

Reserves and production must use the same unit (for example billion barrels and billion barrels per year). The result assumes production stays constant and no new reserves are found.

REAL-WORLD EXAMPLE

A region has 150 billion barrels of proven oil reserves and produces 3.0 billion barrels per year. R/P = 150 ÷ 3.0 = 50 years. If production rises to 5.0 billion barrels per year, the same reserves last 30 years. Faster use shortens the life of a fixed resource. R/P is not a prediction of "the day oil runs out": reserves change with technology and price.

CLASSROOM EXAMPLE SG-U6-E004

A country has proven natural gas reserves of 240 billion cubic meters (bcm) and produces 4.0 bcm per year. (a) Calculate the R/P ratio in years. (b) Production rises by 25%. Calculate the new annual production and the new R/P ratio. (c) Explain, using geologic history, why a country with no ancient sedimentary basins is unlikely to have natural gas reserves.

Fossil fuels: combustion and power generation

A fossil-fuel power plant is a giant kettle. **Combustion** is a chemical reaction between a fuel and oxygen that releases energy. For methane:

$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ + energy

Impurities and air make burning fossil fuels also produce SO_2, NO_x, particulates and, when oxygen is limited, carbon monoxide (CO).

PROCESS & CAUSE-AND-EFFECT

Fuel burned in boiler → heat boils water → high-pressure steam → steam spins turbine → turbine spins generator → electricity → steam cooled in condenser and returned as water (waste heat released to air or cooling water)

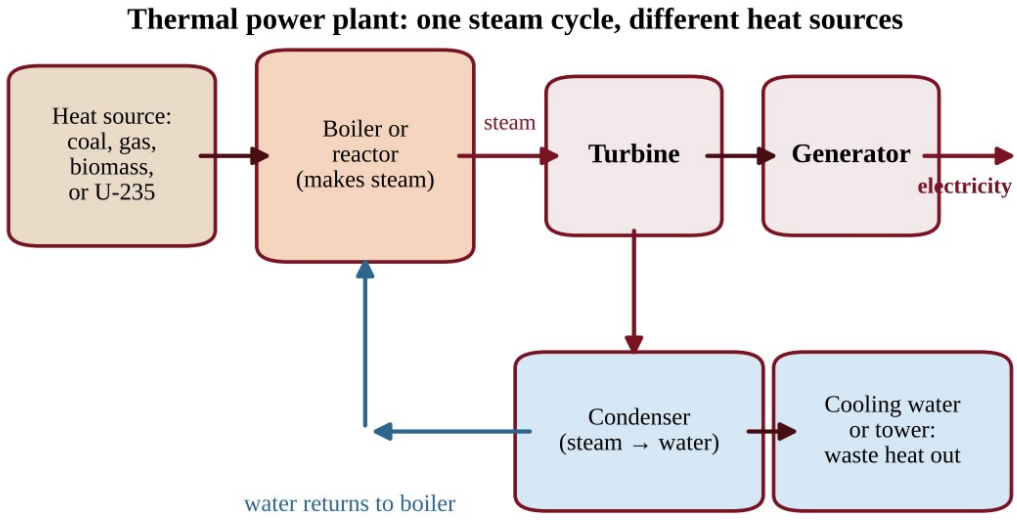


Figure 8.4 — The steam cycle used by coal, gas, biomass and nuclear plants. Only the heat source differs.

Natural gas plants can also use a **combined cycle**: hot exhaust gas first spins a gas turbine, then heats water to run a steam turbine, giving higher efficiency than a coal plant.

Extraction methods

- **Surface (strip) mining** removes soil and rock above shallow coal. It is cheaper and safer for workers but destroys habitat, erodes soil and can send acidic, metal-rich drainage into streams. *Mountaintop removal* is an extreme form.
- **Subsurface mining** uses shafts and tunnels for deep coal. It is more expensive and dangerous (collapses, gas explosions, lung disease) and can cause land subsidence and acid mine drainage.
- **Drilling** pumps oil and gas from wells on land or offshore; spills, leaks and blowouts are risks.
- **Hydraulic fracturing (fracking)** injects water, sand and chemicals at high pressure to crack tight rock and free oil or gas; the sand **props the cracks open**. It can **contaminate groundwater** if wells or wastewater storage fail, release **volatile organic compounds (VOCs)** and methane to the air, use large volumes of water, and wastewater injected deep underground can trigger small earthquakes.
- **Tar sand extraction** uses mining or heating of the deposit and is energy- and water-intensive.

Environmental effects of fossil fuels

- **Climate:** combustion releases CO_2; natural gas leaks release methane.
- **Air quality:** SO_2 and NO_x cause acid deposition and smog; particulates harm lungs; coal burning releases mercury.
- **Water:** thermal pollution from cooling water, oil spills, mine drainage.
- **Land:** mining scars and habitat loss.

Societies keep using fossil fuels because they are energy dense, easy to store and transport, and supported by existing infrastructure; gas drilling also brings jobs, lease income and domestic supply. A utility bill leaves out the **external costs** (illness, lost workdays, ecosystem damage) that society pays. Coal pollution is cut by **scrubbers** (remove SO_2), filters and precipitators (capture particulates), low-sulfur coal and fuel switching. Natural gas is called a **bridge fuel** because it emits less CO_2 and almost no SO_2 per unit of energy than coal, though methane leaks limit the benefit.

KEY DEFINITIONS

VOCs (volatile organic compounds) — carbon-based chemicals that evaporate easily and can form smog.

CLASSROOM EXAMPLE SG-U6-E005

Methane burns according to $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$. Use atomic masses C = 12, H = 1, O = 16. (a) Calculate the mass of CO_2, in kg, released when 16 kg of methane is burned completely. (b) A gas-fired plant produces electricity with an efficiency of 40%. If it uses 500 MWh of fuel energy in an hour, calculate its electrical output in MWh and the energy released as waste heat. (c) Identify the energy transformation sequence from heat to electricity in the plant.

FORMULA & MATH SKILL

Fuel (thermal) power in = electrical power out ÷ efficiency (decimal)

Waste heat = power in − power out

River warming ΔT = waste heat (kJ/s) ÷ [water flow (kg/s) × 4.18 kJ/(kg·°C)]; 1 MW = 1,000 kJ/s and 1 m³ of water = 1,000 kg

Fuel mass = fuel energy needed ÷ energy content of the fuel (MJ/kg)

CO_2 emitted = energy (kWh) × emission factor (kg CO_2/kWh)

Average grid emission factor = Σ (share of each source × its factor)

CO_2 avoided by switching = energy × (old factor − new factor)

REAL-WORLD EXAMPLE

Worked example. A 300 MW electric plant runs at 40% efficiency on coal containing 25 MJ/kg. Power in = 300 ÷ 0.40 = 750 MW, so waste heat = 750 − 300 = 450 MW. Fuel use = 750 MJ/s ÷ 25 MJ/kg = 30 kg/s = 108 metric tons per hour (30 × 3,600 ÷ 1,000). A grid that is 60% coal (1.0 kg CO_2/kWh), 30% gas (0.45) and 10% wind (0) averages 0.60 × 1.0 + 0.30 × 0.45 + 0 = 0.735 kg CO_2/kWh. 1 GWh = 10^6 kWh; 1 metric ton = 1,000 kg.

CLASSROOM EXAMPLE SG-U6-E014

A coal plant delivers 200 MW of electric power at 32% efficiency; its coal holds 24 MJ/kg and it emits 1.0 kg CO_2 per kWh. (a) Calculate the waste heat in MW. (b) Calculate the coal burned per hour in metric tons. (c) A gas plant emitting 0.45 kg CO_2/kWh replaces it. Calculate the CO_2 avoided in 8,000 operating hours, in metric tons.

COMMON STUDENT MISTAKES

- Saying water is the *source* of energy in a steam plant. Water is only the working fluid that carries heat.
- "Efficiency means energy is destroyed." Waste heat is still energy, just not useful for the task.
- Calling natural gas "pollution-free." It is the cleanest fossil fuel, not a clean one.

AP EXAM STRATEGY

For "describe the environmental effects of X," name a specific pollutant or impact, say where it goes (air, water, land) and what harm it causes. "It pollutes" earns no credit; "SO_2 forms sulfuric acid that acidifies lakes" does. For R/P questions, state the assumption (constant production).

Nuclear power

A nuclear plant is also a giant kettle, but the heat comes from splitting atoms instead of burning anything. Power comes from **fission**: a neutron strikes an atom of **uranium-235 (U-235)**, stored in **fuel rods**. The nucleus splits into smaller nuclei and releases heat plus extra neutrons, which strike other U-235 nuclei, creating a **chain reaction**. The heat boils water and the steam turns a turbine and generator, the same steam cycle as in Figure 8.4.

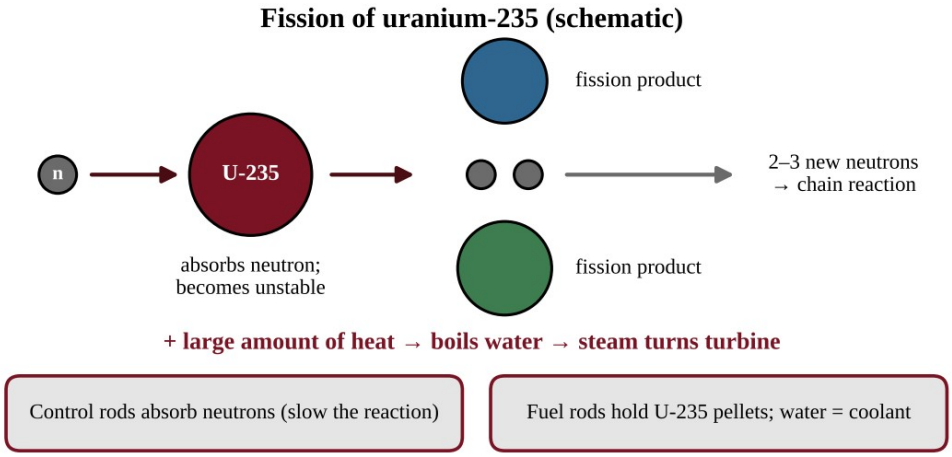


Figure 8.5 — Fission of uranium-235 (schematic, not to scale).

Key parts: **fuel rods** (U-235 pellets), **control rods** (absorb neutrons to slow or stop the reaction), **coolant/moderator** (usually water that removes heat), a **containment structure** (thick concrete and steel) and a cooling system.

Radioactivity and waste. Radioactivity occurs when the nucleus of a radioactive isotope loses energy by emitting radiation. U-235 and some fission products stay radioactive for a very long time, so **waste disposal** is the core challenge: spent fuel is stored first in cooling pools, then in dry casks, and long-term plans call for deep geologic repositories that stay stable for thousands of years. An ideal site has stable rock with no faults or volcanoes, little groundwater, and distance from people. Radiation damages DNA, so exposure can cause mutations and cancer. **Low-level waste** (tools, clothing) is short-lived; **high-level waste** (spent fuel) needs deep geologic storage.

Advantages and disadvantages.

- **Nonrenewable**: uranium is mined and finite. Uranium mining disturbs land and can leave radioactive tailings.
- **No air pollutants or CO_2 from operation**, so it is considered cleaner for air and climate (mining, enrichment and construction still use fossil energy).
- **Thermal pollution**: warm cooling water holds less dissolved oxygen and can exceed organisms' tolerance; intakes can trap fish. Cooling towers reduce the heat released.
- Very high capacity factor (Figure 8.8), but high construction cost.
- Accident risk is low but consequences can be severe.
- Nuclear plants make **electricity only**, so they cannot by themselves replace oil used for transport fuels and petrochemicals.

Accidents.

- **Three Mile Island (USA, 1979):** a partial meltdown from equipment failure and operator error; containment held and little radiation escaped, but regulation changed.
- **Chernobyl (Ukraine, then USSR, 1986):** a reactor explosion and fire released large amounts of radioactive material across Europe. A wide exclusion zone remains, with long-term health effects and contaminated land.
- **Fukushima Daiichi (Japan, 2011):** an earthquake and tsunami knocked out cooling power; reactors overheated and released radiation, requiring evacuation and long cleanup. Radioactive iodine concentrates in the thyroid, so thyroid cancer is a typical health effect.

Half-life math

A radioactive isotope's **half-life** is the time for half of its nuclei to decay. After each half-life the amount remaining is halved, and so is the radioactivity.

FORMULA & MATH SKILL

Remaining amount = initial amount × $(½)^n$, where n = elapsed time ÷ half-life

Elapsed time and half-life must use the same time unit. Example: 800 units, half-life 5 years. After 15 years n = 3, so 800 → 400 → 200 → 100 units.

Radioactive decay: halve the amount for every half-life

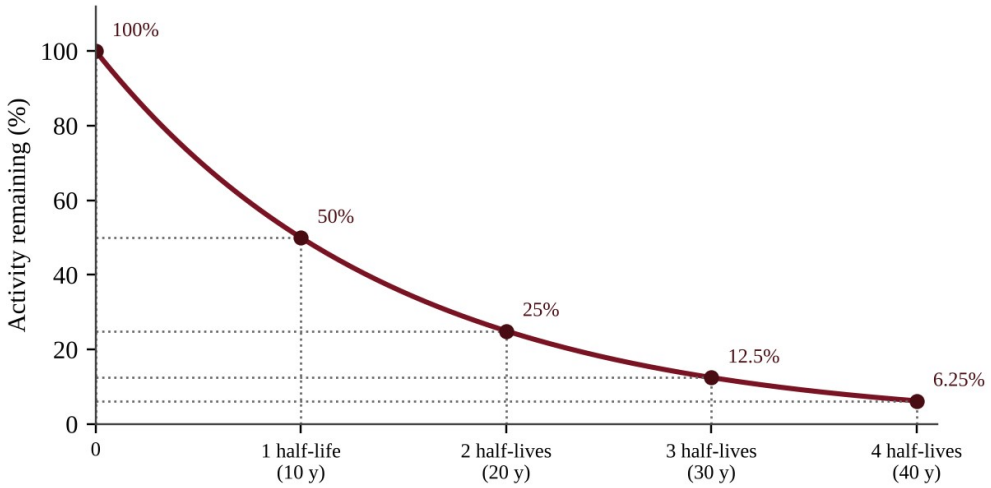


Figure 8.6 — Decay curve for an isotope with a 10-year half-life: every 10 years the amount is cut in half.

Half-life skills. (1) 50%, 25%, 12.5%, 6.25% are 1, 2, 3, 4 half-lives; 1/16 is $(½)^4$. (2) *Find the half-life* as elapsed time ÷ number of halvings, or read the time at which a curve falls to half its start. (3) After about 10 half-lives only about 0.1% is left. The longest-lived isotope controls storage time.

For scale: iodine-131 (about 8 days) is a short-term hazard, cesium-137 (about 30 years) keeps land hazardous for decades, and plutonium and uranium isotopes (thousands to millions of years) are why repositories must work over very long periods.

COMMON STUDENT MISTAKES

- Believing the sample is gone after two half-lives. It is 25% remaining, not zero.
- Thinking "a long half-life means a more intense source." A long half-life decays slowly, so each second is *less* intense but the hazard lasts far longer.

CLASSROOM EXAMPLE SG-U6-E006

A waste sample contains an isotope with a half-life of 8 days. The sample's activity is 640 units today. (a) Calculate its activity after 32 days. (b) Calculate how many days are needed for the activity to fall to 10 units. (c) A second isotope in the same waste has a half-life of 24,000 years. Explain why that isotope, not the first, drives the need for very long-term storage.

CLASSROOM EXAMPLE SG-U6-E015

A strontium-90 sample (half-life 29 years) has an activity of 480 units. (a) Calculate its activity after 87 years. (b) Calculate the time to fall to 6.25% of the start. (c) A different isotope falls from 0.80 mol to 0.10 mol in 21 days. Calculate its half-life.

QUICK CONCEPT SUMMARY

- Fuels: wood and charcoal, peat, coal (lignite < bituminous < anthracite), natural gas (mostly methane, cleanest fossil fuel), crude oil, tar sands, refined fuels. Cogeneration uses waste heat.
- Resource distribution follows geologic history; R/P = reserves ÷ annual production.
- Combustion: fuel + O_2 → CO_2 + H_2O + energy. Heat → steam → turbine → generator. Fracking risks groundwater contamination and VOCs.
- Nuclear: U-235 fission, control rods; no air pollutants but thermal pollution and long-lived waste; Three Mile Island, Chernobyl, Fukushima. Half-life halves the amount per period.

8.4 Renewable Energy Sources

Each renewable source has a different "engine" and a different trade-off. Learn the engine first, then the trade-off.

Biomass

Biomass is organic material from plants and animals used as fuel: wood, crop residues, manure and fuel crops. It is stored sunlight; the carbon released was recently taken from the air, so if plants are regrown the carbon cycle is roughly balanced. It is cheap, so it is the main fuel for cooking and heating for billions of people. But combustion produces **CO_2, carbon monoxide, nitrogen oxides, particulates and VOCs**, and indoor smoke causes respiratory disease. **Overharvesting trees for fuel causes deforestation**, which removes habitat, causes soil erosion and removes carbon sinks. Wood pellets give less fossil CO_2 than natural gas but more particulates and nitrogen oxides.

Ethanol is alcohol made by fermenting sugars from corn, sugarcane or other crops. It can replace some gasoline (blends such as E10 and E85). Burning ethanol adds no *additional* carbon to the atmosphere (a crop absorbed it recently), but the **energy return on energy investment (EROEI)** of ethanol is **low**,

because farming (fertilizer, machinery), transport and distillation use energy, often fossil. Fuel crops can also compete with food crops for land and water and add fertilizer runoff.

FORMULA & MATH SKILL

EROEI = energy delivered by the fuel ÷ energy invested to obtain it

Net energy = energy delivered − energy invested = (EROEI − 1) × energy invested

EROEI > 1 means a net energy gain; the higher, the more attractive. Both quantities must use the same unit (for example MJ per liter).

Land = fuel volume ÷ yield per hectare. **Water** = land × irrigation per hectare. **Blend emissions** = each component's volume × its life-cycle factor, added.

REAL-WORLD EXAMPLE

A distillery uses 8.0 MJ to grow, harvest and process the corn for each liter of ethanol, and the liter contains 21 MJ. EROEI = 21 ÷ 8.0 = 2.6; net energy = 21 − 8.0 = 13 MJ per liter. A good oil field returning 15 MJ per 1 MJ invested has an EROEI of 15.

CLASSROOM EXAMPLE SG-U6-E007

Producing one liter of a biofuel requires 12 MJ of energy inputs, and the liter contains 24 MJ of usable energy. A gasoline-like crude oil source returns 18 MJ for each 1.0 MJ invested. (a) Calculate the biofuel's EROEI and its net energy per liter. (b) State which source returns more energy per unit invested. (c) Describe two environmental effects of replacing forest with fuel crops.

CLASSROOM EXAMPLE SG-U6-E016

A fleet uses 200,000 L of fuel a year. Petroleum diesel emits 3.2 kg CO_2e per liter over its life cycle and biodiesel 1.1. The fleet changes to 30% biodiesel by volume; the crop yields 1,500 L/ha per year and needs 5,000 m³/ha of irrigation. (a) Calculate the percent reduction in emissions. (b) Calculate the hectares and irrigation water needed. (c) A fuel has an EROEI of 3.0; calculate its net gain per 100 MJ invested.

Solar energy

Solar energy is free sunshine caught in three ways.

- **Photovoltaic (PV) cells** capture light and transform it **directly into electrical energy**. Output falls on cloudy days and is zero at night. PV can be rooftop (small, distributed) or in large solar farms.
- **Active solar** systems use mechanical and electrical equipment (pumps, fans, collectors) to heat a liquid and to **collect and store** the energy, for example solar water heaters with storage tanks. Concentrating plants use mirrors to focus sunlight on a fluid to make steam, and store heat in an insulated tank.

- **Passive solar** systems absorb heat directly from the sun **without mechanical or electric equipment**: large south-facing windows (Northern Hemisphere), thick dark thermal-mass floors and overhangs that block summer sun.

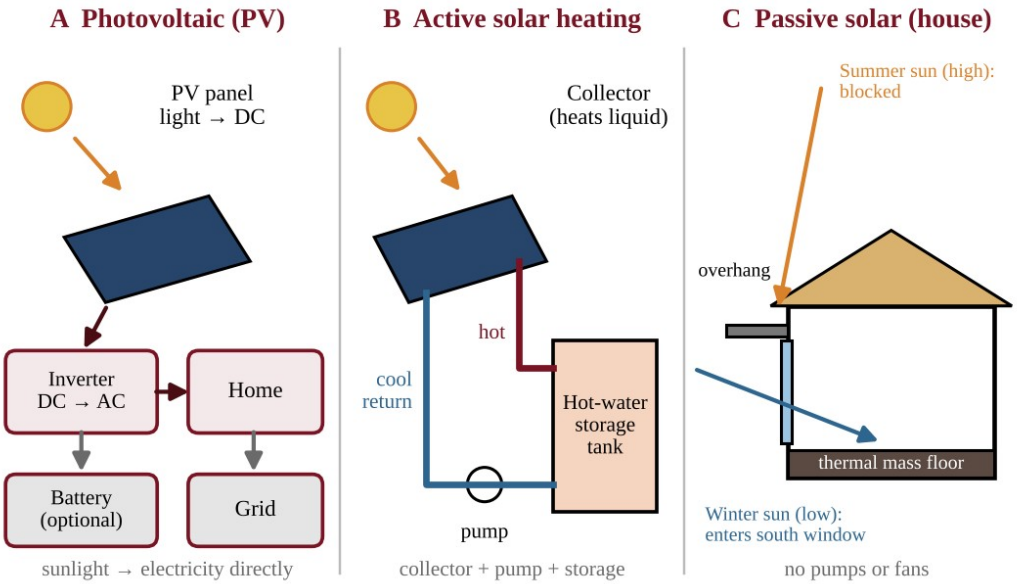


Figure 8.7 — Three kinds of solar systems: (A) a PV system turns light into electricity; (B) an active system pumps liquid through a collector to a storage tank; (C) a passive house uses design alone, with an overhang that admits low winter sun but blocks high summer sun.

A home PV system has **panels** (direct current, DC), an **inverter** (DC to AC) and optionally a **battery**; a grid-connected system sells surplus and buys power at night, so it needs no battery. A stand-alone (off-grid) system needs batteries and a backup, which raises cost. PV output also varies by season: a site with a flatter monthly profile needs less storage or backup. Panels face the equator. PV output peaks near midday but home demand often peaks in the evening.

Environmental effects. Solar has **low environmental impact and produces clean energy** (no air pollution or CO_2 in operation), but equipment can be **expensive**. Manufacturing panels needs mining and energy. Large solar farms can **harm desert ecosystems** by clearing land, disturbing soil and fragmenting habitat of species such as desert tortoises. Rooftop solar avoids this land use.

FORMULA & MATH SKILL

Capacity factor (%) = actual energy produced ÷ (rated capacity × hours) × 100, so energy = capacity × time × CF

Payback period (years) = upfront cost ÷ annual savings

Lifetime savings = annual savings × lifetime − upfront cost

Cost per kWh = total cost ÷ (annual kWh × years of operation)

PV energy (kWh/day) = area (m²) × sunlight (kWh/m²/day) × efficiency (decimal; typically 0.15–0.22)

A plant's **capacity** (kW or MW) is its maximum output; the capacity factor tells you how much it delivers on average. A year has 8,760 hours.

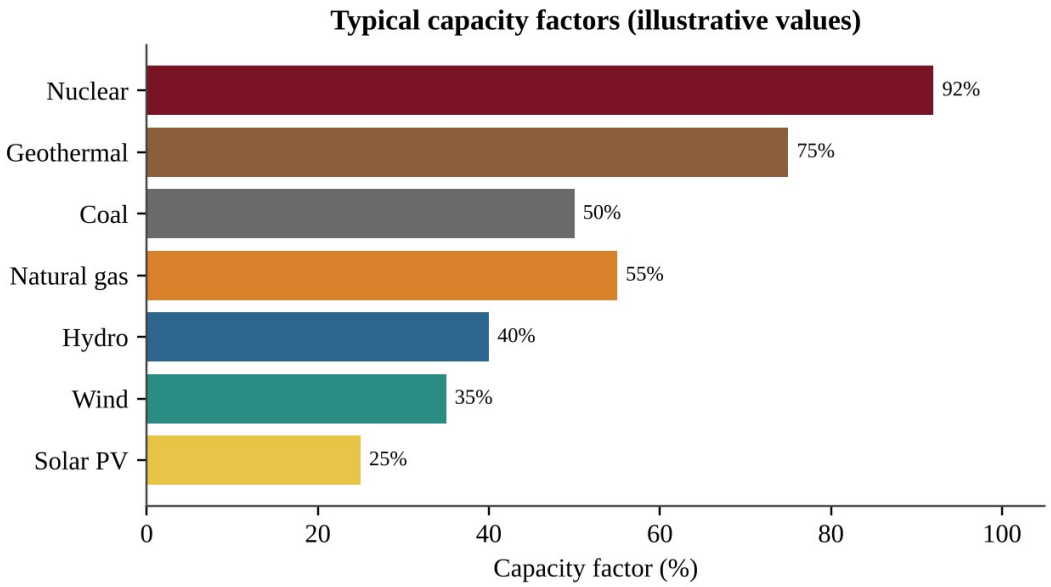


Figure 8.8 — Typical capacity factors of electricity sources (illustrative values). Sun- and wind-dependent sources produce less than their rated maximum on average.

Reading Figure 8.8. A plant that runs around the clock (nuclear, geothermal) has a high capacity factor. Solar PV and wind are lower because the sun sets and the wind varies, not because the equipment is poor. Compare the energy actually produced, not capacity alone.

REAL-WORLD EXAMPLE

Worked example. A rooftop PV array rated 5.0 kW produces 7,300 kWh in a year. Possible output = 5.0 kW × 8,760 h = 43,800 kWh. Capacity factor = 7,300 ÷ 43,800 × 100 = 16.7%. At $0.14 per kWh, savings = 7,300 × 0.14 = $1,022 per year. If the system cost $9,000, payback = 9,000 ÷ 1,022 ≈ 8.8 years.

AP EXAM STRATEGY

For payback questions compute annual savings first, then divide. Note that payback ignores panel degradation and maintenance, and that after payback the energy is essentially free for the rest of the system's life.

CLASSROOM EXAMPLE SG-U6-E008

A 6.0 kW solar PV system installed for $12,000 produces 10,000 kWh in one year. Electricity costs $0.15 per kWh. (a) Calculate the system's capacity factor for the year. (b) Calculate the annual savings and the payback period in years. (c) Express the year's production in megajoules (1 kWh = 3.6 MJ). (d) Distinguish passive solar heating from active solar heating in one sentence.

CLASSROOM EXAMPLE SG-U6-E017

A school installs 800 m² of PV panels at 19% efficiency on a site averaging 4.8 kWh/m² of sunlight per day. The school uses 400,000 kWh per year at $0.13 per kWh, and the grid emits 0.45 kg CO_2 per kWh. (a) Calculate the array's annual output. (b) Calculate the percent of use supplied, the annual savings and the CO_2 avoided in metric tons. (c) Describe one mismatch between PV output and school demand, and propose a response.

Hydroelectric power

Moving water pushes a turbine. A dam stores water high up, like a battery that can be released on demand.

- **Dam and reservoir.** A dam across a river collects water in a **reservoir**. Released water flows through the dam, **spins a turbine** and then a generator: potential energy becomes kinetic energy, then electricity. Operators can release more water when demand is high.
- **Run-of-river. Turbines placed in small rivers** use the flowing water without a large reservoir. They flood much less land but produce less power and depend on river flow.
- **Tidal energy.** Turbines turned by the in-and-out movement of tides. Predictable, but suitable sites are limited and equipment must survive harsh marine conditions. Wave converters are another ocean source.
- **Pumped storage.** Surplus electricity pumps water uphill; later it runs back down through turbines. It stores energy rather than producing it.

Hydroelectric dam: potential → kinetic → electrical energy

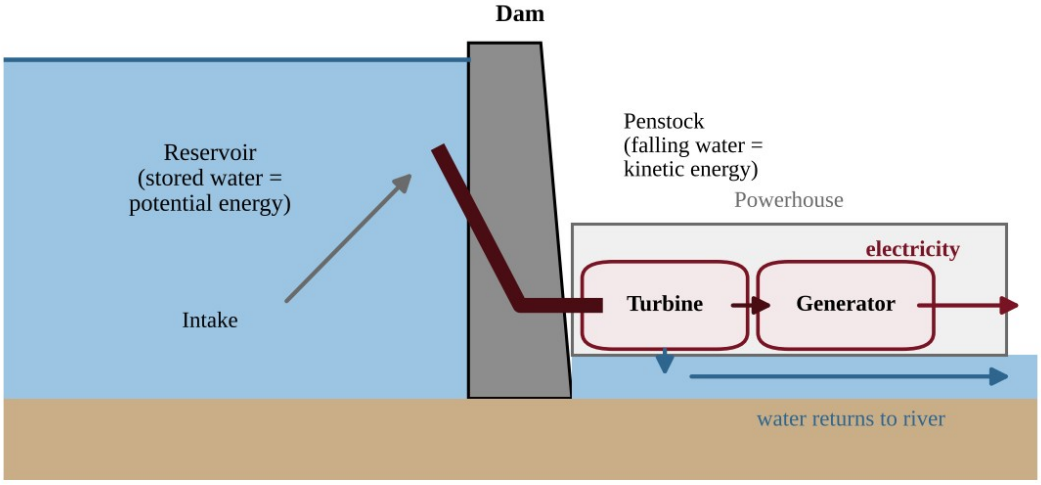


Figure 8.9 — A hydroelectric dam. Stored water (potential energy) falls through the penstock (kinetic energy) and spins a turbine connected to a generator.

Advantages (use these to **justify** hydropower): no air pollution in operation; renewable, driven by the water cycle; low operating cost and long life; flood control, irrigation and recreation; power on demand. **Disadvantages:** construction is **expensive**; reservoirs flood forests, farms and towns, forcing relocation and causing **loss or change of habitat**; dams block fish migration (such as salmon), trap sediment that would nourish downstream floodplains and deltas, and decaying plants in new reservoirs release methane and CO_2. Output falls in droughts. Responses include **fish ladders** and managed flows. **Dam removal** can

restore fish migration, sediment flow and floodplains, but gives up electricity, irrigation water and recreation, and may release stored sediment. A reservoir also loses water by evaporation: volume lost = surface area × depth evaporated, after converting km² to m² (× 10^6).

CLASSROOM EXAMPLE SG-U6-E009

A dam has a rated capacity of 1,200 MW and operates at a capacity factor of 45%. (a) Calculate the electrical energy it generates in a year, in MWh. (b) An average home in the region uses 10,000 kWh per year. Calculate how many homes the dam could supply. (c) Describe two ecological effects of building the dam and identify one advantage that could be used to justify it.

CLASSROOM EXAMPLE SG-U6-E018

A reservoir of 80 km² loses 1.5 m of water to evaporation yearly. A city of 600,000 people uses 120 L per person per day. (a) Calculate the evaporated volume in m³. (b) Calculate the city's annual use in m³ and the evaporation as a percent of it.

Geothermal energy

Geothermal energy uses the **heat stored in Earth's interior**. Wells are drilled to hot rock and water; heated water returns to the surface as steam, which drives a turbine and generator. The cooled water is usually injected back underground. The best sites are near plate boundaries and volcanic areas (Iceland, Kenya's Rift Valley, the western United States).

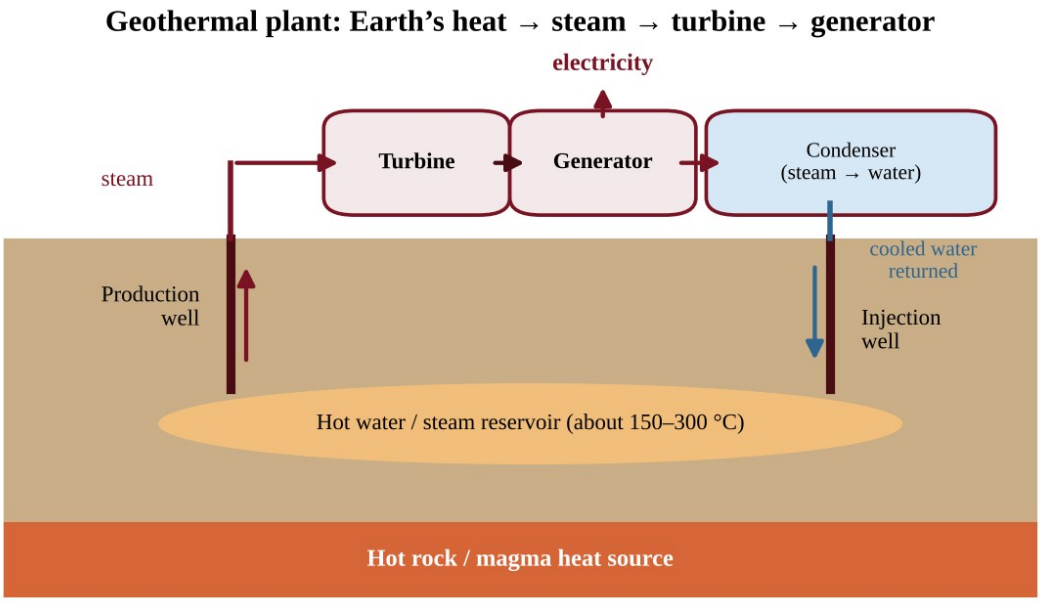


Figure 8.10 — Geothermal power plant. Hot water or steam rises through a production well, drives a turbine, is condensed and returns through an injection well.

Geothermal plants run day and night, giving steady "baseload" power. Designs: **dry steam** (natural steam spins the turbine), **flash** (very hot, high-pressure water turns to steam when pressure drops) and **binary** (hot water heats a second fluid that boils at a lower temperature). **Geothermal heat pumps** use the stable temperature of shallow ground to heat and cool buildings.

Limits and effects. The cost of drilling can be **prohibitively expensive**, and good sites are **not easily accessible in many parts of the world**. Plants can release **hydrogen sulfide (H_2S)**, a toxic gas with a rotten-egg smell. Other issues: corrosive minerals, noise, possible small earthquakes, and local depletion if water is removed faster than it recharges.

Hydrogen fuel cells

A **hydrogen fuel cell** is like a battery that you refill with hydrogen gas. Hydrogen and oxygen go in, electricity and water come out, and no flame is needed:

$2\ H_2 + O_2 \rightarrow 2\ H_2O$ + electrical energy (and some heat)

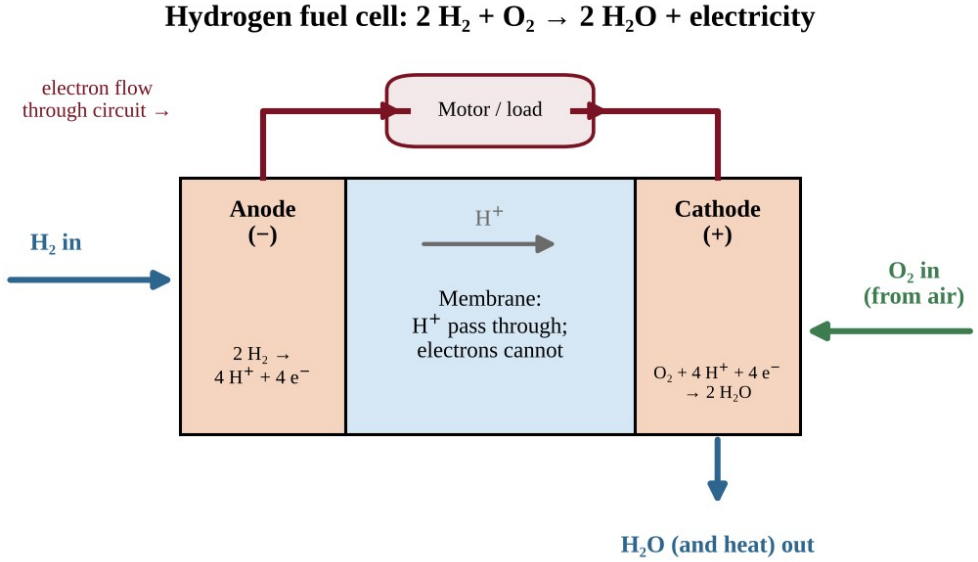


Figure 8.11 — Hydrogen fuel cell. At the anode, hydrogen splits into H^+ and electrons; H^+ crosses the membrane while electrons flow through the external circuit; at the cathode they combine with oxygen to form water.

Water is the only product at the point of use, so a fuel-cell vehicle has no tailpipe CO_2 or air pollutants. **The catch: hydrogen is an energy carrier, not an energy source.** There is almost no free hydrogen on Earth, so it has to be made, and that takes energy. **Electrolysis** (using electricity to split water into H_2 and O_2) emits no CO_2 if the electricity is renewable. Most hydrogen today is made from natural gas (steam reforming) or coal gasification, which release CO_2 at the plant. Other drawbacks: it is **expensive**, hydrogen is hard to store and transport (a very light gas), and refueling networks are limited.

PROCESS & CAUSE-AND-EFFECT

Renewable electricity → splits water (electrolysis) → hydrogen gas stored → fuel cell combines $H_2 + O_2$ → electricity + water → motor or grid

FORMULA & MATH SKILL

Overall efficiency of a chain = $\eta_1 \times \eta_2 \times \ldots$ For electrolysis (η_1) then fuel cell (η_2), overall = $\eta_1 \times \eta_2$ = (useful energy delivered ÷ energy originally put in) × 100.

COMMON STUDENT MISTAKES

"Hydrogen fuel cells are always zero-emission." The cell emits only water, but if the hydrogen comes from natural gas or fossil-fueled electricity, CO_2 is released upstream.

CLASSROOM EXAMPLE SG-U6-E010

Producing 1.0 kg of hydrogen by electrolysis uses 55 kWh of electricity. A fuel cell converts that 1.0 kg of hydrogen back into 20 kWh of electricity. (a) Calculate the overall efficiency of the electricity-to-hydrogen-to-electricity chain. (b) The electricity for electrolysis comes from a coal plant that emits 0.9 kg of CO_2 per kWh. Calculate the CO_2 emitted to produce the 1.0 kg of hydrogen. (c) Explain why this fuel cell is not "zero-emission" in this case, even though its only exhaust is water.

Wind energy

Wind turbines use the **kinetic energy of moving air** to spin blades connected to a shaft and a **generator**. Wind is driven by uneven heating of Earth by the Sun. Farms are placed where winds are strong and steady: plains, ridges, coasts and offshore (stronger winds, higher cost).

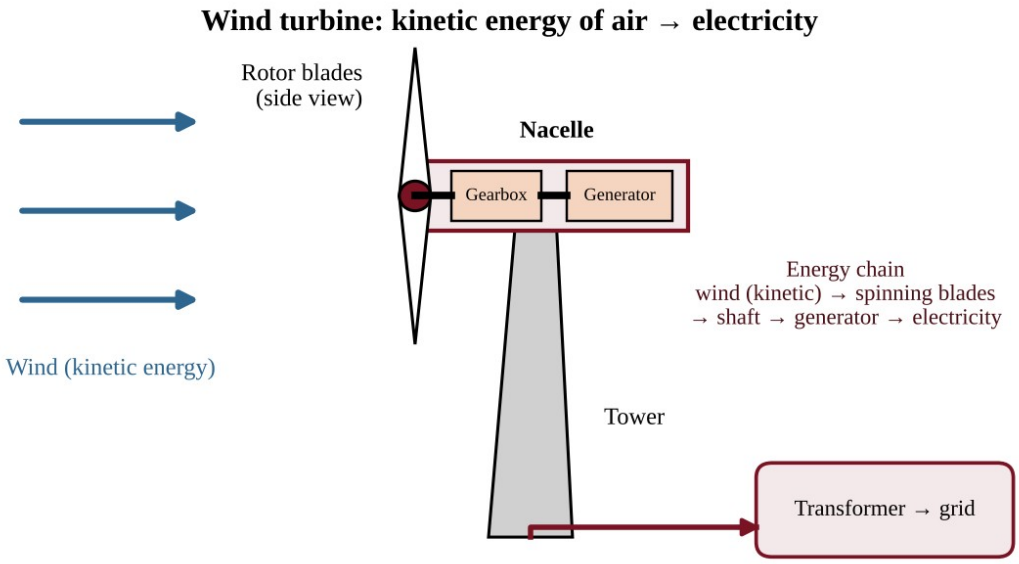


Figure 8.12 — Wind turbine. Moving air turns the blades; the shaft drives a generator in the nacelle; a transformer sends electricity to the grid.

Advantages: renewable and clean (no air pollution or CO_2 in operation), no fuel cost, and land between turbines can still be farmed or grazed. **Disadvantages:** wind is **intermittent**, so backup or storage is needed; birds and bats may be **killed if they fly into spinning blades**; noise and visual impact. Siting away from migration routes and slowing blades at certain times reduce bird and bat deaths.

REAL-WORLD EXAMPLE

Worked example. A wind farm has 20 turbines rated 3.0 MW each, so capacity is 60 MW. At a capacity factor of 35%, annual energy = 60 MW × 8,760 h × 0.35 = 183,960 MWh. If this displaces a plant emitting 0.8 tonnes CO_2 per MWh, avoided CO_2 = 183,960 × 0.8 ≈ 147,000 tonnes per year.

CLASSROOM EXAMPLE SG-U6-E011

A wind farm has 40 turbines, each rated at 2.5 MW, and operates at a capacity factor of 35%. (a) Calculate the farm's annual electricity generation in MWh. (b) The farm replaces electricity from a coal plant that emits 0.9 tonnes of CO_2 per MWh. Calculate the CO_2 emissions avoided per year. (c) Identify one environmental drawback of the wind farm and one action that would reduce it.

The power in wind rises with the **cube** of wind speed. Doubling the speed gives $2^3 = 8$ times the power, and 5.0 m/s versus 7.5 m/s gives $(7.5 \div 5.0)^3 \approx 3.4$ times. A turbine's **power curve** shows what it actually delivers.

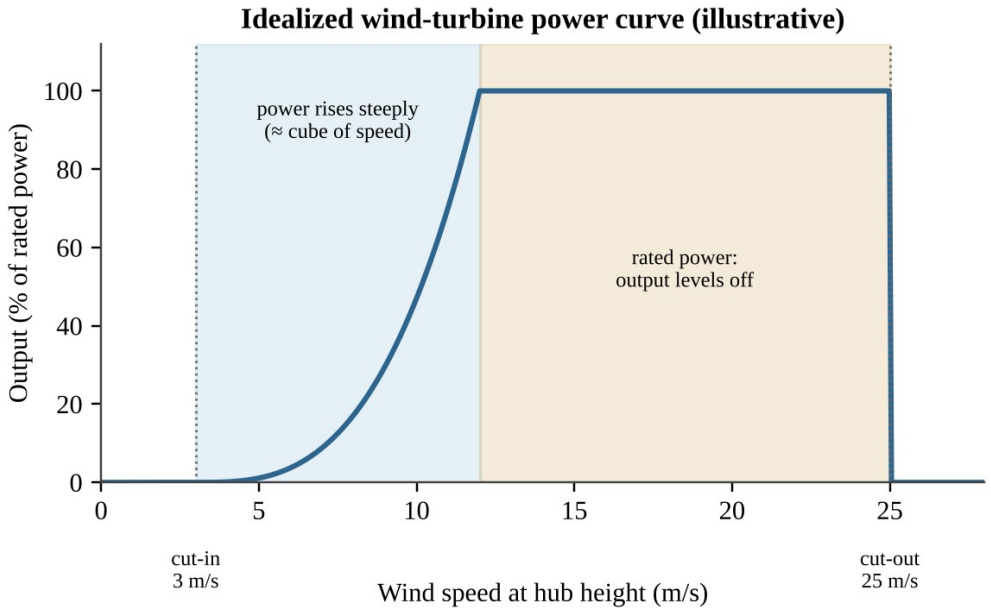


Figure 8.13 — Idealized power curve: no output below cut-in speed, steep rise, a plateau at rated power, then shutdown at cut-out speed to protect the blades.

Reading a curve. Read the output at each speed and divide for a ratio. **Curtailment** means holding blades still at low wind: if bat deaths peak in light winds when little power is made, a higher cut-in speed on migration nights saves bats and costs little electricity.

CLASSROOM EXAMPLE SG-U6-E019

One site averages 6.0 m/s and another 9.0 m/s. (a) Calculate how many times more power the wind carries at the second site. (b) Bat deaths per turbine-night are 0.9 at 3–5 m/s, 0.4 at 5–7 m/s and 0.1 above 7 m/s. Propose an operating rule and justify it with the data. (c) Explain why output stops rising above rated speed.

Comparing the sources

Source	Main advantage	Main environmental drawback
Coal	Cheap, abundant	CO_2, SO_2, particulates, mining damage
Natural gas	Cleanest fossil fuel	CO_2, methane leaks, fracking effects
Oil	Portable liquid fuel	CO_2, spills, air pollutants
Nuclear	No air pollutants, high output	Long-lived waste, thermal pollution, accident risk
Biomass	Low cost, available	Particulates, deforestation, low EROEI (ethanol)
Solar	Clean, low impact	Cost, sunlight-limited, desert land use
Hydro	No air pollution, on demand	Habitat change, expense
Geothermal	Steady output	Site-limited, cost, H_2S
Hydrogen fuel cell	Water-only emission	Expensive, energy needed to make H_2
Wind	Clean, renewable	Intermittent, bird and bat deaths

QUICK CONCEPT SUMMARY

- Biomass: cheap heat but pollutants and deforestation; ethanol has low EROEI. Solar: PV, active and passive; clean but costly, desert land use. Hydro: dams, run-of-river, tidal; habitat loss, no air pollution.
- Geothermal: steady but site-limited, with H_2S. Hydrogen fuel cell: water-only emission, but energy is needed to make H_2. Wind: clean but intermittent; bird and bat deaths (propose siting, curtailment or storage).

8.5 Energy Conservation

CONCEPT MADE SIMPLE

The cheapest, cleanest unit of energy is the one you never use. Conservation means using less energy for the same job (efficiency) or doing without some of the job (behavior change).

Around the home (ENG-3.T.1)

- **Adjust the thermostat**: lower it in winter and raise it in summer. Heat flow depends on the temperature difference between inside and outside, so a smaller difference means less heating or cooling.
- **Conserve water**, especially hot water: heating water takes much energy, and treating and pumping water uses electricity too.
- **Use energy-efficient appliances** (ENERGY STAR refrigerators, LED bulbs, heat pumps). An LED gives the same light as an incandescent bulb with about one-sixth of the power.
- **Insulation, sealing and efficient windows.** Insulation is rated by **R-value**: higher R-value means slower heat flow.
- **Conservation landscaping**: shade trees on the sun side reduce summer cooling needs, windbreaks cut winter heat loss, and native, drought-tolerant plants reduce watering.

On a larger scale (ENG-3.T.2)

- **Improving fuel economy** of vehicles. In the United States, **CAFE** (Corporate Average Fuel Economy) standards set average targets for automakers' fleets.
- **Hybrid vehicles** and **battery electric vehicles (BEVs)** (see Section 8.6).
- **Public transportation**, carpooling and urban planning that shortens trips (see Section 8.7).
- **Green building design**: orientation to the sun, thick insulation, cool or green roofs, daylighting, efficient heating and cooling, and recycled materials. Rating systems such as LEED certify these features.

Policies that push conservation include rebates, tax credits, fuel taxes, efficiency standards such as CAFE, and public charging stations. An efficiency gain saves energy only if use does not rise to cancel it. Some people drive more after buying a more efficient car (the **rebound effect**), so policies pair efficiency with price signals or standards.

FORMULA & MATH SKILL

Fuel economy = distance ÷ fuel used (mpg, or km per liter); **fuel used** = distance ÷ fuel economy

Annual cost savings = (energy or fuel saved) × price

Payback period (years) = extra upfront cost (after any rebate) ÷ annual savings

Energy saved (kWh) = (power_old − power_new in kW) × hours of use

Insulation: heat loss rate ∝ (temperature difference) ÷ (R-value). Doubling the R-value halves the heat flow.

REAL-WORLD EXAMPLE

Worked example. An incandescent bulb uses 60 W and an LED gives the same light with 9 W. Used 4 hours per day for 365 days: energy saved = (0.060 − 0.009) kW × 4 h × 365 = 74.5 kWh per year. At $0.15/kWh this is about $11.17 per year. If the LED costs $3.00, payback = 3.00 ÷ 11.17 ≈ 0.27 year, about 3 months.

COMMON STUDENT MISTAKES

- Comparing cars by miles per gallon *differences*. Going from 15 to 20 mpg saves more fuel than going from 40 to 45 mpg over the same distance. Convert to gallons used before comparing.
- Forgetting that a BEV is only as clean as the grid that charges it.
- Counting only upfront price. The cheapest appliance may cost more over its lifetime.

AP EXAM STRATEGY

For payback and cost questions, set up a three-line plan: (1) energy or fuel saved per year, (2) money saved per year, (3) upfront cost ÷ yearly savings. Label units at every step.

CLASSROOM EXAMPLE SG-U6-E012

A household drives 12,000 miles per year. Its current car gets 25 mpg. A hybrid gets 40 mpg and costs $4,200 more. Gasoline costs $3.50 per gallon. (a) Calculate the gallons of gasoline used per year by each car. (b) Calculate the annual fuel cost savings with the hybrid. (c) Calculate the payback period for the extra cost, in years. (d) Describe one factor not included in the calculation that could make the real payback longer or shorter.

Reading energy data and designing an investigation

Graphs and tables. (a) Read the title and units, (b) state the largest and smallest values, (c) compare using numbers, (d) give a cause.

Designing an energy investigation (fracking and groundwater, panel direction, bats and wind speed).

- *Hypothesis (if–then):* "If methane is measured farther from the well pad, then concentrations will be lower." The **independent variable** is what you change or compare (distance, panel direction); the **dependent variable** is what you measure (methane in mg/L, kWh).
- *Controls:* keep other factors **constant** (same method, depth, tilt, season). Take a **baseline** before the activity and use an untouched **control site**.
- *Reliability:* take **replicates** at each location and repeat over time.
- *Natural variation:* a change counts only if it exceeds the spread of the baseline readings. A distance graph typically falls, then levels off at the background value.

CLASSROOM EXAMPLE SG-U6-E020

A town asks whether a new wind farm raises noise at nearby homes. (a) State an if–then hypothesis and identify the independent and dependent variables. (b) Describe a design with a baseline, a control location and replicates. (c) Explain how you would decide that an increase exceeds normal variation.

AP EXAM STRATEGY

Command words: *Calculate* — show formula, substitution and units. *Describe* — state facts about the source or effect. *Explain* — give a cause-and-effect reason. *Justify* — support a choice with a reason from the data or science. *Propose a solution* — name an action and say how it helps. *Make a claim* — state a position, then support it with two pieces of evidence from the data.

QUICK CONCEPT SUMMARY

- Home: thermostat, water conservation, efficient appliances, insulation and sealing, conservation landscaping.
- Large scale: better fuel economy (CAFE), hybrids and BEVs, public transit, green building design.
- Conservation is usually the cheapest and cleanest "source" of energy; watch for the rebound effect.

8.6 Transportation

CONCEPT MADE SIMPLE

A car is a machine that turns the chemical energy in fuel into motion, and most of it is lost as heat. Better transportation means getting the same trip done with less energy, and ideally with fuel that comes from cleaner sources.

Transportation burns most of the world's oil, so it is a major source of CO_2, nitrogen oxides, particulates and carbon monoxide. Each gallon of gasoline burned releases about 8.9 kg of CO_2; a gallon of diesel about 10.2 kg.

Fuel efficiency

A gasoline engine converts only about 20–30% of the fuel's energy into motion; the rest leaves as heat in the exhaust and radiator. Efficiency improves with lighter vehicles, smaller engines, better aerodynamics, low-rolling-resistance tires and slower, steadier driving. CAFE standards require automakers to meet a fleet-average **fuel economy**. Because fuel used = distance ÷ fuel economy, consumption is clearer in *gallons per 100 miles* (or liters per 100 km) than in mpg.

Hybrids and electric vehicles

- **Hybrid electric vehicles (HEVs)** combine a gasoline engine, an electric motor and a small battery. **Regenerative braking** turns braking energy into electricity instead of heat, and the engine can shut off at stops. They are typically far more efficient in city driving.
- **Plug-in hybrids (PHEVs)** have a larger battery that can be charged from the grid for short trips on electricity.
- **Battery electric vehicles (BEVs)** use only a battery and motor. An electric drivetrain turns about 85–90% of battery energy into motion and has no tailpipe emissions. Their *total* emissions depend on how the electricity is made: a coal-heavy grid gives much higher emissions per mile than a wind-

or hydro-heavy grid. Battery manufacturing needs mined metals (lithium, cobalt, nickel) and energy. Cars are many dispersed sources of pollution; a power plant is one point source, which is easier to regulate and fit with scrubbers.

- **Hydrogen fuel-cell vehicles** (Section 8.4) emit only water at the tailpipe, but hydrogen must be made and distributed.

Biofuels and other fuels

Ethanol (E10 is 10% ethanol in gasoline; E85 is 85%) is made from fermented sugars; **biodiesel** is made from vegetable oils or animal fats. Their EROEI, land and water needs, and life-cycle emissions were covered in Section 8.4. Switching a fleet to a blend lowers fossil CO_2 only in proportion to the share and the biofuel's life-cycle factor.

Emissions per passenger-kilometer

To compare modes fairly, divide a vehicle's emissions per kilometer by the number of people it carries.

FORMULA & MATH SKILL

Emissions per passenger-km = emissions per vehicle-km ÷ number of passengers

CO_2 per year = distance × fuel used per distance × CO_2 per unit of fuel (gasoline ≈ 8.9 kg CO_2 per gallon)

BEV CO_2 = kWh per mile × miles × grid emission factor (kg CO_2/kWh)

REAL-WORLD EXAMPLE

A car that uses 7.8 L of gasoline per 100 km (about 30 mpg) emits roughly 7.8 × 2.3 kg ÷ 100 ≈ 0.18 kg CO_2 per km. With one occupant that is 180 g per passenger-km; with four occupants, 45 g. Occupancy matters as much as the engine.

COMMON STUDENT MISTAKES

- "Electric cars have no emissions." They have no tailpipe emissions; the power plant and battery factory still emit.
- Comparing vehicles per vehicle-km instead of per passenger-km.

CLASSROOM EXAMPLE SG-C8-E003

A household drives 15,000 miles per year. Car A gets 20 mpg and Car B gets 30 mpg. Gasoline releases 8.9 kg of CO_2 per gallon. A battery electric vehicle (BEV) uses 0.30 kWh per mile and charges from a grid that emits 0.45 kg CO_2 per kWh. (a) Calculate the gallons of gasoline each car uses per year. (b) Calculate the CO_2 avoided per year by driving Car B instead of Car A. (c) Calculate the BEV's CO_2 emissions per year. (d) Calculate the grid emission factor, in kg CO_2 per kWh, at which the BEV would emit the same CO_2 per mile as Car B.

QUICK CONCEPT SUMMARY

- Engines are only 20–30% efficient; hybrids recover braking energy; BEVs are efficient but only as clean as the grid.
- Biofuels lower fossil CO_2 only partly, because farming and processing use fossil energy.
- Emissions per passenger-km = emissions per vehicle-km ÷ passengers.

8.7 Mass Transit

CONCEPT MADE SIMPLE

One bus carrying 40 people takes the place of 30 or more cars. Mass transit saves energy because a heavy vehicle's fuel use is shared among many riders, as long as the seats are filled.

Mass transit moves many people together: **buses** (including bus rapid transit, with dedicated lanes), **light rail and subways**, **commuter and intercity rail**, and ferries. Rail runs on steel wheels on steel rails, which have very low rolling resistance, and most trains are electric, so their emissions fall as the grid becomes cleaner.

Ridership and emissions comparison

Energy and emissions per passenger depend on **ridership** (also called load factor): an empty bus is worse than a car, a full bus is far better. Modes with high capacity and steady demand (subways, commuter trains) have the lowest emissions per passenger-km.

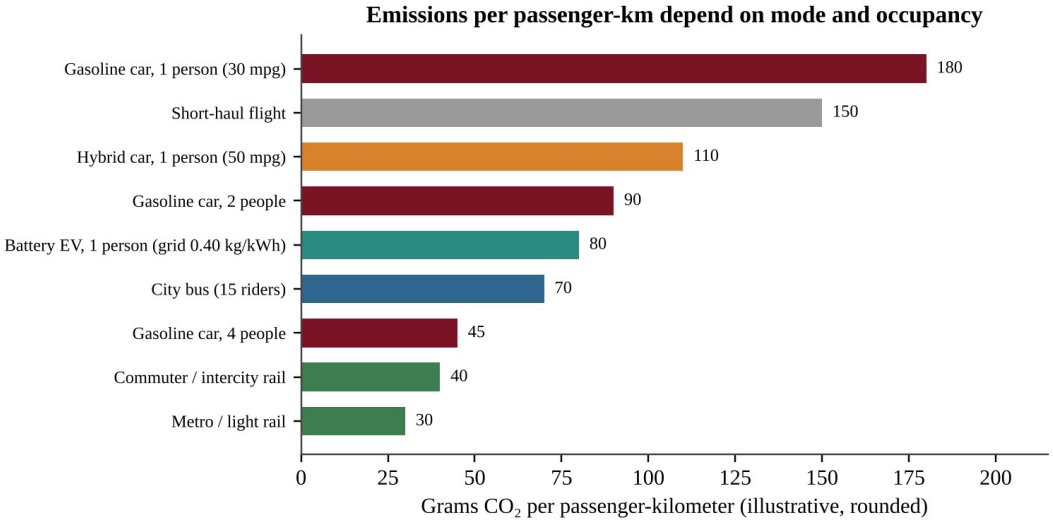


Figure 8.14 — Emissions per passenger-kilometer by mode (illustrative, rounded values). Occupancy changes the same car from the worst to near the best option.

Reading Figure 8.14. The same gasoline car emits 180, 90 or 45 g per passenger-km depending on whether it carries one, two or four people. A bus with only 15 riders already beats a car carrying one person by more than half, and rail is lowest. Short-haul flights emit a large amount per passenger-km and also release emissions high in the atmosphere.

Land use and cities

Transit works best where people live and work near stops. **Dense, mixed-use development** with homes, jobs and shops within walking distance of stations (called transit-oriented development) cuts trip lengths and car ownership. **Urban sprawl**, with low-density housing far from jobs, makes driving almost necessary, paves farmland and habitat and raises energy use per person. Transit also reduces traffic congestion and air pollution, but it needs high upfront cost and enough riders to be worthwhile, and it is less convenient for trips that start and end away from stops. Cycling and walking infrastructure, carpooling and telecommuting reduce energy use further.

KEY DEFINITIONS

Mass transit — shared passenger transport such as buses and trains.

Ridership (load factor) — the average number of passengers carried per vehicle or the share of seats filled.

Transit-oriented development — dense, mixed-use neighborhoods built around transit stations.

AP EXAM STRATEGY

In "compare transportation options" questions, give a number per passenger-km, name the factor that makes it low or high (occupancy, fuel, electricity source) and note one trade-off such as cost, convenience or travel time.

CLASSROOM EXAMPLE SG-C8-E004

A diesel bus emits 1.2 kg of CO_2 per km and carries an average of 20 riders. A car emits 0.20 kg of CO_2 per km and carries an average of 1.25 people. (a) Calculate the emissions per passenger-km of each, in kg. (b) Calculate how many times greater the car's emissions per passenger-km are than the bus's. (c) A commuter travels 25 km each way on 250 days per year. Calculate the CO_2 saved per year by taking the bus instead of the car. (d) Calculate the minimum whole number of riders at which the bus emits no more per passenger-km than the car.

QUICK CONCEPT SUMMARY

- Transit saves energy per passenger because fuel use is shared; it works best with high ridership and dense land use.
- Rail and subway have the lowest emissions per passenger-km; sprawl raises energy use.

8.8 Practice Questions and Answers

Practice is where the chapter's ideas become exam skills, so work through the chapter's examples, then the question sets listed below.

PRACTICE QUESTIONS AND ANSWERS

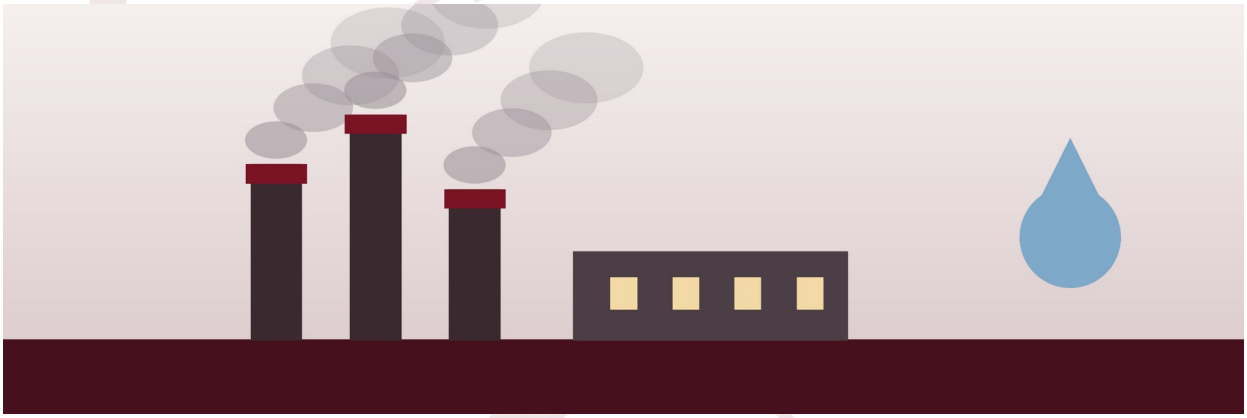
Classroom examples in this chapter (unsolved; space is provided in each box): SG-U6-E001, SG-C8-E001, SG-U6-E002, SG-U6-E013, SG-C8-E002, SG-U6-E003, SG-U6-E004, SG-U6-E005, SG-U6-E014, SG-U6-E006, SG-U6-E015, SG-U6-E007, SG-U6-E016, SG-U6-E008, SG-U6-E017, SG-U6-E009, SG-U6-E018, SG-U6-E010, SG-U6-E011, SG-U6-E019, SG-U6-E012, SG-U6-E020, SG-C8-E003, SG-C8-E004.

Classified practice: Book Two, Chapter 8 — 53 multiple-choice questions (Section A) followed by 4 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 8. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

UNIT VI
POLLUTION



IN THIS UNIT

Chapter 9 Pollution

Chapter 10 Impacts on the Environment and Human Health

CHAPTER
9

UNIT VI — POLLUTION
Pollution

9.1 · Air Pollution
9.2 · Major Air Pollutants
9.3 · Acid Deposition
9.4 · Thermal Pollution
9.5 · Noise Pollution
9.6 · Water Pollution
9.7 · Groundwater Pollution
9.8 · Water Quality—Water Testing
9.9 · Maintaining Water Quality—Water Purification
9.10 · Septic Systems/Sewage Treatment
9.11 · Municipal Solid Waste
9.12 · Practice Questions and Answers

LEARNING OBJECTIVES

- Classify air pollutants as primary or secondary and link each to its sources, effects and control.
- Explain how photochemical smog, thermal inversions and acid deposition form and why some places suffer more.
- Distinguish point from nonpoint sources, and explain oxygen sag, eutrophication, thermal pollution and the services of wetlands.
- Explain why groundwater pollution is slow to appear and slow to clean, and how water is tested and purified.
- Describe sewage treatment, septic systems and the design of landfills, and compare waste-reduction methods.
- Carry out the calculations of the unit: pH and decibel factors, percent change and removal, loads, emissions, BOD, per-capita waste and landfill life.

9.1 Air Pollution

CONCEPT MADE SIMPLE

The air is mostly nitrogen and oxygen. **Air pollution** is anything added in amounts large enough to harm living things or materials. Most of it comes from one habit: burning things. Fuel burns, and the carbon, sulfur and metals in it, plus the nitrogen in the air itself, leave as gases and tiny specks through the chimney or tailpipe.

Why burning makes pollution

A perfect fuel burned with plenty of oxygen would give only CO_2 and water ($CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$). Real burning is not perfect.

- **Impurities in the fuel.** Coal and oil carry sulfur and metals. Sulfur leaves as **SO_2; coal combustion** releases CO_2, SO_2, toxic metals and particulates together. Diesel and other high-sulfur fuels release more SO_2 than low-sulfur fuels.
- **Hot flames.** In engines and furnaces, nitrogen from the air combines with oxygen to form **NO_x** (NO and NO_2).
- **Incomplete burning.** Too little oxygen or too low a temperature leaves **carbon monoxide (CO)**, a colorless, odorless asphyxiant that blocks oxygen delivery by the blood, plus unburned hydrocarbons (VOCs) and soot (particulate matter).

Primary and secondary pollutants

KEY DEFINITIONS

Primary pollutant — a pollutant released straight into the air from its source (CO, SO_2, soot, lead). **Secondary pollutant** — a pollutant that forms in the air when primary pollutants react, often with sunlight or water (ozone, PAN, nitric acid, sulfuric acid, fine sulfate and nitrate particles). **Precursor** — a chemical that reacts to form a secondary pollutant (NO_x and VOCs are precursors of ozone).

A quick test: can you point to the pipe it came out of? If so it is probably primary. NO from an engine is primary; the O_3 it helps create is secondary.

Regulation works: lead and the Clean Air Act

Lead was once added to gasoline to raise octane and stop engine knock, and cars spread it through every city. Under the **Clean Air Act (CAA)** the EPA ordered lead removed from fuel, and atmospheric lead and children's blood-lead levels fell dramatically. The lesson: **regulating a pollutant at its source works.**

The CAA lets the EPA set health-based limits for six common "criteria" pollutants and regulate power plants, factories and vehicles. Its 1990 amendments capped total SO_2 emissions from power plants to fight acid rain. CO_2 is not a criteria pollutant.

Trend graphs often use an **index**: the starting year is set to 100 and each later value is a percentage of it. An index of 40 means 40% of the starting level, a 60% decrease. Falling lines *support* the effectiveness of a law but do not prove it, because other changes (cleaner engines, fuel switching) happened too. A pollutant the law does not target, such as atmospheric CO_2, can keep rising.

CLASSROOM EXAMPLE SG-U7-E010

An index graph (1990 = 100) shows these values for 2020: SO_2 emissions 35, NO_x emissions 52, lead emissions 6, atmospheric CO_2 concentration 118.

(a) Calculate the percent decrease in SO_2 emissions.

(b) Identify the pollutant that fell the most and the one that rose.

(c) Explain why the graph alone does not prove regulation caused the SO_2 decline.

CLASSROOM EXAMPLE SG-U7-E001

Five air pollutants are listed: (1) carbon monoxide from a car exhaust pipe, (2) ground-level ozone measured on a hot afternoon, (3) sulfur dioxide from a coal plant stack, (4) nitric acid dissolved in rainwater, (5) soot from a diesel truck.

(a) Classify each as a primary or a secondary pollutant.

(b) For the two you classified as secondary, name one primary pollutant that is a precursor to each.

(c) Identify which of the five is an asphyxiant and explain, in one sentence, how it harms the body.

Thermal inversion

CONCEPT MADE SIMPLE

Normally the ground warms the air above it, and warm air rises like a chimney, carrying pollution away. In an **inversion** the layers are upside down: cool air sits under a layer of warm air. The warm layer is a lid, and pollution stays trapped where we live.

Normally, temperature falls with height through the troposphere, so warm, light surface air rises, mixes and dilutes pollutants. In a **thermal (temperature) inversion** the normal gradient is altered: the air at the surface is **cooler** than the air higher up. The cool surface air is denser than the warm air over it, so it will not rise. Vertical mixing stops and pollutants, especially **smog and particulates**, are trapped near the ground.

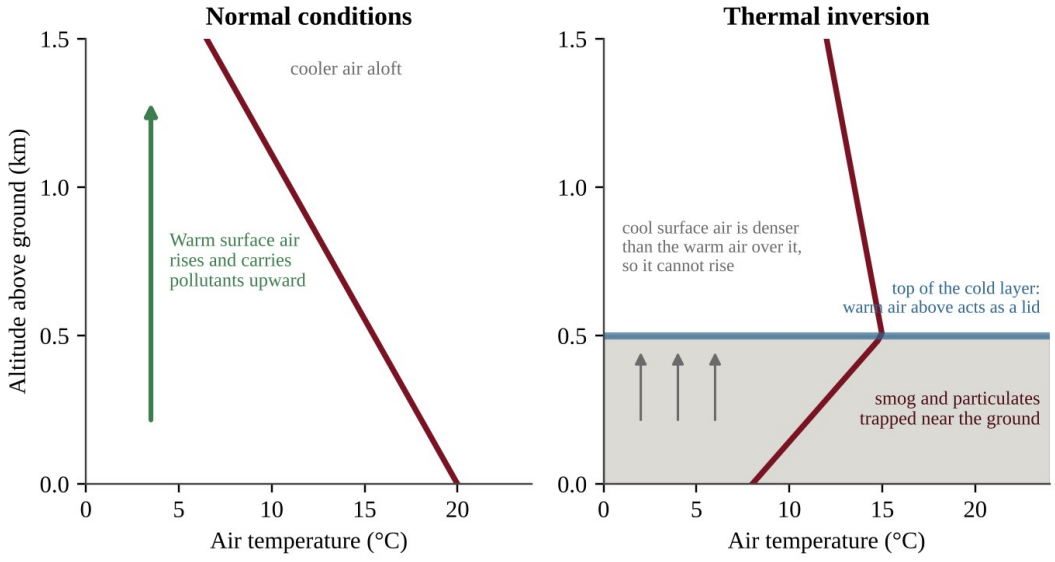


Figure 9.1 — Temperature versus altitude in normal air (left) and during a thermal inversion (right)

How to read Figure 9.1: temperature is on the x-axis, altitude on the y-axis. A line sloping to the upper left (cooler with height) is normal. A stretch bending to the right (warmer with height) is the inversion layer. Pollutants are trapped in the cold air *below* the top of that layer; where the line again falls with height, normal mixing resumes.

PROCESS & CAUSE-AND-EFFECT

Clear, calm night → ground loses heat and chills the air near it (or cold air drains into a valley) → cool dense air under warmer air → no vertical mixing → pollutants trapped → smog and particulates build up → health problems

What favors an inversion: long, clear, calm nights (clouds act as a blanket and prevent it); cold winter ground or snow; valleys and basins ringed by mountains, where cold air pools and slopes block the wind. Inversions usually break up when the sun heats the ground, but a strong one can last for days. Winter inversions raise **particulates** (heating smoke plus trapped air); **ozone** is mainly a summer problem.

KEY DEFINITIONS

Thermal inversion — a layer of warm air above cooler air near the ground, reversing the normal drop in temperature with height and trapping pollutants. **Temperature gradient** — the change in air temperature with altitude.

CLASSROOM EXAMPLE SG-U7-E003

A weather balloon measured the air temperature above a valley town on two mornings.

Altitude above ground (m)	Day X temperature (°C)	Day Y temperature (°C)
0	14	6
250	12	9
500	10	12
750	8	11

(a) Identify which day had a thermal inversion and state the altitude range in which it occurred.

(b) Calculate the temperature change from 0 m to 500 m on Day Y.

(c) The town measured fine particulate matter of 15 µg/m³ on one of the days and 55 µg/m³ on the other. Predict which day had which value and explain.

Reducing air pollution

CONCEPT MADE SIMPLE

Three ways to attack air pollution: **use less** of the dirty thing (conservation), **switch** to something cleaner (alternative fuels), or **capture** what is made before it escapes (control devices). Laws make people actually do it.

Methods to reduce air pollutants include **regulatory practices** (the CAA, emission and fuel standards, permits), **conservation** (driving less, transit, efficient buildings), **alternative fuels** (natural gas instead of coal, renewable electricity, electric vehicles, low-sulfur fuels) and **control devices**.

KEY DEFINITIONS

Vapor recovery nozzle — a device on a gasoline pump that stops fuel vapors (VOCs) escaping while a vehicle is refueled. **Catalytic converter** — a device in a vehicle's exhaust that uses metal catalysts (platinum, palladium, rhodium) to convert CO, NO_x and hydrocarbons into less harmful CO_2, N_2, O_2 and H_2O. **Wet scrubber** — sprays water, or a water and limestone mixture, through exhaust to dissolve gases such as SO_2 and trap particles. **Dry scrubber** — injects a dry powder (such as lime) that reacts with acidic gases and is then filtered out. **Electrostatic precipitator** — charges particles so they stick to oppositely charged plates, which are shaken to collect the ash.

PROCESS & CAUSE-AND-EFFECT

Catalytic converter: $CO + O_2 \rightarrow CO_2$; $NO \rightarrow N_2 + O_2$; hydrocarbons + $O_2 \rightarrow CO_2 + H_2O$. Wet scrubber: SO_2 in flue gas + water–limestone spray → calcium sulfite/sulfate sludge, cleaner gas out the stack. Precipitator: charged particles → attracted to plates → collected as ash.

Which device for which job? A catalytic converter treats the *gases* in vehicle exhaust, not particulates. Scrubbers remove SO_2 (and some other gases and particulates) from power-plant exhaust. An electrostatic precipitator removes *particulates only*. Coal plants often use scrubbers and precipitators together. Vapor recovery nozzles stop VOC evaporation at the pump.

Trade-offs. Devices create waste (scrubber sludge, ash), use energy and cost money, and a catalytic converter does nothing about CO_2. Using less fuel is usually cheaper and cleaner than capturing pollutants afterward. Fuel switching has costs too: low-sulfur coal is dearer, natural gas still emits CO_2, wind and solar are intermittent.

Calculations with emissions

FORMULA & MATH SKILL

Sulfur to SO_2: $S + O_2 \rightarrow SO_2$; atomic mass S = 32, SO_2 = 64, so 1 t of S gives 2 t of SO_2. mass of S = mass of fuel × (% sulfur ÷ 100) uncontrolled SO_2 = mass of S × (64 ÷ 32) SO_2 removed = uncontrolled SO_2 × (% removal ÷ 100); SO_2 emitted = uncontrolled − removed **Cost per metric ton removed** = annual cost ($/yr) ÷ SO_2 removed (t/yr) **Inventory** = number of sources × activity each (miles/yr) × emission factor (g/mile) **Per capita** = total emissions ÷ population (1 metric ton = 1000 kg)

For an inventory, write units as fractions so unwanted units cancel (dimensional analysis), using conversion fractions equal to 1, such as 1 t ÷ 10^6 g. Compare options by cost per ton *removed*: a smaller bill can still cost more per ton if it removes much less.

CLASSROOM EXAMPLE SG-U7-E012

A city runs 400 diesel buses. Each travels 30,000 miles per year and emits 0.35 g of particulates per mile. A filter on every bus cuts particulates by 80% and costs $9,000 per bus, spread over 10 years.

(a) Calculate the fleet's yearly particulate emissions in kilograms.

(b) Calculate the kilograms removed per year.

(c) Calculate the yearly cost per kilogram removed.

CLASSROOM EXAMPLE SG-U7-E013

A power plant burns 5.0×10^5 metric tons of oil per year. The oil is 0.8% sulfur by mass and all of the sulfur becomes SO_2. A dry scrubber removes 85% of the SO_2 and costs $15 million per year.

(a) Calculate the mass of sulfur burned each year.

(b) Calculate the SO_2 the plant would release with no scrubber.

(c) Calculate the SO_2 emitted with the scrubber.

(d) Calculate the scrubber's cost per metric ton of SO_2 removed.

CLASSROOM EXAMPLE SG-U7-E006

A region has two coal-burning power plants and a population of 400,000. Plant A emits 180,000 metric tons of SO_2 per year; Plant B emits 70,000 metric tons per year. Plant A installs wet scrubbers that remove 90% of its SO_2. Plant B makes no change.

(a) Calculate the region's total SO_2 emissions after the installation.

(b) Calculate the percent decrease in the region's total SO_2 emissions.

(c) Calculate the region's SO_2 emissions per person after the installation, in kilograms per person per year.

(d) Identify one by-product of operating a wet scrubber and one pollutant the scrubber does not reduce.

COMMON STUDENT MISTAKES

- Calling ozone "bad" everywhere. In the stratosphere it blocks UV light; at ground level it is a pollutant.
- Thinking "secondary" means less harmful. Ozone and acids are often worse than the pollutants that made them.
- Saying a precipitator removes SO_2 (particulates only), or that a catalytic converter removes CO_2 (it *makes* CO_2 from CO).

QUICK CONCEPT SUMMARY

- Burning fuel gives CO_2, CO, SO_2, NO_x, VOCs, particulates, toxic metals and (historically) lead; primary pollutants are emitted directly, secondary ones form in the air.
- Regulating lead at its source (Clean Air Act) cut atmospheric lead sharply.
- A thermal inversion puts cool air under warm air and traps smog and particulates near the ground.
- Reduce pollution by regulation, conservation and alternative fuels, and with vapor recovery nozzles, catalytic converters, scrubbers and precipitators.

9.2 Major Air Pollutants

The pollutants and the six criteria pollutants

In the United States the EPA tracks six **criteria pollutants** under the Clean Air Act: carbon monoxide, lead, nitrogen dioxide, ground-level ozone, particulate matter and sulfur dioxide (marked * below).

Pollutant	Main sources	Main effects	Main control
Carbon dioxide (CO_2)	Burning any carbon fuel; respiration; decomposition	Traps heat (Chapter 11); not toxic at outdoor levels	Efficiency, renewables
Carbon monoxide (CO)*	Incomplete combustion: vehicles, furnaces, stoves, wildfires	Binds to hemoglobin, blocking oxygen; headache, death	Catalytic converters, tuned appliances
Sulfur dioxide (SO_2)*	Coal plants, diesel, smelters, volcanoes	Lung irritation; acid rain; sulfate particles	Scrubbers, low-sulfur fuel
Nitrogen oxides (NO_x)*	Vehicle engines, power plants	Lung irritation; ozone and smog; nitric acid	Catalytic converters, fewer vehicle miles
Particulate matter (PM)*	Coal and diesel combustion, dust, construction, wildfires	Lung and heart disease; haze	Electrostatic precipitators, filters
Hydrocarbons / VOCs	Unburned fuel, fuel vapor, solvents, trees	Smog ingredient; some toxic	Vapor recovery nozzles, low-VOC products
Lead (Pb)*	Leaded gasoline (historic), old paint, smelters	Nervous-system damage, especially in children	Fuel regulation, paint removal
Toxic metals (e.g., mercury)	Coal combustion, waste incineration	Nerve damage; methylmercury in food webs	Scrubbers, less coal
Ground-level ozone (O_3)*	Secondary: NO_x + VOCs + sunlight	Lung damage, asthma, plant damage	Cut NO_x and VOCs

FORMULA & MATH SKILL

Concentrations of gases: 1 ppm = 1 part per million (1 in 10^6); 1 ppb = 1 part per billion (1 in 10^9); so 1 ppm = 1000 ppb. Multiply ppm by 1000 to get ppb; divide ppb by 1000 to get ppm. Particulates are reported as µg/m³. **Percent above a limit** = (measured − limit) ÷ limit × 100

CLASSROOM EXAMPLE SG-C9-E001

A city's 8-hour average ozone reading is 0.082 ppm. A health-based limit for ozone is 70 ppb.

(a) Convert the reading to ppb.

(b) Calculate the percent by which the reading exceeds the limit. Round to the nearest whole percent.

(c) Name the two groups of primary pollutants that are the precursors of ozone and state whether ozone is a primary or a secondary pollutant.

Photochemical smog

CONCEPT MADE SIMPLE

Smog is a **recipe**. Ingredients: nitrogen oxides from engines and VOCs from fuel vapor and plants. Cooking conditions: strong sunlight and warm air. Product: ground-level ozone and a brown, stinging haze. No sun, no smog; few engines, little smog.

Photochemical smog forms when nitrogen oxides and volatile organic compounds react with heat and sunlight to produce a variety of pollutants. The chemistry you need:

1. Vehicles and power plants emit **NO**, which converts in air to **NO_2**.
2. Sunlight splits NO_2 into NO and a free oxygen atom (O).
3. The oxygen atom joins O_2 to make **ozone (O_3)**.
4. In clean air, ozone would quickly react back with NO. VOCs convert NO back to NO_2 *without* using up ozone, so the cycle repeats and **ozone builds up**.
5. VOCs and NO_x also form other irritants (PAN, aldehydes, fine particles) that give smog its brownish haze.

KEY DEFINITIONS

Photochemical smog — secondary pollutants (mainly ozone) formed when sunlight and heat drive reactions between nitrogen oxides and VOCs. **Volatile organic compound (VOC)** — a carbon-based chemical that evaporates or sublimates at room temperature, such as gasoline vapor and formaldehyde. Trees give off natural VOCs. **Tropospheric (ground-level) ozone** — O_3 in the air we breathe, a harmful secondary pollutant and a greenhouse gas, unlike the protective stratospheric ozone layer.

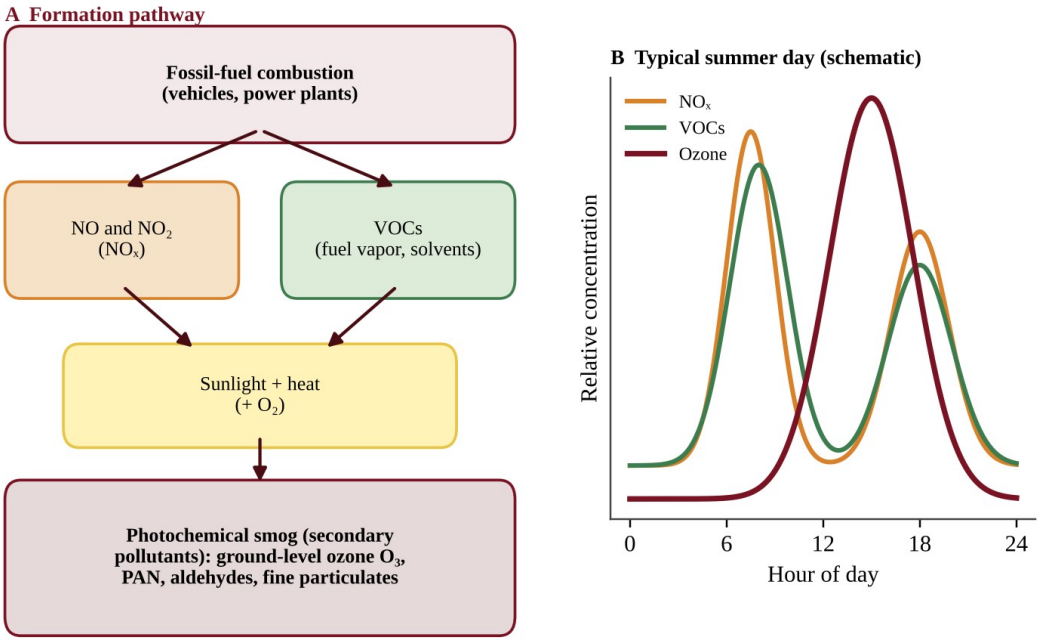


Figure 9.2 — Photochemical smog forms when NO_x and VOCs react in sunlight (A); on a summer day NO_x rises with the morning rush while ozone peaks in the afternoon (B)

Reading the daily pattern. In panel B, NO_x is produced early, in the morning commute. Ozone lags and peaks in the **afternoon**, when sunlight and heat have had hours to drive the reactions, then falls after sunset because the reactions need light. Ozone is also higher in **summer**: stronger sunlight, longer days and warmer air that speeds reactions and makes fuels and plants release more VOCs. Exercise in the morning on bad-ozone days.

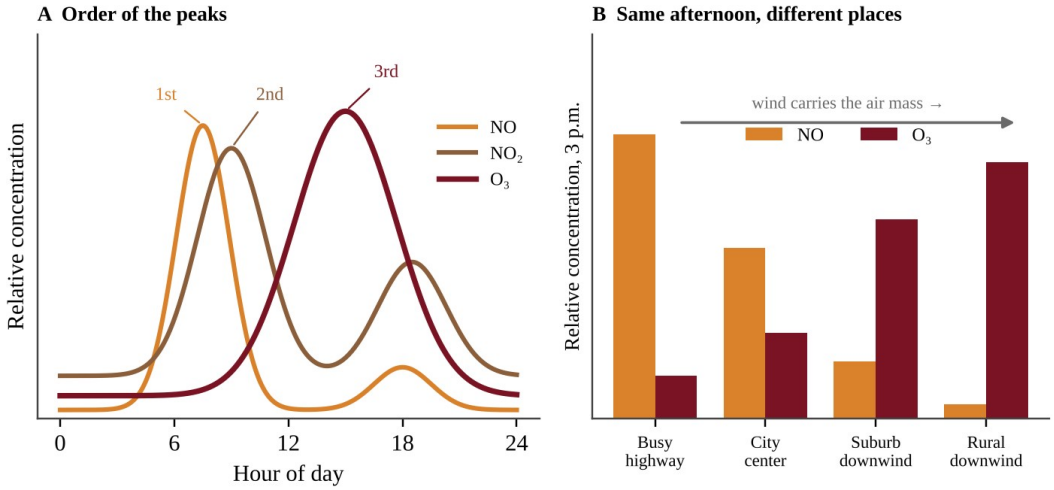


Figure 9.3 — The three gases peak in order (A); where the air has traveled matters as much as when (B). Schematic values

Telling NO, NO_2 and ozone apart. NO comes straight from engines, peaks first, then falls as it turns into **NO_2**, a reddish-brown gas that peaks a little later. **Ozone** peaks last. Place matters too: fresh NO reacts with ozone and removes it, so beside a busy highway NO is high and O_3 low. Downwind, the NO is used up while sunlight keeps driving the reactions, so ozone is highest in suburbs and rural areas **downwind** of the city. Low ozone at a busy road does not mean clean air.

Factors: many vehicles (so mainly urban), strong sunlight and heat, basins ringed by mountains, windless weather and inversions. **Effects and reduction.** Smog causes **respiratory problems** (coughing, asthma, reduced lung function) and **eye irritation**; ozone also injures crop and forest leaves and cracks rubber. Since the precursors are NO_x and VOCs, smog is reduced by **cutting NO_x and VOCs**: catalytic converters, vapor recovery nozzles, low-VOC paints, public transit, electric vehicles and limits on driving on high-ozone days.

COMMON STUDENT MISTAKES

- Saying NO_x and ozone peak at the same time. NO_x peaks with the morning rush; ozone peaks later.
- Describing smog as "smoke plus fog". It is a chemical product of sunlight and pollutants.

CLASSROOM EXAMPLE SG-U7-E002

A monitoring station in a city recorded these hourly averages on a hot, sunny summer day.

Time	NO_2 (ppb)	Ozone (ppb)
6:00	18	12
8:00	40	20
10:00	25	45
12:00	15	70
14:00	12	95
16:00	14	88
18:00	30	60
20:00	20	35

(a) Identify the time of the NO_2 peak and the time of the ozone peak.

(b) Calculate how many hours separate the two peaks.

(c) Explain, using the chemistry of smog, why the ozone peak comes later than the NO_2 peak.

(d) Predict how the ozone curve would change on a cool, cloudy day with the same traffic, and justify your prediction.

Natural sources of CO_2 and particulates

Not all air pollution is man-made; it becomes a problem when people add so much that the balance changes.

Natural CO_2 comes from **respiration** (every plant, animal, fungus and microbe), **decomposition** of dead organisms, and **volcanic eruptions** (which also vent SO_2 and ash). Photosynthesis and ocean uptake balance them (Chapter 5); burning fossil fuels adds extra CO_2 (Chapter 11).

Particulate matter (PM) is tiny solid and liquid particles suspended in air. **PM10** is 10 µm or smaller; **PM2.5** is 2.5 µm or smaller. The smaller the particle, the deeper it penetrates the lungs; PM2.5 can reach the bloodstream. Fine particles also scatter light, causing **haze** (reduced visibility) even in remote parks. Natural sources include volcanic ash, wildfire smoke, dust storms and wind-blown soil, sea-spray salt, pollen and mold spores, and sulfate and organic particles formed from natural gases.

Designing and judging an air-quality investigation

- **Hypothesis:** an if–then statement that names the direction (*If new carpet releases VOCs, then the renovated wing will have more formaldehyde*).
- **Variables:** the *independent variable* is what you compare or change (location, distance, time); the *dependent variable* is what you measure, with a unit (µg/m³). **Controlled variables** stay the same: instrument, height, hours, ventilation.
- **Control or background site:** an upwind, outdoor or unaffected-room sensor measures what is there anyway. Source contribution ≈ downwind reading − background reading.
- **Replication:** many sites, days or samples.

- **Confounding variable:** something else that changes with the independent variable (wind direction when comparing distances from a plant). Measure it or hold it constant.
- **Correlation is not proof of cause.** Say data are *consistent with* a cause, give the mechanism, and name a factor not ruled out. To judge a policy, compare before and after in the same season and note other causes.

CLASSROOM EXAMPLE SG-U7-E011

A town suspects that a new gravel quarry raises PM10 in a neighborhood 1 km to its east; the wind usually blows from the west.

(a) State a testable hypothesis with direction.

(b) Identify the independent and dependent variables and two controlled variables.

(c) Describe where and when to place monitors and what an upwind site is for.

(d) Higher PM10 is found at the east site. Explain why this alone does not prove the quarry is the cause.

CLASSROOM EXAMPLE SG-U7-E004

A valley town recorded PM2.5 (µg/m³) on six winter days. Days 3, 4 and 6 had a thermal inversion; Days 1, 2 and 5 did not.

Day	1	2	3	4	5	6
PM2.5	14	11	46	58	17	52

(a) Describe the relationship between inversion and PM2.5 shown by the data.

(b) Calculate the mean PM2.5 for the inversion days and for the non-inversion days.

(c) Calculate the percent by which the inversion-day mean exceeds the non-inversion mean. Round to the nearest whole percent.

(d) Propose one action the town could take to lower winter PM2.5 and justify it.

Indoor air pollutants

Indoor air can be more polluted than outdoor air because there is less dilution.

Source type	Examples
Natural	Radon, mold, dust
Human-made (building materials, furnishings)	Insulation; VOCs from furniture, paneling, carpets; formaldehyde from building materials and upholstery; lead from paint; asbestos

Source type	Examples
Combustion	Carbon monoxide, nitrogen oxides, sulfur dioxide, particulates, tobacco smoke

KEY DEFINITIONS

Asphyxiant — a gas that starves the body of oxygen. **Carbon monoxide** is an odorless, colorless asphyxiant that binds to hemoglobin far more strongly than oxygen does. **Indoor particulates** — asbestos, dust and smoke. **Formaldehyde** — a VOC released from building materials, furniture, upholstery and carpets; an eye and lung irritant and a carcinogen. **Radon-222** — a naturally occurring radioactive gas produced by the decay of uranium in some rocks and soils.

- **Carbon monoxide** comes from faulty furnaces, gas stoves and heaters, wood stoves and idling cars in garages. High levels cause headaches, dizziness and death. Humans cannot sense it, so a CO detector is the safeguard.
- **Asbestos** fibers from old insulation, tiles and roofing are dangerous when disturbed and inhaled: lung scarring, lung cancer and mesothelioma.
- **VOCs and formaldehyde** off-gas from new carpet, paneling, glues, paints and cleaners and cause irritation and headaches.
- **Lead** from old paint chips and dust harms the nervous system, especially in young children.
- **Mold** (damp places) triggers allergies and asthma. **Combustion products** irritate lungs; cooking over open indoor fires is a leading cause of smoke illness in many developing regions.

Radon is the unusual one. Uranium in some rocks and soils decays through radium to **radon-222**, a gas that moves up through soil. It enters homes through cracks in foundations and walls and gaps around pipes, and it is also **dissolved in well water**. It cannot be seen or smelled. Once inhaled, radon and its decay products emit radiation inside the lungs. Radon-induced lung cancer is the **second leading cause of lung cancer in the United States**, after smoking. Homeowners can test with a kit; the EPA recommends action at 4 pCi/L (picocuries per liter) or more.

PROCESS & CAUSE-AND-EFFECT

Uranium in rock and soil → radon-222 gas → rises through soil → enters basement through cracks (or in well water) → builds up in a closed house → inhaled → radiation damages lung tissue → higher lung-cancer risk

Cutting exposure: remove the source first, then ventilate outdoors, then detect or filter (CO detectors, radon test and sub-slab fan, licensed asbestos removal, humidity control, low-VOC products, intact lead paint, vented stoves). Children are at greater risk: they breathe more air per kilogram, their lungs and nervous systems are still developing, and they put hands in their mouths.

COMMON STUDENT MISTAKES

- Saying radon is a factory chemical. It is natural and radioactive.
- Thinking a tightly sealed new home is automatically safe. Poor ventilation traps indoor pollutants.
- Classing carbon monoxide as a particulate. It is a gas and an asphyxiant.
- Masking a pollutant (air fresheners, incense) instead of removing the source or venting it outdoors.

CLASSROOM EXAMPLE SG-U7-E005

A health department tested the basements of five homes for radon. The EPA action level is 4.0 pCi/L.

Home	1	2	3	4	5
Radon (pCi/L)	2.1	5.6	3.9	8.4	3.0

(a) Identify which homes exceed the action level and calculate the percentage of tested homes that do.

(b) Calculate the mean radon level for the five homes.

(c) Explain how radon gas enters a house from the ground.

(d) Propose one way to reduce radon in Home 4 and justify why it works.

QUICK CONCEPT SUMMARY

- Know the six criteria pollutants, the table of sources and effects, and which control matches which pollutant.
- Smog: NO_x + VOCs + sunlight and heat → ozone; NO_x peaks in the morning, ozone in the afternoon and in summer; cut NO_x and VOCs.
- Natural CO_2 sources: respiration, decomposition, volcanoes; natural particulates: ash, smoke, dust, sea spray, pollen, spores. PM2.5 is the most dangerous to health.
- Indoor pollutants are natural, human-made or from combustion; CO is an asphyxiant; radon-222 is the second leading cause of lung cancer in the U.S.

9.3 Acid Deposition

CONCEPT MADE SIMPLE

Burning coal and fuels releases SO_2 and NO_x. In the air these turn into **sulfuric and nitric acid**, which fall in rain, snow, fog or dry particles, often hundreds of kilometers from the source. The land and water that receive them become more acidic.

Acid deposition includes **wet deposition** (acid rain, snow, fog) and **dry deposition** (acidic gases and particles settling out). It starts with two gases from anthropogenic and natural sources:

- **Sulfur oxides.** SO_2 from coal-burning power plants (and volcanoes) reacts with oxygen and water to form sulfuric acid (H_2SO_4).
- **Nitrogen oxides.** NO_x from motor vehicles and coal-burning power plants (and lightning) forms nitric acid (HNO_3).

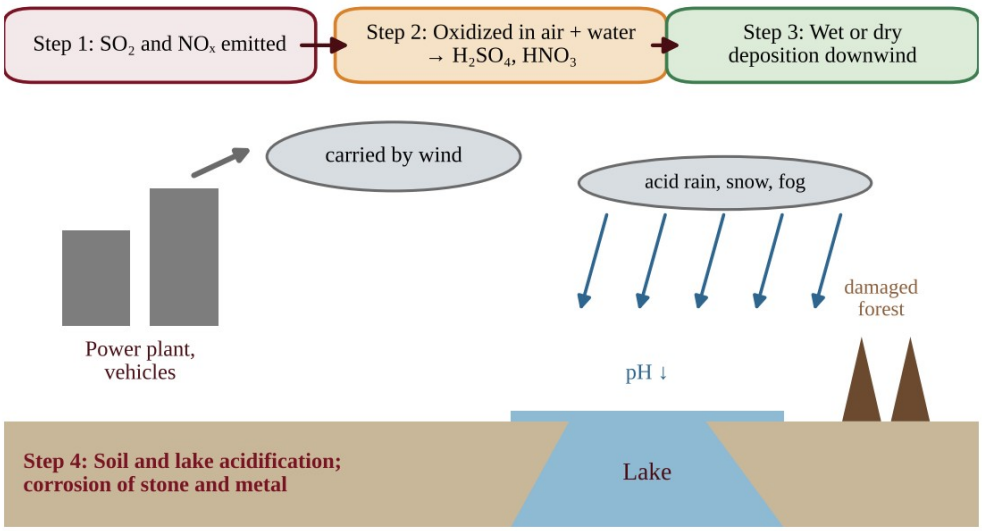


Figure 9.4 — Pathway of acid deposition from sources to damaged lakes, forests and structures

The pH scale

FORMULA & MATH SKILL

pH = $-\log_{10}[H^+]$, where $[H^+]$ is the hydrogen ion concentration in moles per liter. The scale runs 0 to 14: below 7 acidic, 7 neutral, above 7 basic. **Each whole pH unit is a factor of 10 in H^+.** factor = 10^(difference in pH): 1 unit = 10×, 2 units = 100×, 3 units = 1000×. Lower pH means more acidic.

Clean rain is already slightly acidic, about **pH 5.6**, because CO_2 dissolves in it to form weak carbonic acid. Rain is **acid rain** below about 5.6; typical acid rain is pH 4.2–4.4. Differences need not be whole numbers: pH 4.3 versus 5.6 is 1.3 units, so 10^1.3 ≈ 20 times the H^+ (use the 10^x key; never say "1.3 times"). From a concentration, $[H^+] = 1 \times 10^{-5}$ mol/L is pH 5.

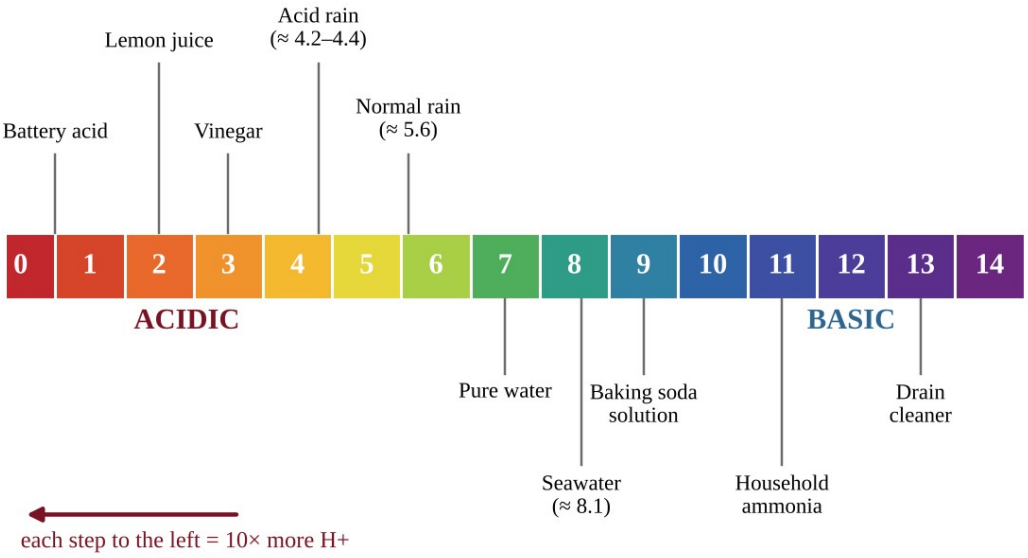


Figure 9.5 — The pH scale with example substances; each step is a tenfold change in H^+

Effects

1. **Soils.** Acid leaches calcium and magnesium away and frees toxic aluminum and other metals, harming roots and weakening forests, especially at high elevations bathed in acid fog.
2. **Lakes and streams.** Falling pH harms fish eggs, amphibians and invertebrates; many fish decline below about pH 5, and released aluminum is toxic to fish.

3. **Structures.** Acid dissolves limestone and marble and corrodes metals in buildings, statues and bridges.
4. **Human health.** The acids form fine particles that reach the lungs.

Why some places are hit harder

Acid deposition mainly affects communities **downwind** of coal-burning power plants; in North America, winds carry Midwest emissions to the Northeast and eastern Canada. **Bedrock and soil matter.** Limestone (calcium carbonate) and carbonate-rich soils **buffer** acid: they neutralize it and keep lake pH near neutral. Lakes on granite and thin, sandy soils have little neutralizing capacity and acidify quickly.

KEY DEFINITIONS

Acid deposition — acids and acid-forming pollutants falling from the air as wet or dry deposition. **Buffering capacity** — the ability of water or soil to neutralize acids and resist a drop in pH; limestone supplies it. **Downwind** — in the direction the wind blows, away from the source.

PROCESS & CAUSE-AND-EFFECT

Burning coal and fuels → SO_2 and NO_x → oxidized in air and dissolved in water → H_2SO_4 and HNO_3 → carried downwind → wet and dry deposition → soil and lake pH falls → nutrient loss, aluminum release, fish and tree damage, stone corrosion

Solutions cut the gases at the source: scrubbers, low-sulfur fuels, natural gas and renewables, catalytic converters, and cap-and-trade (the CAA's SO_2 cap). Adding crushed limestone ("liming") to a lake treats the symptom, not the cause. Recovery lags behind emission cuts because soils have lost much of their calcium and magnesium.

COMMON STUDENT MISTAKES

- Saying pH 4 is "twice as acidic" as pH 5. It is ten times.
- Calling pure rain "pH 7". Clean rain is about 5.6.
- Assuming acid rain only falls near the source, or forgetting dry deposition and natural sources.
- Writing that taller smokestacks solve acid rain. They only carry the acid farther.

CLASSROOM EXAMPLE SG-U7-E007

Three rain samples were collected: Sample A has pH 5.6, Sample B has pH 4.6, and Sample C has pH 3.6.

(a) Rank the samples from most acidic to least acidic.

(b) Calculate how many times greater the hydrogen ion concentration of Sample C is than that of Sample A.

(c) Identify the two gaseous pollutants most responsible for rain with a pH below 5.6 and name one human source of each.

(d) Explain why clean, unpolluted rain has a pH below 7.

CLASSROOM EXAMPLE SG-U7-E014

Rain over a downwind forest has a pH of 4.2; rain over a remote ocean has a pH of 5.6. A lake sample has an H^+ concentration of 1×10^{-6} mol/L.

(a) Calculate how many times greater the H^+ concentration of the forest rain is than that of the ocean rain.

(b) Calculate the pH of the lake sample and state whether it is acidic.

(c) A lake has pH 6.3. Calculate how many times lower its H^+ concentration is than the forest rain's.

CLASSROOM EXAMPLE SG-U7-E008

Two lakes receive acid rain of the same pH from the same distant power plants. Lake P lies on granite bedrock with thin soil; its pH fell from 6.0 to 5.0 over 30 years. Lake Q lies on limestone bedrock and its pH stayed at 7.1.

(a) Calculate the factor by which the hydrogen ion concentration of Lake P increased.

(b) Explain why Lake Q's pH did not fall.

(c) Predict which lake has the greater risk of fish loss and justify.

(d) Propose one action at the source and one local action that would help Lake P, and state which treats the cause.

QUICK CONCEPT SUMMARY

- Acid deposition comes from SO_2 (coal plants, volcanoes) and NO_x (vehicles, coal plants) converting to H_2SO_4 and HNO_3.
- Each pH unit is a tenfold change in H^+; clean rain is about 5.6.
- Effects: acidified soils and lakes, aluminum release, forest decline, corrosion. Downwind areas are hit most; limestone buffers, granite does not.

9.4 Thermal Pollution

Warm water holds less oxygen, as a warm soda goes flat. **Thermal pollution** occurs when heat released into water produces negative effects on the organisms in that ecosystem. The usual source is **cooling water**: power plants and factories take in river water, use it to carry off waste heat and return it warmer. Heat also enters streams when clear-cutting removes streamside shade and when runoff crosses hot pavement.

Why warm water is a problem. Warm water does not contain as much dissolved oxygen as cold water. And the metabolic rates of cold-blooded aquatic animals rise with temperature, so they need more oxygen just when less is available. Cold-water species such as trout have narrow tolerance ranges and die or leave; sudden temperature changes can also shock fish.

Water temperature (°C)	5	10	20	30
Maximum DO at saturation (mg/L, fresh water, sea level)	12.8	11.3	9.1	7.5

Remedies: cooling towers, recycling cooling water, reusing waste heat, and shade trees along streams.

KEY DEFINITIONS

Thermal pollution — heat added to water by human activities that harms organisms.

CLASSROOM EXAMPLE SG-U8-E004

A power plant takes in river water at 10°C and discharges it at 30°C. Use the saturation table above. (a) Calculate the percent decrease in the maximum DO the water can hold when it warms from 10°C to 30°C. (b) State two reasons, other than the lower oxygen capacity, that the warmed discharge could harm cold-water fish. (c) Propose one engineering and one landscape solution that would reduce the thermal impact.

QUICK CONCEPT SUMMARY

- Thermal pollution is heat added to water, mainly power-plant cooling water.
- Warm water holds less oxygen and speeds up metabolism, so stress and fish kills occur.
- Cooling towers, shade and heat reuse reduce the problem.

9.5 Noise Pollution

In cities the main sources are **transportation** (road traffic, trains, aircraft), **construction**, and **domestic and industrial activity** (machinery, power tools, lawn equipment, loud music).

Loudness is measured in **decibels (dB)** on a logarithmic scale, like pH. **Every 10 dB is ten times the sound intensity**: 20 dB is 100 times, 30 dB is 1000 times.

Sound	Approx. level (dB)
Whisper	30
Normal conversation	60
Busy city traffic	80
Chain saw	110
Jet taking off nearby	130

Hearing damage depends on both level and time: long exposure to about 85 dB or more can cause permanent hearing loss, and very loud sounds damage hearing in minutes.

FORMULA & MATH SKILL

Intensity ratio = 10^(Δ dB ÷ 10), where Δ dB is the difference between two sound levels. A 30 dB difference = 10^3 = 1000 times; a 25 dB difference = 10^2.5 ≈ 320 times.

Effects on people: physiological stress (higher heart rate and blood pressure, disturbed sleep), hearing loss, and trouble concentrating and learning.

Effects on animals: stress hormones rise; **masking** of sounds used to communicate or hunt (birds sing louder or higher near roads; ship noise interferes with whale and dolphin calls and sonar); **damaged hearing** from blasts and sonar; and **changed migration routes** as animals avoid noisy zones.

Reducing noise at the source works best: quieter engines and mufflers, sound barriers, buffer zones, speed limits, quiet pavement, restricting night flights and ship speeds, and ear protection for workers.

KEY DEFINITIONS

Decibel (dB) — unit of sound level; every 10 dB is a tenfold change in intensity. **Masking** — noise covering up sounds animals use to communicate, find prey or detect predators.

CLASSROOM EXAMPLE SG-U7-E009

A factory floor measures 95 dB with all machines running. Workers wearing no hearing protection spend 8 hours a day there. A quiet office elsewhere in the building measures 55 dB. Near the factory, a wetland supports a population of frogs that call at night to attract mates.

(a) Calculate how many times greater the sound intensity is on the factory floor than in the quiet office.

(b) Explain why a worker is at risk of hearing loss at 95 dB.

(c) Describe how noise from the factory could reduce frog breeding success.

(d) Propose one source-level and one worker-level control to reduce harm.

QUICK CONCEPT SUMMARY

- Noise pollution is sound loud enough to cause stress and hearing loss; sources are transportation, construction, and domestic and industrial activity.
- Decibels are logarithmic: +10 dB = 10× the intensity.
- Wildlife are stressed, masked, hearing-damaged and pushed to change migration routes.

9.6 Water Pollution

Sources: point and nonpoint

CONCEPT MADE SIMPLE

Find a puddle of dirty water. If you can follow it back to one pipe, that is a **point source**. If the dirt arrived from the whole neighborhood at once, washed in by rain from lawns, streets and fields, that is a **nonpoint source**. One pipe can be fixed with a filter or a fine. A whole neighborhood is much harder.

A **pollutant** is any substance, or form of energy such as heat or noise, added to the environment faster than natural processes can remove it, harming living things.

KEY DEFINITIONS

Point source — a single, identifiable place where a pollutant enters the environment, such as a smokestack or a waste discharge pipe. **Nonpoint source** — a diffuse source spread over a wide area with no single outlet, such as pesticide spraying or urban runoff.

A point source has a location, an owner and a measurable flow, so regulators can monitor the pipe, set a permit limit and fine violators. A nonpoint source has thousands of small contributors, such as leaking

driveways, fertilized farms and over-treated lawns, and rain carries their contributions to the nearest stream. It is harder to identify, measure and control, and in many watersheds it is now the larger problem. In the United States the **Clean Water Act** works mainly through permits limiting what a point source may discharge; the **Safe Drinking Water Act** sets limits for contaminants in public drinking water.

COMMON STUDENT MISTAKES

- Thinking "nonpoint" means "natural". Lawn fertilizer is entirely human-caused.
- Calling a very large source nonpoint. A giant factory is a point source if the pollutant leaves through an identifiable outlet; identifiability, not size, is the test.

CLASSROOM EXAMPLE SG-U8-E001

For each source, state whether it is a point source or a nonpoint source and justify your choice in one sentence: (i) exhaust leaving the stack of a coal-fired power plant; (ii) oil and tire residue washed from a city's streets into storm drains during a rainstorm; (iii) treated wastewater leaving a municipal plant through a single underwater pipe; (iv) fertilizer washing off 200 separate cornfields into a river. Then explain why a regulator could set a measurable discharge limit for only some of these.

Range of tolerance

Every organism has a **range of tolerance** for each pollutant and physical factor. Within an **optimum range** it keeps its internal conditions steady (homeostasis). Outside that range it suffers physiological stress, limited growth, reduced reproduction and, in extreme cases, death. Species differ: a trout needs cold, well-oxygenated water; a carp tolerates warm, low-oxygen water. That is why the species present indicate water quality: stoneflies and mayflies suggest clean water; only sludge worms suggest heavy organic pollution.

Impacts on aquatic ecosystems

- **Coral reefs** are damaged by rising ocean temperature (corals expel their symbiotic algae: bleaching), by sediment runoff that smothers them and blocks light, and by destructive fishing such as dynamite or bottom trawling.
- **Oil spills.** Hydrocarbons are toxic and kill organisms directly. Floating oil coats the feathers of birds and fur of marine mammals, destroying insulation and buoyancy. Some components sink and kill bottom-dwelling organisms. Oil on beaches harms fishing and tourism. Cleanup uses floating **booms** and skimmers to contain and collect surface oil, **dispersants** that break slicks into droplets (sparing birds and shores but spreading toxic oil through the water), and oil-eating microbes (bioremediation), which work slowly.
- **Heavy metals** (lead, arsenic, cadmium) from industry, especially mining and burning fossil fuels, can reach groundwater and contaminate drinking supplies. Lead, from old paint, pipes and past leaded gasoline, damages the nervous system, and children are most vulnerable.
- **Litter** is unsightly, can cause intestinal blockage and choking in wildlife, and carries toxic substances into the food chain.
- **Sediment** reduces light penetration, harming producers and visual predators (animals that hunt by sight); when it settles it smothers spawning gravel and coral.
- **Mercury.** Elemental mercury from coal burning and mining enters water, where bacteria convert it to highly toxic **methylmercury**, the form that accumulates in organisms (Chapter 10).

- **Oceanic dead zones** are areas of low oxygen caused by increased nutrient pollution (see eutrophication below).

Dissolved oxygen, BOD and the oxygen sag curve

Dissolved oxygen (DO) is oxygen gas dissolved in water, in mg/L. Fish and most aquatic animals need roughly 5 mg/L or more; below about 2 mg/L most cannot survive. Oxygen enters from the atmosphere and from photosynthesis and is removed by respiration and decomposition.

When organic waste such as sewage or manure enters a stream, bacteria decompose it using oxygen. The oxygen microbes need to break down the organic matter in a sample is its **biochemical oxygen demand (BOD)**. High BOD means a heavy load of decomposable material and predicts a drop in DO. Clean streams have low BOD; raw sewage has very high BOD.

An **oxygen sag curve** is a plot of dissolved oxygen versus distance from a source of pollution, usually excess nutrients and biological refuse.

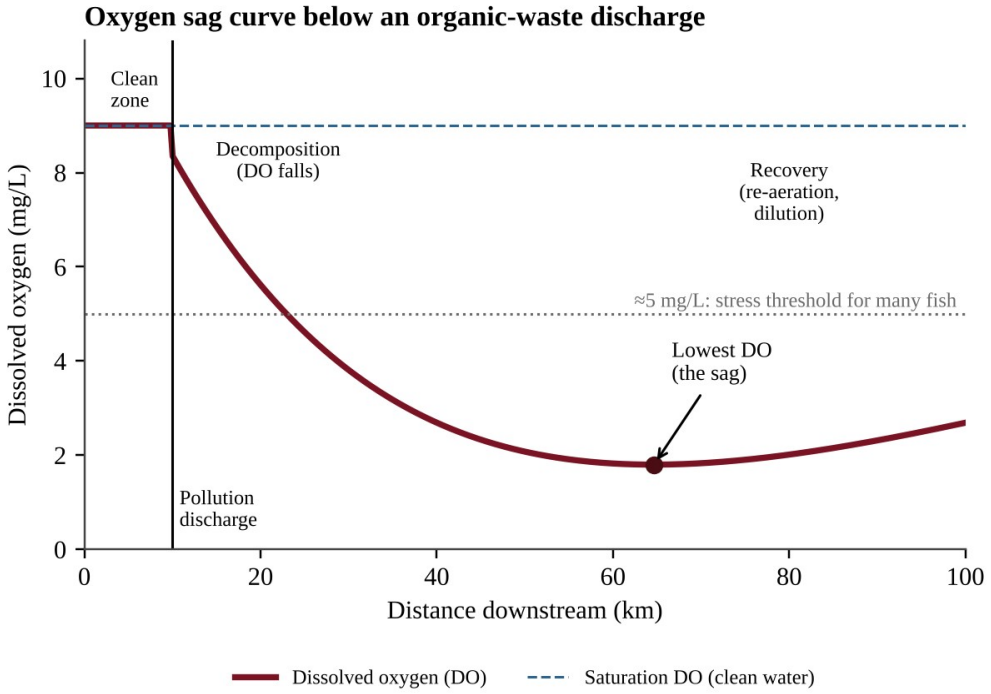


Figure 9.6 — Oxygen sag curve: DO is high upstream, falls as bacteria decompose waste below the discharge, reaches a minimum, then recovers as the stream re-aerates and the waste is used up

PROCESS & CAUSE-AND-EFFECT

Organic waste enters stream → bacteria decompose it and consume oxygen (BOD) → DO falls downstream → minimum DO (the sag) where decomposition outpaces re-aeration → waste is used up while oxygen enters from air and photosynthesis → DO recovers toward saturation

How to read the graph. The horizontal axis is distance downstream. Before the discharge the water sits near its saturation DO. The steepest fall is just below the outfall; the lowest point is where oxygen is scarcest (fish kills occur here); the rise afterward is recovery. A larger waste load, warmer water or slow flow makes the sag deeper and longer; a treated discharge (lower BOD) makes it shallower. If the minimum falls below a species' tolerance level, that species disappears from the affected stretch.

BOD is often plotted on the same axes. The curves mirror each other: BOD jumps at the outfall and falls as microbes consume the waste; DO is lowest a little farther downstream, where waste is still being broken down faster than oxygen enters.

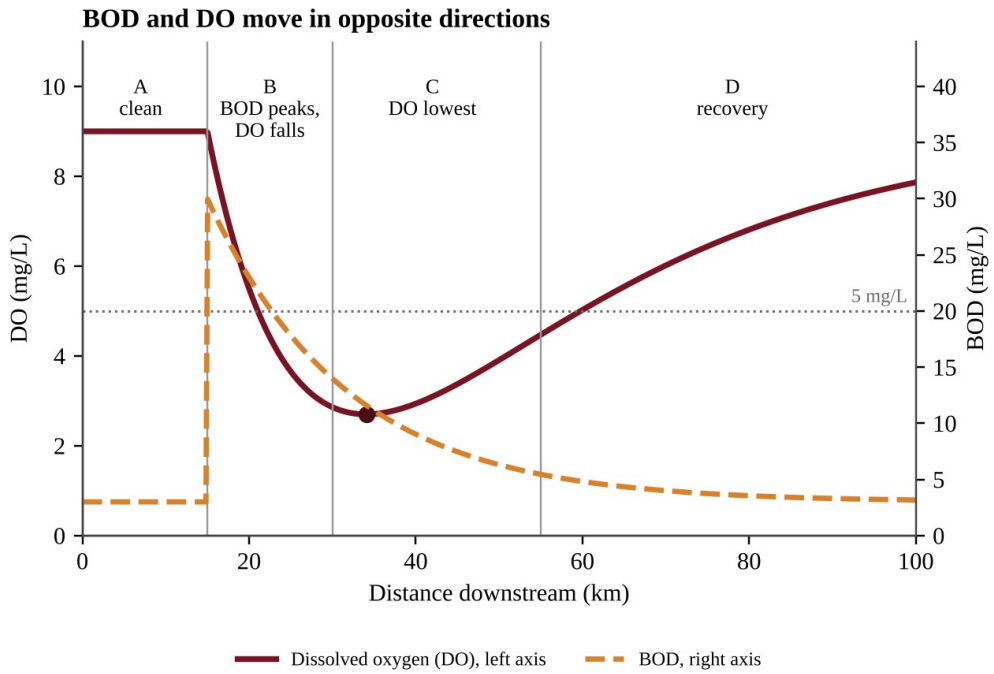


Figure 9.7 — BOD and DO plotted together: BOD peaks at the discharge, DO is lowest farther downstream, then both recover. The dotted line marks the 5 mg/L most fish need

KEY DEFINITIONS

Dissolved oxygen (DO) — oxygen gas dissolved in water, in mg/L. **BOD** — biochemical oxygen demand; oxygen consumed by microbes decomposing organic matter in water. More organic waste means higher BOD and lower DO. **Dead zone** — water so low in oxygen (hypoxic) that most animals cannot live there.

COMMON STUDENT MISTAKES

- Saying the sag is caused by the pollutant being "poisonous". In an organic-waste sag, bacteria feeding on the waste use up the oxygen.
- Thinking DO keeps falling forever. It recovers as waste is used up and the water re-aerates.
- Mixing up BOD and DO: high BOD goes with low DO.

CLASSROOM EXAMPLE SG-U8-E002

A stream receives treated effluent from a cannery at kilometer 10. Dissolved oxygen was measured at several sites:

Distance (km)	DO (mg/L)
0	8.8
10	8.8
15	6.5
20	4.2
30	2.9
40	3.6
60	6.1
80	8.0

(a) Identify the site with the lowest DO and give its value. (b) List the sampled sites where DO was below the 5 mg/L level needed by most fish. (c) Calculate the percent decrease in DO from the discharge point (km 10) to the lowest value. (d) Explain, using BOD, why DO falls between km 10 and km 30, and explain why it rises between km 40 and km 80.

Wetlands and mangroves

CONCEPT MADE SIMPLE

A wetland is nature's sponge and kidney. It soaks up floodwater, filters pollutants and shelters a huge number of species. Fill in or drain a wetland and all those free services are lost at once.

Wetlands are areas where water covers or saturates the soil for all or part of the year. Three features define them: water (hydrology), waterlogged **hydric soils** and plants adapted to wet conditions. Examples are marshes, swamps, bogs and **mangroves**, salt-tolerant coastal forests of tropical and subtropical shores with tangled roots standing in the water.

Ecological services: *water purification and filtration* (slow water lets solids settle, as in primary treatment, while microbes digest organic waste and plants take up nutrients, as in secondary and tertiary treatment), *flood protection* (storing floodwater and, for mangroves and coastal marshes, absorbing wave energy and storm surge) and *habitat* (nurseries for fish and shellfish; mangrove roots shelter juvenile fish that later live on reefs). They also have economic value: fisheries, tourism and lower storm damage.

Threats include commercial development (filling land for buildings, roads and aquaculture ponds), dams, levees and channelized rivers (which cut off the sediment that rebuilds delta marshes, so the land sinks and the marsh drowns), overfishing, and pollutants from agriculture and industrial waste. When a wetland is lost, communities must build levees and treatment plants at great cost. **Responses:** laws limiting wetland filling, restoring drained wetlands, replanting mangroves, buffer zones and cutting upstream fertilizer runoff.

A pollutant **load** is the mass moving past a point in a given time: concentration × volume of water. Because 1 mg/L = 1 g/m³, mg/L × m³ gives grams.

FORMULA & MATH SKILL

Load (g/day) = concentration (mg/L) × flow (m^3/day); load (kg/day) = load in g ÷ 1000 Mass removed = load × (percent removed ÷ 100)

Illustration. A canal carries 10,000 m^3/day with 3 mg/L of phosphate: load = 3 × 10,000 = 30,000 g = 30 kg/day. A marsh removing 50% removes 15 kg/day; if it is drained, that extra 15 kg/day reaches the lake.

CLASSROOM EXAMPLE SG-U8-E011

A drainage ditch carries 25,000 m^3 of water per day containing 8 mg/L of nitrate-nitrogen. A wetland along the ditch removes 45% of the nitrate-nitrogen before the water reaches a lake. (a) Calculate the nitrate-nitrogen load entering the wetland, in kg per day. (b) Calculate the mass the wetland removes each day, in kg, and the mass that reaches the lake. (c) The wetland is drained for a parking lot. Calculate the additional nitrate-nitrogen, in kg per year, that reaches the lake. (d) Name two other services the community loses.

Eutrophication

CONCEPT MADE SIMPLE

Fertilizer is plant food. Dump it into a lake and the algae feast, covering the surface in green. But algae live briefly. When they die, bacteria eat them and use up the oxygen. Fish suffocate.

Eutrophication occurs when a body of water is enriched in nutrients, mainly nitrogen and phosphorus. A **limiting nutrient** is the one in shortest supply, so adding it raises growth the most: phosphorus in most freshwater lakes, nitrogen in many coastal waters. To find which nutrient limits growth, fill identical, replicated containers of lake water and add nothing, N only, P only, or N + P; the nutrient that boosts growth is limiting.

Eutrophication cascade

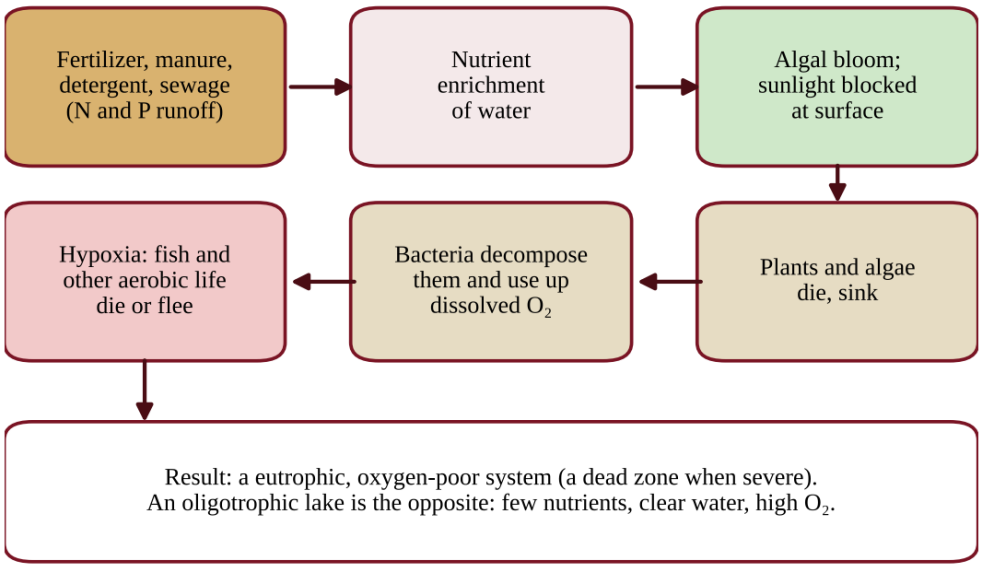


Figure 9.8 — The eutrophication cascade, from nutrient input to oxygen depletion and die-offs

PROCESS & CAUSE-AND-EFFECT

Fertilizer, manure, detergent or sewage runs into water → N and P rise → algal bloom → algae shade plants below and die → decomposers digest the dead algae and use oxygen → dissolved oxygen falls (hypoxia) → fish and other organisms die

The bloom may also block sunlight, killing submerged plants, and some algae release toxins. **Anthropogenic causes** are agricultural runoff (fertilizer and animal manure) and wastewater release (sewage, phosphate detergents). Natural eutrophication takes centuries as a lake fills with sediment; human-caused eutrophication can take a few years.

KEY DEFINITIONS

Eutrophic — nutrient-rich, with frequent algal blooms and low dissolved oxygen. **Oligotrophic** — very low nutrients, stable algae populations, clear water, high dissolved oxygen. **Hypoxic** — low in dissolved oxygen.

Solutions: apply fertilizer at the right time and amount, plant buffer strips and keep wetlands along streams, ban phosphate detergents, upgrade wastewater plants to remove N and P (tertiary treatment), and manage manure. Treating a lake's symptoms (aerating the water, harvesting weeds, dredging nutrient-rich sediment) is costly and temporary.

Designing an investigation of a pollution problem

FRQ 1 follows one pattern.

1. **Hypothesis.** An "if … then …" prediction: *If runoff from fertilized lawns adds nutrients to the pond, then nitrate and phosphate will be higher at sites closer to the lawns.*
2. **Variables.** The **independent variable** is what you change or compare (distance from the source, nutrient added); the **dependent variable** is what you measure (nitrate, chlorophyll, DO).
3. **Control and constants.** A **control** receives no treatment (a site above the source, or a container with no added nutrient). **Constants** are held the same or recorded: depth, time of day, temperature, recent rainfall, method.
4. **Procedure.** Say where (above and below the source), when and how often (same time of day, weekly, and after rain) and name the measurement and tool (nitrate kit, DO meter).
5. **Replication.** Take several samples per site and date, then average them.
6. **Analysis.** Graph the averaged dependent variable against the independent variable and state the pattern (nitrate highest at the outfall and falling downstream). A control and a plausible mechanism strengthen a claim of cause.

CLASSROOM EXAMPLE SG-U8-E012

A farm pond turns green each July after a nearby field is fertilized. A student suspects that phosphorus in the runoff is the cause. Four sampling sites are available: S1 in the drainage ditch next to the field, S2 at the pond inlet, S3 at the pond center, and S4 in a similar pond with no farm nearby. (a) Write an "if … then …" hypothesis. (b) Identify the independent and dependent variables and the site that serves as the control. (c) Describe a sampling plan that states when and how often samples are taken and how replication is achieved. (d) Name two variables that must be kept constant or recorded at every site. (e) Describe results that would support the hypothesis.

CLASSROOM EXAMPLE SG-U8-E003

In an outdoor experiment, four identical tanks received lake water and the same number of fish. Different amounts of phosphate were added once at the start. After 14 days:

Tank	Phosphate added (mg/L)	Algal density (cells per mL)	DO at the bottom (mg/L)
1	0	2,000	8.1
2	0.05	18,000	6.4
3	0.20	95,000	3.2
4	0.50	210,000	1.1

(a) Identify the control tank. (b) Describe the trend in algal density and in bottom DO as phosphate increases. (c) Explain, step by step, how higher phosphate led to lower bottom DO. (d) Classify tank 1 and tank 4 as oligotrophic-like or eutrophic-like.

COMMON STUDENT MISTAKES

- Saying the *living* algae use up the oxygen. They produce oxygen in daylight; bacteria decomposing the dead algae consume it.
- Calling green, productive-looking eutrophic water "clean".

QUICK CONCEPT SUMMARY

- A point source has one outlet; a nonpoint source is diffuse and harder to regulate.
- Pollutants push organisms outside their tolerance range; organic waste raises BOD and lowers DO (oxygen sag).
- Reefs, wetlands and mangroves are damaged by warming, sediment, development and runoff; wetlands purify, protect and shelter.
- Eutrophication: nutrients → bloom → decomposition → hypoxia → die-off; oligotrophic water is the opposite.

9.7 Groundwater Pollution

Where groundwater is and how it moves

Water that seeps down passes through the **unsaturated zone** and reaches the **water table**, the top of the **saturated zone** where every pore is full. A layer of sand, gravel or fractured rock that holds and yields water is an **aquifer**; a layer of clay that blocks water is an **aquitard**. Groundwater is rain and snowmelt that soaked into the ground; it is not an underground river. It creeps through pores, typically a few centimeters to a few meters per day, from high ground toward low ground, and it is pumped out through wells or discharges to streams and springs.

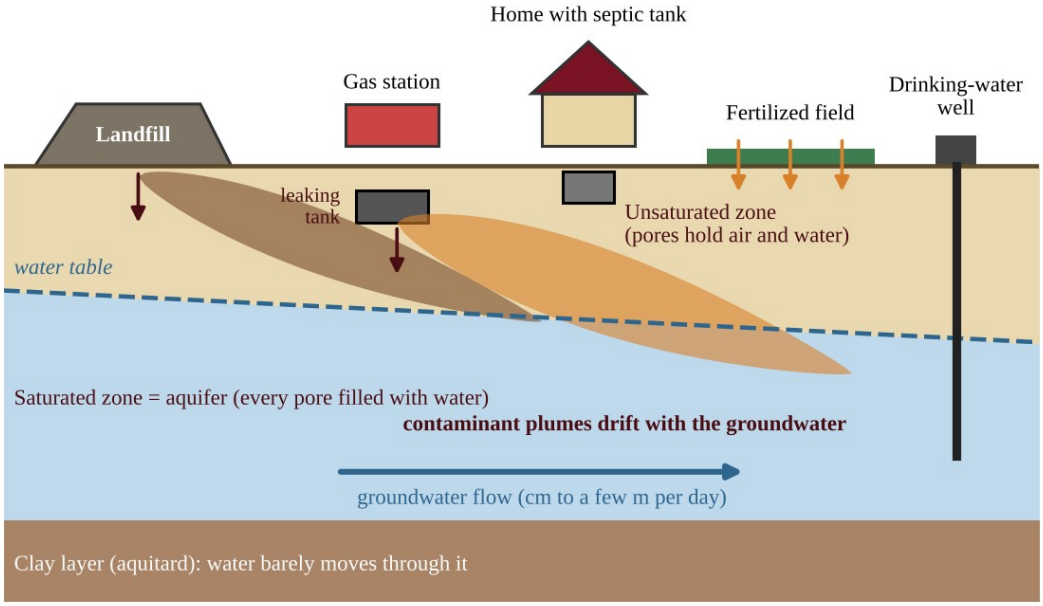


Figure 9.9 — Pollution sources above an aquifer. Contaminants seep to the water table and form plumes that drift in the direction of flow, toward the drinking-water well

Sources of groundwater pollution

- **Leaking underground storage tanks** (gasoline stations) release benzene and other fuel chemicals.
- **Landfills and dumps** produce leachate when a liner fails or is absent (Section 9.11).
- **Septic systems** that are failing or too dense add nitrate, pathogens and household chemicals (Section 9.10).
- **Fertilizers and pesticides** carry nitrate and chemicals through soil; nitrate above 10 mg/L (as nitrogen) in drinking water can cause blue-baby syndrome in infants.
- **Industrial spills and mining** add solvents and heavy metals such as lead and arsenic.
- **Road salt** and over-pumping near coasts (salt-water intrusion) make water too salty.
- **Natural contaminants** such as arsenic, fluoride and radon dissolve from some rocks.

Why groundwater is slow to clean

A river flushes a spill downstream in days. An aquifer does not, for several reasons.

1. **Slow flow.** Water moves only centimeters to meters per day, so a plume may take years to reach a well and decades to clear.
2. **Little dilution or mixing.** Water in the pores does not mix quickly, so the plume stays concentrated.
3. **Little oxygen, no sunlight, cold temperature.** Microbes that break down pollutants work slowly, and sunlight cannot break pollutants down as it does at the surface.
4. **Sticking to soil.** Many chemicals attach to soil particles and release slowly for years.
5. **Hard to reach and see.** Pollution is found only by drilling monitoring wells, and heavy liquids such as chlorinated solvents can sink and pool at the bottom of an aquifer.

PROCESS & CAUSE-AND-EFFECT

Spill or leak at the surface → seeps through the unsaturated zone → reaches the water table → forms a plume that moves with the flow → reaches a well or stream years later → source is removed but contamination lingers

Prevention and cleanup

Prevention is far cheaper than cleanup. Use double-walled tanks with leak detection, lined and capped landfills, properly sited and maintained septic systems, careful fertilizer use, protected zones around public wells, and monitoring wells. **Cleanup** options include *pump-and-treat* (pump polluted water up and clean it), air sparging and bioremediation (adding air or nutrients so microbes degrade the pollutant), permeable barriers that treat water as it passes, and *monitored natural attenuation*. All are slow and costly. Underground tanks fall under RCRA, and abandoned contaminated sites under CERCLA (Section 9.11).

KEY DEFINITIONS

Aquifer — underground layer of rock or sediment that holds and yields usable groundwater. **Water table** — the top of the saturated zone. **Plume** — a body of contaminated groundwater that spreads from a source in the direction of flow. **Leachate** — liquid that has percolated through waste and carries dissolved contaminants.

FORMULA & MATH SKILL

Travel time = distance ÷ velocity of groundwater (same length unit, e.g., m and m/day)

CLASSROOM EXAMPLE SG-C9-E002

A leaking gasoline tank lies 600 m upgradient of a town well. Groundwater in the aquifer moves toward the well at 0.25 m per day.

(a) Calculate the travel time to the well, in days.

(b) Convert your answer to years and round to the nearest 0.1 year.

(c) The tank is dug up and removed today. State whether the gasoline already in the aquifer disappears at once, and give one reason.

QUICK CONCEPT SUMMARY

- Groundwater flows slowly through pores; sources of pollution include tanks, landfills, septic systems, fertilizers, industry and natural minerals.
- It is slow to clean because of slow flow, little mixing or oxygen, sticking to soil and poor access.
- Prevent first; cleanup (pump-and-treat, bioremediation) is slow and expensive.

9.8 Water Quality—Water Testing

What we measure

Test	Unit	What it tells us	Warning sign
Temperature	°C	Oxygen capacity, thermal pollution	Warm water, low saturation DO
Dissolved oxygen (DO)	mg/L	Whether animals can breathe	Below 5 mg/L (stress); below 2 (hypoxic)
BOD_5	mg/L	Load of decomposable waste	Over about 5 mg/L in a stream; raw sewage about 200–300

Test	Unit	What it tells us	Warning sign
pH	no unit	Acidity (acid rain, mine drainage)	Outside about 6.5–8.5
Turbidity	NTU, or Secchi depth (cm)	Suspended sediment and algae	High NTU, shallow Secchi depth
Nitrate and phosphate	mg/L	Nutrient pollution	High values and algal blooms; nitrate over 10 mg/L (as N) unsafe to drink
Coliform bacteria / *E. coli*	count per 100 mL	Possible fecal contamination	Any detectable in drinking water
Conductivity or TDS	µS/cm or mg/L	Dissolved salts and ions	Sudden rise (road salt, runoff)
Macroinvertebrates	taxa count	Long-term health of the stream	Few sensitive taxa

Dissolved oxygen is read with a probe or by chemical titration (the Winkler method). It changes through the day: photosynthesis makes it highest in the afternoon and lowest just before dawn, so samples must be taken at the same time of day. **Percent saturation** compares DO with the maximum the water could hold at that temperature (Section 9.4): a reading of 6.0 mg/L at 20°C, where saturation is 9.1 mg/L, is 66% saturated.

BOD_5 is found by measuring DO in a sealed bottle, keeping it in the dark at 20°C for five days, then measuring again. The drop in DO is the oxygen used by microbes. When the sample is diluted with oxygen-rich water, multiply by the dilution factor.

FORMULA & MATH SKILL

BOD_5 (mg/L) = (DO initial − DO final) × (bottle volume ÷ sample volume) Dilution factor = bottle volume ÷ sample volume (assume the dilution water uses no oxygen) Percent saturation = DO measured ÷ DO at saturation × 100

Coliform bacteria (including *E. coli*) live in intestines. They are harmless indicators: if they are present, feces and so pathogens may be present too, and testing for them is cheaper than testing for each pathogen. **Nitrate** and **phosphate** are measured with color-change kits or lab instruments. **Turbidity** is measured with a turbidity meter or a black-and-white Secchi disk lowered until it disappears; the shallower that depth, the cloudier the water.

Living indicators

Stream animals that you can see without a microscope (**benthic macroinvertebrates**: insect larvae, snails, worms, crustaceans) integrate conditions over months. Sensitive taxa need clean, oxygen-rich water; tolerant ones survive pollution.

Group	Examples	Indicates
Sensitive	Stonefly, mayfly and caddisfly larvae	Clean, well-oxygenated water
Moderately tolerant	Crayfish, dragonfly larvae, scuds	Fair water
Tolerant	Sludge worms, midge larvae, leeches	Heavy organic pollution

Sampling well and reading results

Sample above and below a suspected source plus a reference site, take replicates, and match the pattern of results with the likely source.

Likely source	Turbidity	Nutrients (N, P)	DO	Other signs
Construction site or eroding bank	Much higher	About the same	About the same or slightly lower	Settled silt, fewer sight-feeding fish

Likely source	Turbidity	Nutrients (N, P)	DO	Other signs
Fertilizer or manure runoff	Little change	Much higher	Lower	Algal bloom; manure also adds pathogens
Raw or poorly treated sewage	Higher	Higher	Much lower	High BOD; coliforms; few sensitive species
Heated cooling water	No change	No change	Lower	Temperature clearly higher

Moving downstream from an organic-waste outfall, the readings trace the oxygen sag (Figures 9.6 and 9.7): BOD jumps at the outfall, DO falls and bottoms out farther downstream, and sensitive macroinvertebrates vanish and then return as the stream recovers.

CLASSROOM EXAMPLE SG-C9-E003

A 15 mL sample of treated effluent was placed in a 300 mL bottle and filled with oxygen-rich dilution water that uses no oxygen itself. The dissolved oxygen at the start was 8.6 mg/L. After 5 days in the dark at 20°C it was 4.4 mg/L.

(a) Calculate the dilution factor.

(b) Calculate the drop in dissolved oxygen in the bottle.

(c) Calculate the BOD_5 of the effluent, in mg/L.

(d) A clean stream has a BOD_5 of about 2 mg/L. Calculate how many times greater the effluent's BOD_5 is.

CLASSROOM EXAMPLE SG-C9-E004

Students sorted stream animals into pollution-tolerance groups with equal sampling effort at two sites.

Site	Sensitive	Moderately tolerant	Tolerant
Upstream of an outfall	48	30	22
Below the outfall	3	12	60

(a) Calculate the percent of all animals that are sensitive at each site.

(b) Calculate the percent decrease in the number of sensitive animals from the upstream site to the site below the outfall. Round to the nearest whole percent.

(c) Predict which site has the lower dissolved oxygen and justify your prediction using the data.

(d) Identify one measurement that would support your prediction and the result you would expect.

QUICK CONCEPT SUMMARY

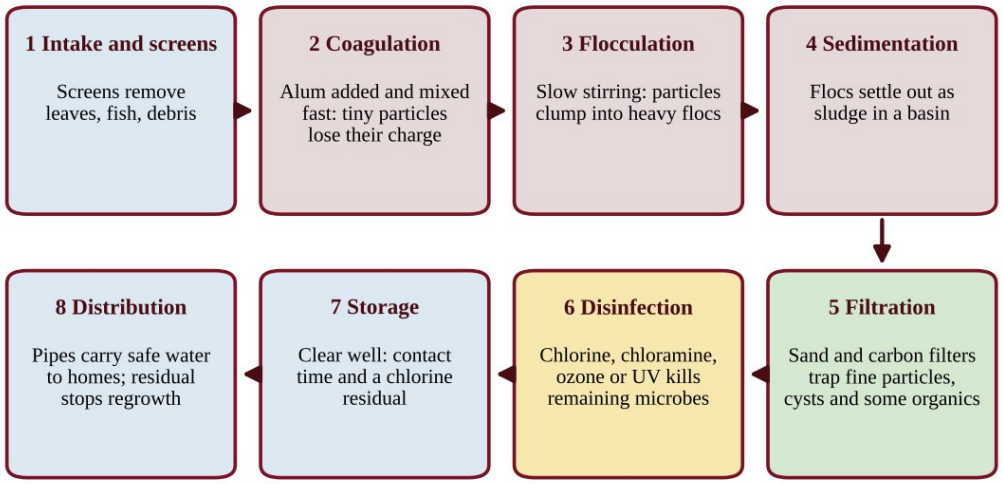
- Test temperature, DO, BOD_5, pH, turbidity, nutrients, coliforms and macroinvertebrates; together they diagnose the problem.
- BOD_5 = DO drop × dilution factor; high BOD goes with low DO downstream of a discharge.
- Sensitive macroinvertebrates indicate clean water; tolerant ones indicate pollution.

9.9 Maintaining Water Quality—Water Purification

Drinking-water treatment

Public water systems follow the **Safe Drinking Water Act**, which sets a maximum allowed level for each regulated contaminant. Surface water (rivers and lakes) is usually the dirtiest and goes through the full train in Figure 9.10; groundwater is often clearer and may need only disinfection, or softening and removal of iron, arsenic or radon.

Conventional surface-water treatment plant



Particles are removed first (steps 1 to 5); dirty water would use up the disinfectant.

Figure 9.10 — Conventional treatment of surface water: remove particles first, then kill microbes

1. **Screening.** Screens catch debris.
2. **Coagulation.** A chemical such as alum (aluminum sulfate) is mixed in fast. Tiny suspended particles carry a surface charge that keeps them apart; the coagulant cancels it.
3. **Flocculation.** Slow stirring lets the destabilized particles collide and clump into **flocs**.
4. **Sedimentation.** In a quiet basin the heavy flocs settle out as sludge.
5. **Filtration.** Water passes through sand and anthracite (and often activated carbon, which adsorbs organic chemicals, tastes and odors). Filters trap fine particles and protozoan cysts such as *Giardia* and *Cryptosporidium*.
6. **Disinfection.** Chlorine, chloramine, ozone or UV light kills bacteria and viruses.
7. **Storage and distribution.** Water sits in a clear well long enough for the disinfectant to work. A small **chlorine residual** is kept so germs cannot regrow in the pipes.

Filtration comes before disinfection because particles shield microbes and use up disinfectant. Some plants also adjust pH, add fluoride, or use **membrane filtration** with pores small enough to block bacteria.

Chlorination: benefits and costs

Chlorine is cheap and leaves a residual that protects water to the tap. Organic matter and metals use up some (the **chlorine demand**); the rest is the residual. Chlorine can form **disinfection by-products**

(trihalomethanes) with organic matter, and *Cryptosporidium* resists it, so filtration or UV is needed; ozone and UV leave no residual.

FORMULA & MATH SKILL
Chlorine dose (mg/L) = chlorine demand (mg/L) + residual (mg/L) Mass of chemical per day (kg/day) = dose (mg/L) × flow (m³/day) ÷ 1000, because 1 mg/L = 1 g/m³

Desalination and other sources

Where fresh water is scarce, **desalination** removes dissolved salt from seawater or brackish water. **Distillation** boils the water and condenses the steam. **Reverse osmosis (RO)** pushes water at high pressure through a membrane that blocks salt, against the direction in which osmosis would move it. RO uses less energy than distillation but still a lot. The fraction of feed water that becomes fresh water is the **recovery**. The rest leaves as **brine** with a higher salt concentration, which can harm marine life near the outfall.

FORMULA & MATH SKILL
Recovery (%) = fresh water out ÷ feed water in × 100 Brine volume = feed volume − fresh volume Brine concentration = (feed concentration × feed volume) ÷ brine volume, if the membrane blocks all salt

CLASSROOM EXAMPLE SG-C9-E005
A plant treats 20,000 m³ of water per day. The chlorine dose is 2.5 mg/L and the chlorine demand of the water is 1.9 mg/L.

(a) Calculate the chlorine residual, in mg/L.

(b) Calculate the mass of chlorine used per day, in kg.

(c) Explain why the plant keeps a residual in the water instead of dosing only the amount of the demand.

CLASSROOM EXAMPLE SG-C9-E006
A reverse-osmosis plant treats 100,000 m³ of seawater per day. The seawater contains 35,000 mg/L of dissolved salts, the recovery is 40%, and the membrane blocks all of the salt.

(a) Calculate the volume of fresh water produced per day.

(b) Calculate the volume of brine discharged per day.

(c) Calculate the salt concentration of the brine, in mg/L, to two significant figures.

(d) Calculate how many times saltier the brine is than the seawater, and state one way the discharge could harm marine life.

QUICK CONCEPT SUMMARY

- Drinking-water treatment: screening, coagulation, flocculation, sedimentation, filtration, disinfection, storage with a chlorine residual.
- Chlorine is cheap and leaves a residual but can form by-products; UV and ozone leave no residual.
- Desalination (distillation, reverse osmosis) supplies fresh water but needs energy and produces brine.

9.10 Septic Systems/Sewage Treatment

CONCEPT MADE SIMPLE

A sewage plant is a series of cleaning steps. First, take out what you can screen or settle. Second, let bacteria eat the dissolved waste. Third, polish the water to remove leftovers. Finally, kill germs before letting the water go.

Sewage (wastewater from toilets, sinks and industry) carries organic matter, nutrients and disease-causing microbes. Without treatment it would cause high BOD, oxygen sags, eutrophication and disease. Wastewater from sinks, showers and laundry is **gray water**; toilet waste is **black water**. Gray water can be reused for irrigation after simple treatment.

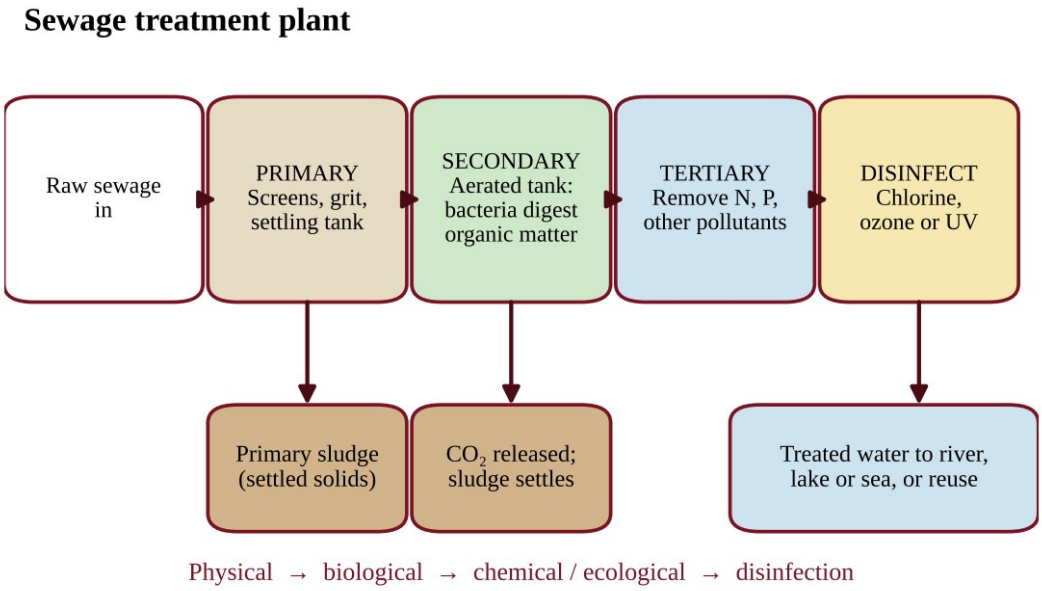


Figure 9.11 — Stages of sewage treatment from raw sewage to disinfected effluent

- **Primary treatment** is the physical removal of large objects, often through screens and grates, followed by the settling of solid waste in the bottom of a tank. It removes solids and some BOD without using biology.
- **Secondary treatment** is a biological process in which bacteria break down organic matter into carbon dioxide and inorganic sludge, which settles in a tank. The tank is **aerated** to speed the aerobic bacteria. This stage removes most of the remaining BOD.
- **Tertiary treatment** uses ecological or chemical processes to remove pollutants left after primary and secondary treatment: constructed wetlands, filters, chemical precipitation of phosphorus or biological removal of nitrogen.
- **Sludge and plastics.** Screens and settling catch most plastic litter, but some microplastics such as microbeads pass through and are swallowed by animals, and drug residues pass through. Digested sludge is often spread on farmland as **biosolids**, a cheap fertilizer that may carry heavy metals, pathogens or drug residues.
- **Disinfection.** Before discharge the water is exposed to one or more disinfectants (usually chlorine, ozone or UV light) to kill bacteria.

PROCESS & CAUSE-AND-EFFECT

Raw sewage → screens and grates remove large objects → settling tank removes solids (primary sludge) → aerated tank where bacteria digest organic matter (secondary) → sludge settles → chemical or ecological polishing removes nutrients (tertiary) → chlorine, ozone or UV kills pathogens → release or reuse

CLASSROOM EXAMPLE SG-U8-E008

A treatment plant measured BOD at three points: raw sewage entering, 240 mg/L; after the primary settling tank, 168 mg/L; after the aerated secondary tank, 21 mg/L. (a) Calculate the percent of BOD removed by primary treatment, relative to the raw sewage. (b) Calculate the percent of the remaining BOD removed by the secondary stage, relative to the water entering the secondary tank. (c) Calculate the overall percent removal. (d) Explain why the secondary stage removes BOD by a different mechanism than the primary stage, and why aeration is required.

Septic systems

About one in five U.S. homes is not connected to a sewer; many use an on-site **septic system**, which is a small version of the same idea.

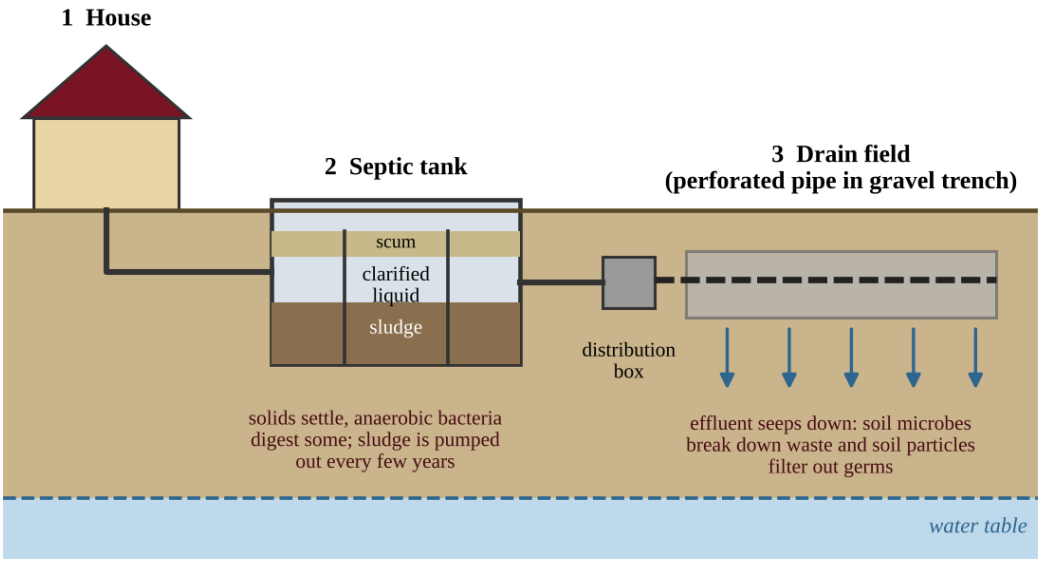


Figure 9.12 — A septic system: wastewater settles in the tank, and the clarified liquid is treated by soil in the drain field

1. Wastewater flows from the house into a buried, watertight **septic tank**. Heavy solids sink as **sludge**; grease and oil float as **scum**; anaerobic bacteria digest some of the solids.
2. The clarified liquid in the middle flows out through a distribution box into a **drain field** (leach field) of perforated pipes in gravel trenches.
3. The liquid seeps down through the soil, where aerobic microbes break down organic matter and soil particles filter out germs. By the time it reaches the water table it is much cleaner.

A septic system works only if it is **maintained**: pump the tank every few years or sludge will flow into the drain field and clog it; do not pour grease, chemicals or non-degradable materials down drains; and keep heavy vehicles off the field. Failures cause sewage to surface or reach groundwater. Septic systems also

fail where soils drain too fast (little treatment before the water table) or too slowly, where the water table is high, and where houses are packed too closely. Their nitrate and pathogens are a common source of groundwater and well pollution (Section 9.7).

CLASSROOM EXAMPLE SG-C9-E007

A household of 5 people uses 140 L of water per person per day, and all of it flows to a septic system. The soil of the drain field can absorb 25 L per square meter per day. The septic tank holds 3,000 L.

(a) Calculate the daily wastewater flow, in L.

(b) Calculate the drain-field area needed, in m².

(c) Calculate the average time wastewater stays in the tank (tank volume ÷ daily flow), in days, to the nearest 0.1 day.

QUICK CONCEPT SUMMARY

- Primary: physical (screens, settling). Secondary: aerated bacteria convert organics to CO_2 and sludge. Tertiary: chemical or ecological removal of leftovers such as nutrients. Disinfection (chlorine, ozone, UV) kills pathogens.
- A septic system uses a settling tank and a soil drain field; it needs regular pumping and good soil.
- Percent removal = (in − out) ÷ in × 100.

9.11 Municipal Solid Waste

CONCEPT MADE SIMPLE

Throw something "away" and it does not vanish; it goes somewhere. Most places bury it in a lined landfill, some burn it, and some dump it where they should not. Each choice trades one problem for another: buried trash can leak liquid and gas; burned trash makes smoke and ash. The best trash is trash that never exists.

Solid waste is any discarded material that is not a liquid or a gas, generated in domestic, industrial, business and agricultural sectors. **Municipal solid waste (MSW)** is what households, businesses and institutions throw away: paper, food and yard waste, plastics, metals, glass, wood and textiles. Paper and paperboard is the largest category by weight, followed by food scraps, yard waste and plastics. A person in the United States throws away about 2 kg a day. **Electronic waste (e-waste)** is discarded devices such as televisions, cell phones and computers; it holds valuable metals but also hazardous lead and mercury. Some is exported to countries with cheaper labor and weaker rules, where crude recycling exposes workers and pollutes.

Landfills

Solid waste is most often disposed of in landfills. A **sanitary municipal landfill** is engineered to limit harm. It has a **bottom liner** (plastic or clay), a **storm-water collection system**, a **leachate collection system**, a **cap** (cover) and a **methane collection system**. It is sited on low-permeability soil, well above the water table and away from streams, wells and floodplains.

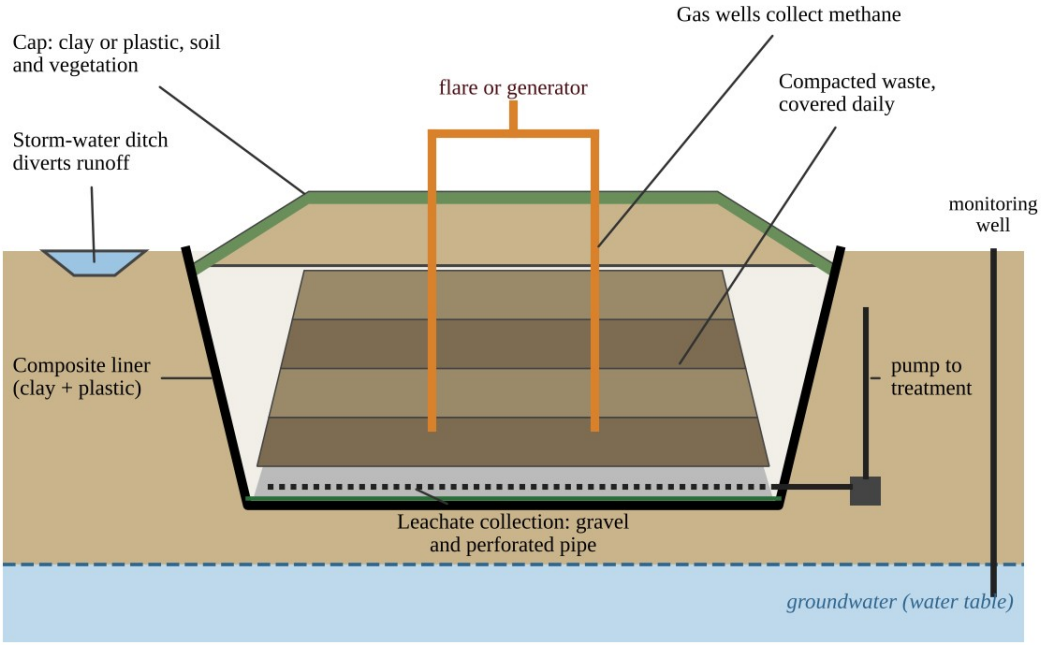


Figure 9.13 — A sanitary municipal landfill: the liner and leachate system protect groundwater, the cap keeps rain out and gas wells collect methane

PROCESS & CAUSE-AND-EFFECT

Rain falls on the waste → water seeps down and picks up dissolved contaminants (leachate) → liner stops it reaching groundwater → collection pipes pump it to treatment. Meanwhile microbes decompose organic waste → methane and CO_2 form → gas wells collect the methane for flaring or energy

Methane is a potent greenhouse gas and can explode, so it is collected. Landfills can still contaminate groundwater and release harmful gases when liners fail or when they predate modern standards. Crushed waste releases more contaminants than large pieces because more surface area is exposed. **Factors in decomposition** are the **composition of the trash** and the **conditions microbes need**: moisture, temperature and oxygen. Tightly compacted, covered waste is deprived of oxygen and moisture, so even paper and food can last for decades and plastics far longer.

Incineration and illegal dumping

- **Incineration** burns waste at high temperatures. It cuts the volume by about 90%, but releases air pollutants (particulates, NO_x, SO_2, sometimes toxic metals and dioxins) and leaves ash that must be disposed of, often as hazardous waste. When the heat generates electricity it is called waste-to-energy.
- **Illegal dumping.** Items not accepted in sanitary landfills may be dumped illegally. Used tires left in piles breed mosquitoes that spread disease, and tire piles can burn and release toxic smoke.
- **Ocean dumping.** Some countries dump waste at sea. With other sources of plastic, this has formed large floating islands of trash. Wildlife become entangled in the waste and eat it. Plastic persists for decades to centuries and only breaks into smaller pieces, so cutting plastic at its source (using less, reusing, recycling incentives) works better than cleaning up afterward.

Hazardous waste (toxic, flammable or corrosive: batteries, paint, motor oil, pesticides) needs special handling. **RCRA** regulates hazardous waste from creation to disposal ("cradle to grave"); **CERCLA** (Superfund) pays for and organizes cleanup of abandoned hazardous waste sites.

KEY DEFINITIONS

Solid waste — discarded material that is not a liquid or a gas. **E-waste** — discarded electronic devices such as televisions, phones and computers. **Leachate** — liquid that has percolated through waste and carries dissolved contaminants.

Waste reduction methods

Work up the ladder: **reduce** and **reuse** first, then **recycle** and **compost**, then recover energy, and bury only what is left. **Source reduction** (for example, less packaging) is preferred because the waste never has to be collected or processed.

- **Recycling** processes waste materials into new products. It reduces demand for minerals (aluminum, copper) and usually takes far less energy than mining and refining new aluminum, but collection, sorting and reprocessing are energy-intensive and costly.
- **Composting** is the decomposition of organic matter such as food scraps, paper and yard waste. The product serves as fertilizer or a soil conditioner. Drawbacks are odor and rodents, which good management reduces.
- **E-waste** is reduced by recycling and reuse. Its heavy metals such as lead and mercury can leach from landfills into groundwater if not disposed of properly.
- **Landfill mitigation** ranges from burning waste for energy to restoring habitat on former landfills for use as parks. Burning landfill gas from decomposing organic matter to turn turbines generates electricity and reduces landfill volume.

Method	Main benefit	Main drawback
Landfill	Cheap; methane can be captured	Land use; leachate and gas risks
Incineration	Volume cut about 90%; energy	Air pollutants; toxic ash
Recycling	Saves minerals and landfill space	Energy-intensive, costly
Composting	Makes fertilizer; diverts organics	Odor, rodents

Calculations

FORMULA & MATH SKILL

Per capita waste = total mass of waste ÷ number of people (kg per person per day or per year) Percent diversion = (mass recycled + mass composted) ÷ total mass generated × 100 Volume (m^3) = mass (kg) ÷ density (kg/m^3) Landfill life (years) = remaining capacity ÷ amount buried per year (same units) Electricity from landfill gas (MJ) = gas volume (m^3) × energy content (MJ/m^3) × efficiency (decimal); kWh = MJ ÷ 3.6

Capacity is a volume but waste is a mass, so use the density of the compacted waste. *Illustration.* 30,000 t/yr at 500 kg/m^3 is 3.0×10^7 kg ÷ 500 = 6.0×10^4 m^3/yr; with 1.2×10^6 m^3 left, life = 20 years. Per capita: 14,600 t × 1000 kg/t ÷ 20,000 people = 730 kg per person per year (2.0 kg per day).

CLASSROOM EXAMPLE SG-U8-E006

A city of 125,000 people sends 82,125 metric tons of solid waste to its landfill in one year (365 days). (a) Calculate the waste per person per day, in kilograms. (b) The city sets a goal of reducing the per capita rate by 20%. Calculate the goal in kilograms per person per day and the annual mass of waste, in metric tons, if the population stays the same. (c) Name two methods, one for plastics and metals and one for food scraps, that could help meet the goal.

CLASSROOM EXAMPLE SG-U8-E007

A county landfill has 1,200,000 metric tons of remaining capacity. At present 40,000 t of waste is buried each year. The county considers a plan that would recycle 25% and compost 15% of the 40,000 t generated each year, burying only the remainder. (a) Calculate the percent diversion under the plan. (b) Calculate the number of years the landfill will last under the current practice and under the plan. (c) Describe one benefit and one drawback of the plan other than landfill life.

CLASSROOM EXAMPLE SG-U8-E013

A town of 80,000 people produces 1.6 kg of solid waste per person per day. It recycles 20% and composts 10%; the rest goes to a landfill and is compacted to 550 kg/m³. The landfill has 2.0×10^6 m³ of space left. (a) Calculate the mass landfilled in one year (365 days), in metric tons. (b) Calculate the volume this occupies, in m³ per year. (c) Calculate how many years the landfill will last. (d) The landfill also captures 4.0×10^5 m³ of methane per year at 36 MJ/m³ and burns it in a generator that is 35% efficient. Calculate the electricity produced, in kWh per year.

COMMON STUDENT MISTAKES

- Treating all diversion as free. Recycling and composting need energy, money and clean sorting.
- Dividing in the wrong order for per capita (waste ÷ people, not people ÷ waste).
- Forgetting to convert tons to kilograms or years to days.

QUICK CONCEPT SUMMARY

- A sanitary landfill has a liner, storm-water system, leachate system, cap and methane collection; incineration cuts volume but releases air pollutants; illegal and ocean dumping cause serious harm.
- Reduce and reuse first; recycle, compost, recover energy; landfill last.
- Recycling saves minerals but is energy-intensive; composting makes fertilizer but can cause odor and rodents; e-waste contains lead and mercury.
- Per capita = waste ÷ people; diversion % = diverted ÷ generated × 100; landfill life = capacity ÷ annual volume.

9.12 Practice Questions and Answers

Welcome to the practice set for Chapter 9: work through the classroom examples and the question sets below to turn what you have read into exam skill.

PRACTICE QUESTIONS AND ANSWERS

Classroom examples in this chapter (unsolved; space is provided in each box): SG-U7-E010, SG-U7-E001, SG-U7-E003, SG-U7-E012, SG-U7-E013, SG-U7-E006, SG-C9-E001, SG-U7-E002, SG-U7-E011, SG-U7-E004, SG-U7-E005, SG-U7-E007, SG-U7-E014, SG-U7-E008, SG-U8-E004, SG-U7-E009, SG-U8-E001, SG-U8-E002, SG-U8-E011, SG-U8-E012, SG-U8-E003, SG-C9-E002, SG-C9-E003, SG-C9-E004, SG-C9-E005, SG-C9-E006, SG-U8-E008, SG-C9-E007, SG-U8-E006, SG-U8-E007, SG-U8-E013.

Classified practice: Book Two, Chapter 9 — 64 multiple-choice questions (Section A) followed by 7 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 9. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

CHAPTER
10

UNIT VI — POLLUTION
Impacts on the Environment and Human Health

10.1 · Hazards to Human Health
10.2 · Dose-Response Relationships
10.3 · Hazardous Chemicals in the Environment
10.4 · Treatment, Disposal, and Cleanup of Contaminated Sites
10.5 · Perpetual Storage
10.6 · Bioaccumulation—Biomagnification
10.7 · Environmental Economics
10.8 · Sustainability—Resiliency
10.9 · Practice Questions and Answers

LEARNING OBJECTIVES

- Link pollution and pathogens to human health problems and explain how scientists test those links.
- Use LD50 and dose-response curves to compare toxicity and to calculate doses.
- Explain why endocrine disruptors and persistent organic pollutants (POPs) harm ecosystems, and calculate bioaccumulation and biomagnification.
- Describe how hazardous waste, contaminated sites, and nuclear waste are treated, stored, and cleaned up.
- Use basic environmental economics (externalities, cost-benefit analysis, discounting) to judge pollution policies.
- Explain sustainability and resilience, and read indicators of whether we are moving toward or away from them.

10.1 Hazards to Human Health

CONCEPT MADE SIMPLE

Many things can make people sick: air, water, food, jobs, genes, habits. Scientists untangle them by comparing exposed groups with similar unexposed groups. Germs are stopped by breaking their route from one host to the next.

A **hazard** is anything that can cause harm: **chemical** (toxins, pollutants), **physical** (radiation, noise), **biological** (pathogens), or **cultural** (smoking, diet, occupation). **Risk** is the chance that harm actually happens, so it depends on the hazard *and* on exposure. A very toxic chemical sealed in a drum has low risk; a mildly toxic one in daily drinking water may have high risk. **Acute** exposure is short and intense; **chronic** exposure is long and low-level and may cause disease only after a delay called the **latency period**.

Pollution and health

It can be hard to establish cause and effect between a pollutant and a health problem because people are exposed to many chemicals at once.

Health issue	Linked pollutant or cause	Key facts
Dysentery	Untreated sewage in streams and rivers	Intestinal infection with severe diarrhea; spreads when drinking water is contaminated with feces

Health issue	Linked pollutant or cause	Key facts
Mesothelioma	Asbestos	A cancer of the lining of the lungs and chest, caused mainly by inhaled asbestos fibers (old insulation and building materials); appears decades after exposure
Respiratory problems	Elevated tropospheric (ground-level) ozone	Ozone irritates the airways and reduces lung function, worsening asthma and other lung disease

Research methods

A **controlled experiment** (lab animals or cell cultures) changes one variable, the dose, while everything else is held constant. For people, researchers use **epidemiological studies** that compare disease rates among groups with different exposures and adjust statistically for age, smoking, and other factors. These show *correlation*. Evidence for *causation* builds up from a dose-response pattern (more exposure, more disease), a plausible biological mechanism, and consistent results in several studies. Large samples and control groups are essential.

KEY DEFINITIONS

Risk — the probability that a hazard causes harm; depends on toxicity and exposure. **Epidemiology** — the study of how often diseases occur in different groups and what is associated with them. **Latency period** — the delay between exposure and the appearance of disease.

Pathogens and infectious diseases

A **pathogen** is an organism or virus that causes disease. A **vector** is an animal, such as a mosquito, that carries a pathogen from one host to another. Pathogens **adapt** to take advantage of new opportunities to infect and spread through human populations (crowding, travel, land-use change, contact with animals). Specific pathogens occur in many environments **regardless of the appearance of sanitary conditions**: clear-looking water can carry bacteria, so water must be tested or treated, not judged by eye.

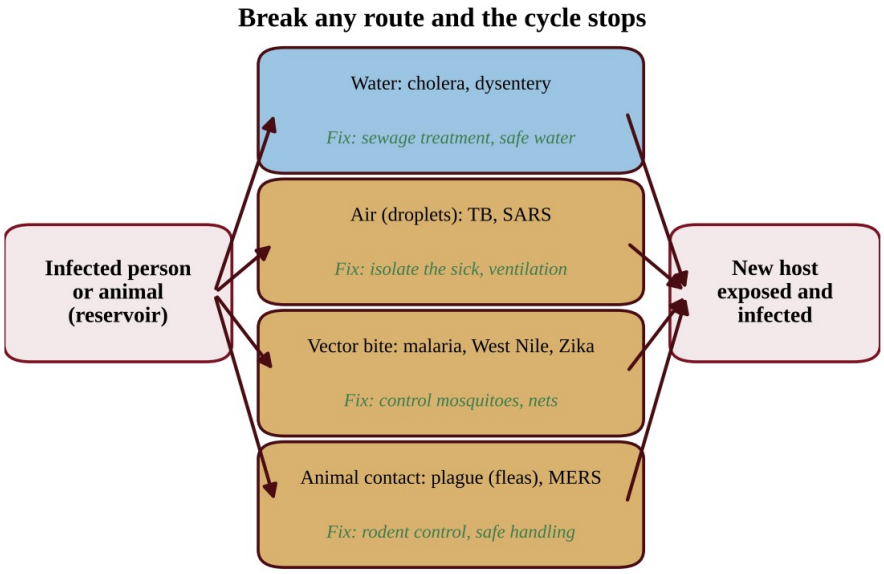


Figure 10.1 — Routes by which pathogens pass from an infected reservoir to a new host, with the public-health measure that breaks each route.

Climate. As equatorial-type climate zones spread north and south into what are now subtropical and temperate zones, pathogens, infectious diseases, and their vectors are spreading into areas where the disease was not previously known. Warmer conditions let mosquitoes survive in new regions for more months of the year.

Poverty. Poverty-stricken, low-income areas often lack sanitary waste disposal and have contaminated drinking water, creating havens and opportunities for infectious diseases to spread. Crowding, poor nutrition, and flooding (which mixes sewage with drinking water) add to the problem. Clean water and sewage treatment are among the most powerful public-health measures.

Disease	Type of pathogen	Spread / source
Plague	Bacterium	Carried by infected animals such as rodents and their fleas; transferred to humans by the bite of an infected organism or contact with contaminated fluids or tissues
Tuberculosis (TB)	Bacterium	Typically attacks the lungs; spread by breathing in bacteria from the bodily fluids of an infected person
Malaria	Parasite	Bites from infected mosquitoes; most often found in sub-Saharan Africa
West Nile virus	Virus	Bites from infected mosquitoes
SARS	Virus; a form of pneumonia	Transferred by inhaling or touching infected fluids
MERS	Virus; respiratory illness	Transferred from animals to humans
Zika	Virus	Bites from infected mosquitoes; can also be transmitted through sexual contact
Cholera	Bacterium	Contracted from infected water

PROCESS & CAUSE-AND-EFFECT

Infected host sheds the pathogen (feces, fluids, blood) → pathogen enters the environment (water, air, or a vector) → new person exposed (drinking, breathing, bite, contact) → infection → more shedding. Break any arrow and the cycle stops: treat sewage, purify water, control vectors, isolate the sick.

COMMON STUDENT MISTAKES

- Assuming clean-looking water is free of pathogens.
- Saying a correlation between a pollutant and a disease *proves* the pollutant caused it.

CLASSROOM EXAMPLE SG-U8-E010

Three villages along one river were surveyed for cholera during one rainy season:

Village	Drinking water	Sewage handling	Cases per 10,000 people
A	Chlorinated, from a protected well	Latrines connected to a treatment plant	2
B	Untreated river water	Latrines connected to a treatment plant	14
C	Untreated river water	Open disposal upstream of the intake	61

(a) Calculate how many times higher the case rate is in village C than in village A. (b) Make a claim about the factor most strongly associated with the difference between A and B, and support it with data. (c) Explain the transmission route that links open sewage disposal to cases in village C. (d) State one reason this survey alone cannot prove that sewage handling caused the difference in cases, and propose one low-cost intervention for village C.

QUICK CONCEPT SUMMARY

- Hazards are chemical, physical, biological, or cultural; risk depends on hazard and exposure.
- Untreated sewage causes dysentery, asbestos causes mesothelioma, and ground-level ozone harms the lungs. Cause and effect is hard to prove, so studies use controls, large samples, and dose-dependent patterns.
- Pathogens adapt, occur even in clean-looking places, spread into new regions as climate zones shift, and thrive where poverty means poor sanitation. Know each disease's route.

10.2 Dose-Response Relationships

CONCEPT MADE SIMPLE

"The dose makes the poison." To compare poisons, scientists ask how much it takes to kill half of a test group: the LD50. Tiny LD50 means very toxic. A dose-response curve shows the whole picture: dose on the x-axis, how many are affected on the y-axis.

LD50

Lethal dose 50% (LD50) is the dose of a chemical that is lethal to 50% of the population of a particular species. It is found in controlled tests on groups of test animals given different doses and is usually reported in milligrams of chemical per kilogram of body mass (mg/kg) so that animals of different sizes can be compared.

FORMULA & MATH SKILL

Total dose (mg) = dose (mg/kg) × body mass (kg) Times more toxic = larger LD50 ÷ smaller LD50 The **lower** the LD50, the **more toxic** the chemical. Units: mg of chemical per kg of body mass (mg/kg). Convert grams to kilograms first.

Illustration. An LD50 of 40 mg/kg in rats: a 0.25 kg rat given 10 mg has received 10 ÷ 0.25 = 40 mg/kg, so about half of such rats would die.

How many times more toxic? Divide, do not subtract: LD50 of 12 mg/kg (A) versus 480 mg/kg (B) gives 480 ÷ 12 = 40, so A is 40 times more toxic.

The LD50 measures death only (not illness or cancer), applies to one species and one exposure route, and says nothing about long-term low-level exposure. It is not a safe dose: some animals die below it. A rat LD50 cannot simply be applied to humans, because body size, metabolism, and exposure route differ, so animal tests give only estimates (testing people is unethical).

The dose-response curve

A **dose-response curve** describes the effect on an organism, or the mortality rate in a population, based on the dose of a particular toxin or drug. Dose goes on the horizontal axis (often logarithmic, because doses span a wide range) and response (percent affected or dying) on the vertical axis.

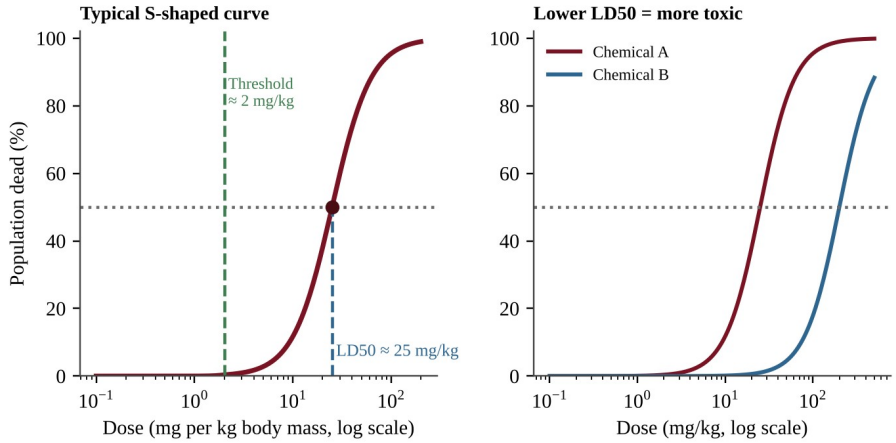


Figure 10.2 — Left: an S-shaped dose-response curve with a threshold dose and the LD50 marked. Right: the chemical whose curve lies farther left (lower LD50) is more toxic.

Reading the curve

1. **Flat start (threshold).** At very low doses there may be no measurable effect. The **threshold dose** is the highest dose with no effect above the control group's background response (a few control animals may die anyway). Some substances, such as some carcinogens, may have no safe threshold in practice.
2. **Steep middle, then plateau.** Over a narrow range the response climbs rapidly, then levels off near 100%.
3. **LD50.** Draw a horizontal line from 50% on the y-axis to the curve, then drop down to the x-axis.
4. **Comparing chemicals.** A curve shifted left reaches 50% at a smaller dose, so that chemical is more toxic. A steeper curve leaves a narrow gap between "no effect" and "high effect."

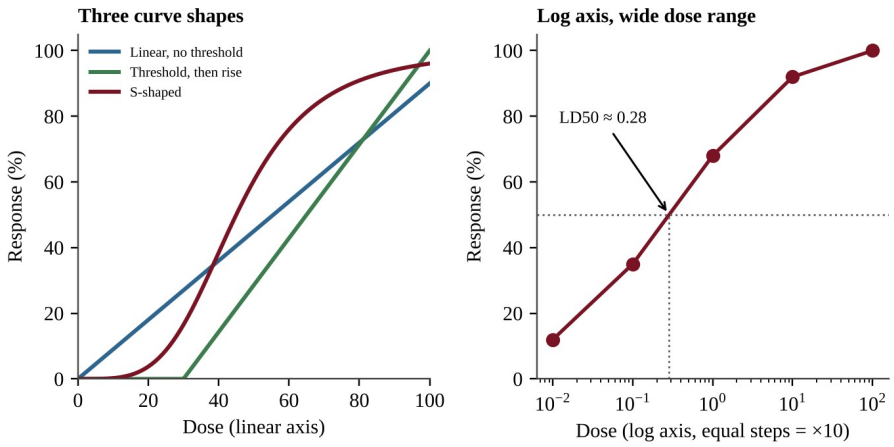


Figure 10.3 — Left: three curve shapes on a linear axis. Right: the same kind of data on a log dose axis, with the LD50 found by interpolation.

- **Shapes.** A *threshold* curve is flat until a dose is reached. A *linear, no-threshold* curve rises from the first dose (the working assumption for many carcinogens). Most mortality curves are S-shaped. A curve can also fall: if a drug *reduces* an effect as concentration rises, the relationship is inverse, and "the dose makes the poison" does not describe it. Read the axis labels first.
- **Log axis.** Each equal gap means multiplication by 10. To interpolate an LD50 between two plotted doses, interpolate the exponent: if 35% die at 0.1 and 68% at 1, then 50% is (50 − 35) ÷ (68 − 35) = 0.45 of the way from exponent −1 to 0, so the exponent is −0.55 and the dose is $10^{-0.55} \approx 0.28$.
- **Linear interpolation.** LD50 ≈ low dose + [(50 − low response) ÷ (high response − low response)] × (high dose − low dose).
- **Expected deaths.** Percent × number exposed (30% of 40,000 is 12,000).

Setting up a toxicity test. Expose several identical groups of one species to different doses, plus a **control group** that receives none. Keep temperature, age, food, and exposure time the same. Dose is the independent variable; percent responding is the dependent variable.

KEY DEFINITIONS

LD50 — the dose of a chemical lethal to 50% of a test population of one species. **Dose-response curve** — a graph of the effect on an organism, or mortality in a population, versus the dose of a toxin or drug. **Threshold dose** — the highest dose at which no effect is observed.

COMMON STUDENT MISTAKES

- Reading the dose on the y-axis. Dose is on the x-axis; response is on the y-axis.
- Looking for 50 on the x-axis to find the LD50. The 50 is on the y-axis.
- Saying the curve farther right is more toxic. It is less toxic.

AP EXAM STRATEGY

To compare toxicity, quote the LD50 values (smaller means more toxic). On graphs, draw reading lines on the figure and give the value with units.

CLASSROOM EXAMPLE SG-U8-E009

Two insecticides were tested on separate groups of identical laboratory mice, with the following percent mortality after 24 hours:

Insecticide X dose (mg/kg)	5	10	20	40	80
Mice dead (%)	8	22	46	72	94

Insecticide Y dose (mg/kg)	100	200	400	800
Mice dead (%)	30	52	80	96

(a) For each insecticide, estimate the LD50 by linear interpolation between the two tested doses whose mortalities bracket 50%. Give the answer to the nearest 1 mg/kg. (b) State which insecticide is more toxic and justify your answer using the LD50 values. (c) Using your rounded LD50 for insecticide X, calculate the total dose, in mg, for a mouse of body mass 0.025 kg. (d) Give one limitation of using LD50 to decide whether either insecticide is safe in a river.

CLASSROOM EXAMPLE SG-U8-E014

Brine shrimp (100 per dish) were exposed to a salt of a heavy metal. A control dish received no metal. After 48 hours:

Concentration (mg/L)	Shrimp dead (of 100)
0 (control)	3
0.01	8
0.1	24
1	61
10	93
100	100

(a) Explain why a log scale is appropriate for the concentration axis and describe how you would set up the axes. (b) Estimate the concentration that kills 50% by interpolating the exponent between the two doses that bracket 50%. (c) Explain why the control dish matters.

QUICK CONCEPT SUMMARY

- LD50 is the dose lethal to 50% of a test population; lower LD50 means more toxic, and total dose = mg/kg × body mass.
- A dose-response curve is typically S-shaped with a threshold and a plateau; a curve farther left is more toxic.
- Risk depends on both toxicity (the curve) and exposure.

10.3 Hazardous Chemicals in the Environment

CONCEPT MADE SIMPLE

Hormones are the body's chemical text messages. An **endocrine disruptor** sends fake messages or blocks real ones, so cells do the wrong thing even at very low concentrations. **POPs** are like plastic: man-made molecules microbes cannot break down. They last for years, dissolve in fat so animals store them, and ride wind and ocean currents far from where they were used.

Endocrine disruptors

The **endocrine system** is the set of glands that release hormones controlling growth, development, metabolism, and reproduction. **Endocrine disruptors** are chemicals that can interfere with the endocrine system of animals. Some mimic natural hormones (for example, imitating estrogen), some block hormone receptors, and some change how much hormone the body makes. Examples include certain pesticides, PCBs, some plastics chemicals, and medication residues in sewage effluent.

Because hormones control development and reproduction, disruption shows up there first. Endocrine disruptors can lead to **birth defects, developmental disorders, and gender imbalances** in fish and other species. In a documented pattern, male fish downstream of wastewater outfalls develop female traits, which skews sex ratios, lowers breeding success, and can shrink the population. Amphibians absorb chemicals through thin skin and develop in water, so they are especially vulnerable. If the share of affected males is highest at the outfall and falls downstream, the effluent is the likely source, but chemical analysis is needed to name the chemical.

Persistent organic pollutants (POPs)

POPs do not easily break down in the environment because they are synthetic, carbon-based molecules (such as DDT and PCBs). Decomposers have no enzymes evolved for these structures, so POPs remain in soil, sediment, and water for years or decades. "Organic" here means carbon-based, not "grown without pesticides."

POPs can be toxic to organisms because they are **soluble in fat** (lipid-soluble) rather than water. Water-soluble chemicals are excreted in urine; fat-soluble ones are retained and accumulate in fatty tissues such as body fat, liver, and eggs (Section 10.6).

POPs can **travel over long distances** via wind and water before being redeposited. They evaporate in warm regions, ride air currents, and condense in cold regions, so they contaminate places where they were never used, such as Arctic wildlife and the people who eat it.

KEY DEFINITIONS

Endocrine disruptor — a chemical that interferes with the endocrine system of animals by imitating, blocking, or altering hormone signals. **POP** — a synthetic, carbon-based pollutant that resists breakdown, dissolves in fat, and can be transported long distances. Examples: DDT, PCBs.

QUICK CONCEPT SUMMARY

- Endocrine disruptors interfere with animals' hormones, causing birth defects, developmental disorders, and gender imbalances in fish and other species.
- POPs are persistent (synthetic, carbon-based), fat-soluble (stored in fatty tissue), and mobile (long-distance transport by wind and water).

10.4 Treatment, Disposal, and Cleanup of Contaminated Sites

CONCEPT MADE SIMPLE

Hazardous waste needs special handling from the day it is made until it is made harmless or sealed away. When land or groundwater is already contaminated, we can pump the water out and clean it, let microbes or plants do the work, or dig up the soil. The cheapest cleanup is not making the waste.

Hazardous waste and the laws that govern it

Hazardous waste is toxic, corrosive, flammable, or reactive (solvents, pesticides, acids, heavy-metal sludge, used oil, lead batteries). Two U.S. laws are central:

Law	What it does
RCRA (Resource Conservation and Recovery Act, 1976)	Regulates hazardous waste from **cradle to grave**: generator, transporter, treatment, storage, and disposal facility must be tracked and permitted. Also sets rules for ordinary landfills. Prevents *new* problems.
CERCLA (Comprehensive Environmental Response, Compensation, and Liability Act, 1980, "**Superfund**")	Cleans up *old* contaminated sites. Creates a trust fund and makes the responsible parties pay (**strict, joint, and retroactive liability**: owners, operators, generators, and transporters can all be charged, whatever the date of the dumping). Sites are ranked on the National Priorities List.

Treating and disposing of hazardous waste

The preferred order is: **reduce** (use less toxic materials), **reuse or recycle** (recover solvents or metals), **treat** (neutralize acids, use microbes, incinerate at high temperature), and only then **dispose**. Disposal options include **secure hazardous-waste landfills** with double liners and leachate collection, **deep-well injection** into rock far below drinking-water aquifers, and lined **surface impoundments**. Toxic metals such as lead and mercury are elements: they never decay or biodegrade, so they must be isolated indefinitely.

Landfill liners and leachate

Rain that percolates through buried waste picks up dissolved chemicals and becomes **leachate**, a toxic liquid that can contaminate groundwater. A modern landfill sits on a **liner** of compacted clay topped by a plastic **geomembrane**, with a drainage layer and pipes that collect leachate for treatment. Wells monitor the groundwater. Waste is covered with soil daily, a final cap seals the top, and pipes collect methane (a greenhouse gas) that can be burned for energy.

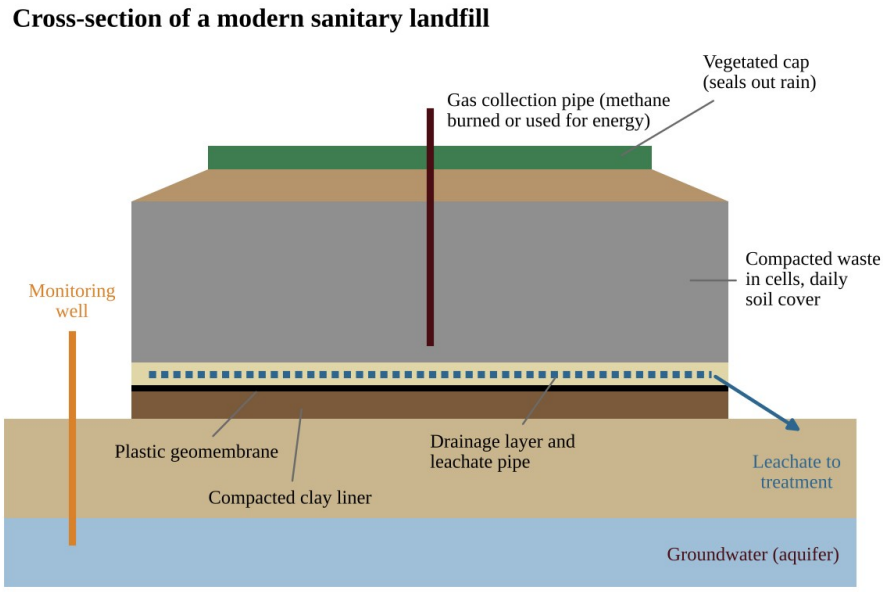


Figure 10.4 — Cross-section of a modern sanitary landfill: cap, waste cells, gas collection, leachate collection above a geomembrane and clay liner, and a groundwater monitoring well.

Cleaning up contaminated sites

A cleanup starts with **site assessment**: sampling soil, water, and air to find the contaminant, how far the **plume** (the moving cloud of contaminated groundwater) has spread, and who might be exposed. Then a method is chosen.

- **Pump-and-treat.** Wells pull contaminated groundwater to the surface, where carbon filters or air strippers remove the chemical before the water is returned or discharged. It also keeps the plume from spreading. It is reliable but slow, because contaminant stuck to soil keeps releasing; it may run for decades.
- **Bioremediation.** Microbes break down contaminants (oil, solvents, fuels) into less harmful products; adding oxygen and nutrients speeds them up. It works only for biodegradable chemicals.
- **Phytoremediation.** Plants take up, store, or break down contaminants through their roots (some take up heavy metals, which are then harvested). It is low-cost and gentle on the land but slow and limited by root depth.
- **Excavation and capping.** Soil is dug out and hauled to a secure landfill, or covered with a cap. It is fast but expensive and moves the problem.

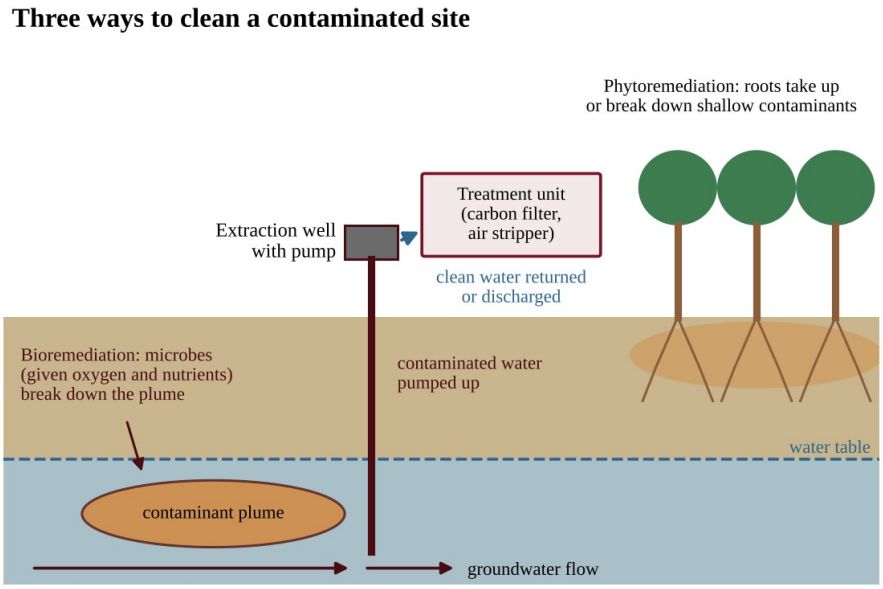


Figure 10.5 — Three cleanup approaches: pump-and-treat for a groundwater plume, bioremediation, and phytoremediation.

Brownfields are abandoned or underused industrial and commercial properties where reuse is complicated by real or suspected contamination (old factories, gas stations, dry cleaners). Redeveloping them reduces sprawl, restores tax revenue, and uses land that already has roads and utilities.

FORMULA & MATH SKILL

Mass removed = volume treated × concentration Mass (mg) = volume (L) × concentration (mg/L); then ÷ 1,000,000 for kg (1 m³ = 1,000 L) Percent of contaminant removed = mass removed ÷ total mass × 100

REAL-WORLD EXAMPLE

Love Canal. In the 1970s, homes at Love Canal, New York, were found on buried chemical waste. The outcry helped lead to CERCLA (Superfund) in 1980.

KEY DEFINITIONS

Leachate — liquid that has percolated through waste and carries dissolved contaminants. **Plume** — a spreading body of contaminated groundwater. **Brownfield** — an abandoned or underused site whose reuse is complicated by contamination.

COMMON STUDENT MISTAKES

- Mixing up the two laws: RCRA manages waste being made now (cradle to grave); CERCLA cleans up old sites and makes polluters pay.
- Thinking a liner makes a landfill leak-proof. Liners slow leachate; they need collection and monitoring.
- Assuming one cleanup method fits everything. Bioremediation fails on non-biodegradable chemicals; phytoremediation fails on deep contamination.

AP EXAM STRATEGY

For "propose a solution," match the method to the problem (groundwater plume: pump-and-treat; oil or fuel: bioremediation; shallow, light contamination: phytoremediation) and add prevention or monitoring.

CLASSROOM EXAMPLE SG-C10-E001

A pump-and-treat system removes contaminated groundwater at 25 L per minute. The water contains 40 mg/L of benzene, and the treatment unit removes all of it. Estimate that the plume holds 3,000 kg of benzene in total. (a) Calculate the mass of benzene removed in one year (365 days), in kilograms. (b) Calculate the percent of the 3,000 kg removed in one year, to the nearest 0.1%. (c) Explain why the real cleanup will take longer than the simple division suggests.

QUICK CONCEPT SUMMARY

- RCRA tracks hazardous waste from cradle to grave; CERCLA (Superfund) cleans up old sites and makes responsible parties pay.
- Liners, leachate collection, gas collection, caps, and monitoring wells make landfills safer.
- Cleanup methods: pump-and-treat, bioremediation, phytoremediation, excavation and capping; brownfields are contaminated sites reused after cleanup.

10.5 Perpetual Storage

CONCEPT MADE SIMPLE

Some wastes stay dangerous longer than any human institution has lasted. We cannot make high-level nuclear waste harmless, so we isolate it deep underground and keep watch for thousands of years.

High-level nuclear waste

Nuclear power plants produce **spent nuclear fuel** (used fuel rods), a **high-level radioactive waste**. It is intensely radioactive and hot at first, so it is stored in water-filled pools for some years and then often in dry steel-and-concrete casks. Its radiation damages tissue and DNA, causing cancer and genetic damage. Lower-level waste (gloves, tools, medical waste) is buried in shallow, controlled sites.

Half-life

A radioactive isotope decays at a fixed rate. Its **half-life** is the time for half of the atoms to decay. After one half-life 50% remains, after two 25%, after three 12.5%. After about ten half-lives only about 0.1% remains. Isotopes with long half-lives therefore set the storage time.

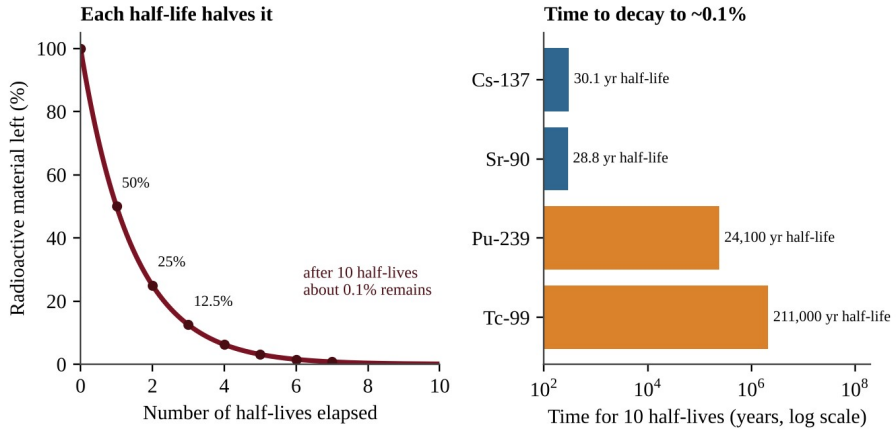


Figure 10.6 — Left: radioactive decay, fraction remaining against number of half-lives. Right: time for ten half-lives for four isotopes found in nuclear waste (log scale).

FORMULA & MATH SKILL

Amount remaining = initial amount × (½)^n, where n = elapsed time ÷ half-life Time to decay to a given fraction: t = half-life × log(initial ÷ remaining) ÷ log 2 Units: any unit of amount (kg, Ci, or %); time and half-life in the same unit (years).

Strontium-90 and cesium-137 (half-lives about 29 and 30 years) fade within a few centuries. Plutonium-239 (about 24,100 years) and technetium-99 (about 211,000 years) stay hazardous for tens to hundreds of thousands of years, far longer than recorded human history.

Repositories and long-term stewardship

The accepted plan is a **deep geologic repository**: tunnels in stable rock hundreds of meters underground, in a dry place with little groundwater movement. Waste is sealed in corrosion-resistant canisters with several barriers.

- **WIPP (Waste Isolation Pilot Plant), New Mexico.** An operating U.S. repository, in a deep salt bed, for long-lived radioactive waste from nuclear weapons programs. Salt is attractive because it is dry and slowly creeps shut around the waste.
- **Yucca Mountain, Nevada.** A proposed repository for commercial spent fuel in volcanic rock. Studied for decades but never opened, it was held up by scientific questions (water movement, earthquakes, volcanism) and political opposition. Spent fuel still sits at power plants.
- **Finland's Onkalo.** A repository in crystalline bedrock being built for spent fuel, with local consent.

Long-term stewardship means records, markers, monitoring, and funding must last for centuries. How do we warn future people who may not read our languages, who pays for guarding the site, and is it fair to leave them a waste we benefited from? Siting also raises **environmental justice** concerns when the burden falls on a community with little political power.

KEY DEFINITIONS

Half-life — the time for half of the atoms of a radioactive isotope to decay. **Spent nuclear fuel** — used fuel rods removed from a reactor; high-level radioactive waste. **Geologic repository** — a deep underground facility built for permanent isolation of waste. **Long-term stewardship** — monitoring and maintenance of a waste site for as long as it stays hazardous.

COMMON STUDENT MISTAKES

- Believing the waste is safe after one half-life. Half remains, and the half-life is only the first step.
- Assuming a long half-life means high intensity. Long half-life isotopes are less intense but last far longer.

CLASSROOM EXAMPLE SG-C10-E002

A batch of waste contains cesium-137, half-life 30 years, with an initial activity of 640 units. (a) Calculate the activity remaining after 90 years. (b) Calculate how many years it takes for the activity to fall to 1% of the initial value, to the nearest year. (c) Explain why plutonium-239, with a half-life of 24,100 years, needs a repository designed for a much longer time than cesium-137.

QUICK CONCEPT SUMMARY

- High-level nuclear waste stays dangerous for thousands of years; half-life tells how fast it fades (about ten half-lives to reach roughly 0.1%).
- Deep geologic repositories (WIPP operating, Yucca Mountain proposed, Onkalo under construction) aim for permanent isolation.
- Stewardship means centuries of record keeping, monitoring, and funding, and raises ethical and justice questions.

10.6 Bioaccumulation—Biomagnification

CONCEPT MADE SIMPLE

Bioaccumulation happens inside one animal: it takes in a pollutant faster than it gets rid of it, so the amount in its body keeps rising over its lifetime. **Biomagnification** happens up a food chain: a predator eats many contaminated prey, so the concentration climbs at each step. An osprey ends up with far more toxin per gram than the water it fishes in.

Bioaccumulation is the selective absorption and concentration of elements or compounds by cells in a living organism, most commonly fat-soluble compounds. **Biomagnification** is the increase in concentration of substances per unit of body tissue that occurs in successively higher trophic levels of a food chain or food web.

Why does it magnify? Only about 10% of energy passes up each trophic level, so a predator eats many prey over its lifetime. If the toxin is stored and not excreted, the predator collects the whole load. A common rule of thumb is about **10× per trophic level**, though real factors vary.

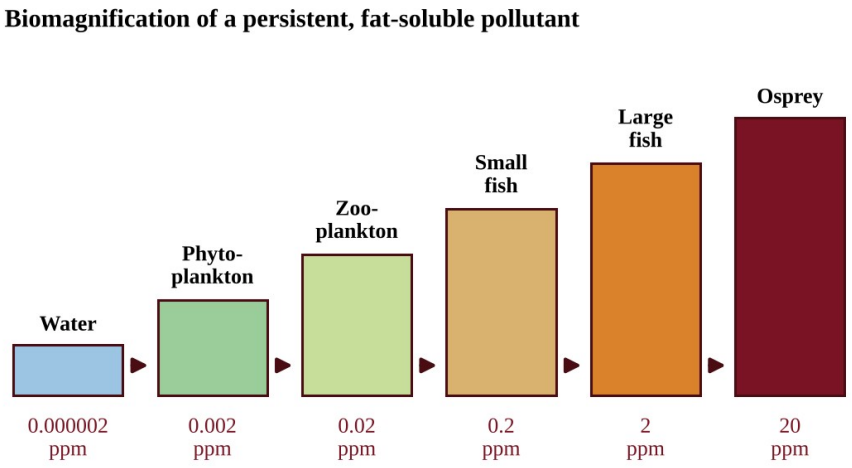


Figure 10.7 — Biomagnification of a persistent, fat-soluble pollutant in an aquatic food chain (illustrative values).

FORMULA & MATH SKILL

Biomagnification factor (BMF) = concentration in the consumer ÷ concentration in its food (the trophic level below) Concentration at level n ≈ starting concentration × $(\text{BMF})^n$; over several steps multiply the factors Units: ppm, ppb, or mg/kg of tissue (the same units for both levels).

For example, if plankton hold 0.04 ppm and each step multiplies by 10, small fish hold 0.4 ppm, larger fish 4 ppm, and a fish-eating bird 40 ppm after three steps ($0.04 \times 10^3 = 40$).

Effects

Effects in an ecosystem include **eggshell thinning** and **developmental deformities** in top carnivores at the higher trophic levels. DDT famously thinned the eggshells of birds of prey such as eagles, ospreys, and peregrine falcons, so eggs broke in the nest and populations crashed. Humans also experience harmful effects from biomagnification, including problems of the **reproductive, nervous, and circulatory** systems.

Three substances to know: **DDT** (insecticide), **PCBs** (industrial chemicals), and **mercury** (converted to methylmercury by aquatic bacteria; it harms the nervous system). All three bioaccumulate and have major environmental impacts.

Mercury follows a clear path: coal burning releases it to the air, wind carries it, rain or dry fallout deposits it on lakes and oceans, bacteria convert it to methylmercury, and it biomagnifies in fish. People are advised to limit large predatory fish, since lower trophic levels carry less toxin. Cutting coal emissions lowers mercury input, but lake fish recover slowly because mercury persists in sediment. A persistent chemical banned in one country can still turn up in its people through imported food.

KEY DEFINITIONS

Bioaccumulation — the buildup of a substance, especially a fat-soluble one, in the tissues of a single organism over time. **Biomagnification** — the increase in a substance's concentration per unit of tissue at successively higher trophic levels.

COMMON STUDENT MISTAKES

- Using the terms interchangeably. Accumulation is within one organism over time; magnification is across trophic levels.
- Assuming the producers have the highest concentrations. They have the lowest tissue concentration among organisms in the chain; the top consumer has the highest.
- Saying the pollutant "disappears" at each step. The food chain concentrates it.

AP EXAM STRATEGY

Biomagnifiers are persistent, fat-soluble, and not easily excreted. Explain the mechanism (eaten in large numbers, stored in fat) and pair it with an effect.

CLASSROOM EXAMPLE SG-U8-E005

Tissue samples from a pond food chain contained the following concentrations of a persistent pesticide:

Organism	Trophic level	Concentration (ppm)
Algae	Producer	0.8
Minnows	Primary consumer	6.4
Trout	Secondary consumer	52
Heron	Tertiary consumer	416

(a) Calculate the biomagnification factor for each step (consumer ÷ food), rounded to one decimal. (b) Calculate the overall factor from algae to heron. (c) Explain why the heron's concentration is much higher than the algae's even though the pond water is only slightly contaminated. (d) Identify one effect this pesticide might have on the heron population.

QUICK CONCEPT SUMMARY

- Bioaccumulation: buildup in one organism; biomagnification: increase up trophic levels.
- Fat-soluble, persistent, non-excretable substances such as DDT, PCBs, and methylmercury do both.
- Effects: eggshell thinning and deformities in top predators; reproductive, nervous, and circulatory harm in humans.

10.7 Environmental Economics

CONCEPT MADE SIMPLE

A product's price normally covers only what it costs the company to make. If a factory also damages air, water, and health, the neighbors pay for that. Environmental economics puts a price on that damage and compares costs and benefits that arrive at different times.

Externalities

An **externality** is a cost or benefit of an activity that falls on people who did not choose it. A **negative externality** is a cost to others (pollution, noise); a **positive externality** is a benefit to others (a wetland that cleans water downstream). Because producers ignore external costs, the **private cost** is lower than the **social cost** (private + external), and the market produces *more* of the polluting good than is best for society. This is a **market failure**, related to the **tragedy of the commons**: a shared resource such as air or a fishery is overused because no one pays for harming it.

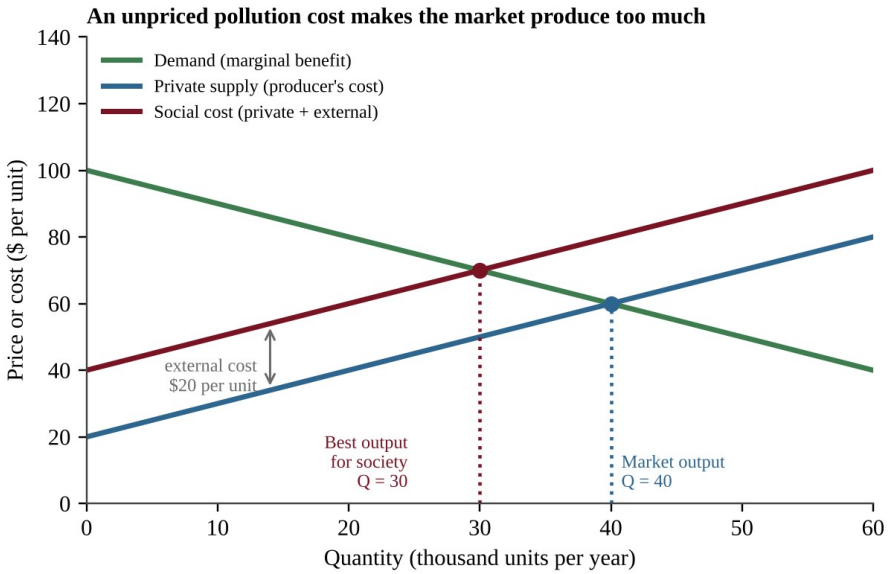


Figure 10.8 — Supply and demand with a negative externality. The market produces where demand meets private cost; society's best output is lower, where demand meets social cost.

Internalizing external costs

Internalizing a cost means making the polluter pay it. Main tools:

Tool	How it works	Example
Pollution tax or fee	A tax per unit of pollution or product raises private cost to match social cost	Carbon tax; landfill fee
Cap-and-trade (tradable permits)	A cap limits total pollution; permits are issued and firms that cut emissions cheaply can sell spare permits	The U.S. sulfur dioxide trading program for power plants
Subsidies	Payments that reward clean choices	Tax credits for solar panels
Regulation and liability	A legal standard, ban, or duty to pay for damage	Emission limits; Superfund

A tax fixes the *price* of pollution; a cap fixes the *amount*.

Cost-benefit analysis

Cost-benefit analysis (CBA) adds up all costs and benefits of a project in dollars and compares them; it is worthwhile if benefits exceed costs.

FORMULA & MATH SKILL

Net benefit = total benefits − total costs Marginal cost = change in total cost ÷ change in amount (e.g., per extra percentage point removed) Benefit-cost ratio = total benefits ÷ total costs (greater than 1 means benefits exceed costs) Pigouvian tax per unit = external cost per unit

Marginal cost usually rises steeply: removing the first 50% of a pollutant is cheap, the last few percent cost far more. CBA has limits: some benefits are hard to price (a species, a view), the future is uncertain, and it does not say who gains and who loses, which raises environmental justice questions.

Valuing ecosystem services

Ecosystem services are the benefits nature provides to people: *provisioning* (food, timber, water), *regulating* (flood control, water purification, pollination), *supporting* (soil formation, nutrient cycling), and *cultural* (recreation). Economists value them by **replacement cost** (what a water treatment plant would cost instead of a forested watershed), by the market value of what they produce, by the cost of damage avoided, or by surveys of **willingness to pay**. This lets CBA count the benefits of keeping a

wetland or forest, not just the profit from developing it. New York City, for example, protected its upstate watersheds instead of building a very costly filtration plant.

Discounting

A dollar today is worth more than a dollar in the future, because it can be invested and people prefer benefits sooner. **Discounting** converts a future amount into its **present value** using a discount rate. A higher rate makes distant benefits (a recovered forest, avoided climate damage) look smaller and favors quick payoffs. Choosing the rate is partly an ethical question about fairness to future generations.

FORMULA & MATH SKILL

Present value: $PV = FV \div (1 + r)^t$ FV = future amount ($), r = annual discount rate as a decimal, t = years until it arrives. Method: write r as a decimal, raise (1 + r) to the power t, divide. Example: $1,000 in 10 years at 5% → 1,000 ÷ 1.629 ≈ $614 today.

REAL-WORLD EXAMPLE

SO_2 trading. The U.S. acid rain program capped sulfur dioxide from power plants and let plants trade permits. Emissions fell at a lower cost than many predicted.

KEY DEFINITIONS

Externality — a cost or benefit of an activity that falls on others who did not choose it. **Internalizing a cost** — making the producer pay for the external cost so the price reflects it. **Cost-benefit analysis** — comparing the monetary value of a project's benefits and costs. **Ecosystem services** — benefits that ecosystems provide to people. **Discounting** — converting future amounts into present value.

COMMON STUDENT MISTAKES

- Saying a higher discount rate makes future benefits look larger. It makes them look *smaller*.

AP EXAM STRATEGY

In economics answers, name the tool (tax, cap-and-trade, subsidy), say whose cost it internalizes, and give one drawback. In CBA, show the ratio, then justify your claim with it.

CLASSROOM EXAMPLE SG-C10-E003

A factory sells a product, with demand given by price P = 100 − Q and marginal private cost P = 20 + Q (P in dollars per unit, Q in thousands of units). Each unit also causes $20 of pollution damage to neighbors that the factory does not pay. (a) Find the market quantity Q and price P where demand equals private cost. (b) Find the quantity that is best for society, where demand equals private cost plus the external cost. (c) State the per-unit tax that makes the factory choose the quantity in (b). (d) Calculate the annual tax revenue, in thousands of dollars, at the quantity in (b).

CLASSROOM EXAMPLE SG-C10-E004

A wetland restoration costs $400,000 now and is expected to deliver ecosystem-service benefits worth $1,000,000 twenty years from now (flood protection and water purification). (a) Calculate the present value of the benefit at a 4% discount rate, to the nearest $1,000. (b) Calculate the benefit-cost ratio at 4%, to two decimals. (c) Repeat the benefit-cost ratio at a 7% discount rate and state whether the project still passes the test of benefits exceeding costs. (d) Explain how a higher discount rate affects projects whose benefits come far in the future.

QUICK CONCEPT SUMMARY

- An externality is a cost or benefit borne by others; when costs are external, the market over-produces the polluting good.
- Internalize with taxes, cap-and-trade, subsidies, regulation, or liability. Cost-benefit analysis compares monetized benefits and costs, including ecosystem services.
- Discounting: $PV = FV \div (1 + r)^t$; higher rates shrink the value of future benefits.

10.8 Sustainability—Resiliency

CONCEPT MADE SIMPLE

Sustainability means living on Earth's income, not on its savings. We can use the interest, the forests that regrow and the fish that breed, but we cannot keep spending the principal. **Resilience** is the ability to take a hit, such as a drought, a fire, or a pollution spill, and bounce back.

Sustainability refers to humans living on Earth and using resources without depleting them for future generations. It has **environmental, economic, and social** sides: a practice that wrecks the soil, bankrupts the farmer, or ignores the community will not last.

Environmental indicators

An **environmental indicator** is a measurement that tells us whether we are moving toward or away from sustainability. Indicators named in the course:

- **Biological diversity:** falling diversity signals ecosystem stress.
- **Food production:** falling per-person output signals degraded land and water.
- **Average global surface temperature and CO_2 concentration:** rising values indicate climate change.
- **Human population:** growth increases resource demand.
- **Resource depletion:** declining stocks of fish, water, forests, soil, or minerals.

Sustainable yield

Sustainable yield is the amount of a renewable resource that can be taken without reducing the available supply. If a forest grows 20 m^3 of wood per hectare per year, cutting more than 20 m^3 per hectare per year will shrink the forest. Non-renewable resources such as ore have no sustainable yield; sustainable use means reducing demand, recycling, and substituting.

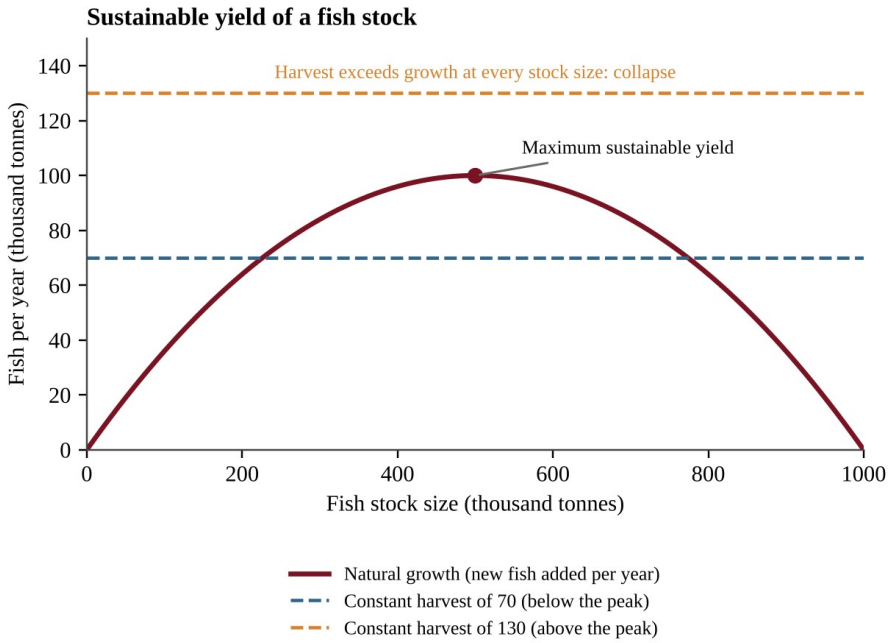


Figure 10.9 — Sustainable yield of a fish stock. A harvest below the natural-growth curve can continue indefinitely; a harvest above it at every stock size drives the stock to collapse. Tonnes are metric tons.

FORMULA & MATH SKILL

Sustainable yield (per year) = natural growth (or regeneration) per year Overharvest (%) = (harvest − sustainable yield) ÷ sustainable yield × 100 Total growth = area × growth rate per unit area

Resilience of ecosystems and communities

Resilience is the capacity of a system to absorb a disturbance and return to its earlier state without shifting into a different, often degraded state. Ecosystems are more resilient with high biodiversity (many species can fill the same role), connected habitats, healthy soils, and little other stress. Pushed past a **tipping point**, a system can flip into a state that is hard to reverse, such as a clear lake that becomes a murky algae-dominated lake, or a forest that turns to grassland after repeated fires.

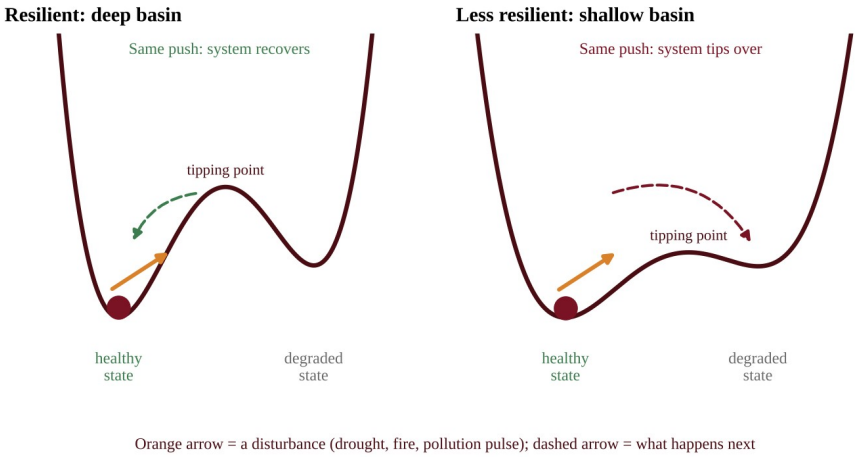


Figure 10.10 — The same disturbance leaves a resilient system in its healthy state, but tips a less resilient system into a degraded state.

Communities are resilient when they can absorb shocks such as floods, hurricanes, or droughts and recover quickly. Helpful features are diverse local economies, reserves of food and water, backup power, flood-wise land use (wetlands and floodplains left open), strong social networks, and good planning. Protecting mangroves, wetlands, and forests builds resilience and supports sustainability.

KEY DEFINITIONS

Sustainability — using resources to meet present needs without depleting them for future generations. **Sustainable yield** — the amount of a renewable resource that can be harvested without reducing the supply. **Environmental indicator** — a measure of environmental condition such as biodiversity or atmospheric CO_2. **Resilience** — the ability of an ecosystem or community to absorb a disturbance and recover. **Tipping point** — a threshold beyond which a system shifts to a different state that is hard to reverse.

COMMON STUDENT MISTAKES

- Believing renewable means unlimited. A fish stock or forest collapses if harvest exceeds regeneration.
- Thinking sustainability is only environmental. It is also economic and social.

CLASSROOM EXAMPLE SG-C10-E005

A managed forest of 1,200 hectares grows 8 m^3 of wood per hectare per year. Loggers harvest 11,000 m^3 per year. (a) Calculate the sustainable yield of the forest, in m^3 per year. (b) Calculate by what percent the harvest exceeds the sustainable yield, to one decimal. (c) Identify one environmental indicator that would show the forest is being used unsustainably and state the direction it would change. (d) Propose one change that would make the harvest sustainable.

QUICK CONCEPT SUMMARY

- Sustainability means meeting present needs without depleting resources for the future; indicators are biodiversity, food production, temperature and CO_2, population, and resource depletion.
- Sustainable yield is the harvest a renewable resource can supply without shrinking; exceeding it leads to depletion.
- Resilience is the ability to absorb disturbance and recover; diversity and healthy habitats raise it, and passing a tipping point can lock in a degraded state.

10.9 Practice Questions and Answers

Now that you have the ideas, test yourself with the classroom examples and exam-style questions listed below.

PRACTICE QUESTIONS AND ANSWERS

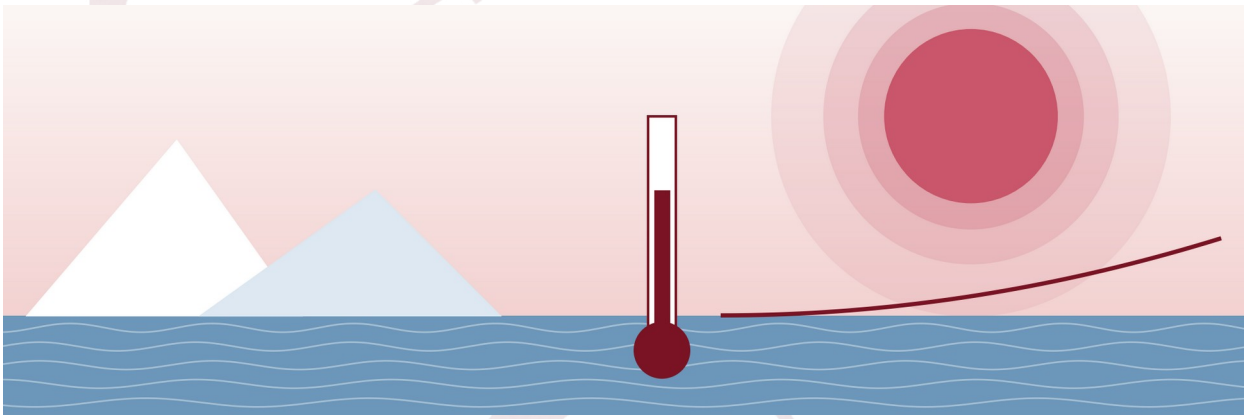
Classroom examples in this chapter (unsolved; space is provided in each box): SG-U8-E010, SG-U8-E009, SG-U8-E014, SG-C10-E001, SG-C10-E002, SG-U8-E005, SG-C10-E003, SG-C10-E004, SG-C10-E005.

Classified practice: Book Two, Chapter 10 — 23 multiple-choice questions (Section A) followed by 3 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 10. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

UNIT VII
GLOBAL CHANGE



IN THIS UNIT

Chapter 11 Stratospheric Ozone and Global Warming

CHAPTER
11

UNIT VII — GLOBAL CHANGE
Stratospheric Ozone and Global Warming

11.1 · Stratospheric Ozone
11.2 · Ozone Depletion
11.3 · Global Warming
11.4 · Impacts and Consequences of Global Warming
11.5 · Reducing Climate Change
11.6 · Loss of Biodiversity
11.7 · Practice Questions and Answers

LEARNING OBJECTIVES

- Explain why stratospheric ozone matters and how CFCs, chlorine and the polar vortex deplete it.
- Describe how the Montreal Protocol and its substitutes worked, and what they left unsolved.
- Explain the greenhouse effect, the principal greenhouse gases, and how GWP compares them.
- Interpret ice-core and Keeling-curve data and explain positive and negative feedbacks.
- Describe the effects of warming on sea level, oceans, weather, health and ecosystems, and evaluate mitigation and adaptation.
- Use HIPPCO to explain biodiversity loss, and evaluate ways to control invasive species and protect endangered ones.

Two global problems share this chapter: a thinning ozone layer (largely solved) and a warming climate (still growing). Both show one big idea: **local and regional human activities can have impacts at the global level.** The last part shows the second big idea: **the health of a species is tied to its ecosystem, so small environmental changes can have large effects.**

11.1 Stratospheric Ozone

CONCEPT MADE SIMPLE

Ozone in the stratosphere is a thin sunscreen wrapped around the planet. It absorbs most of the Sun's harmful ultraviolet (UV) light. If the sunscreen thins, more UV reaches the ground.

Where ozone is, and why it matters

Ozone (O_3) is a molecule of three oxygen atoms. The same molecule plays opposite roles in two places. **Tropospheric (ground-level) ozone** is an air pollutant that irritates lungs and damages plants; it forms when sunlight drives reactions of NO_x and VOCs (Chapter 9). **Stratospheric ozone** is the "good" ozone. It sits mostly between about 15 and 35 km above the surface, with the highest concentration near 20–30 km.

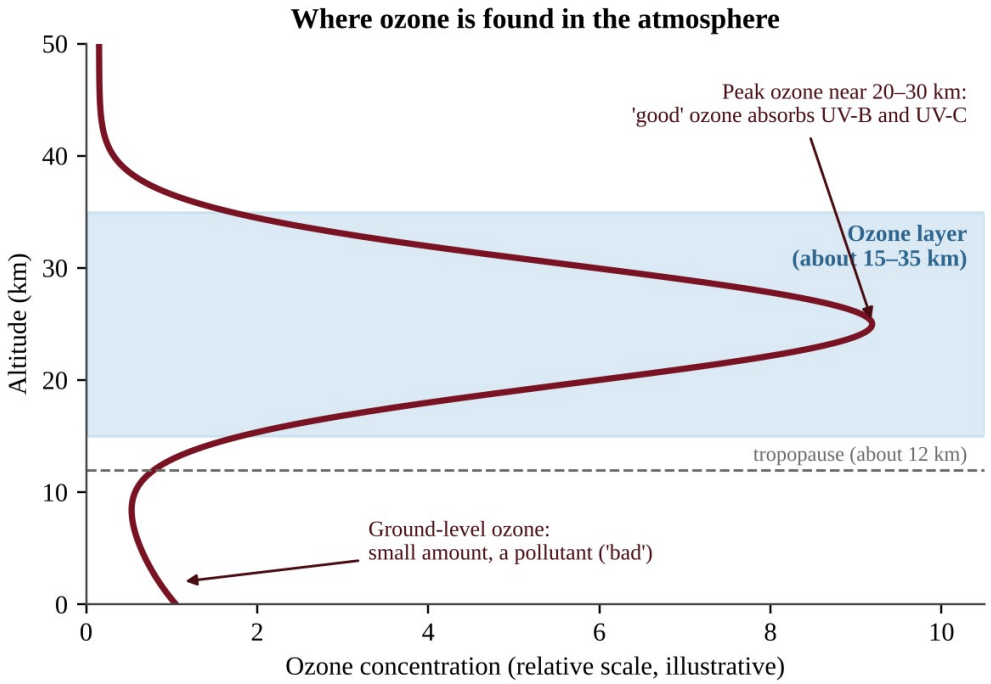


Figure 11.1 — Ozone is a pollutant near the ground but forms a protective layer in the stratosphere

The layer is made and destroyed continuously by sunlight:

- $O_2 + UV \rightarrow O + O$ (high-energy UV splits oxygen molecules)
- $O + O_2 \rightarrow O_3$ (an oxygen atom joins another O_2)
- $O_3 + UV \rightarrow O_2 + O$ (ozone absorbs UV and is split again; the energy becomes heat)

Natural creation and destruction roughly balance, so the layer keeps a steady thickness. Because ozone turns UV into heat, the stratosphere gets warmer with height, the reverse of the troposphere.

Why it matters. UV-B and UV-C damage DNA and proteins in living cells. Early life stayed in the oceans, where water shielded it from UV. As photosynthesis added oxygen to the air, an ozone layer formed, and that helped make life on land possible.

Effects of more UV at the surface

When stratospheric ozone decreases, **more UV reaches Earth's surface**. In humans this raises the risk of **skin cancer** and **cataracts** (clouding of the eye's lens) and can weaken the immune system. Higher UV also harms phytoplankton (the base of marine food webs), fish and amphibian larvae and crops, and it breaks down plastics and paints. In tank experiments phytoplankton under a UV-blocking filter photosynthesize more than those under natural or boosted UV. Regions near the ozone hole, such as southern Australia, show the clearest rise in UV-related skin cancer.

Measuring ozone

Ozone is measured as a **column**: all the O_3 in a vertical strip of air, in **Dobson units (DU)**. A typical global value is about 300 DU. Values below about 220 DU are called an **ozone hole**.

KEY DEFINITIONS

Ozone layer — the part of the stratosphere with high O_3 concentration that absorbs most UV-B and UV-C. **Ultraviolet (UV) radiation** — short-wavelength light from the Sun that can damage living tissue. **Dobson unit (DU)** — a measure of the total ozone in a column of air.

COMMON STUDENT MISTAKES

- Mixing up the two ozones. Ground-level ozone is a pollutant; stratospheric ozone protects. Always name the location.
- Saying UV causes skin cancer "by heat." UV damages DNA directly.
- Saying ozone depletion causes global warming. It is a different problem (CFCs are also greenhouse gases, see 11.3).

CLASSROOM EXAMPLE SG-U9-E001

A city's meteorological office reports that over ten years, the average stratospheric ozone concentration above the region fell, while a separate monitor shows ground-level ozone in the same city rose. (a) Identify which of the two changes increases the UV that reaches residents and explain why. (b) State one human health effect of the increased UV exposure and one effect of the increased ground-level ozone.

CLASSROOM EXAMPLE SG-C11-E001

A polar monitoring station records a lowest yearly ozone column of 280 DU in 1975 and 105 DU in 2000. (a) Calculate the percent decrease. (b) State whether the 2000 value is below the 220 DU hole threshold and by how many DU. (c) Name one human health effect expected from the extra UV that reaches the surface.

QUICK CONCEPT SUMMARY

- Stratospheric ozone absorbs harmful UV and made life on land possible.
- Ground-level ozone is a pollutant; stratospheric ozone is protective.
- Less ozone means more UV, which means more skin cancer and cataracts and damage to plankton and crops.

11.2 Ozone Depletion

CONCEPT MADE SIMPLE

Humans released long-lived chemicals (CFCs) that drift up, get split by sunlight and free chlorine atoms. One chlorine atom destroys ozone again and again. We fixed the problem by banning the chemicals, but the substitutes are not all harmless.

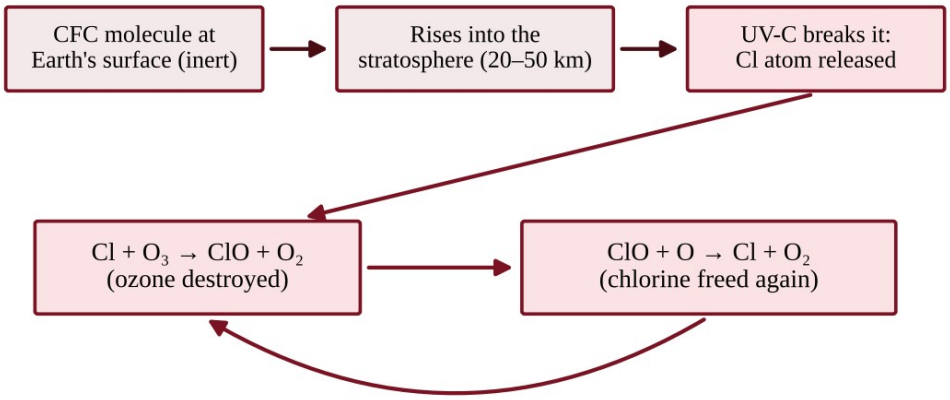
What depletes ozone

Human causes. The main culprits are **chlorofluorocarbons (CFCs)**, synthetic compounds of carbon, chlorine and fluorine. They were used as refrigerants, aerosol propellants, foam-blowing agents and solvents. CFCs are unreactive in the lower atmosphere, which is why they seemed safe. It is also why they

last decades to a century and drift up into the stratosphere. Halons (fire suppressants with bromine) and methyl bromide also deplete ozone.

PROCESS & CAUSE-AND-EFFECT
CFC at the surface (stable) → winds carry it up into the stratosphere → UV-C breaks a C–Cl bond and frees a chlorine atom (Cl) → $Cl + O_3 \rightarrow ClO + O_2$ (ozone destroyed) → $ClO + O \rightarrow Cl + O_2$ (Cl regenerated) → the same Cl attacks another ozone molecule → one Cl atom can destroy thousands of ozone molecules.

CFC-driven ozone destruction



One Cl atom repeats this cycle thousands of times before it is removed from the stratosphere.

Net: $2\ O_3 \rightarrow 3\ O_2$ (UV-absorbing ozone is lost; Cl acts as a catalyst)

Figure 11.2 — Chlorine is a catalyst: it is released, destroys ozone, and is released again to repeat the cycle

A **catalyst** speeds up a reaction without being used up. Chlorine leaves the cycle unchanged, so a trace gas measured in parts per trillion can thin a layer that spans the globe.

Natural factors. During the long Antarctic winter, air circulates in an isolated **polar vortex** and becomes extremely cold, so thin ice crystals form in polar stratospheric clouds. Reactions on the surface of these crystals change stable chlorine compounds into reactive forms. Little ozone is lost in the dark, because freeing chlorine atoms needs sunlight. When sunlight returns at the start of the Antarctic spring, UV splits the reactive chlorine, and ozone is destroyed quickly until the air warms and the ice crystals melt. That is why the Antarctic **ozone hole** appears each September–October. Cold-vortex conditions set the timing; the chlorine that makes the hole so severe comes from human-made chemicals.

The Montreal Protocol

In the 1970s and 1980s scientists linked CFCs to ozone loss and documented the Antarctic ozone hole. In 1987 nations signed the **Montreal Protocol**, an agreement to phase out CFCs, halons and other ozone-depleting substances. It was strengthened several times and nearly every country joined. It is widely seen as the most successful international environmental treaty because the science was clear, affordable substitutes existed, and developing countries got a slower schedule with financial help.

Because CFCs persist for decades, recovery is slow. Stratospheric chlorine peaked around 2000 and has fallen since. Ozone responds with a lag, so the ozone minimum bottoms out *after* chlorine peaks. "Recovered" means back to the earlier (about 1980) level, so a minimum that rose from 100 to 125 DU has improved but not recovered. The layer is expected to recover around the middle of this century.

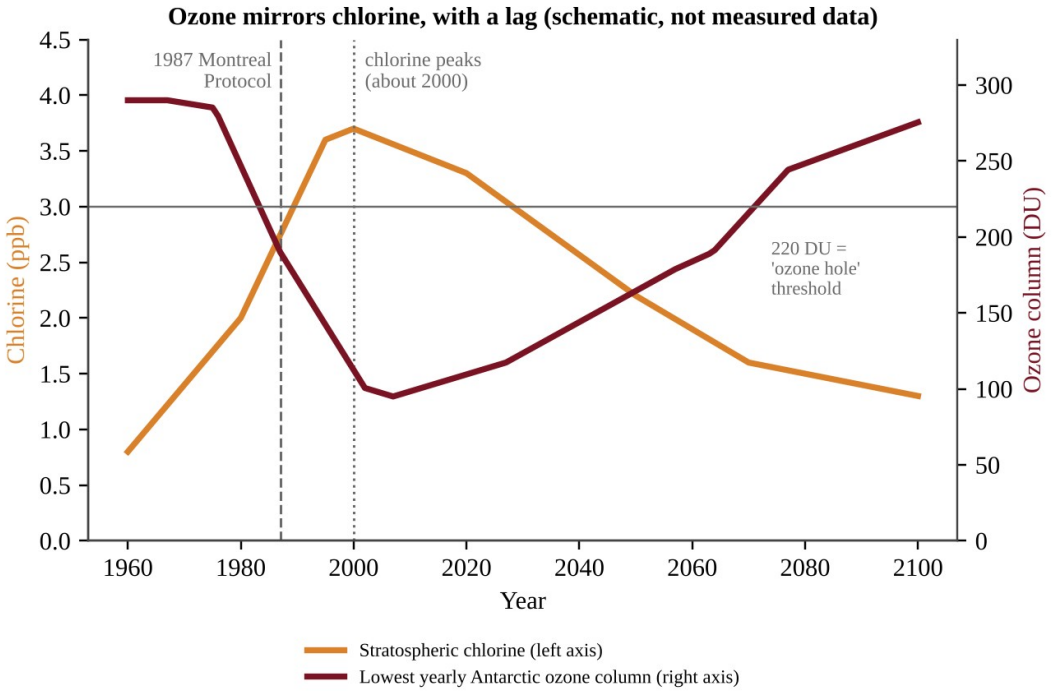


Figure 11.3 — Ozone mirrors chlorine, with a lag (schematic)

Substitutes for CFCs

Substitute	Contains chlorine?	Ozone effect	Other concern
HCFCs (hydrochlorofluorocarbons)	Yes, but less	Much less depleting; interim step	Greenhouse gases; also being phased out
HFCs (hydrofluorocarbons)	No	Do not deplete ozone	Many are strong greenhouse gases
Hydrocarbons, ammonia, CO_2	No	None	Some flammable or toxic; need careful design

HFCs contain no chlorine, so they do not harm the ozone layer. But some HFCs trap heat hundreds to thousands of times more strongly than CO_2 per unit mass, so their growing use in air conditioning worsens climate change. A 2016 amendment to the Montreal Protocol now phases HFCs down too. **A solution to one environmental problem can create an unintended consequence elsewhere**, so judge a substitute on more than one impact.

Two separate scales rate a chemical. **Ozone depletion potential (ODP)** compares its ozone-destroying power with a CFC reference of 1. **GWP** compares its heat trapping with CO_2 (see 11.3). An HFC can have ODP 0 and GWP in the thousands, so never use one scale to answer a question about the other.

KEY DEFINITIONS

Montreal Protocol — the 1987 treaty that phased out ozone-depleting substances. **Catalyst** — a substance that speeds a reaction without being consumed. **Polar vortex** — the isolated, very cold winter air over a pole. **HFC (hydrofluorocarbon)** — a CFC substitute with no chlorine; ozone-safe but often a potent greenhouse gas. **Phase-out** — a scheduled reduction of production and use until a substance is eliminated.

COMMON STUDENT MISTAKES

- Writing that the Montreal Protocol "fixed" the ozone layer. It stopped new damage; recovery takes decades.
- Calling HFCs "harmless." They are ozone-safe, not climate-safe.
- Expecting the ozone hole to disappear the year chlorine peaks. Recovery lags the chlorine decline.

AP EXAM STRATEGY

To describe how CFCs deplete ozone, name the sequence: CFCs reach the stratosphere, UV frees chlorine, chlorine breaks down ozone, chlorine is regenerated (catalyst). Finish with the impact: more UV, skin cancer and cataracts.

CLASSROOM EXAMPLE SG-U9-E002

Three coolants are being considered for a new refrigeration plant. Coolant X contains chlorine and has low heat-trapping ability. Coolant Y contains no chlorine and has a very high heat-trapping ability per molecule. Coolant Z contains no chlorine and has low heat-trapping ability but is flammable. (a) Which coolant would harm the stratospheric ozone layer most? Explain using the chlorine cycle. (b) Which coolant would contribute most to the greenhouse effect per kilogram released? (c) Make a claim for which coolant should be chosen and justify it, naming one precaution you would take.

CLASSROOM EXAMPLE SG-U9-E013

A monitoring station reports stratospheric chlorine (ppb) and the lowest yearly ozone column over a polar site (DU): 1985: 2.2 ppb, 190 DU; 1995: 3.6 ppb, 120 DU; 2005: 3.5 ppb, 110 DU; 2015: 3.0 ppb, 120 DU. A refrigerant has an ODP of 0.04 and a GWP of 900. (a) Calculate the percent decrease in chlorine from 1995 to 2015. (b) Explain why ozone in 2015 was still below its 1985 value although chlorine had been falling for 20 years. (c) Calculate the CO_2e of releasing 50 kg of the refrigerant and state whether it harms ozone more or less than a CFC (ODP 1).

QUICK CONCEPT SUMMARY

- CFCs reach the stratosphere, UV frees chlorine, and chlorine catalytically destroys ozone. Cold polar vortex conditions set the timing of the Antarctic ozone hole.
- The Montreal Protocol (1987) phased out ozone-depleting substances; recovery is slow because CFCs persist.
- Substitutes: HCFCs (interim), HFCs (no chlorine, often high GWP), and non-halogen options.

11.3 Global Warming

CONCEPT MADE SIMPLE

Sunlight warms the ground. The ground sends the energy back out as invisible infrared heat. Greenhouse gases absorb some of that heat and re-emit it in all directions, including back down, like a blanket. Without the blanket Earth would be frozen; a thicker blanket makes it hotter.

The greenhouse effect

Earth receives most of its energy as **shortwave radiation** (visible light and some UV). About 30% is reflected straight back by clouds, aerosols, ice and bright surfaces (**albedo** is the fraction reflected). The rest warms the atmosphere and surface. A warm surface emits **longwave (infrared) radiation**. To keep a steady temperature, Earth must send out as much energy as it absorbs.

Greenhouse gases let sunlight in but absorb outgoing infrared. They re-emit it in every direction, and part goes back down, warming the surface and lower atmosphere. Nitrogen and oxygen (about 99% of air) do not absorb infrared significantly, so they are not greenhouse gases.

The **natural greenhouse effect is essential**: it gives an average surface temperature near 15°C instead of about −18°C. The environmental problem is the **enhanced greenhouse effect**: human activities add greenhouse gases, strengthen the blanket and raise the temperature at which energy in equals energy out.

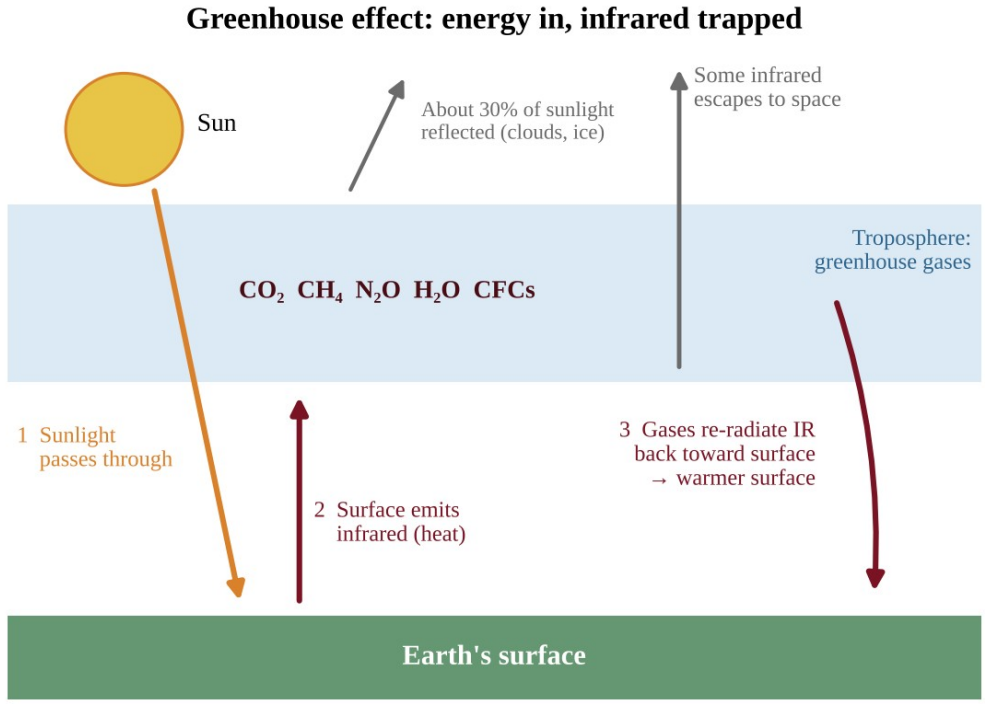


Figure 11.4 — Shortwave sunlight arrives, infrared leaves the surface, and greenhouse gases return part of it downward

PROCESS & CAUSE-AND-EFFECT

More greenhouse gas → more outgoing infrared absorbed and re-emitted downward → less heat escapes to space → Earth absorbs more than it emits → atmosphere, land and oceans warm → warming continues until outgoing radiation again matches incoming energy, at a higher temperature.

Following the energy in numbers. Energy budgets use 100 units of arriving sunlight. Two checks always work: (1) at the **surface**, sunlight absorbed + infrared returned by the atmosphere = infrared emitted by the surface; (2) at the **top of the atmosphere**, reflected sunlight + infrared escaping = sunlight arriving. In Figure 11.5, 45 + 75 = 120 at the surface, and 30 + 15 + 55 = 100 balances the sunlight. The surface gets more energy from the atmosphere (75) than from the Sun (45): the greenhouse effect in one number. More greenhouse gas raises the 75 and lowers the 15 that escapes directly, so the budget is briefly unbalanced. A warmer surface then emits more infrared, a **negative feedback** that restores balance. **Aerosols** such as volcanic sulfate droplets scatter sunlight and cool the surface for a few years, the opposite push to CO_2.

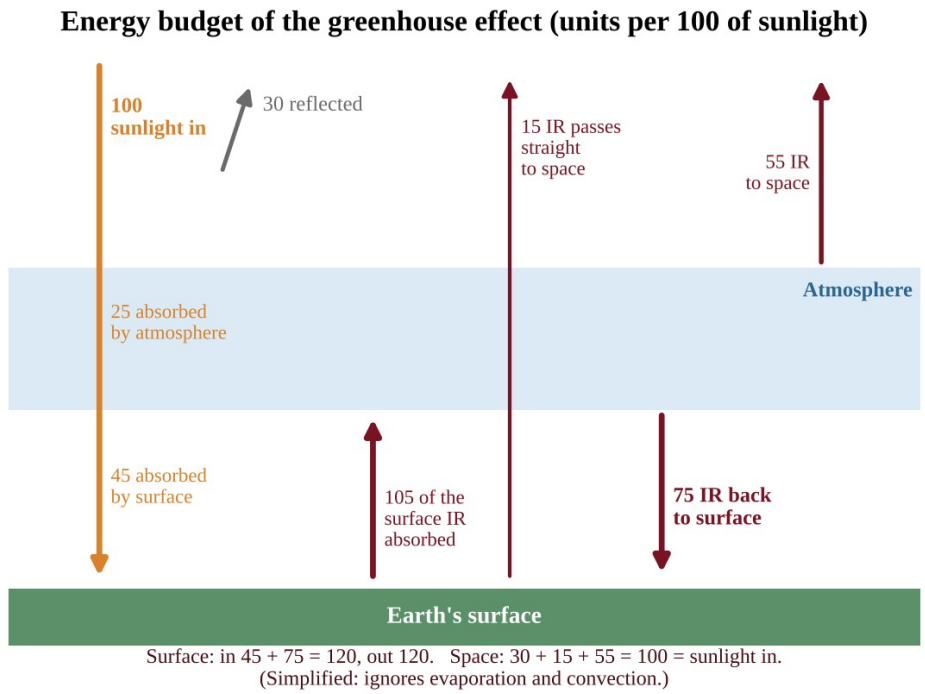


Figure 11.5 — A simplified energy budget: the surface absorbs more energy from the atmosphere than directly from the Sun

Greenhouse gases: sources and potency

The principal greenhouse gases are carbon dioxide (CO_2), methane (CH_4), water vapor (H_2O), nitrous oxide (N_2O) and chlorofluorocarbons (CFCs).

Water vapor is the most abundant greenhouse gas, yet it is not a major cause of human-driven climate change. Its **residence time** is short (days), so it rains out. Its amount is set by temperature, not by emissions: warming raises it (a feedback, below).

Global warming potential (GWP) compares the heat a gas traps with CO_2 over a set period (usually 100 years), per unit mass. CO_2 is the reference: **GWP = 1. CFCs have the highest GWP, followed by nitrous oxide, then methane.**

Gas	Main sources	Approx. GWP (100 yr)	Notes
CO_2	Fossil fuels, deforestation, cement	1	Largest total effect: huge emissions, long-lived
CH_4	Livestock, rice paddies, landfills, gas leaks, wetlands, thawing permafrost	about 25–30	Lives about a decade; strong per molecule
N_2O	Synthetic fertilizer, manure, fuel combustion	about 265–300	Also depletes stratospheric ozone
CFCs	Old refrigerants, foams, aerosols	thousands (up to 10,000+)	Very potent, small amounts, declining

CO_2, CH_4, N_2O and water vapor have natural and human sources. **CFCs and HFCs have none**, so any in the air is human-made. Ground-level ozone is also a greenhouse gas. Methane forms where organic matter decays **without oxygen**: wetlands, rice paddies, landfills, manure lagoons and the guts of cattle and termites. Burning fossil fuel adds carbon that was locked underground for millions of years. Burning recently grown plant material returns carbon that plants took from the air recently, so it is often treated as part of the existing cycle, though it still releases CO_2.

Potency versus total. GWP shows strength per kilogram. A gas's total contribution depends on potency, how much is emitted and how long it stays. CO_2 has the lowest GWP yet is the largest driver, because humans emit billions of tonnes a year and part of it stays for centuries.

FORMULA & MATH SKILL

CO_2-equivalent (CO_2e) = mass of gas × GWP Mass in tonnes (or kg); GWP is a unitless multiplier; CO_2e has the same mass unit. Use CO_2e to add different gases: total = ($mass_1$ × GWP_1) + ($mass_2$ × GWP_2) + … *Demonstration:* a landfill emits 200 t CH_4 (GWP 25) and a plant emits 1,000 t CO_2. CO_2e = 200 × 25 + 1,000 × 1 = 6,000 t; methane is 5,000 ÷ 6,000 ≈ 83% of the effect.

Increases in greenhouse gases: reading the record

Ice cores. In Antarctica and Greenland, snow compresses into ice that traps bubbles of ancient air. Deeper ice is older, so cores give a record of about 800,000 years. Oxygen-isotope ratios in the ice estimate past temperatures. Over that time CO_2 cycled between about 180 ppm in cold glacial periods and about 280 ppm in warm interglacials, and **temperature and CO_2 rose and fell together**. Slow changes in Earth's orbit and tilt started these swings and feedbacks amplified them.

The Keeling curve. Direct measurements at Mauna Loa, Hawaii, since 1958 show CO_2 rising from about 315 ppm to over 420 ppm. A small yearly zigzag appears: in the Northern Hemisphere growing season plants take up CO_2 and the level dips; in winter it climbs. Under the zigzag every year is higher than the last. Today's CO_2 is about 50% above the pre-industrial level of about 280 ppm and higher than anything in the ice-core record. Methane and N_2O have also risen well above pre-industrial levels.

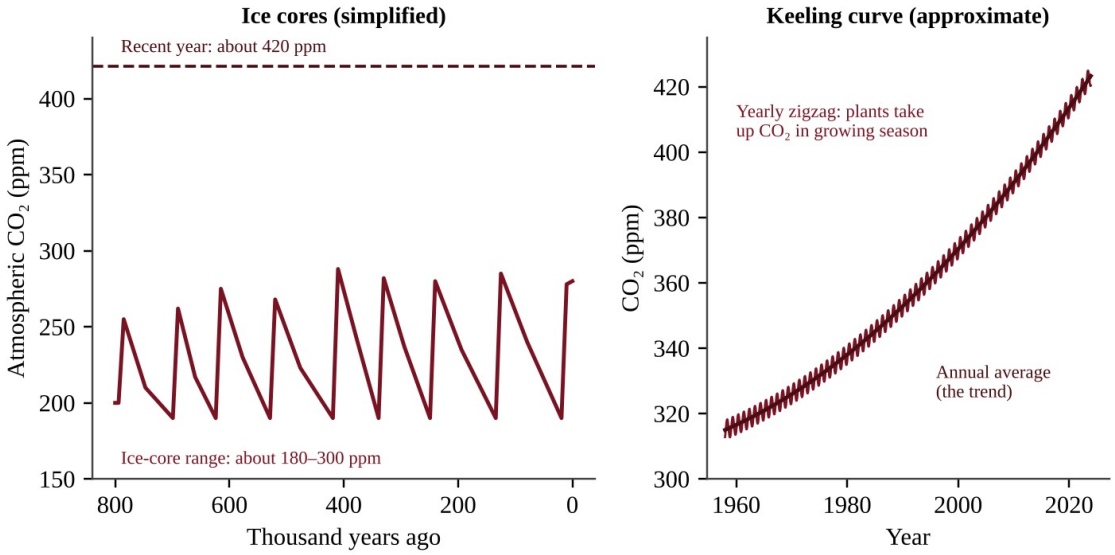


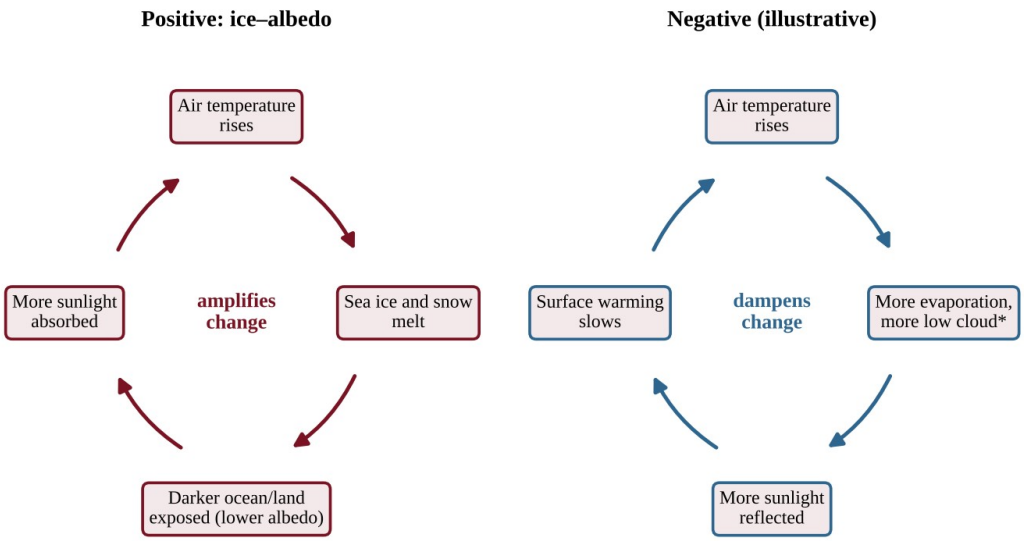
Figure 11.6 — Ice cores show 180–300 ppm for 800,000 years; direct measurement shows a steep recent rise (both panels simplified)

Reading an ice-core graph. Match the peaks of CO_2 and temperature, quote values (180–300 ppm) and compare with today's line. Use careful verbs ("closely linked," not "CO_2 alone caused every change").

Climate change through time and feedback loops

Earth's climate has swung between ice ages and warm periods, recorded in ice cores, ocean sediments and tree rings, with natural drivers such as orbit, solar output, volcanoes and greenhouse gases. What is new is the **speed** and the **cause**: average global temperature has risen about 1.1–1.2°C since the late 1800s, far faster than natural transitions, and it matches the rise in human-emitted greenhouse gases. **Weather** is the state of the atmosphere at a place and time; **climate** is the long-term average (30 years or more). One cold winter does not disprove warming.

A **positive feedback loop** amplifies a change. A **negative feedback loop** counteracts it. Positive and negative describe direction, not good or bad.



*Cloud feedbacks are uncertain; shown only as an example of a stabilizing loop.

Figure 11.7 — A positive feedback loop amplifies warming; a negative loop dampens it

- **Ice–albedo.** Bright snow and ice reflect sunlight. Warming melts them, exposing darker water and land that absorb more sunlight, causing more warming and melting. Dark shrubs and trees standing above the snow act the same way. Polar regions respond **faster** than most places because of this loop; the Arctic warms two to four times faster than the global average.
- **Permafrost thaw.** Frozen tundra soil stores vast amounts of dead plant matter. When it thaws, microbes release CO_2 and **methane**, adding greenhouse gas and more warming. The **active layer** is the surface soil that thaws each summer; as it deepens, more carbon becomes available. Thawing also destabilizes ground under buildings, roads and pipelines (adaptations: pile foundations, insulated pads, relocation).
- **Water vapor.** Warmer air holds more vapor, a greenhouse gas.
- **Declining ocean uptake.** Warmer water dissolves less CO_2, leaving more in the air.

Negative feedbacks also exist: a warmer surface radiates more infrared to space.

PROCESS & CAUSE-AND-EFFECT

Warming → sea ice and snow melt → albedo falls → more sunlight absorbed by dark ocean and land → more warming → more melt. Warming → permafrost thaws → decomposition releases CO_2 and CH_4 → stronger greenhouse effect → more warming → more thaw.

KEY DEFINITIONS

Greenhouse effect — warming of the lower atmosphere and surface because gases absorb and re-emit outgoing infrared. **Global warming potential (GWP)** — a gas's heat trapping relative to the same mass of CO_2. **Residence time** — the average time a molecule stays in the atmosphere. **Albedo** — the fraction of incoming sunlight a surface reflects. **Permafrost** — ground frozen for two or more years, holding large stores of carbon. **Positive / negative feedback loop** — a process whose effects amplify / reduce the original change.

COMMON STUDENT MISTAKES

- "The greenhouse effect is bad." The natural effect is necessary; the **enhanced** effect is the problem.
- "Greenhouse gases trap sunlight." They absorb outgoing **infrared**.
- Naming water vapor as the main cause of climate change because it is abundant. Short residence time and temperature control are the key points.
- Reading the seasonal zigzag in a CO_2 graph as a trend. The annual average is the trend.
- Treating the energy budget as "energy trapped forever." At equilibrium energy out equals energy in, at a higher temperature.
- Calling every feedback "good" or "bad." Positive means amplifying.

AP EXAM STRATEGY

If a question gives amounts of several gases and asks which has the biggest climate impact, compute CO_2e (mass × GWP) for each; do not pick the highest GWP automatically. In a feedback FRQ, write the chain one link at a time, give the mechanism (albedo, methane release), then label it "positive, because the change is amplified."

CLASSROOM EXAMPLE SG-U9-E003

A dairy and crop farm reports its annual emissions: 600 tonnes of CO_2, 40 tonnes of CH_4 and 3 tonnes of N_2O. Use GWP values of 1 for CO_2, 28 for CH_4 and 265 for N_2O. (a) Calculate the farm's total emissions in tonnes of CO_2e. (b) Calculate the percentage of the total that comes from methane. (c) Identify one source of each of the three gases on a farm.

CLASSROOM EXAMPLE SG-U9-E014

In a simplified budget, 100 units of sunlight arrive. Clouds and ice reflect 28; the atmosphere absorbs 22 and the surface absorbs 50. The surface emits 130 units of infrared, of which 10 pass directly to space. (a) Calculate how many units of infrared the atmosphere returns to the surface. (b) Calculate how many units the atmosphere emits to space. (c) Extra greenhouse gas now absorbs 4 of the 10 units that used to pass directly to space. Predict the immediate effect on the total energy leaving Earth compared with the 100 units arriving, the long-term effect on surface temperature, and the feedback that finally restores balance.

CLASSROOM EXAMPLE SG-U9-E004

Atmospheric CO_2 was about 280 ppm before the Industrial Revolution and 420 ppm in a recent year. (a) Calculate the percent increase. (b) An ice-core record shows that CO_2 over the last 800,000 years never exceeded 300 ppm. Explain in one or two sentences what this comparison suggests about the present concentration. (c) A graph of CO_2 measured monthly at one observatory shows a yearly zigzag. Explain what causes the zigzag and why it does not contradict the long-term increase.

CLASSROOM EXAMPLE SG-U9-E006

In a high-latitude lake region, a scientist records that each year the ice-free period lengthens, exposing dark water. (a) Describe how this is a positive feedback loop, including the role of albedo. (b) Explain why a loop in which warmer air increases evaporation, forming bright low clouds that reflect more sunlight, would be a negative feedback loop. (c) Identify one effect on an ice-dependent species.

CLASSROOM EXAMPLE SG-C11-E002

Annual-average CO_2 at an observatory was 370 ppm in 2000 and 410 ppm in 2020. (a) Calculate the average rate of increase in ppm per year. (b) Assuming the same rate continues, calculate the year in which CO_2 reaches 450 ppm. (c) State one reason the real future rate could differ from this straight-line projection.

QUICK CONCEPT SUMMARY

- Greenhouse gases absorb outgoing infrared; the natural effect keeps Earth habitable, the enhanced effect warms it.
- Principal gases: CO_2, CH_4, H_2O, N_2O, CFCs. Water vapor is not a major driver (short residence time). GWP (CO_2 = 1): CFCs > N_2O > CH_4 > CO_2.
- Ice cores (800,000 years, 180–280 ppm, tracking temperature) and the Keeling curve (continuous rise since 1958) show today's CO_2 is unprecedented.
- Positive feedbacks (ice–albedo, permafrost methane) amplify warming, especially at the poles.

11.4 Impacts and Consequences of Global Warming

CONCEPT MADE SIMPLE

A warmer world melts ice, swells the oceans, shifts weather and stresses species. The oceans soak up most of the extra heat and about a quarter to a third of our CO_2, so they warm and turn less basic. Each effect also feeds back on people.

Sea level and coasts

Sea level rises for two reasons: (1) **thermal expansion**, because water expands as it warms, and (2) **melting land ice** (mountain glaciers and the Greenland and Antarctic ice sheets), which adds new water. Melting floating sea ice does not raise sea level noticeably, like an ice cube in a full glass, though it feeds the albedo loop.

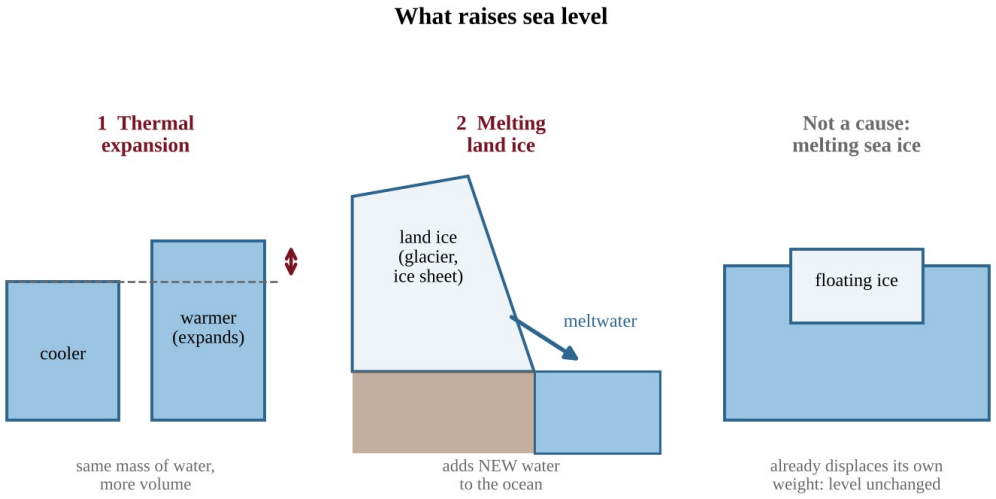


Figure 11.8 — Sea level rises from thermal expansion and melting land ice, not from floating sea ice

Flooding, storm surge and saltwater intrusion into groundwater make low-lying land uninhabitable, so **coastal populations are displaced**. Island nations and river deltas are exposed first. Salt water moves farther up estuaries and kills species with narrow salt tolerance. Marshes and mangroves cannot migrate inland when seawalls or development block them.

Health, population and movement

As winters shorten, **disease vectors spread from the tropics toward the poles** and uphill. Mosquitoes and ticks that carry malaria, dengue and Lyme disease are limited by cold, so new areas face people with no immunity and prepared health systems. Heat waves raise heat illness and deaths, and hotter air worsens smog and ground-level ozone. Drought, failed crops, flooded coasts and heat lower a region's carrying capacity, so **population dynamics and movements change**: people migrate (climate refugees), altering birth, death and migration patterns where they leave and where they arrive.

Weather, circulation and soil

- **Extreme weather.** Warmer air holds more water vapor, so heavy rain and flooding become more intense. Heat waves, droughts and wildfire seasons lengthen. Warmer ocean surfaces give tropical cyclones more energy, so the strongest storms get stronger and carry more rain; sea-level rise adds to storm surge.
- **Atmospheric circulation.** Winds carry heat around the planet. Warm air rises at the equator and flows poleward in **Hadley cells**; the **jet stream** is a fast air current near the tropopause that steers storms. Warming changes the temperature contrast driving them, so cells can widen and the jet stream can become wavier, shifting rainfall belts and storm tracks.
- **Ocean currents.** Currents, the ocean conveyor belt, carry heat across the globe. Cold, salty water sinks in the North Atlantic and drives the circulation. Melting ice adds fresh, less dense water that

sinks less readily, so the conveyor may slow, which could change climates, especially in coastal regions such as northwestern Europe.

- **Soils.** Higher temperatures speed evaporation and decomposition. Drought leaves soil dry and bare; intense storms increase erosion. Soil viability for crops can fall and erosion can rise.

Effects on ecosystems

- **Ranges shift** toward the poles and uphill. Mountaintop species run out of higher ground, and cold-water fish lose streams that warm. Timing mismatches arise when insects emerge earlier than the birds that feed on them.
- **Ice-dependent species** such as polar bears and seals lose hunting and breeding platforms, and under-ice algae at the base of the food web are lost.
- **Marine ecosystems and sea level.** Rising seas create new shallow habitat over flooded continental shelves, a gain for some species. But communities that need light, such as seagrass and corals, may end up deeper than the **photic zone** (the sunlit layer where photosynthesis is possible) and decline. Coastal wetlands are squeezed against development.

Ocean warming and coral bleaching

Ocean warming is caused by **greenhouse gases**. Water has a very high heat capacity, so the oceans absorb more than 90% of the extra heat. Surface water warms first and mixes slowly downward, so oceans keep warming long after emissions fall.

- **Habitat loss.** Species with narrow temperature tolerance lose range. Cold-water fish, kelp forests and sea-ice communities shrink and many species move poleward, changing fisheries.
- **Metabolic changes.** Most marine animals are ectotherms. Warming speeds their metabolism and oxygen needs, while **warmer water holds less dissolved oxygen**. This stresses fish and invertebrates and can create low-oxygen zones.
- **Reproductive changes.** Spawning and larval survival are timed to temperature. In sea turtles, warm nests produce more females. Larvae can miss their plankton food.

Coral bleaching. Reef-building corals are animals that host microscopic algae (zooxanthellae). The algae photosynthesize and supply most of the coral's food and color; the coral gives shelter and nutrients (a mutualism). When water stays a degree or two above the normal summer maximum for weeks, the algae produce harmful compounds and the coral expels them, leaving transparent tissue over a white skeleton. Bleached corals are starving, not dead. **Some corals recover** if temperatures drop soon enough and algae return; if the stress lasts, **corals die**, and the reef loses the habitat of about a quarter of marine species plus fisheries and coastal protection.

Ocean acidification

Ocean acidification is the **decrease in ocean pH, primarily from increased atmospheric CO_2**. The oceans absorb roughly a quarter to a third of the CO_2 humans emit. In seawater:

$CO_2 + H_2O \rightleftharpoons H_2CO_3$ (carbonic acid) $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ (bicarbonate) $H^+ + CO_3^{2-} \rightleftharpoons HCO_3^-$ (carbonate is used up)

More CO_2 gives more H^+ **and less carbonate**. Shell and skeleton builders make calcium carbonate: $Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3$. With less carbonate, building costs more energy, and in sufficiently acidic water calcium carbonate can **dissolve**.

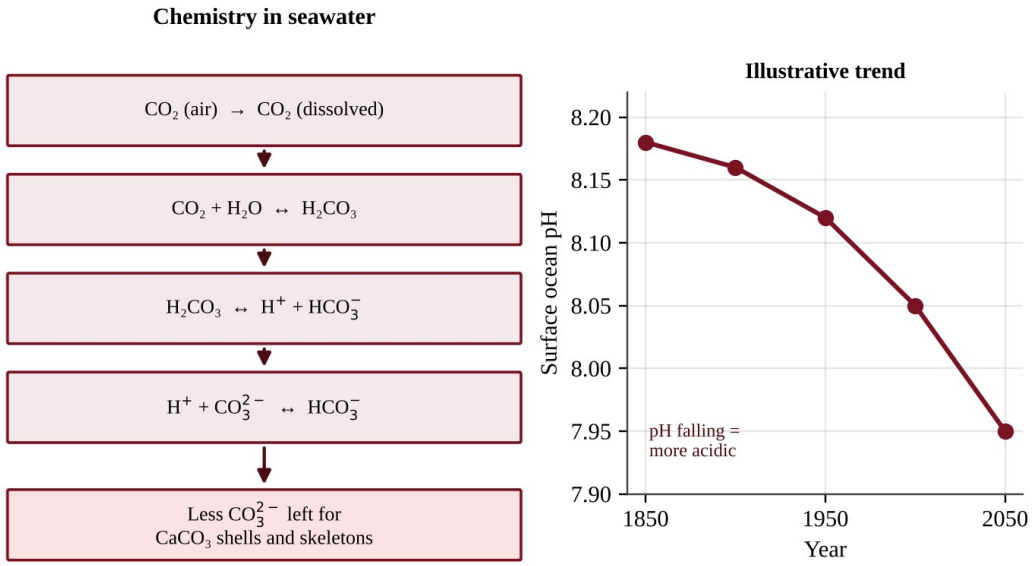


Figure 11.9 — CO_2 dissolves and releases H^+, lowering pH and removing carbonate; the pH values shown are illustrative

Understanding pH. The pH scale is **logarithmic**: a change of 1 unit is a 10-fold change in H^+. Average surface pH has fallen from about 8.2 to about 8.1. That 0.1 drop means H^+ rose by 10^0.1 ≈ 1.26, about **26%**. The ocean is still slightly basic (pH above 7); "acidification" means becoming *less basic*.

pH drop	Factor 10^drop	Increase in H^+
0.05	1.12	12%
0.10	1.26	26%
0.20	1.58	58%
0.30	2.00	100%
1.00	10	900%

A CO_2 graph and a pH graph for the same years mirror each other (inverse relationship). **Local pH can swing more than the global average.** In coastal **upwelling**, winds draw up cold deep water that holds extra CO_2 from decaying organic matter and has a lower pH. Upwelled water can kill oyster larvae building their first shells.

Causes and effects. Activities that raise atmospheric CO_2, and so acidify the ocean, are **burning fossil fuels, vehicle emissions and deforestation**, the same cause as global warming. Corals, oysters, clams, sea urchins and planktonic snails (pteropods) have calcium carbonate shells or skeletons, so growth slows, larvae weaken and structures dissolve. Pteropods feed fish such as salmon, and reefs lose the structure that shelters fish and shields shorelines. Heat plus acidification harms coral more than either alone.

FORMULA & MATH SKILL

Change in H^+ concentration = 10^(pH_1 − pH_2) (new $[H^+]$ ÷ old $[H^+]$) pH_1 is the higher (earlier) value; pH_2 the lower (later) value. Each 1.0 decrease means 10 times more H^+; each 0.3 means about 2 times more. Percent increase = (factor − 1) × 100. **Sea-level rise = rate × time**, with units converted (4 mm/yr × 50 yr = 200 mm = 0.2 m). **Volume = area × thickness; mass = volume × density.** Convert units first (3 mm = 0.003 m).

KEY DEFINITIONS

Thermal expansion — the increase in volume of water as it warms. **Ectotherm** — an organism whose body temperature follows its surroundings. **Coral bleaching** — loss of symbiotic algae from coral under stress, leaving it white. **Ocean acidification** — the fall in ocean pH caused mainly by absorbed atmospheric CO_2. **Calcification** — building shells or skeletons from calcium carbonate.

COMMON STUDENT MISTAKES

- Saying melting sea ice raises sea level the way melting land ice does, or forgetting thermal expansion.
- Saying bleached coral is always dead, or that acidification causes bleaching. Bleaching is mainly heat stress; acidification harms coral by removing carbonate.
- Saying the ocean is "now acidic," or that a drop from 8.2 to 8.1 is "small" (26% more H^+).
- Forgetting that warm water holds less dissolved oxygen.

AP EXAM STRATEGY

To "explain how burning fossil fuels harms coral reefs," write the chain: CO_2 rises → more dissolves in the ocean → H_2CO_3 forms → H^+ rises, pH falls → carbonate decreases → calcium carbonate skeletons are harder to build. Add warming and bleaching as a second effect.

CLASSROOM EXAMPLE SG-U9-E005

The table shows global mean sea level rise measured by satellites for a small island nation's region.

Year	Sea level above 2000 baseline (mm)
2000	0
2010	32
2020	66

(a) Calculate the average rate of rise per year for 2000–2010 and for 2010–2020. (b) State whether the rate is speeding up or slowing and by how many mm/yr. (c) Identify the two main physical processes that raise sea level in a warming world.

CLASSROOM EXAMPLE SG-U9-E007

A coral reef's summer maximum sea temperature is normally 29.0°C. During one summer the water stays at 30.8°C for six weeks, and many corals turn white. Three months later, survey teams find that corals in a shaded, deeper part of the reef have regained their color while corals on the shallow, sunny flat are covered in algae growth and have no living tissue. (a) Calculate by how many degrees the water exceeded the normal maximum. (b) Explain what the corals lost to turn white and why this harms them. (c) State which part of the reef shows recovery and which shows mortality, and cite the evidence from the survey.

CLASSROOM EXAMPLE SG-U9-E008

Average surface-ocean pH in a monitored bay was 8.20 in 1990 and 8.10 in 2020. (a) By what factor did the hydrogen ion concentration change between those years? Express it as a percent increase. (b) Write the chemical equation showing how dissolved CO_2 combines with water to form carbonic acid. (c) Explain why this change makes it harder for oysters in the bay to build shells.

CLASSROOM EXAMPLE SG-C11-E003

A 1,000 m deep column of ocean water warms by 0.5°C. Seawater expands by about 2×10^{-4} of its height for each 1°C of warming. Assume the whole column warms evenly. (a) Calculate the rise in the height of the column, in meters and centimeters. (b) Identify whether this is a thermal expansion or a melting-ice effect, and name one other process that raises sea level.

QUICK CONCEPT SUMMARY

- Sea level rises from thermal expansion plus melting land ice; coasts flood and populations move; disease vectors, heat, storms and drought spread.
- Circulation (Hadley cells, jet stream, conveyor belt) and soils change; species shift ranges and ice-dependent species lose habitat.
- Oceans absorb most excess heat: habitat loss, less oxygen, reproductive changes and coral bleaching.
- $CO_2 + H_2O \rightleftharpoons H_2CO_3$ lowers pH and removes carbonate; 0.1 pH unit ≈ 26% more H^+.

11.5 Reducing Climate Change

CONCEPT MADE SIMPLE

There are two ways to respond to a warming world. **Mitigation** attacks the cause by cutting emissions or removing carbon. **Adaptation** reduces the damage from changes already under way. We need both.

Mitigation and adaptation

Mitigation reduces the cause: renewable energy, efficiency, public transit, fewer methane leaks, protecting forests, and carbon removal. **Adaptation** reduces harm: sea walls, raised buildings, beach nourishment, drought-tolerant crops, early-warning systems, wildlife corridors, cooling centers, and managed retreat (zoning limits on shoreline building, buyouts, ending subsidized flood insurance). Mitigation limits how bad things get; adaptation helps societies cope with what is already locked in.

Where emissions come from and how to cut them

Most CO_2 comes from burning coal, oil and natural gas for electricity, heating, transport and industry. Methane comes from livestock, landfills, rice and gas leaks; N_2O from fertilizer. Major levers:

- **Switch fuels.** Replace coal with natural gas (about half the CO_2 per unit of electricity, but methane leaks offset some of the gain), and then with wind, solar, hydro, geothermal and nuclear, which emit little CO_2 in operation.
- **Use less energy.** Efficient buildings, appliances and vehicles; rail and public transit burn less fuel per passenger. Efficiency is often the cheapest lever: the fuel saved can repay the cost, giving a negative cost per tonne avoided.
- **Capture methane.** Landfill and manure digesters turn CH_4 (GWP about 28) into CO_2 (GWP 1) when burned. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, so 16 g of CH_4 gives 44 g of CO_2, and **1 t CH_4 gives 44 ÷ 16 = 2.75 t CO_2.**
- **Cut fertilizer and manure N_2O** with precise application.

Net effect of an action is what it avoids minus what it adds. Flaring 100 t of CH_4 avoids 100 × 28 = 2,800 t CO_2e but creates 100 × 2.75 = 275 t CO_2, so net = 2,525 t CO_2e avoided.

Carbon sinks

A **carbon sink** takes in more carbon than it releases. About 45% of human CO_2 emissions stays in the air; the rest (about 55%) is absorbed by the ocean and by land.

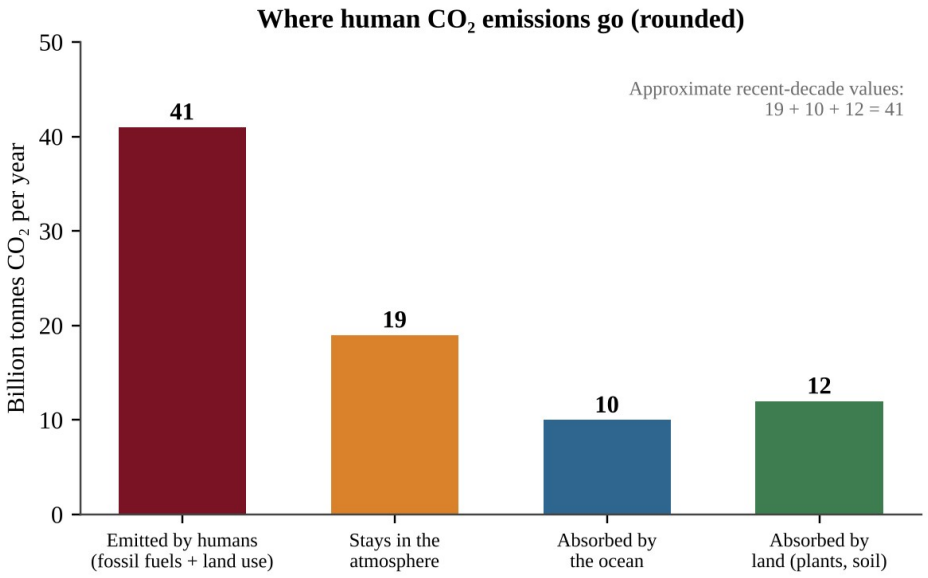


Figure 11.10 — About half of human CO_2 emissions is absorbed by the ocean and land (approximate, rounded)

- **Forests and soils.** Growing trees store carbon in wood and roots. Protecting forests and replanting (afforestation, reforestation) keep carbon out of the air; deforestation and burning release it. Farming that disturbs soil less and keeps it covered builds soil carbon.
- **Oceans.** They absorb CO_2, but this causes acidification (11.4), and warmer water holds less.
- **Coastal "blue carbon".** Mangroves, salt marshes and seagrass bury carbon in waterlogged mud for centuries.
- **Carbon capture and storage (CCS).** CO_2 is captured at a power plant or factory and injected into deep rock. It adds cost and energy use, and stored CO_2 must not leak.

To convert **carbon mass to CO_2 mass**, multiply by 44 ÷ 12 ≈ 3.67.

Energy policy

Policy	How it works	Trade-off
Carbon tax	Puts a price on each tonne of CO_2e	Raises energy costs; can be softened by returning revenue
Cap-and-trade	Caps total emissions; firms buy and sell permits	Needs strict, falling caps and monitoring

Policy	How it works	Trade-off
Subsidies and tax credits for renewables	Lower the cost of clean energy	Cost taxpayers; can be hard to end
Efficiency standards, renewable portfolio standards	Require efficient products or a share of clean power	Slow to turn over equipment
Ending fossil-fuel subsidies	Removes cost advantage of fossil fuels	Politically difficult; affects low-income households

International agreements and personal action

Climate change crosses borders, so countries negotiate. The **Kyoto Protocol (1997)** set binding emission targets for developed countries. The **Paris Agreement (2015)** aims to hold warming well below 2°C and pursue 1.5°C above pre-industrial levels. Each country sets its own **nationally determined contribution (NDC)** and reports progress, with targets meant to be strengthened every five years. Because pledges are voluntary, success depends on trust, finance for developing countries and transparent reporting.

Personal action adds up when many people act: driving less or choosing electric or transit, flying less, insulating homes, using renewable electricity, eating less beef and wasting less food.

FORMULA & MATH SKILL

Percent change = (new − old) ÷ old × 100 (the old value is the denominator). **Cost per tonne avoided** = total cost ÷ (annual reduction × years). **Target and yearly cut:** target = baseline × (1 − x ÷ 100); yearly cut = (current − target) ÷ years left. **Payback time** = cost ÷ annual savings. **Total cost** over n years = one-time cost + annual cost × n. **Percentage points vs percent:** a fall from 90% to 60% is 30 percentage points but 30 ÷ 90 = 33% percent change. Projections assume a constant trend; feedbacks can speed or slow it.

KEY DEFINITIONS

Mitigation — action to reduce the causes of climate change. **Adaptation** — action to adjust to the effects of climate change. **Carbon sink** — a reservoir that absorbs more carbon than it releases. **Carbon tax** — a fee per tonne of CO_2e emitted. **Nationally determined contribution (NDC)** — a country's own emission pledge under the Paris Agreement.

COMMON STUDENT MISTAKES

- Treating mitigation and adaptation as the same thing, or as alternatives. A sea wall is adaptation; solar power is mitigation.
- Counting only the CO_2 saved and forgetting the CO_2 created when methane is burned (unless told to ignore it).
- Dividing a percent change by the new value instead of the old value.

AP EXAM STRATEGY

When asked to justify a solution, compare cost per tonne avoided, then give one advantage and one disadvantage and say whether it is mitigation or adaptation. Show every step of a unit conversion.

CLASSROOM EXAMPLE SG-U9-E012

A city is choosing between two ten-year plans to reduce greenhouse gases. Plan A (replace coal heating with a district heat network) costs $18 million and cuts emissions by 1,500 t CO_2e per year. Plan B (capture methane at the landfill) costs $6 million and captures 400 t of CH_4 per year; use a GWP of 25 for CH_4 to convert captured methane to CO_2e. Assume each plan works for the whole ten years and ignore interest. (a) Calculate the total CO_2e avoided by each plan over ten years. (b) Calculate the cost per tonne of CO_2e avoided for each plan. (c) Make a claim for which plan the city should fund first and give one advantage and one disadvantage of the plan you chose. (d) Classify each plan as mitigation or adaptation.

CLASSROOM EXAMPLE SG-U9-E015

A farm captures 300 t of CH_4 a year from manure and burns it. Use GWP = 28 for CH_4. (a) Calculate the CO_2 produced by burning it, using 44 ÷ 16, and the net change in CO_2e per year. (b) The farm's emissions were 20,000 t CO_2e before the project. Calculate the percent reduction. (c) The farm pledges to end 40% below its original emissions in 8 years. Calculate the target and the average yearly cut still needed, starting from its emissions after the project. (d) The system cost $420,000 and earns $35,000 a year from selling the gas. Calculate the payback time.

CLASSROOM EXAMPLE SG-C11-E004

A household drives 12,000 km a year in a gasoline car that emits 0.20 kg of CO_2 per km. It switches to an electric car that uses 0.15 kWh per km, charged from a grid that emits 0.40 kg of CO_2 per kWh. (a) Calculate the household's yearly CO_2 emissions from driving before and after the switch, in kg. (b) Calculate the percent reduction. (c) State whether this is mitigation or adaptation and name one way the electricity source could make the reduction larger.

CLASSROOM EXAMPLE SG-C11-E005

A company plants 50 ha of forest that stores 4.0 t of carbon per hectare per year. The company emits 5,000 t of CO_2 per year. (a) Calculate the carbon stored per year, in tonnes of carbon. (b) Convert it to tonnes of CO_2 using 44 ÷ 12. (c) Calculate the percent of the company's yearly emissions this offsets. (d) Identify one reason the forest may offset less than calculated over several decades.

QUICK CONCEPT SUMMARY

- Mitigation cuts emissions or removes carbon; adaptation reduces damage. Both are needed.
- Levers: cleaner fuels, efficiency, methane capture, sinks (forests, soils, blue carbon, CCS), and policy (carbon tax, cap-and-trade, standards, subsidies).
- Paris-style agreements rely on voluntary national pledges that are reviewed and tightened.
- Compare options by CO_2e, cost per tonne avoided, net effect and payback.

11.6 Loss of Biodiversity

CONCEPT MADE SIMPLE

Biodiversity is falling because of six human pressures, remembered as **HIPPCO**. Most species are not killed directly. Their home is removed, cut into pieces, polluted or changed faster than they can adjust. Invasive species and a low, falling population make it worse.

HIPPCO

Biodiversity includes genetic diversity within species, species diversity within ecosystems and ecosystem diversity across a region. It gives stability and resilience, and provides food, medicines, materials, pollination, water purification and recreation.

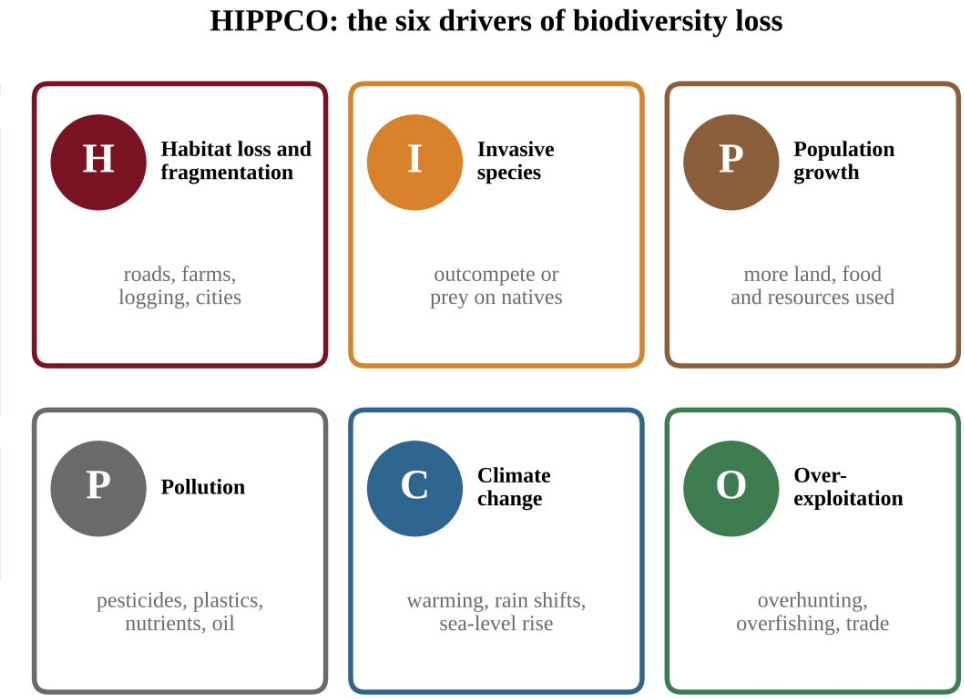


Figure 11.11 — HIPPCO: the six main factors reducing biodiversity

Letter	Factor	How it reduces biodiversity
H	Habitat destruction	Conversion to farms, cities, mines; loss of forests, wetlands, reefs; fragmentation
I	Invasive species	Outcompete or prey on natives
P	Population growth (human)	More people use more land, water and resources, magnifying every other pressure
P	Pollution	Pesticides, plastics, nutrients, oil and heavy metals poison or alter habitats
C	Climate change	Warming, shifting rainfall, sea-level rise, acidification
O	Over-exploitation	Overhunting, overfishing, bycatch, overharvesting, illegal wildlife trade

Invasive species

A **native** species evolved in an ecosystem. A **non-native** (introduced) species arrives from elsewhere; most do no harm. An **invasive species** can live, and sometimes thrive, outside its normal habitat and spreads so that it **threatens native species**. Invasive species can sometimes be beneficial (food, erosion control), but they are classed as invasive when they threaten natives.

Pathways. Ballast water of cargo ships, shipping crates and packing wood, the pet and aquarium trade, deliberate introduction for farming or ornament, and accidental transport on boats, vehicles and clothes. Trailers and bait water move aquatic invaders between lakes; roads and trails carry seeds into disturbed ground, so invasive plants are densest near them.

Why they succeed. Invasive species are often **generalist, r-selected species** that may **outcompete natives for resources**:

- **Generalists** tolerate wide conditions and eat many foods.
- **r-selected traits:** rapid growth, early maturity, many offspring, effective dispersal.
- **Few natural enemies** in the new range (enemy release).

- Aggressive behavior or chemicals that suppress other plants (allelopathy).

Reading invasion data. Linear growth adds the same amount each year. **Exponential** growth adds more each year and doubles every fixed time: size = $N_0 \times 2^n$ after n doublings (25 ha doubling every 5 years covers $25 \times 2^3 = 200$ ha after 15 years). **Logistic** growth levels off near carrying capacity. While an invader is on the upward bend, control is cheap and each year of delay costs more.

Examples. Zebra mussels (Great Lakes; they filter plankton, clearing the water and starving native filter feeders), kudzu (smothers plants), Burmese pythons (Everglades), lionfish (Caribbean) and cane toads (Australia; brought in to eat crop beetles, they poisoned native predators, a biological control that backfired). Consequences: loss of native biodiversity, altered food webs and nutrient cycles, economic costs, loss of ecosystem services.

Control strategy	Example	Advantage	Disadvantage
Prevention	Cargo inspection, boat cleaning, ballast rules	Cheapest, most effective	Needs monitoring and cooperation
Physical removal	Hand-pulling, trapping, hunting	Targeted, no chemicals	Labor-intensive
Chemical control	Herbicides, pesticides	Fast, large areas	Harms non-target species; resistance
Biological control	Release a predator or pathogen from the home range	Self-sustaining	May attack natives or itself become invasive
Habitat management	Prescribed burns, restoring native plants	Strengthens natives	Ongoing effort
Legislation and education	Laws on transport and release; awareness campaigns	Broad impact	Enforcement

Endangered species

Conservation groups rank species by closeness to extinction. The IUCN Red List runs from Least Concern through Near Threatened, **Vulnerable, Endangered, Critically Endangered**, to Extinct in the Wild and Extinct. In the United States the **Endangered Species Act (1973)** protects species listed as endangered (in danger of extinction across all or a large part of their range) or threatened (likely to become endangered). **CITES** regulates international trade in wild plants and animals.

How species become endangered. A species is at risk when it is:

- **extensively hunted or collected** (poaching for ivory, horn, fur, pets);
- limited to a **narrow diet** (pandas and bamboo, koalas and eucalyptus);
- **outcompeted by invasive species**;
- tied to **specific, limited habitat** (an island or single mountaintop).

Other risk factors: K-selected traits (large body, slow maturity, few young, so slow recovery), small population size, and being a top predator. **Small populations** are fragile: chance events remove a large share, a bottleneck leaves little genetic diversity so inbreeding follows, and mates are hard to find. A disease seldom kills every host, because some individuals resist it and fewer hosts slow its spread, but a small, genetically similar population may have no resistant members.

Not all species are in danger of extinction when exposed to the same changes. Species that can **adapt** (variation in behavior or genes) or **move** (disperse) to a new environment are less likely to go extinct. An island specialist can do neither.

Selective pressures are any factors that change the behaviors and fitness of organisms: predation, disease, climate change, food shortage, a new competitor, hunting. Individuals with helpful traits leave more offspring, so the population evolves, but evolution needs time and genetic variation. Species also **compete for resources** (territory, food, mates, habitat). If human change intensifies competition, the weaker competitor may become endangered or extinct.

Protecting species. (1) **Criminalizing poaching** and trade, with enforcement; (2) **protecting habitats** through parks, reserves and conservation easements, because habitat loss is the leading cause; (3) **legislation** such as the ESA and CITES. Match the strategy to the threat: given data on threats, find the largest category and pick the action that targets it. Each has trade-offs: reserves can limit local land use, captive breeding is costly and may produce animals that cannot live in the wild, and laws work only if enforced. For a population whose births add 5% a year and deaths (including poaching) remove 9%, net change is −4% a year; the Rule of 70 gives 70 ÷ 4 ≈ 18 years to halve.

Habitat fragmentation and edge effects

Habitat fragmentation breaks a large habitat into smaller, isolated areas. Causes: **roads and pipelines**, **clearing for agriculture or development**, and **logging**. Even with habitat left, fragmentation harms:

- Each patch holds fewer individuals, so populations are small and more vulnerable to chance and inbreeding.
- Animals cannot move between patches for mates, food or territory, so gene flow is cut.
- **Edge effects.** Patch edges have more light, wind and temperature extremes, more predators and parasites (cowbirds lay eggs in forest birds' nests) and more invasive species. A smaller patch has proportionally more edge. A 10 × 10 km square has a 40 km perimeter; four 5 × 5 km squares with the same total area have 80 km.
- Roads kill animals that try to cross.

The scale at which fragmentation harms **varies from species to species**. A road barely affects a bird but blocks a salamander or large carnivore. Species needing large territories or interior habitat are hit hardest.

Counting interior habitat. If the edge zone is w wide on every side, the interior of an a × b rectangle is **(a − 2w) × (b − 2w)**; if a side is 2w or less, the interior is zero. A road creates new edges, so apply the rule to each piece and add. Percent interior = interior ÷ total area × 100.

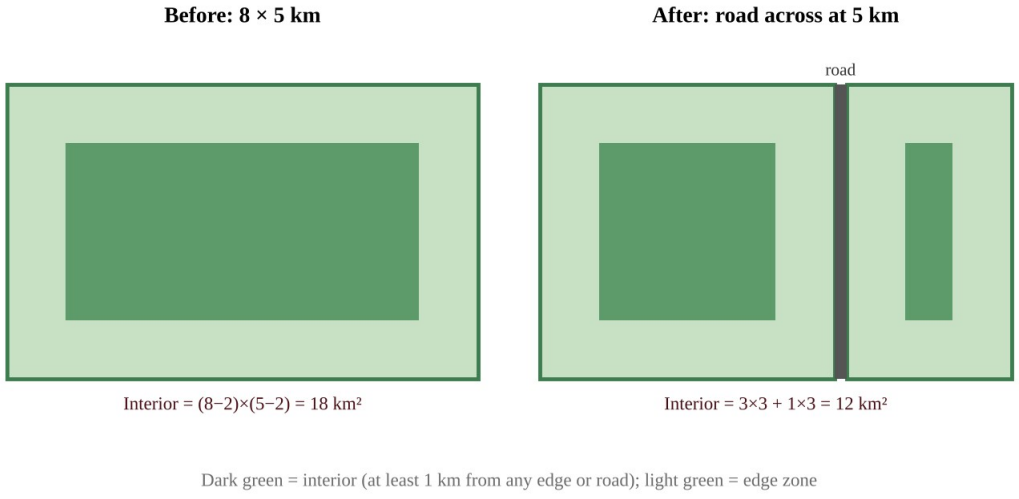


Figure 11.12 — A road cuts an 8 × 5 km forest in two; with a 1 km edge zone, interior habitat falls from 18 km² to 12 km²

Climate change, domestication and conservation

Climate change causes habitat loss through changes in temperature, precipitation and sea level: mountain species run out of higher ground, reefs bleach, coastal wetlands drown, and arid regions expand.

Domestication. Some organisms are partly or fully domesticated and **managed for economic returns**, such as honeybee colonies and livestock. Farmers select a few high-yield breeds and replicate them, so local varieties disappear and the remaining population is vulnerable to one disease or climate stress. Wild relatives can be displaced or hybridize with domestic forms, and land converted for livestock and monocultures adds to habitat loss.

Strategies to mitigate biodiversity loss: protected areas (parks, reserves, marine protected areas); **habitat corridors** (wildlife overpasses and underpasses, riparian strips) that let animals move and genes

flow; **sustainable land use** (agroforestry, selective logging, rotational grazing, sustainable fisheries, compact cities); **restoring lost habitat** (replanting, rewetting wetlands, removing dams, restoring reefs); and addressing the other HIPPCO factors. Evaluate them: protected areas conserve the most but can restrict local livelihoods and be too small; corridors cost money and can help invasives or disease spread; restoration is slow and rarely recreates the original; sustainable-use rules need enforcement.

REAL-WORLD EXAMPLE

A small island lizard lives on one 12 km² island, eats a single berry and lays two eggs a year. Introduced rats eat the eggs and a hurricane destroyed part of the berry forest: small range, specialist diet, low reproduction, a new predator and habitat loss are stacked together. The recovery plan removes the rats, fences and replants the forest and starts a captive-breeding colony as insurance.

Designing and judging an investigation

Free-response questions often ask you to design or criticize an investigation.

- **Hypothesis.** A testable "If … then …" that names the direction: *if distance from a trail increases, invasive shrub density will decrease and native wildflower richness will increase.*
- **Variables.** The **independent variable** is what you change or compare; the **dependent variable** is what you measure; **controlled variables** are kept the same (temperature, salinity, light, plot size, season).
- **Control group.** A comparison with no treatment or the normal condition, such as tanks at present-day pH or plots without removal.
- **Replication.** Several tanks, plots or sites per treatment, averaged, so chance does not decide the result.
- **Random and matched sampling.** Assign treatments randomly and match plots on soil, slope, cover and elevation, so a hidden variable does not explain the difference.
- **Observation versus experiment.** Surveys show association; showing cause needs a manipulation such as fencing or removing the species, with a control.
- **Results and limits.** State the pattern that supports the hypothesis and one limit: small sample, one site, short time or an unmeasured variable.

Reading an article. The *claim* is its main conclusion, the *evidence* is the data and examples it cites, and an *assumption* is a step it leaves unstated. Ask whether the evidence really supports the claim.

Reading graphs. Identify axes and units, quote two values, describe the shape, and say what a trend shows and cannot show (correlation versus cause).

KEY DEFINITIONS

Biodiversity — the variety of life: genetic, species and ecosystem diversity. **HIPPCO** — habitat destruction, invasive species, population growth, pollution, climate change, over-exploitation. **Invasive species** — a species living outside its normal habitat that spreads and threatens natives. **Generalist / specialist** — a species with a wide / narrow niche. **Endangered species** — a species at risk of extinction across all or a large part of its range. **Selective pressure** — an environmental factor that affects survival and reproduction. **Habitat fragmentation / edge effect / corridor** — a habitat divided into isolated patches / changed conditions at a patch boundary / a strip of habitat joining patches.

COMMON STUDENT MISTAKES

- Equating "non-native" with "invasive," or saying invaders are always predators. Many are plants, insects or pathogens.
- Recommending biological control without noting its risk.
- Claiming all species go extinct under the same change. Vulnerability varies with traits.
- Saying habitat loss is the same as extinction (it is a leading cause), or that "endangered" means none remain.

AP EXAM STRATEGY

To explain why species A is more at risk than species B, compare trait by trait (range size, diet breadth, reproduction rate, population size, hunting pressure, dispersal) using "because" statements. To choose a conservation action, find the largest threat in the data and target it.

CLASSROOM EXAMPLE SG-U9-E009

A shrub introduced as a garden ornamental has spread into a wetland. A 2-hectare study plot was surveyed twice.

Year	Native sedge cover (% of plot)	Introduced shrub cover (% of plot)
2018	80	5
2024	44	38

(a) Calculate the percent change in native sedge cover from 2018 to 2024. (b) Identify two traits of the shrub that could explain its spread. (c) Propose one control method and state one disadvantage of it.

CLASSROOM EXAMPLE SG-U9-E010

Two mammals live in the same mountain forest that is being logged and warmed by a changing climate. Species P is a generalist omnivore that breeds at one year of age, produces five young per year and can move to nearby valleys. Species Q eats only the bark of one tree species, breeds at five years of age with one young every two years, and lives only above 2,000 m. (a) Identify which species is more at risk of extinction and give three traits that support your claim. (b) Define selective pressure and give one example from this scenario. (c) Propose one strategy to protect Species Q and describe one limitation of it.

CLASSROOM EXAMPLE SG-U9-E011

A square reserve measures 10 km on each side. Assume that within 1 km of any edge the habitat is "edge habitat," and the remaining area is "interior habitat." A new road and pipeline cross the reserve and divide it into four equal squares, each 5 km on a side, with the same edge-zone width of 1 km on every side of every square. (a) Calculate the interior area of the original reserve and of the four squares combined. (b) Calculate the percentage of the total area that is interior habitat in each case. (c) Explain how this change would affect a forest-interior bird and name one measure that could help it.

CLASSROOM EXAMPLE SG-U9-E016

A lake association wants to know whether a floating invasive plant lowers dissolved oxygen (DO) and the number of native fish species. They can remove the plant from some bays. (a) Write an "if … then …" hypothesis. (b) Identify the independent and dependent variables. (c) Describe a design that includes a control group, replication and a method of choosing bays. (d) Identify two variables to keep similar across bays. (e) Describe the pattern of results that would support the hypothesis and name one limitation of the study.

CLASSROOM EXAMPLE SG-C11-E006

A tiger population of 3,200 declines by 4% each year. (a) Calculate the population after 10 years, to the nearest 100 animals. (b) Use the Rule of 70 to estimate the years needed for the population to halve. (c) Identify one trait of tigers that makes recovery slow and one conservation action that targets poaching.

QUICK CONCEPT SUMMARY

- HIPPCO: habitat destruction, invasive species, population growth, pollution, climate change, over-exploitation.
- Invaders are often generalist, r-selected species with few enemies; prevention is the cheapest control, and every method has trade-offs.
- Risk rises with hunting, narrow diet or habitat, small population and K-selected traits; adaptable, mobile species fare better.
- Fragmentation isolates small populations and creates edge effects; domestication narrows genetic diversity.
- Solutions: protected areas, corridors, sustainable land use, restoration and law, each with trade-offs.

11.7 Practice Questions and Answers

Work through the chapter's classroom examples, then test yourself with the question sets listed below.

PRACTICE QUESTIONS AND ANSWERS

Classroom examples in this chapter (unsolved; space is provided in each box): SG-U9-E001, SG-C11-E001, SG-U9-E002, SG-U9-E013, SG-U9-E003, SG-U9-E014, SG-U9-E004, SG-U9-E006, SG-C11-E002, SG-U9-E005, SG-U9-E007, SG-U9-E008, SG-C11-E003, SG-U9-E012, SG-U9-E015, SG-C11-E004, SG-C11-E005, SG-U9-E009, SG-U9-E010, SG-U9-E011, SG-U9-E016, SG-C11-E006.

Classified practice: Book Two, Chapter 11 — 60 multiple-choice questions (Section A) followed by 6 free-response questions (Section B).

Answers and complete worked solutions: Book Three — Part I (classroom examples) and Part II (practice questions), Chapter 11. Answers are deliberately not printed in this guide.

- Try every example without notes, then check your reasoning against Book Three.
- For each miss, re-read the section named in the solution and rewrite the idea in your own words.
- Redo missed questions after two days to lock in the learning.

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